

Stanwell

Stanwell New Projects Office  
Switchyard Road,  
Stanwell, QLD, 4702

25010 / cd Spec / T1

Architectural Specification

| Revision | Date       | Description | Issued by |
|----------|------------|-------------|-----------|
| T1       | 11/03/2026 | For Tender  | JB        |

PRELIMINARY  
NOT FOR CONSTRUCTION



# Stanwell New Projects Office

Switchyard Road  
Stanwell QLD 4702

25010 / T1 Spec

Architectural Specification

| Revision | Date    | Approved by |
|----------|---------|-------------|
| T1       | 11/3/26 | BH          |

|                          |
|--------------------------|
| <b>TABLE OF CONTENTS</b> |
|--------------------------|

|                                                          |     |
|----------------------------------------------------------|-----|
| Table of contents.....                                   | 2   |
| 0121 Tendering .....                                     | 4   |
| 0144 Preliminaries - AS 4000 (2025) .....                | 5   |
| 0160 Quality .....                                       | 14  |
| 0164 Commissioning.....                                  | 17  |
| 0171 General requirements .....                          | 19  |
| 0172 Environmental management .....                      | 52  |
| 0181 Adhesives, sealants and fasteners.....              | 57  |
| 0183 Metals and prefinishes.....                         | 62  |
| 0184 Termite management .....                            | 66  |
| 0185 Timber products, finishes and treatment.....        | 69  |
| 0191 Sundry items.....                                   | 73  |
| 0193 Building access safety systems .....                | 75  |
| 0201 Demolition .....                                    | 80  |
| 0221 Site preparation .....                              | 86  |
| 0222 Earthwork .....                                     | 90  |
| 0223 Service trenching .....                             | 97  |
| 0224 Stormwater - site .....                             | 104 |
| 0274 Concrete pavement .....                             | 109 |
| 0334 Block construction.....                             | 121 |
| 0342 Light steel framing .....                           | 126 |
| 0345p DULUX protective coatings.....                     | 130 |
| 0382 Light timber framing.....                           | 137 |
| 0423 Roofing - profiled sheet metal.....                 | 143 |
| 0424 Roofing - seamed sheet metal .....                  | 148 |
| 0436 Cladding - profiled and seamed sheet metal.....     | 153 |
| 0451 Windows and glazed doors .....                      | 157 |
| 0453 Doors and access panels .....                       | 164 |
| 0455 Door hardware .....                                 | 169 |
| 0456 Louvre windows .....                                | 176 |
| 0461 Glazing.....                                        | 181 |
| 0467 Glass components .....                              | 185 |
| 0511 Lining.....                                         | 188 |
| 0522 Partitions - framed and lined.....                  | 194 |
| 0524 Partitions - glazed.....                            | 201 |
| 0525 Cubicle systems .....                               | 207 |
| 0532 Suspended ceilings - flush lined .....              | 211 |
| 0551 Joinery .....                                       | 216 |
| 0552 Metalwork - fabricated.....                         | 223 |
| 0573 Fire extinguishers and blankets .....               | 228 |
| 0574 Window coverings.....                               | 230 |
| 0581 Signage.....                                        | 233 |
| 0612 Cementitious toppings.....                          | 237 |
| 0621 Waterproofing - wet areas .....                     | 243 |
| 0632 Stone and terrazzo tiling INCL Concrete pavers..... | 246 |

|                                         |            |
|-----------------------------------------|------------|
| <b>0641 Applied wall finishes .....</b> | <b>253</b> |
| <b>0651 Resilient finishes .....</b>    | <b>258</b> |
| <b>0652 Carpets .....</b>               | <b>263</b> |
| <b>0671 Painting .....</b>              | <b>269</b> |
| <b>0673 Powder coatings.....</b>        | <b>272</b> |
| <b>0702 Mechanical Services .....</b>   | <b>274</b> |
| <b>0802 Hydraulic Services .....</b>    | <b>276</b> |
| <b>0811s Sanitary fixtures .....</b>    | <b>279</b> |
| <b>0812s Tapware .....</b>              | <b>280</b> |
| <b>0902 Electrical Services.....</b>    | <b>281</b> |

|                       |
|-----------------------|
| <b>0121 TENDERING</b> |
|-----------------------|

---

**1 TENDERING INFORMATION**

---

**1.1 GENERAL**

General: Refer to the Stanwell Corporation Tender Documents for all details relating to the Tender.

**1.2 PROJECT INFORMATION**

**Contact person**

Inquiries: Refer inquiries to the following:

- Name: Kaitlyn Feeney
- Telephone: +61 423 479 607
- Email: Kaitlyn.Feeney@stanwell.com

|                                            |
|--------------------------------------------|
| <b>0144 PRELIMINARIES - AS 4000 (2025)</b> |
|--------------------------------------------|

---

## **1 GENERAL**

### **1.1 GENERAL**

#### **General conditions**

Requirement: To AS 4000 (2025), General Conditions of Contract.

Cross reference: **INTERPRETATION** in 0171 *General requirements* also applies.

### **1.2 DESCRIPTION OF WORKS**

#### **General**

Background: Stanwell Corporation Ltd (Stanwell) is building a new workplace for Stanwell staff located at Stanwell Power Station (SPS). The New Projects Office (NPO) involves the construction of a new single story workplace facility, suspended structure encompassing approximately 1,543 square meters of Gross Floor Area, designed around a central garden.

Purpose of this Document: This Document outlines the Scope of Work and provides detailed technical specifications for the construction of the NPO. It is intended to be read in conjunction with all architectural drawings, engineering plans and all further supporting sub-consultant Documentation that have been included as a part of this tender.

#### **Site**

Location: The New Projects Office will be located at Stanwell Power Station, 22km west of Rockhampton in Central Queensland. The NPO will replace the current demountable office on Site due to its practical location. See Figure 1.



Figure 1 Site of Work location (detailed battery limits illustrated in Site Precinct Plan: A-040001)

Site conditions: Site conditions are based upon historical climate data (1939-2022) at the Rockhampton Aero weather station (039083).

**Table 1**      *Site conditions*

| Climate statistic                                   | Unit    | Value                       |
|-----------------------------------------------------|---------|-----------------------------|
| Minimum temperature (039083)                        | °C      | -1.0                        |
| Maximum temperature (039083)                        | °C      | 45.3                        |
| Mean relative humidity (039083)                     | %       | 40 – 73                     |
| Site elevation (Site of Work)                       | M (ASL) | 60                          |
| Wind speed region (according to AS/NZS 1170.2:2021) | -       | Region C                    |
| Corrosion region (to AS 4312-2019)                  |         | C3 – Very High (Industrial) |
| Importance level                                    |         | 2                           |
| Terrain category                                    |         | 2                           |

Integration with Stanwell Power Station (SPS): As the Works will be situated at SPS, the Contractor must comply with all relevant policies, procedures and rules of the Principal in relation to the Site or the Works. This is to ensure that the Works do not have the potential to compromise the SPS Operations.

Site access and deliveries to the Site will utilise shared private roads between Power Station Road (at entry to SPS), and the proposed location of the Works. Access should be coordinated with SPS as per Site procedures, including ensuring necessary inductions and approvals have been acquired prior.

The Contractor shall be the *Work Health and Safety* (WHS) Principal Contractor (PC) of the Site with battery limits defined in the Architectural Drawings in the Site Precinct Plan: A-040001.

### Scope of Works

Overview: The project comprises a single-storey, suspended lightweight building with a skillion roof, approximately 46.2m x 46.2m in size. The design is centred around an elevated landscaped courtyard, which serves as a focal point and includes a link bridge, water feature, and stormwater collection system.

The total built area is 1,543.2m<sup>2</sup>, accommodating a modern workplace environment with the following key functional spaces:

#### Workplace Facilities:

- Open-plan office space for up to 107 staff
- Meeting rooms and associated breakout areas
- Boardroom
- First aid room
- Kitchen and dining facilities
- Staff kitchen and outdoor kitchen
- Lockers and amenities
- Training rooms
- Storage areas
- Covered outdoor area
- Associated support services

#### External and Site Works:

- Landscaping works including planting and hardscape paving
- Steel-framed mesh fencing
- External access ramps and stairs
- Relocation and refurbishment of two existing covered shade structures
- External mechanical plant enclosure for chilled water systems
- Dedicated mechanical plant zones for air handling units, generally located above meeting spaces

#### Infrastructure and Civil Works:

- Demolition of approximately 1,700m<sup>2</sup> of existing demountable office buildings
- Installation of a new Fire Indicator Panel
- Site regrading to support overland flow and integration of a new retaining wall for the courtyard

#### Acoustic Design Considerations:

- Acoustic treatments to manage external ambient noise
- Internal acoustic privacy measures for meeting rooms, training spaces, and open-plan work areas
- Compliance with sound rating requirements to ensure a productive and comfortable workplace environment

### Methods of Construction

The design has been developed to allow the opportunity for proposals to include a variety of construction methods. This may include a fully traditional, in-situ construction methodology, or a combination of in-situ and Modern Methods of Construction techniques.

### Notable conditions

The Works includes elements requiring particular attention, including:

- A continuous folded metal sheet cladding with associated flashings and trims.
- Aluminium window frames with integrated shade fins.
- Internal linings with cover battens and coordinated services and fixtures, requiring coordination with structural members.
- Central landscaped courtyard including a water feature, sculpted ground plane, rocky swale, and elevated walkway with seat.

#### Deviations from *Deemed to Satisfy* design

The Works includes a number of deviations from *Deemed to Satisfy* design that require conforming Performance Solutions for approval, including:

- Façade cladding designed as a rainscreen weatherproofing construction.
- Roof valley gutter replaced by a box gutter to mitigate internal water ingress risks.
- Energy efficiency verified with a comparison model.

For compliance details refer to the Building Certifier's Compliance Report and associated Performance Based Design Briefs prepared by relevant specialists.

### 1.3 ASSIGNMENT AND SUBCONTRACTING

#### Prequalified subcontractor schedule

| Works                      | Subcontractor                                  |
|----------------------------|------------------------------------------------|
| Metal cladding and roofing | Sustainable Cladding and Roofing (Ryan Finlay) |
| Building Access System     | Kattsafe                                       |

#### Preferred suppliers schedule

| Works                      | Suppliers                                      |
|----------------------------|------------------------------------------------|
| Metal cladding and roofing | Sustainable Cladding and Roofing (Ryan Finlay) |
| Metal cladding and roofing | Stratco Qld (Cindy Lee)                        |
| Waterproofing              | Tremco (Justin Allingham)                      |

#### Separate contracts

Work concurrent to this Contract: The following works on Site, to be carried out by persons engaged by the Principal, are not included in this Contract: Audio Visual Scope of Works, including auxiliary cabling. Contractor to coordinate timing for rough-in and fit-off for concurrent contract. Allow a minimum 3 weeks notice for those works to be conducted on Site.

Work before this Contract: relocation of existing training module.

Work subsequent to this Contract: Western parking bay on secure side of Site.

### 1.4 LEGISLATIVE REQUIREMENTS

#### Compliance

General: The Principal, before entering into the Contract, has given the notices, paid the fees, and obtained the permits, approvals and other authorisations, as Documented in the **Prior applications and approvals schedule**.

#### Prior applications and approvals schedule

| Prior notices given and applications made                   | Fees paid | Permits, approvals and authorisations received |
|-------------------------------------------------------------|-----------|------------------------------------------------|
| Local Authority Plumbing and Drainage application (pending) | Pending   | Pending                                        |
|                                                             |           |                                                |
|                                                             |           |                                                |

#### Authority conditions schedule

| Authority | Document | Condition |
|-----------|----------|-----------|
|           |          |           |
|           |          |           |
|           |          |           |

## 1.5 PROTECTION OF PEOPLE AND PROPERTY

### Occupied premises

General: For the parts of the Site Documented in the **Occupied premises schedule**:

- Allow occupants to continue in secure possession and occupancy of the premises for the required period.
- Maintain safe access for occupants.
- Arrange work to minimise nuisance to occupants and for their safety.
- Protect occupants against weather, dust, dirt, water or other nuisance.

Proposals: Submit details of proposed methods.

- Purpose of submission: For information.
- Timing of submission: Before commencement of WUC.

### Occupied premises schedule

| Occupants                          | Occupied premises      | Period of occupancy   |
|------------------------------------|------------------------|-----------------------|
| Stanwell Staff and Sub Contractors | Stanwell Power Station | Duration of the Works |
|                                    |                        |                       |
|                                    |                        |                       |

### Safety

Accidents: Promptly notify the Superintendent of the occurrence of the following:

- Accidents involving death or personal injury.
- Accidents involving loss of time.
- Incidents with accident potential such as equipment failure, slides and cave-ins.

Accident reports: Submit reports of accidents.

- Purpose of submission: For information.
- Timing of submission: As soon as reasonably practical.

### Protective clothing

Requirement: Make available protective clothing for the use of visitors, as follows:

- Safety helmets: Type 1 to AS/NZS 1801 (2024).
- High visibility safety vests: To AS 4602.1 (2024).

Certification: Required.

- Certification provider: An organisation accredited by the Joint Accreditation System of Australia and New Zealand (JASANZ).

Number of helmets: Six

Number of high visibility safety vests: Six

## 1.6 CARE OF THE WORK AND REINSTATEMENT OF DAMAGE/DAMAGE TO PERSONS AND PROPERTY OTHER THAN THE WORKS

### Reinstatement

General: Clean and repair damage caused by installation or use of Temporary Work and restore existing facilities used during construction to original condition.

Accessways and services: Do not obstruct or damage roadways and footpaths, drains and watercourses and other existing services in use on or adjacent to the Site. Determine the location of such services. Rectify immediately any obstruction or damage to such services and provide temporary services whilst repairs are carried out.

Trees and properties: Do not interfere with or damage trees and properties that are to remain on or adjacent to the Site, including adjoining property encroaching onto the Site. Rectify immediately any interference or damage to such trees and properties.

### Existing services

Service to be continued: Repair, divert or relocate service, as Documented.

Trenches: If the existing service crosses the line of a required trench or will lose support when the trench is excavated, provide permanent support for the existing service.

Redundant services: Remove redundant parts and make safe.

Interruptions to services: Minimise the number and duration of interruptions.

Changes to existing services: Submit proposals.

- Purpose of submission: For review.
- Timing of submission: Before starting WUC to existing services.

## 1.7 SITE

### Site definitions

Site: as described in Item 28 of Annexure Part A of the General Conditions of Contract.

Laydown Area: as defined on the Site Precinct Plan: A-040001

Secure Area: the part of the Stanwell Power Station location requiring security clearance from Stanwell to enter, indicated by the Secure Area Boundary on the Site Precinct Plan: A-040001.

### Site restrictions

Site limitations: Comply with the following restrictions on the use of the Site: The Site is limited to the area identified in the Documents. Access is to be maintained to the footpath allowing access to the bus drop off areas, unless approved in advance by the Superintendent and the Principal.

Major works to adjoining Site/s: Major works for power generation unit outages and maintenance are scheduled for:

1. Unit 3 outage: 30/05/26 to 24/07/26
2. Unit 4 outage: 03/07/27 to 27/08/27

During these times the total population on the Power Station location will increase. This may impact coordination of access to and around the construction Site.

Access area restrictions: Access onto and within the Site, and use of the Site for Temporary Works and constructional plant, including working and storage areas, location of offices, workshops, sheds, roads and parking, is restricted to the following areas: as Documented; the Construction Area and Lay Down Area.

Lay Down Area: The exact extent of the Lay Down Area is to be determined in consultation with the Principal on commencement of the WUC on Site.

Limitations to Delivery Times: Deliveries should be avoided during shift start and end periods; 06:30 to 07:30 and 16:00 to 17:00, Monday to Friday.

### Secure area

Conditions of entry: Security clearance and inductions are required for access to the Secure Area. For Works within the Secure Area, a secondary temporary line of security is to be established in coordination with the Principal.

Entry permits: Application is to be made to the Principal for entry to the Secure Area.

Personnel: Submit the full name, address, and date and place of birth of persons required to enter designated secure areas.

- Purpose of submission: For review.
- Timing of submission: At least 14 days before entry is required.

## 1.8 SETTING OUT THE WORKS

### Surveys

Setting out: as Documented. Refer also to: Additional Detail Survey of Part of Stanwell Power Station: 9555-40-DTL: 02/06/2025 by Capricorn Survey Group CQ.

Check surveys: required; particularly for all services.

Final survey: Required.

## 1.9 CLEANING UP

### Final cleaning

General: Before the Date for Practical Completion, clean throughout, including interior and exterior surfaces exposed to view. Vacuum carpeted and soft surfaces. Clean debris from the Site, roofs, gutters, downpipes and drainage systems. Remove waste and surplus materials.

Samples: Remove non-incorporated samples, prototypes and sample panels.

#### **Pest eradication**

Requirement: Employ suitably qualified pest exterminators. At the Date of Practical Completion, verify that completed Works are free of pests.

#### **Removal of Temporary Works and Construction Plant**

Time for removal: before The Date of Practical Completion.

### **1.10 MATERIALS, LABOUR AND CONSTRUCTION PLANT**

#### **General**

Temporary Works: Provide and maintain required hoardings, barricades, guards, fencing, shoring, temporary roadways, footpaths, signs, lighting and traffic management.

#### **Access roads**

Temporary roads: none.

Principal's existing roads: Use only designated roads.

Location: Access by Gatehouse Road and part of Administration Road.

#### **Parking**

Principal's existing parking areas: Use only designated parking areas.

Location: Contractor to establish within the Construction Area.

Overflow: Carpark B at the approval of the Principal.

No access: Carpark A.

#### **Temporary fence**

Requirement: by the Contractor.

#### **Use of existing services**

General: Existing services may be used as temporary services for the performance of the Contract.

#### **Existing services schedule**

| <b>Service</b> | <b>Conditions of use</b>                                                                               |
|----------------|--------------------------------------------------------------------------------------------------------|
| Electricity    | To be isolated for demolition. Contractor to provide temporary connection infrastructure as required.  |
| Potable water. | To be isolated for demolition. Contractor to provide temporary connection infrastructure as required.  |
| Raw water      | Accessible within secure part of the Site, to be accessed by water truck. Security induction required. |

**Note:** the Contractor is required to provide their own telecommunications and internet services. No ablution facilities will be made available for the Contractor on Site.

#### **Project signboards**

General: Provide project specific signboards and as follows:

- Locate where directed.
- Maintain in good condition for duration of the WUC.
- Obtain permission for removal.
- Remove on completion.

Other signboards: Obtain approval before display of advertisements or provision of other signboards.

#### **Contractor's representative**

General: Must be accessible, and fluent in English and technical terminology.

Contacts: Submit names and telephone numbers of responsible persons who may be contacted after hours during the course of the Contract.

- Purpose of submission: For information.
- Timing of submission: At the first Site meeting.

**Items supplied by Principal**

General: Materials and other items supplied free of charge to the Contractor for installation in the execution of the Works. Unload and take delivery, inspect for Defects and take care of the items. If Defects are found, advise. Return unused items to the Principal.

**Items supplied by Principal schedule**

Refer to Appendix A – T-Sheet: G2 items.

**Changes to existing items**

General: At least 14 days before changing existing items, give notice.

Existing items requiring notification: Security Fence.

**1.11 WORKING HOURS****General**

Working hours that can be accommodated by the Principal: 6am to 6pm

Working days that can be accommodated by the Principal: 7 days per week.

Work to be undertaken outside of these times is to be approved by the Principal.

**1.12 PROGRAMMING****Program of Work**

Construction Program: Submit a construction Program showing the following:

- Sequence of WUC.
- Critical paths of activities related to the WUC.
- Allowance for holidays.
- Activity inter-relationships.
- External dependencies including provision of access, Document approvals and work by others.
- Periods within which various stages or parts of the WUC are to be executed.

Time scale: Working days.

Updated Program: Submit revised construction Programs as required by the progress of the WUC. Identify changes since the previous issue, and show the estimated percentage of completion for each item of WUC.

Purpose of submission: For information.

Timing of submission: Within 14 days after the date for possession of the Site and with each progress claim.

Program chart: Display in the Contractor's Site office an up-to-date bar chart and network diagram based on the construction Program.

**Site meetings**

General: Hold and attend Site meetings throughout the Contract and arrange for the attendance of appropriate subcontractors, Superintendent and appropriate consultants.

Frequency: weekly.

Minutes: Make a record of Site meetings. Distribute a copy of the minutes to each party.

- Purpose of submission: For review.
- Timing of submission: Within 5 working days after each meeting.

**Progress photographs**

General: Take colour progress photographs within 7 days before each Site meeting. Submit 2 sets of prints and the digital files. Identify the project, date, time, location and orientation.

- Purpose of submission: For information.
- Timing of submission: At each Site meeting.

Format: jpeg, png or pdf.

**1.13 PAYMENT****Progress claims**

Progress claim breakdown: Submit a statement of amounts claimed in respect of each worksection or trade heading designated in the specification, together with Variations included in the claim.

- Purpose of submission: For review.

- Timing of submission: With each progress claim.

**Method of measurement**

General: In conformance with the principles of the Australian and New Zealand standard method of measurement of building works (ANZSMM) (2022).

|                     |
|---------------------|
| <b>0160 QUALITY</b> |
|---------------------|

## **1 GENERAL**

### **1.1 RESPONSIBILITIES**

#### **General**

Requirement: Provide a project quality management system, as Documented.

### **1.2 CROSS REFERENCES**

#### **General**

Requirement: Conform to the following:

- 0171 General requirements.

### **1.3 STANDARDS**

#### **General**

Standard: To AS/NZS ISO 9001 (2016).

### **1.4 INTERPRETATION**

#### **Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS ISO 9000 (2016) and the following apply:

- Quality package: A designated part of the Works, which may include the whole Works, for which an individual quality system is required.
- Service: After sales service, repairs, maintenance.

### **1.5 SUBMISSIONS**

#### **Quality plan**

Quality package: Submit a quality plan for each quality package, at least 10 Business Days before work on that package commences. Keep a copy of each approved quality plan on Site.

#### **Calculations**

Statistical techniques: Provide the methodology for statistical evaluation.

#### **Execution details**

Requirement: Provide the procedure for sign-off and audit.

### **1.6 INSPECTION**

#### **Notice**

Inspection: Give notice so inspection may be made of the following:

- As determined by the Quality Plan

## **2 PROJECT QUALITY MANAGEMENT SYSTEM REQUIREMENTS**

### **2.1 DOCUMENTATION REQUIREMENTS**

#### **Quality plan**

Standard: Conform to the recommendations of AS ISO 10005 (2018). Include inspection and Test plans.

#### **Documented procedures**

Review: Provide evidence of revision(s) including dates, approval and status of each procedure.

Register: Maintain a register of Documented procedures including the title, identifier and revision status.

### **2.2 DESIGN AND DEVELOPMENT OF PRODUCTS AND SERVICES**

#### **General**

Plan and control of product design and development: As Documented.

## **2.3 CONTROL OF EXTERNALLY PROVIDED PROCESSES, PRODUCTS AND SERVICES**

### **General**

External audits: Perform pre-tender surveys and audits of subcontractors and suppliers, as necessary with an approved auditor. Include audit and surveillance proposals in the quality plan along with results of pre-tender surveys.

Verification: The Superintendent may verify at source, or upon receipt, that purchased products conform to Documented requirements.

## **2.4 PRODUCTION AND SERVICE PROVISION**

### **Product identification and traceability**

Requirement: As Documented.

Identification: Identify all items of Work, samples and Site records by lot/number/barcode.

Traceability: Provide and maintain records of components for audit.

### **Service**

Requirement: As Documented.

---

## **3 MONITORING AND MEASUREMENT**

### **3.1 DOCUMENT CONTROL**

#### **Changes to Documents**

Revision: Review and approve changes to Documents using the same functions of the organisations that performed the original review and approval of the Documents.

Retention: As Documented.

Evaluation: Make quality records available to the Superintendent for evaluation, within 2 days of the initial request.

Period of evaluation: Agree for each request, but at least 5 Business Days.

### **3.2 INSPECTION AND TEST PLANS**

#### **Content**

Plan: Include the following:

- Detail all inspections and Tests required including Hold points.
- Identify acceptance criteria and frequency of sampling and Testing.
- Identify responsibilities for inspection and Testing and product/service approval criteria.

#### **Control of non-conforming product**

Acceptance of concession: Before the provision or repair of a non-conforming product, obtain permission to use the product.

#### **Hold points**

Stages: Hold points during the construction/manufacturing process require release by the Superintendent.

Release: Requirements for release of a Hold point may include the following:

- Provision of information required by the technical specifications.
- Certification of design/construction or installation.
- Submission of any checklists or non-conformance forms as required.
- Inspection/demonstration of Works.

#### **Frequency of Testing**

General: Conform to the Test requirements of the relevant standards and worksections, as Documented.

### **3.3 AUDITING**

#### **General**

Audit plan: Conform to the recommendations of AS/NZS ISO 19011 (2019) clauses 6.4 and 6.5.

Initial systems audit: Carry out before date of Site possession.

### 3.4 CORRECTIVE ACTION

#### General

Review: Provide a procedure to review the various control methods to minimise non-conformance. Record amendments to the project quality management system resulting from corrective action.

Non-conforming Works: Include in the quality plan the procedure for reporting any non-conforming Works to the Superintendent and any corrective action requests.

## 4 SELECTIONS

### 4.1 SCHEDULES

#### Project quality management system schedule

| Quality package        | Activities included | Standard               | Options   |
|------------------------|---------------------|------------------------|-----------|
| The whole of the Works | All                 | AS/NZS ISO 9001 (2016) | See below |

|                           |
|---------------------------|
| <b>0164 COMMISSIONING</b> |
|---------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide commissioning and related activities as Documented.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.

### 1.3 STANDARDS

#### General

Standard: To SA TS 5342 (2021).

### 1.4 OTHER DOCUMENTS

#### Informative

Requirement: The Documents listed in **COMMISSIONING INFORMATION, Documents provided for information only** are provided for information only.

### 1.5 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the definitions given in SA TS 5342 (2021) apply.

### 1.6 SUBMISSIONS

#### General

Requirement: To 0171 General requirements and the following:

- Names and contact details of persons nominated as representatives.
- Evidence of nominated representatives' qualifications and expertise in relation to commissioning.

## 2 EXECUTION

### 2.1 REPRESENTATIVES

#### Qualifications

Requirement: For each role in the **Project roles schedule** within the control of this Contract, provide representatives with qualifications, expertise and authority to act with respect of that role.

#### Functions

Requirement: Provide representatives to undertake commissioning related activities including the following:

- Attendance at commissioning meetings to **MEETINGS**.
- Review Documents, procedures and methodologies for commissioning.
- Evaluate performance deficiencies identified during commissioning and recommend corrective action.
- Facilitate resolution of commissioning issues.

### 2.2 MEETINGS

#### Attendance

Requirement: Attend commissioning meetings to facilitate the commissioning process including to review, plan, coordinate and schedule commissioning

Frequency: To **Commissioning meetings schedule**.

Additional commissioning meetings: Attend additional commissioning related meetings as required throughout construction.

### 3 SELECTIONS

#### 3.1 COMMISSIONING INFORMATION

#### 3.2 ROLES AND RESPONSIBILITIES

##### Project roles schedule

| Role                  | Organisation | Person responsible in that organisation |
|-----------------------|--------------|-----------------------------------------|
| Principal             | TBC          | TBC                                     |
| Commissioning manager | TBC          | TBC                                     |

#### 3.3 MEETINGS

##### Commissioning meetings schedule

| Meeting                    | Frequency                     |
|----------------------------|-------------------------------|
| Commissioning team meeting | To the commissioning schedule |

|                                  |
|----------------------------------|
| <b>0171 GENERAL REQUIREMENTS</b> |
|----------------------------------|

## **1 GENERAL**

### **1.1 PRECEDENCE**

#### **General**

Responsibility: All obligations within this specification are the responsibility of the Contractor, unless expressly stated otherwise.

Order of precedence: If there is conflict or inconsistency between the worksections of this specification, the requirements of worksections take the following order of precedence:

- All worksections other than those listed below.
- *0701 Mechanical systems, 0801 Hydraulic systems, 0901 Electrical systems and 1001 Fire services systems.*
- *0171 General requirements.*

### **1.2 SUPPORTING SPECIFICATIONS**

#### **General**

Cross reference: Documents to be referenced as supporting this specification include:

- *Structural Engineering Specification.*
- *Electrical Engineering Specification.*
- *Mechanical Engineering Specification.*
- *Hydraulic Engineering Specification.*
- *Façade Performance Specification.*
- *Acoustic Engineering Performance Specification.*

### **1.3 CROSS REFERENCES**

#### **Common requirements**

Requirement: Conform to the following worksections:

- *0181 Adhesives, sealants and fasteners.*
- *0182 Fire-stopping.*
- *0183 Metals and prefinishes.*
- *0184 Termite management.*
- *0185 Timber products, finishes and treatment.*
- *0193 Building access safety systems.*

#### **Cross referencing styles**

General: Within the text, titles are cross referenced using the following styles:

- Worksection titles are indicated by *Italicised* text.
- Subsection titles are indicated by CAPITAL text.
- Clause titles are indicated by **BOLD CAPITAL** text.
- Subclause titles are indicated by **Bold Sentence case** text.

### **1.4 REFERENCED DOCUMENTS**

#### **General**

Precedence: The requirements of worksections override conflicting requirements of their referenced Documents. The requirements of the referenced Documents are minimum requirements.

Contractual relationships: Responsibilities and duties of the Principal, Contractor and Superintendent are not altered by requirements in the Documents referenced in this specification.

Current editions: All referenced Documents are the editions, with amendments, current on 1st September 2025.

Exception to current editions: If statutory requirements reference other editions or standards, conform to those other editions or standards. If the NCC (2022) references editions other than the current edition, the same editions cited in the NCC (2022) are referenced in each worksection.

Maintenance and repair works: If statutory requirements applicable to the maintenance or repair works reference other editions or standards, conform to those other editions or standards.

European standards: Any national European Standard (e.g. IS EN or DIN EN) may be used in place of the equivalent referenced European Standard (EN).

## 1.5 CONTRACT DOCUMENTS

### Services diagrammatic layouts

General: Layouts of service lines, plant and equipment shown on the drawings are diagrammatic only, except where figured dimensions are provided or calculable.

Before commencing WUC on-Site:

- Obtain measurements and other necessary information.
- Coordinate the design and installation in conjunction with all trades.

### Levels

General: Spot levels take precedence over contour lines and ground profile lines.

### Drawings and manuals for existing services

Subsurface services: All underground services are to be confirmed by the Contractor before commencing WUC on-Site. Underground services shown on Documents have not been verified.

Warranty: No warranty is given as to the completeness or accuracy of drawings and/or manuals of existing services.

## 1.6 COORDINATION OF TRADES

General: The Contractor is responsible for coordinating the worksections of this specification with the relevant trades, including the determination of responsibility at trade interfaces.

## 1.7 INTERPRETATION

### Abbreviations

General: For the purposes of this specification the following abbreviations apply:

- BCA: National Construction Code Series Volume One: Building Code of Australia Class 2 to 9 Buildings and Volume Two: Building Code of Australia Class 1 and Class 10 Buildings.
- NATA: National Association of Testing Authorities.
- NCC: National Construction Code.
- PCA: National Construction Code Series Volume 3: Plumbing Code of Australia.
- PVC: Polyvinyl Chloride.
- PVC-U: Unplasticised Polyvinyl Chloride. Also known as UPVC.
- SDS: Safety data sheets.
- VOC: Volatile Organic Compound.
- WHS: *Work Health and Safety*.

### Definitions

AS4000(2024) definitions: terms defined in the AS4000(2024) General Conditions of Contract are Capitalised in this specification.

General: For the purposes of this specification, the following definitions apply:

- Access for maintenance: Includes access for maintenance, inspection, measurement, operation, adjustment, repair, replacement and other maintenance related tasks.
- Accessible, readily: Readily accessible, easily accessible, easy access and similar terms mean capable of being reached quickly and without the use of a tool, without hazard, climbing over or removing obstructions, using a movable ladder and in any case not more than 2.0 m above the ground, floor or platform.
- Accredited Testing Laboratory: One of the following:
  - . An organisation accredited by the National Association of Testing Authorities (NATA) to undertake the relevant Tests.

- . An organisation outside Australia accredited to undertake the relevant Tests by an authority recognised by NATA through a mutual recognition agreement.
- . An organisation recognised as being an Accredited Testing Laboratory under legislation at the time the Test was undertaken.
- . An organisation accredited for compliance with AS ISO/IEC 17025 (2018) to undertake the relevant Tests.
- Adjust: To align, support, straighten, reposition, secure, recalibrate, reset, tighten, loosen or otherwise perform work to restore to proper working order.
- Appropriately qualified person: To NCC (2022) Schedule 1.
- Attendance: Attendance, provide attendance and similar expressions mean give assistance for examination and Testing.
- Baseline data: Data derived from the final design, installation and commissioning, which serve as a basis for verification of results of routine servicing.
- Check: To examine to determine accuracy, quality or condition.
- Commissioning: Advancement of an installation from static completion to full working order, including verification that the systems, subsystems, and their components meet the project requirements. This includes all work described as commissioning in referenced Documents, even if carried out before static completion.
- Concealed: Obstructed from view, including below ground, in walls, cavities or roof spaces.
- Consumable: Materials or components intended to be depleted, worn out, replaced or replenished within the service life of the associated systems, plant or equipment as part of normal operation and maintenance.
- Contract administrator: Has the same meaning as architect, Superintendent or Principal's authorised person and is the person appointed by the owner or Principal under the Contract.
- Contractor: Has the same meaning as builder and is the person or organisation bound to carry out and complete the WUC.
- Default: Specified value, product or installation method that is to be provided unless otherwise Documented.
- Design life: The period of time for which it is assumed, in the design, that an asset will be able to perform its intended purpose with only anticipated maintenance but no major repair or replacement being necessary.
- Design parameters: Information used as the basis for design. It includes design requirements, performance criteria, performance parameters and similar terms.
- Disconnect: Includes all preparation required to safely detach, isolate, unfasten, unplug or break the connection between a supply of electricity, gas or water and a device or an appliance.
- Documented: Documented, as Documented and similar terms mean contained in the Contract Documents. (Where specific references are provided they are to assist with locating the primary reference information, but are not an exhaustive list of locations of reference information within the Contract Documents. The Contractor is to become familiar with the full extent of Contract Documents).
- Economic life: The period of time from the acquisition of an asset to the time when the asset, while still physically capable of fulfilling its function and with only anticipated maintenance, ceases to be the lowest cost alternative for satisfying that function.
- Electricity distributor: Any person or organisation that provides electricity from an electricity distribution system to one or more electrical installations. Includes distributor, supply authority, network operator, local network service provider, electricity retailer or electricity entity, as may be appropriate in the relevant jurisdiction.
- Errors and omissions: For the design prepared by the Contractor, errors and omissions have the same meaning as Defects.
- Fastener: Device used for fixing an item to another item, base material or substrate, by mechanical means.
- Fastening: Assembly used to secure an item to another item, base material or substrate, comprising one or more fasteners and accessories.
- Fire hazard properties: To NCC (2022) Schedule 1.

- Fixing: The action of securing an item to another item, base material or substrate.
- Gas Network Operator: Has the same meaning as network operator in AS/NZS 5601.1 (2022).
- Geotechnical Site investigation: The process of evaluating the geotechnical characteristics of the Site in the context of existing or proposed construction.
- Give notice: Give notice, submit, advise, inform and similar expressions mean give notice (submit, advise, inform) in writing to the Superintendent.
- High level interface: Systems transfer information in a digital format using an open system interface.
- Hold point: The activity cannot proceed without the approval of the Superintendent.
- Hot dip galvanized: Zinc coated or zinc iron alloy coated to AS/NZS 4680 (2025) after fabrication with coating thickness and mass conforming to AS/NZS 4680 (2025) Tables 3 and 4.
- Include: Include or including means to contain or comprise and are not words of limitation.
- Ingress protection: IP, IP code, IP rating and similar expression have the same meaning as IP Code in AS 60529 (2025).
- Joints:
  - . Construction joint: A joint with continuous reinforcement provided to suit construction sequence.
  - . Contraction joint: An opening control joint with a bond breaking coating separating the joint surfaces to allow independent and controlled contraction of different parts or components, induced by shrinkage, temperature changes or other causes. It may include unbound dowels to assist vertical deflection control.
  - . Control joint: An unreinforced joint between or within discrete elements of construction that allows for relative movement of the elements.
  - . Expansion joint: A closing control joint with the joint surfaces separated by a compressible filler to allow axial movement due to thermal expansion or contraction with changes in temperature or creep. It may include unbound dowels to assist vertical deflection control.
  - . Sealant joint: A joint filled with a flexible synthetic compound that adheres to surfaces within the joint to prevent the passage of dust, moisture and gases.
  - . Structural control joint: A control joint (contraction, expansion and isolation) in structural elements when used with applied material and finishes.
  - . Substrate joint: A joint in the substrate, which includes construction joints and joints between different materials.
  - . Weakened plane joint: A contraction joint created by forming a groove, extending at least one-quarter the depth of the section, either by using a grooving tool, by sawing or by inserting a premoulded strip.
- Local authority (local council): A body established for the purposes of local government by or under a law applying in a state or territory.
- Low level interface: Systems transfer information via terminals and voltage free contacts.
- Maintenance: Work that is carried out to preserve an asset, to allow for its continued use and function over its designed service life.
  - . Corrective maintenance: Maintenance initiated as a result of plant, equipment, systems, elements, before or after functional failure. This work can be planned or unplanned and includes repairs and replacement.
  - . Preventive maintenance: Planned maintenance of plant, equipment and other systems or elements, including cyclical or periodic maintenance, fire safety measures and statutory requirements.
- Manufacturer's and supplier's recommendations: Recommendations, instructions, requirements, specifications (and similar expressions) provided in written or other form by the manufacturer and/or supplier relating to the suitability, use, installation, storage and/or handling of a product.
- Metallic-coated: Steel coated with zinc or aluminium-zinc alloy as follows:
  - . Metallic-coated steel sheet: To AS 1397 (2021). Metal thicknesses specified are base metal thicknesses.
  - . Ferrous open sections zinc coated by an in-line process: To AS/NZS 4791 (2006).
  - . Ferrous hollow sections zinc coated by a continuous or specialised process: To AS/NZS 4792 (2006).

- Network Utility Operator: To NCC (2022) Schedule 1. A person who undertakes the piped distribution of drinking water or non-drinking water for supply; or is the operator of a sewerage system or a stormwater drainage system.
- Obtain: Obtain, seek and similar expressions mean obtain (seek) in writing from the Superintendent.
- Pipe: Includes pipe and tube.
- Practical completion or Defects free completion: refer to the General Conditions of Contract.
- Pre-commissioning: Verifying that the installation of a system is complete and ready for commissioning.
- Principal: Principal has the same meaning as owner, client and proprietor and is the party to whom the Contractor is legally bound to construct the Works.
- Professional engineer: To NCC (2022) Schedule 1.
- Proprietary: Identifiable by naming the manufacturer, supplier, installer, trade name, brand name, catalogue or reference number.
- Prototype: A full size mock-up of components, systems or elements to demonstrate or Test construction methods, junctions and finishes, and to define the level of quality.
- Provide: Provide and similar expressions mean supply and install and include development of the design beyond that Documented.
- Record drawings: Record drawings has the same meaning as as-installed drawings, as-built drawings and work-as-executed drawings. It includes technical drawings and digital models.
- Recovered/reclaimed materials: Material previously used in a building or project that is then re-used in another project. The material may be altered, re-sized, refinished, or adapted, but is not reprocessed in any way, and remains in its original form.
- Referenced Documents: Standards and other Documents whose requirements are included in this specification by reference.
- Required: Required by the Contract Documents, the local or statutory authorities.
  - . If required: A conditional specification term for Work that may be shown in the Documents or is a Legislative Requirement.
- Sample: A physical example that illustrates workmanship, materials or equipment, and establishes standards by which the Work will be judged. It includes samples and sample panels.
- Static completion: The state of a system when installation works are complete but have not been commissioned.
- Statutory authority: A public sector entity created by legislation, that is, a specific law of the Commonwealth, State or Territory.
- Subcontractor: A person or organisation, other than the principal, having a contract with the Contractor for the provision of part of the Works.
- Supply: Supply, furnish and similar expressions mean supply only.
- Tests - integrated system: Tests conducted on the project as a complete, integrated system to verify successful integration, interaction, and operation of all interrelated systems to the project requirements.
- Tests - production: Tests carried out on an item, before delivery to the Site.
- Tests - Site: Tests carried out on Site.
- Tests - type: Tests carried out on an item identical with a production item, including with respect to materials, material suppliers, manufacturing processes, dimensions and marking.
- Tolerance: The permitted difference between the upper limit and the lower limit of dimension, value or quantity.
- Utility service provider: Includes Electricity distributor, Network Utility Operator, Gas Network Operator and organisations providing other reticulated utilities including data and telecommunications services.
- Verification: Provision of evidence or proof that a performance requirement has been met or a default exists.

## 2 SUBMISSIONS AND INSPECTIONS

### 2.1 SUBMISSIONS

#### General

Requirement: Make submissions, as Documented.

Submit to: Superintendent or Superintendent's Representative.

Contractor review: Before submitting, review each submission item, and check for coordination with other Work of the Contract and conformance to Contract Documents.

#### Submission times

Default timing: Submit information or other material for information, comment or approval at least 5 Business Days before ordering products or starting installation of the respective portion of the Works.

Submission response times: Allow in the construction Program for at least the following times:

- Shop drawings: 10 Business Days
- Samples and prototypes: 10 Business Days
- Manufacturer's or supplier's recommendations: 10 Business Days
- Product data: 10 Business Days
- Product/design substitution or modification: 10 Business Days

Proposed products schedules: Submit a schedule of proposed products that have not been specified as proprietary items within 3 weeks of starting WUC on Site.

#### Identification

Requirement: Identify the project, Contractor, subcontractor or supplier, manufacturer, applicable product, model number and options, as appropriate and include relevant Contract Document references. If the submission covers more than one item, identify the item in the Contract Documents the submitted items relate to.

Non-conformance: Identify proposals that do not conform with project requirements, and characteristics that may be detrimental to successful performance of the completed WUC.

#### Errors

Requirement: If a submission contains errors, make a new or amended submission as appropriate, indicating changes made since the previous submission.

#### Electronic submissions

Electronic copies file format: .pdf

CAD file format: .dwg (version 2022 compatible)

Quantity: 1

Transmission medium: electronic file transfer or email

#### Project requirements

General: Submit the following, as Documented:

- Authority approvals: Notes of meetings with regulatory authorities and utility service providers whose requirements apply to the WUC and evidence that notices, fees and permits have been sought and paid, that utility service provider connections are complete and that statutory approvals by the authorities whose requirements apply to the Work have been received.
- Baseline data: To **BASELINE DATA**.
- Building penetrations: Details of the methods to maintain the required structural, fire and other properties to **BUILDING PENETRATIONS**.
- Certification: Certificates of conformance to Documented and statutory requirements.
- Commissioning plan: For the whole of the Work to **COMMISSIONING**.
- Commissioning program: For the whole of the Work to **COMMISSIONING**.
- Design Documentation: Drawings, calculations and specifications as Documented.
- Electronic facility and asset management information: For the whole of the Work to **ELECTRONIC FACILITY AND ASSET MANAGEMENT INFORMATION**.
- Execution details: Execution programs, schedules and details of proposed methods and equipment. For building services include the following:

- Embedded services: Proposed method for embedding services in concrete walls or floors or chasing into concrete or masonry walls.
- Fixing of services: Typical details of locations, types and methods of fixing services to the building structure.
- Inaccessible services: If services will be enclosed and not accessible after completion, submit proposals for location of service runs and fittings.
- Fire performance: Evidence of conformity to requirement for non-combustibility, fire hazard properties and fire-resistance of building elements.
- Marking and labelling: Samples and schedules of proposed marking and labels to **MARKING AND LABELLING**.
- Operation and maintenance manuals: For the whole of the Work to **OPERATION AND MAINTENANCE MANUALS**.
- Products and materials: Products and materials data, including manufacturer's technical specifications and drawings, product data sheets, type Tests results, evidence of conformity to Documented requirements, product certification, performance and rating tables, service connection requirements and installation and maintenance recommendations.
- Prototypes: Prototypes of components, systems or elements.
- Records: Record drawings to **RECORD DRAWINGS**, reports of Testing or work, photographs, system diagrams, schedules and logbooks.
- Safe Work Method Statement: For high risk construction works.
- Safety in design report: For the proposed work to **DESIGN DEVELOPMENT, Safety in design**.
- Samples: Representative of proposed products and materials and including proposals to incorporate samples into the Works, if any to **SAMPLES AND PROTOTYPES**.
- Shop drawings: To **SHOP DRAWINGS**.
- Substitutions: To **SUBSTITUTIONS**.
- Tests: Test reports for Testing performed under the Contract.
- Warranties: To **WARRANTIES**.

## 2.2 INSPECTION

### Notice

Concealment: If notice of inspection is required for parts of the Works that are to be concealed, give notice when the inspection can be made before concealment.

### Notification times

Minimum notice: As Documented.

### Light levels

Lighting levels for inspection: To AS/NZS 1680.2.4 (2017).

### Attendance

General: Provide attendance for Documented inspections and Tests.

---

## 3 PERFORMANCE

### 3.1 CORROSION RESISTANCE

#### Atmospheric corrosivity category

General: Atmospheric corrosivity category as defined in AS 4312 (2019):

- Exterior atmospheric corrosivity category: C3, as determined by Dulux Protective Coatings
- Interior atmospheric corrosivity category: C2, as determined by Dulux Protective Coatings

#### Galvanizing

Requirement: Galvanize mild steel components (including fasteners) to AS/NZS 1214 (2016) or AS/NZS 4680 (2025) as appropriate, if:

- Exposed to weather.
- Embedded in masonry.
- Exposed to or in air spaces behind the external leaf of masonry walls.
- In contact with chemically treated timber, other than copper chrome arsenate (CCA).

### 3.2 NOISE LEVELS

#### General

Requirement: Install systems to operate within the noise level limits, as Documented for the Contract design and Documented equipment performance.

### 3.3 STRUCTURE

#### General

Requirement: If provision of the Works requires structural design, provide structures, installations and components as follows:

- Fixed accessways: To AS 1657 (2018).
- Structural design actions: To the AS/NZS 1170 series.

Importance level: 2

## 4 DESIGN

---

### 4.1 DESIGN DEVELOPMENT

#### General

Requirement: Complete the design of the Work, including development of the design beyond that Documented.

Conflict with the Documents: If it is believed that a conflict exists between statutory requirements and the Documents, notify the Superintendent immediately and provide a recommendation to resolve the conflict.

#### Stanwell Engineering Requirements

Design provided by the Contractor is to satisfy the following Stanwell Engineering Requirements, Appendix Y – Principal Supplied Documents:

- ASM-STD-ENG-MAN-01 - Documentation Drawings and Manuals.
- T-3640 - Earthing.
- T-3643[2] - General Electrical & Engineering Standards.
- T-3645 - Lighting & Small Power.
- T-3646[3] - Low Voltage Switchgear.
- T-3655 - Equipment Identification Labelling Signage & Marking Physical Hazards.
- T-3685 - Noise and Vibration.
- T-3686 - Painting & Corrosion Prevention.
- HSE-PROC-18 - Principal Contractor HSE Requirements.

Where a conflict exists between the Stanwell Engineering Requirements and the project Documented design, the Superintendent shall be notified immediately.

#### Building Approval and Certificate of Occupancy

GOC Exemption: this project will be certified under a Government Owned Corporation exemption so no formal IDAS Building Development Application is required. Notwithstanding the GOC exemption, all legislative requirements for design and construction compliance must be met to the satisfaction of the Building Certifier, including the provision of all required certificates. Building works are not to proceed without the approval of the Building Certifier. The Contractor shall satisfy the requirements of the Building Certifier for sign off of a Form 11 - Certificate of Occupancy.

Form 15 submissions: Complete the design of the Work, including development of the design beyond that Documented, and provide a Form 15 from an RPEQ registered engineer for:

- F3P1 Performance Solution Report; refer to Façade Engineer's Documents.
- Roof Water Drainage Performance Solution Report; refer to Hydraulic Engineer's Documents.
- Building Access System; refer to Appendix G – Building Access System: proposal by Kattsafe.
- Timber Trusses; refer to Structural Engineer's Documents.
- Verified Mechanical design following coordination with final truss design; refer to Mechanical Engineer's Documents.
- Light steel framing systems such as walls, ceilings and the like.

- Seismic restraint for non-structural elements such as suspended services and the like.
- Screw piles in lieu of bored piers if approved by the Superintendent.

The Form 15 is to be submitted prior to commencement of the Works, at the time of shop drawing submission. Refer also to 6.4 *Support of Plant and Equipment* and 6.5 *Restraint of Non-Structural Parts and Components*.

Form 12 submission: submit all Form 12s required to certify the installation of the design elements to the satisfaction of the certifying authority.

Local Authority Charges: Contractor is responsible for payment of all local authority inspections and charge associated with the project including plumbing and drainage.

#### **Traditional Delivery of Design Provided**

If the Contractor chooses to deliver the project as Documented, including the project's verified design elements, then the remaining verification of supply items, systems, and installation to the design parameters, required by the certifier for the certificate of classification remain as Documented.

#### **Alternative Delivery Methods Requiring Amended Design Verification**

If the Contractor elects to amend the design, taking a different approach to procurement and/or delivery (for example utilising Modern Methods of Construction) then the Contractor will be responsible for providing verification to the satisfaction of the certifier for the design, as well as verification of supply items, systems, and installation to the approved (amended) design.

#### **Time and Cost for Amended Design, Design Verification and Alternative Delivery**

The extent of any amended process, including any additional time, fees, or services (including for the Principal's certifier) are deemed to be included in the Contractor's lump sum tender offer.

#### **Safety in design**

Requirement: Provide a design that allows for safe construction, operation and maintenance, and demolition in conformance with statutory requirements.

## **4.2 DESIGNER**

### **General**

Design by Contractor: If the Contractor provides design, use only appropriately qualified and registered persons to undertake design work. If requested, provide Documents verifying the qualification and experience.

Engineering design by Contractor: If the Contractor provides engineering design:

Provide design carried out by an RPEQ registered engineer.

Evidence: Signature of a nominated professional engineer in the respective discipline to each design Document submitted.

Insurance: Provide evidence of currency of insurance of the engineering designer for professional indemnity insurance and public liability insurance.

### **ALTERNATIVE DESIGN SOLUTIONS**

#### **Conforming alternative design solutions**

General: Proposals for alternative designs that conform to Documented requirements are permitted.

Requirement: If proposing alternative design solutions, submit the proposed alternative solution with a comparison to the original. Include the following factors in the comparison:

Safety, health and amenity of building occupants, including maintenance personnel.

Cost.

Anticipated life.

Energy efficiency.

Environmental impact, including greenhouse gas emissions.

Effect on construction Program.

Effect on other trades.

Availability of products.

Certification requirements.

Maintenance requirements.

Access for maintenance.

Flexibility for future changes.

Safety of equipment.

Sustainability.

Other information required for submission in **SUBSTITUTIONS**.

#### **Non-conforming alternative design solutions**

Requirement: If the proposed design solution does not conform to the Documented requirements, conform to **SUBSTITUTIONS**.

#### **Costs**

Requirement: Pay the cost of submissions and evaluation and/or Testing of proposed alternatives, whether subsequently adopted or not. The costs will be calculated at the current charge-out rates of the relevant consultant.

### **DESIGN DOCUMENTS**

#### **General**

Requirement: Provide calculations, drawing, shop drawings, digital models and specifications to Document a design conforming to the Documented design parameters and statutory requirements, as required.

#### **Drawings**

Requirement: To **SHOP DRAWINGS**.

Drawing registers: Submit drawing registers with the drawings showing current and previous drawing issues.

#### **Specifications**

Requirement: Provide project specific specification using the NATSPEC National Master Specification and incorporating this specification by reference.

Variations from this specification: If it is proposed to depart from the wording of this specification, submit the proposed changes as Documented in **ALTERNATIVE DESIGN SOLUTIONS**.

Licence: Provide specifications prepared by a current subscriber to the applicable NATSPEC package.

Worksections: Use worksections from the current update of NATSPEC and customise for the project.

Additional material: If the specification requires material not included in this specification, use worksections from the current update of NATSPEC and customise for the project.

Additional content: If additional content is required for the project specification not covered in either format.

Redundant content: Delete redundant content from the NATSPEC worksection *Templates*.

### **ERRORS IN AND OMISSIONS FROM THE DESIGN**

#### **General**

Requirement: If the design contains errors and/or omissions, correct the errors and make a new or amended submission, as appropriate, indicating changes made since the previous submission.

Notice: Immediately on detecting a design error and/or omission, submit notification in writing describing the error and/or omission, anticipated consequences and proposed corrective action.

Consequences of errors and/or omissions: Pay the costs associated with correcting design errors and/or omissions and minimise the effects of the errors.

---

## **5 PRODUCTS AND MATERIALS**

### **5.1 GENERAL**

#### **Sources policy**

General: Locally sourced materials and suppliers are preferred.

#### **Consistency**

General: For each material or product use the same source or manufacturer and provide consistent type, size, quality and appearance.

#### **Low VOC emitting paints**

Paint types: To the recommendations of AS/NZS 2311 (2017) Table 4.2.

#### **Prohibited materials**

General: Do not provide the following:

- Materials exceeding the limits of those listed in the Safe Work Australia *Hazardous Chemical Information System* (HCIS) Workplace exposure standards.
- Blowing agents:
  - . Materials that use chlorofluorocarbon (CFC) or hydrochlorofluorocarbon (HCFC) in the manufacturing process.
  - . A blowing agent with a Global Warming Potential (GWP) not less than 700.

## 5.2 PROPRIETARY ITEMS

### Manufacturer's or supplier's recommendations

General: Provide manufactured items to the manufacturer's or supplier's recommendations.

Proprietary items/systems/assemblies: Deliver, handle, store, assemble, install or fix to substrate to the manufacturer's or supplier's recommendations.

Project modifications: Advise of activities that supplement, or are contrary to the manufacturer's or supplier's recommendations.

### Identification of proprietary items

Sealed containers: If items are supplied by the manufacturer in closed or sealed containers or packages, bring them to point of use in the original containers or packages.

Other items: Marked to show the following, as applicable:

- Manufacturer's identification.
- Brand name.
- Product type.
- Quantity.
- Reference code and batch number.
- Date of manufacture.

## 5.3 SUBSTITUTIONS

### Substitution Request Form

Identified proprietary items: Identification of a proprietary item does not necessarily imply exclusive preference for the identified item, but indicates the necessary properties of the item.

Alternatives: If alternatives to the Documented products, methods or systems are proposed, submit a completed Substitution Request Form (Appendix Z). The following conditions must be met:

1. Contractor must submit a fully completed Substitution Request Form at least 5 Business Days before ordering products for, or starting installation of, the respective portion of the Works. Any substitution requests made without submitting a completed form will not be considered. A copy of the required form is enclosed within the appendices of this specification.
2. State whether the use of the substitution will require alteration to any part of the Works. If the substitution is adopted, carry out any such alteration without extra charge. In the case where the affected Works are part of another subcontract, obtain cost from the other subcontractor for such alteration and include it in the cost of the substitution.
3. No claim shall arise from any rejection, nor, unless otherwise agreed, shall adoption of a substitution be grounds for any claim for Variation to cost or time.

Conditions: Contractor's request for substitutions will be received and considered when extensive revisions to Contract Documents are not required and changes are in keeping with general intent of Contract Documents, and when timely, fully Documented and properly submitted, all as judged by the Architect.

Approval: No work involving any request for substitution shall be commenced, or materials ordered, until the Contractor has received written evidence of the approval of the substitution provided by the Architect.

Costs: Pay the cost of submissions and of evaluations and Tests of proposed alternatives, whether subsequently adopted or not. The costs will be calculated at the current charge-out rates of the relevant consultant(s).

## 5.4 SAMPLES AND PROTOTYPES

### General

Incorporation into Works: Only incorporate samples and prototypes that have been endorsed for inclusion in the Works. Do not incorporate other samples or prototypes.

Retention: Keep endorsed samples and prototypes in good condition, until the Date for Practical Completion.

Unincorporated samples and prototypes: Remove on completion.

## 5.5 SHOP DRAWINGS

### General

Standard: To AS 1100.101 (1992), AS 1100.201 (1992), AS 1100.301 (2008), AS 1100.401 (1984) and AS/NZS 1100.501 (2002) as applicable.

Documentation: Include dimensioned drawings showing details of the fabrication and installation of structural elements, building components, services and equipment, including relationship to building structure and other services, cable type and size, and marking details.

Diagrammatic layouts: Coordinate work shown diagrammatically in the Contract Documents, and prepare dimensioned set-out drawings.

Services coordination: Coordinate with other building and service elements. Show adjusted positions on the shop drawings.

Space requirements: Check space and access for maintenance requirements of equipment and services indicated diagrammatically in the Contract Documents.

Commissioning requirements: Show provisions for Testing and commissioning on the drawings.

Access for maintenance: Show space and provisions for access for maintenance.

Building work drawings for building services: On dimensioned drawings show the following:

- Access doors and panels.
- Conduits to be cast in slabs.
- Hold down bolts and other fasteners required complete with loads to be imposed on the structure during installation and operation.
- Openings, penetrations and blockouts.
- Sleeves.
- Plinths, kerbs and bases.
- Required external openings.

Submission medium: Electronic

Drawing size: A1 or A3

CAD base drawings: Refer to consultant File Transfer Disclaimer

Record drawings: Amend all Documented shop drawings to include changes made during the progress of the Work and up to the end of the Defects Liability Period.

## 6 ANCILLARY BUILDING WORK

### 6.1 WALL CHASING

#### Holes and chases

General: If holes and chases are required in masonry walls, make sure structural integrity of the wall is maintained. Do not chase walls with a fire-resistance level or an acoustic rating.

Parallel chases or recesses on opposite faces of a wall: Not closer than 600 mm to each other.

Chasing blockwork: Only chase core-filled hollow blocks or solid blocks that are not structural.

#### Concrete blockwork chasing table

| Block thickness (mm) | Maximum depth of chase (mm) |
|----------------------|-----------------------------|
| 190                  | 35                          |
| 140                  | 25                          |
| 90                   | 20                          |

## 6.2 FIXING

### General

Suitability: If equipment is not suitable for fixing to non-structural building elements, fix directly to structure and trim around penetrations in non-structural elements.

### Fasteners

General: Use proprietary fasteners capable of transmitting the loads imposed, and sufficient for the rigidity of the assembly.

## 6.3 BUILDING PENETRATIONS

### Penetrations

Requirement: Maintain the required structural integrity, fire performance, waterproofing performance and other properties when penetrating or fixing to the following:

- Structural building elements including external walls, fire walls, fire doors and access panels, other Tested and rated assemblies or elements, floor slabs and beams.
- Membrane elements including damp-proof courses, waterproofing membranes and roof coverings. If penetrating membranes, provide a waterproof seal between the membrane and the penetrating component.

### Sealing

Fire-resisting building elements: Seal penetrations with a system conforming to AS 4072.1 (2005).

Non fire-resisting building elements: Seal penetrations around conduits and sleeves. Seal around cables within sleeves. If the building element is acoustically rated, maintain the rating.

### Sleeves

General: If piping, cables or conduits penetrate building elements, provide metal or PVC-U sleeves formed from pipe sections as follows:

- Movement: Arrange to permit normal pipe or conduit movement.
- Diameter (for non fire-resisting building elements): Sufficient to provide a ring shaped space around the pipe or pipe insulation of at least 12 mm.
- Ferrous surfaces: Prime paint.
- Sealing: Seal between pipes or conduits and sleeves to prevent the entry of vermin.
- Terminations:
  - . Cover plates fitted: Flush with the finished building surface.
  - . Fire-resisting and acoustic rated building elements: 50 mm beyond finished building surface.
  - . Floors draining to floor wastes: 50 mm above finished floor.
  - . Other locations: 5 mm beyond finished building surface.
  - . Termite management: To AS 3660.1 (2014).
- Thickness:
  - . Metal: 1 mm or greater.
  - . PVC-U: 3 mm or greater.

## 6.4 SUPPORT OF PLANT AND EQUIPMENT

### Concrete plinths

General: Provide concrete plinths as Documented and under all equipment located on concrete floor slabs as follows:

- Surround: Zinc (hot-dipped) coated steel, at least 75 mm high and 1.6 mm thick. Fix to the floor with masonry anchors. Fill with concrete.
- Height: 75 mm or greater, as Documented.
- Reinforcement: Single layer of F62 fabric.
- Concrete: Grade N20.
- Finish: Steel float, flush with top edge of the surround.

### Support of ground level plant and equipment

Ground level: Conform to the following:

- If the ground slope is 15° or more, or the area of the plant and equipment is extensive, obtain the advice of a professional engineer for the Documentation of a suitable slab or platform.

- In all other cases, provide proprietary plastic or concrete supports installed with falls that achieve a raised, impervious and water shedding bearing surface.

Balustrades: If balustrades or screening are required, obtain the advice of a registered architect.

#### **Support of plant and equipment mounted on roofs or elevated platforms**

Platforms: If a platform is required, or the area of the plant and equipment mounted on roofs or elevated platforms is extensive, obtain the advice of a professional engineer for the Documentation of a suitable platform.

Balustrades: If balustrades or screening are required, obtain the advice of a registered architect.

Roof level support: If any of the following apply to roof level support, obtain the advice of a professional engineer:

- The total load from any unit of plant or equipment exceeds 500 kg.
- The load from a unit of plant or equipment to any single support point exceeds 100 kg.
- The average loading of plant and equipment over the area extending 1 m on all sides beyond the plant and equipment exceeds 25 kg/m<sup>2</sup>.

### **6.5 RESTRAINT OF NON-STRUCTURAL PARTS AND COMPONENTS**

#### **Seismic restraint**

Standard: To the NCC cited AS 1170.4 (2007).

Earthquake design category: I

Requirement: Conform to the following:

Use parts or components that will not be damaged by earthquake actions.

Protect systems against the adverse effects of parts or components such as tilt switches, that although may not be damaged in an earthquake, may malfunction.

Engineering services: To the recommendations of SA Gov G172 (2021) *Seismic restraint of engineering services* and the following:

Securely fix all plant and equipment to the building structure.

Do not rely on gravity and/or friction to resist seismic forces.

Anti-vibration mounts: Use horizontally restrained type.

#### **Wind restraint**

General: Provide restraints to resist the effect of ultimate limit state wind pressures.

Wind pressure: To AS/NZS 1170.2 (2021).

## **7 BUILDING SERVICES**

---

### **7.1 SERVICES CONNECTIONS**

#### **Connections**

General: Connect to utility service provider services or service points. Excavate to locate and expose connection points. Reinstall the surfaces and facilities that have been disturbed.

#### **Utility service provider requirements**

General: If the utility service provider elects to perform or supply part of the Works, make the necessary arrangements. Install equipment supplied, but not installed, by the utility service provider.

### **7.2 SERVICES INSTALLATION**

#### **General**

Installation: Install equipment and services as follows:

- Plumb and securely fixed.
- Allow for movement in both structure and services.
- Arrange services running together, parallel to each other and adjacent building elements.

Concealment: Conceal all cables, ducts, trays and pipes except where installed in plant spaces, ceiling spaces and riser cupboards or Documented to be exposed. If alternative routes are available, do not locate on external walls.

Lifting: Provide heavy items of equipment with permanent fixtures for lifting to the manufacturer's recommendations.

Suspended ground floors: Keep all parts of services suspended under ground floors at least 150 mm clear of the ground surface. Make sure services do not impede access.

#### **Dissimilar metals**

Jointing: Join dissimilar metals with fittings of electrolytically compatible material.

#### **Temporary capping**

Pipe ends: During construction, protect open ends of pipe with metal or plastic covers or caps.

#### **Piping**

General: Install piping in straight lines at uniform grades without sags. Arrange to prevent air locks. Provide sufficient unions, flanges and isolating valves to allow removal of piping and fittings for maintenance or replacement of plant.

Spacing: Provide at least 25 mm clear between pipes and between pipes and building elements, additional to insulation.

Changes of direction: Provide as follows:

- If practicable, long radius elbows or bends and sets, and swept branch connections.
- If pipes are led up or along walls and then through to fixtures, provide elbows or short radius bends.
- Do not provide mitred fittings.

Vibration: Arrange and support piping to prevent vibration whilst permitting necessary movement. Minimise the number of joints.

Embedded pipes: Do not embed pipes that operate under pressure in concrete or surfacing material.

Embedded pipes: If pipes that operate under pressure are to be embedded in concrete or surfacing material conform to AS 2896 (2021) clause 4.3.3.4. Pressure Test and rectify leaks before the concrete is poured.

Valve groupings: If possible, locate valves in groups.

Pressure Testing precautions: Isolate items not rated for the Test pressure. Restrain pipes and equipment to prevent movement during pressure Testing.

#### **Support and structure**

Requirement: Provide incidental supports and structures to suit the services.

#### **Pipe support systems**

Standard: To AS 4041 (2006) clause 3.28.

General: Provide hangers, brackets, saddles, clips, and support system components to resist live and dead loads and to control pipe movement caused by thermal and water pressure effects. Incorporate provisions for adjustment of spacing, alignment, grading and load distribution. Support pipework from associated equipment or building structure. Support valves, strainers and major line fittings so that no load is placed on connected piping or transmitted to it during operation and maintenance.

Fastenings: Provide fastenings to the associated equipment or building structure designed to withstand the loads imposed by the pipe supports.

Channel section supports: Proprietary channel section with clamps and hangers sized to match external diameter of pipe being supported. Provide all components from the same manufacturer.

Channel and fastening material: Metallic-coated steel.

Vertical pipes: Provide anchors and guides to maintain long pipes in position, and supports designed for the mass of the pipe and its contents.

Saddles: Do not use saddle type supports for pipes larger than DN 20.

Dissimilar metals: If pipe and support materials are dissimilar, provide industrial grade electrically non-conductive material securely bonded to the pipe to separate them. Provide fasteners of electrolytically compatible material.

Fixing to masonry and concrete: Provide metallic-coated steel or non-ferrous metal bolts or screws into chemical or expanding metal masonry anchors.

Uninsulated pipes: Clamp piping supports directly to pipes. Provide electrical isolation of dissimilar metals.

Insulated pipes:

- Spacers: Provide spacers at least as thick as the insulation between piping supports and pipes. Extend either side of the support by at least 20 mm.

- Spacer material: Rigid insulation material of sufficient strength to support the piping and suitable for the temperature application.
- Vapour barriers: For cold pipes, apply aluminium foil tape over the circumference of the spacer to form a vapour barrier. Fit to spacer before installation of the bracket on the pipe.
- Metal sheathing: If metal sheathing is Documented, provide a band of the Documented sheathing materials between the aluminium foil tape and the support for the full width of the spacer.

Hanger sizes: Conform to the following:

- Gas installations: To AS/NZS 5601.1 (2022) Table 5.8.3.
- Other pipes: Provide hangers sized to the manufacturer's recommendations to suit operating conditions and regulatory requirements including the loads due to valves and other attached components, pipe material, pipe contents and temperature and seismic loads.

Support spacing: Provide supports at no greater spacing than the following:

- Cold and heated water: To the NCC cited AS/NZS 3500.1 (2021) Table 5.7.4.
- Sanitary plumbing: To the NCC cited AS/NZS 3500.2 (2021) Table 10.2.1.
- Stormwater: To the NCC cited AS/NZS 3500.2 (2021) clause 4.9.
- Fuel gas: To AS/NZS 5601.1 (2022) Table 5.8.2.
- Fire sprinklers and combined wet suppression systems: To AS 2118.9 (1995) Table 2.6.1.
- Fire hydrants:
  - . Metal piping: To AS 2419.1 (2021) clause 10.6.
  - . Plastic piping: To the NCC cited AS/NZS 3500.1 (2021).
- Gaseous fire suppression systems:
  - . General gaseous fire suppression systems: To AS 4214 (2018) clause 6.3.4.
  - . Carbon dioxide fire suppression systems: To AS 6183 (2025) clause 6.3.4.
- Medical gases: To AS 2896 (2021) Table 4.1.
- Refrigerant: To AS/NZS 5149.2 (2016) Tables 5 and 6.
- Other ferrous pipes under pressure: To AS 4041 (2006) Table 3.28.2.
- Other copper pipes: To AS 4809 (2017) Table 6.2.
- ABS pipes: To AS/NZS 3690 (2009) Table 6.2.
- PVC pipes: To AS/NZS 2032 (2006) Table 6.3.
- Polyolefin including PE and PP pipes: To AS/NZS 2033 (2024) clause 7.3.2.
- Other non-ferrous pipe carrying liquids: To the NCC cited AS/NZS 3500.1 (2021) Table 5.7.4.
- Other pipes carrying air or gases: To AS/NZS 5601.1 (2022) Table 5.8.2.
- Proprietary grooved piping systems: To the manufacturer's recommendations.

Additional supports: Provide additional supports as follows:

- Proprietary grooved piping systems: To the manufacturer's recommendations.
- Valves and other heavy pipe mounted components: Adjacent to the valve or component.
- Adjacent pipe mounted components requiring regular maintenance.
- At changes of direction and adjacent to wall or floor penetrations.
- Where required to anchor piping or control thermal or other movement.

#### **Differential movement**

General: If the geotechnical Site investigation report predicts differential movements between buildings and the ground in which pipes or conduits are buried, provide control joints in the pipes or conduits, as follows:

- Arrangement: Arrange pipes and conduits to minimise the number of control joints.
- Magnitude: Accommodate the predicted movements.

### **7.3 PLANT AND EQUIPMENT**

#### **General**

Location: Locate so failure of plant and equipment (including leaks) does not create a hazard for the building occupants and causes a minimum or no damage to the building, its finishes and contents including water sensitive equipment or finishes.

Safe tray and an overflow pipe: Provide to each tank, hot water heater and storage vessel.

#### 7.4 ACCESS FOR MAINTENANCE

##### General

Requirement: Provide access for maintenance of all items requiring inspection, measurement, operation, adjustment, repair, replacement and other maintenance-related tasks.

Standards: Conform to the relevant requirements of AS 1657 (2018), AS 1892.1 (2018), AS 2865 (2009) and AS/NZS 3666.1 (2011).

*Work Health and Safety*: Conform to the requirements of the applicable *Work Health and Safety* regulations.

Access safety systems: Provide access safety systems to 0193 *Building access safety systems*.

Refrigerated or cooling plant: If the space is a refrigerated or cooling chamber inside a duct, air handling plant or similar, provided with an access door or personnel access panel and of sufficient size for a person to enter, provide the following to BCA (2022) G1D3:

- An access door.
- Internal lighting with external indicator lamp.
- An alarm.

Protection from injury: Protect personnel from injury caused by contact with objects including those that are sharp, hot or protrude at low level.

Plant room flooring surfaces: R10 Slip resistance classification to AS 4586 (2013).

Trip hazards: Do not run small services including drains and conduits across floors where they may be a trip hazard.

Manufacturer's standard equipment: If necessary, modify manufacturer's standard equipment to provide the plant access Documented.

##### Clearances

Minimum clearances for access: Conform to the following:

- Vertical clearance: Not less than 2100 mm, vertically above horizontal floors, ground and platforms.
- Horizontal clearance: Preferably not less than 750 mm clear, but in no case less than 600 mm between equipment or between equipment and building features including walls.
- If tools are required to operate, adjust or remove equipment, provide sufficient space so the tools can be used in their normal manner and without requiring the user to employ undue or awkward force.
- Hinged or removable components: To the manufacturer's recommendations.
- Within plant items: Conform to the preceding requirements, and not less than the clearances recommended in BS 8313 (1997).

##### Elevated services other than in occupied areas

Access classifications:

- Access class A: Readily accessible. Provide clear and immediate access to and around plant items. If plant or equipment is located more than 2.0 m above the ground, floor or platform, provide a platform with handrails accessible by a stair, all to AS 1657 (2018).
- Access class B: If the plant item requiring access is located more than 2.0 m above the ground, floor or platform, provide a platform with handrails accessible by a non-vertical ladder, all to AS 1657 (2018).
- Access class C: Locate plant so temporary means of access conforming to *Work Health and Safety* regulations can be provided.

Temporary means of access: Make sure there is adequate provision in place, which is safe and effective.

Areas in which access is restricted to authorised maintenance personnel: Provide access as follows:

- Instruments, gauges and indicators (including warning and indicating lights) requiring inspection at any frequency: Readily accessible.
- Access required monthly or more frequently: Access class A.
- Access required between monthly and six monthly: Access class A or B.
- Access required less frequently than six monthly: Access class A, B or C.

Other areas: Provide access as follows:

- Locate to minimise inconvenience and disruption to building occupants or damage to the building structure or finishes.
- In suspended ceilings, locate items of equipment that require inspection and/or maintenance above tiled parts. If not possible, provide access panels where located above set plaster or other inaccessible ceilings. Arrange services and plant locations to reduce the number of access panels. Coordinate with other trades to use common access panels where feasible.
- Do not locate equipment requiring access above partitions.
- Instruments, gauges and other items requiring inspection at any frequency: Readily accessible.
- Labelling: If equipment is concealed in ceilings, provide marking to **MARKING AND LABELLING**, equipment concealed in ceilings.

#### **Facilities for equipment removal and replacement**

Requirement: Provide facilities to permit removal from the building and replacement of plant and equipment, including space large enough to accommodate it and any required lifting and/or transportation equipment. Arrange plant so large and/or heavy items can be moved with the minimum changes of direction.

Removal of components: Allow sufficient space for removal and replacement of equipment components including air filters, tubes of shell and tube heat exchangers, removable heat exchanger bundles, coils and fan shafts. Provide access panels or doors large enough to permit the safe removal and replacement of components within air handling units.

#### **Facilities for access**

Equipment behind hinged doors: Provide doors opening at least 150°.

Equipment behind removable panels: Provide panels with quick release fasteners or captive metal thread screws.

Removable panels: Provide handles to permit easy and safe removal and replacement.

Insulated plant and services: If insulation must be removed to access plant and services for maintenance, arrange it to allow for removal and replacement without damage.

#### **Piping**

Requirement: Conform to the following:

- Provide access and clearance at fittings that require maintenance, inspection or servicing, including control valves and joints intended to permit pipe removal.
- Arrange piping so it does not interfere with the removal or servicing of associated equipment or valves or block access or ventilation openings.
- Preferably run piping, conduits, cable trays and ducts at high level and drop vertically to equipment.

#### **Electrical equipment and controls**

Electrical equipment: Provide clearances and access space to AS/NZS 3000 (2018).

Switchboards and electrical control equipment: Locate near the main entrance to plant space and with switchboards visible from the plant being operated.

Control panels: Locate near and visible from the plant being controlled.

### **7.5 VIBRATION SUPPRESSION**

#### **General**

Requirement: Minimise the transmission of vibration from rotating or reciprocating equipment to other building elements.

#### **Standard**

Machinery noise and vibration: Vibration severity in Zone A to ISO 20816-1 (2016) and ISO 20816-3 (2022).

#### **Speeds**

General: If no maximum speed is prescribed, do not exceed 1500 r/min for direct driven equipment.

#### **Connections**

General: Provide flexible connections to rotating machinery and assemblies containing rotating machinery. Isolate pipes by incorporating sufficient flexibility into the pipework or by use of proprietary flexible pipe connections installed to prevent placing stress on pipes due to end reaction.

**Inertia bases**

General: If necessary to achieve the required level of vibration isolation, provide inertia bases having appropriate mass and to the following:

- Construction: Steel or steel-framed reinforced concrete with reinforcing bars welded between base sections. Position foundation bolts for equipment before pouring concrete.
- Supports: Support on vibration isolation mountings using height saving support brackets.

**Vibration isolation mountings**

General: Except for external equipment that is not connected to the structure of any building, support rotating or reciprocating equipment on mountings as follows:

- Static deflections less than 15 mm: Single or double deflection neoprene in-shear mountings incorporating steel top and base plates and a tapped hole for bolting to equipment.
- Static deflections not less than 15 mm: Spring mountings.

Selection: Provide mountings selected to achieve 95% isolation efficiency at the normal operating speeds of the equipment.

Installation: Set and adjust vibration isolation mounting supports to give clearance for free movement of the supports.

Spring mountings: Provide freestanding laterally stable springs as follows:

- Clearances: Not less than 12 mm between springs and other members such as bolts and housing.
- High frequency isolation: 5 mm neoprene acoustic isolation pads between base plate and support.
- Levelling: Provide bolts and lock nuts.
- Minimum travel to solid: Not less than 150% of the designated minimum static deflection.
- Ratio of mean coil diameter to compressed length at the designated minimum static deflection: Not less than 0.8:1.
- Snubbing: Snub the springs to prevent bounce at start-up.
- Vertical resilient limit stops: To prevent spring extension when unloaded, to serve as blocking during erection and which remain out of contact during normal operation.

**7.6 FINISHES TO BUILDING SERVICES****General**

Requirement: If exposed to view (including in plant rooms), paint building services and equipment.

Surfaces painted or finished off Site: Conform to *0183 Metals and prefinishes*.

Exceptions: Do not paint chromium or nickel plating, anodised aluminium, glass-reinforced plastic, stainless steel, non-metallic flexible materials and normally lubricated machined surfaces. Surfaces with finishes applied off Site need not be re-painted on Site provided the corrosion resistance of the finish is not less than that of the respective finish Documented.

Standard: Conform to the recommendations of AS/NZS 2311 (2017) Sections 3, 6 and 7 or AS 2312.1 (2014) Sections 6, 7 and 8, as applicable.

Inaccessible surfaces: If surfaces are inaccessible after installation, complete finish before installation.

**Painting systems**

New unpainted interior surfaces: To AS/NZS 2311 (2017) Table 5.1.

New unpainted exterior surfaces: To AS/NZS 2311 (2017) Table 5.2.

**Paint application**

Coats: Apply the first coat immediately after substrate preparation and before contamination of the substrate can occur. Make sure each coat of paint or clear finish is uniform in colour, gloss, thickness and texture and free of runs, sags, blisters or other discontinuities.

Combinations: Do not combine paints from different manufacturers in a paint system.

Protection: Remove fixtures before starting to paint and refix in position undamaged when painting is complete.

**Underground metal piping**

Requirement: Provide corrosion protection for the following:

- Underground ferrous piping.
- Underground non-ferrous metal piping in chemically aggressive soils and environments.

Corrosion protection: Select from the following:

- Cathodic protection: Sacrificial anodes or impressed current. Incorporate a facility for periodic Testing. Conform to the recommendations of AS 2832.1 (2015).
- Continuous wrapping using proprietary petroleum taping material.
- Impermeable flexible plastic coating.
- Sealed polyethylene sleeve.

Aggressive soils: If metallic piping or components are installed in chemically aggressive soil, provide additional protection as follows:

- Material: Continuous polyethylene sleeve to ASTM D1248 (2016) with a minimum thickness of 0.25 mm.
- Installation: Wrap or sleeve pipes and components. Tape joints between sections of polyethylene and between polyethylene and piping.

### **Repairs to finishes**

Requirement: Repair damaged finishes to restore their corrosion protection, appearance and service life.

Painting of pipe threads: After pipe installation and before other finishes or insulation are applied, paint exposed threads in metallic-coated steel pipe with zinc-rich paint.

## **7.7 MARKING AND LABELLING**

### **General**

Requirement: Mark and label services and equipment for identification purposes as follows:

- Locations exposed to weather: Provide durable materials.
- Pipes, conduits and ducts: To AS 1345 (1995) throughout its length, including in concealed spaces.
- Cables: Label to indicate the origin and destination of the cable.

Consistency: Label and mark equipment using a consistent scheme across all services elements of the project.

### **Label samples and schedules**

Requirement: For each item or type of item, prepare a schedule of marking and labelling, including the following:

- A description of the item or type of item for identification.
- The proposed text for marking or labelling.
- The proposed location of the marking and labelling.

Submission timing: Before marking or labelling.

### **Electrical accessories**

Circuit identification: Label isolating switches and outlets to identify circuit origin.

### **Operable devices**

Requirement: Mark to identify the following:

- Controls.
- Indicators, gauges, meters.
- Isolating switches.

### **Equipment concealed in ceilings**

Location: Provide a label on the ceiling, to indicate the location of each concealed item requiring access for routine inspection, maintenance and/or operation and as follows:

- Tiled ceilings, locate the label on the ceiling grid closest to the concealed item access point.
- Flush lined ceilings, locate adjacent to closest access panel.

Concealed equipment: Label items including the following:

- Fan coil units and terminal equipment (e.g. VAV terminals).
- Fire and smoke dampers.
- Isolating valves not directly connected to items otherwise labelled.
- Motorised dampers.

### **Wall mounted equipment in occupied areas**

Location: Provide labels on wall mounted items in occupied areas including the following:

- Services control switches.

- Temperature and humidity sensors.

### Points lists

Automatic control points: Provide plasticised, fade-free points lists for each automatic control panel and include terminal numbers, point addresses, short and long descriptors in the lists. Store in a pocket on the door of the panel.

### Pressure vessels

General: Mount manufacturer's certificates in glazed frames on a wall next to the vessel.

### Valves and pumps

General: Label to associate pumps with their starters and valves. Screw fix labels to body or attach label to valve handwheels with a key ring.

### Underground services

Survey: Accurately record the routes of underground cables and pipes before backfilling. Include on the record drawings.

Records: Provide digital photographic records of underground cable and pipe routes before backfilling. Include in operation and maintenance manual.

Location marking: Accurately mark the location of underground cables and pipes with route markers consisting of a marker plate set flush in a concrete base, engraved to show the direction of the line and the name of the service.

Markers: Place markers at ground level at each joint, route junction, change of direction, termination and building entry point and in straight runs at intervals of not more than 100 m.

Marker bases: 200 mm diameter x 200 mm deep, minimum concrete.

Direction marking: Show the direction of the cable and pipe run by means of direction arrows on the marker plate. Indicate distance to the next marker.

Plates: Brass, aluminium or stainless steel with black filled engraved lettering, minimum size 75 x 75 x 1 mm thick.

Plate fastening: Waterproof adhesive and 4 brass or stainless steel countersunk screws.

Marker height: Set the marker plate flush with paved surfaces, and 25 mm above other surfaces.

Marker tape: Where electric bricks or covers are not provided over underground wiring, provide a 150 mm wide yellow or orange marker tape bearing the words WARNING – electric cable buried below, laid in the trench 150 mm below ground level.

Plastic pipe: Provide a detectable marker tape with trace wire to identify the route of buried piping. Terminate with 1000 mm coil in a readily accessible location. Tag to match the record drawings.

### Labels and notices

Materials: Select from the following:

- Cast metal.
- For indoor applications only, engraved two-colour laminated plastic.
- Proprietary pre-printed self-adhesive flexible plastic labels with machine printed black lettering.
- Stainless steel or brass minimum 1 mm thick with black filled engraved lettering.

Emergency functions: To AS 1319 (1994).

Colours: Generally to AS 1345 (1995) as appropriate, otherwise black lettering on white background except as follows:

- Danger, warning labels: White lettering on red background.
- Main switch and caution labels: Red lettering on white background.

Edges: If labels exceed 1.5 mm thickness, radius or bevel the edges.

Labelling text and marking: To correspond to terminology and identifying number of the respective item as Documented on the record drawings and in operating and maintenance manuals.

Lettering heights:

- Danger, warning and caution notices: Minimum 10 mm for main heading, minimum 5 mm for remainder.
- Equipment labels within cabinets: Minimum 5 mm.
- Equipment nameplates: Minimum 40 mm.
- Identifying labels on outside of cabinets: Minimum 5 mm.

- Isolating switches: Minimum 5 mm.
- Switchboards, main assembly designation: Minimum 25 mm.
- Switchboards, outgoing functional units: Minimum 10 mm.
- Switchboards, sub assembly designations: Minimum 15 mm.
- Valves:
  - . Not less than DN 65: Minimum 25 mm.
  - . Less than DN 65: Minimum 10 mm.
- Self-adhesive flexible plastic labels:
  - . Labels less than 2000 mm above floor: 5 mm.
  - . Labels minimum 2000 mm above floor: 10 mm.
  - . Other locations: Minimum 5 mm.

Label locations: Locate labels so they are easily seen and are either attached to, below or next to the item being marked.

Fixing: Fix labels securely using screws, rivets, proprietary self-adhesive labels or double-sided adhesive tape and as follows:

- If labels are mounted in extruded aluminium sections, use rivets or countersunk screws to fix the extrusions.
- Use aluminium or monel rivets for aluminium labels.

Vapour barriers: Do not penetrate vapour barriers.

---

## 8 COMPLETION

### 8.1 TOOLS AND SPARE PARTS

#### **Spare parts**

General: Provide spare parts listed as Documented.

Replacement: Replace spare parts used during the maintenance period.

#### **Tools and spare parts schedule**

Submission timing: At least 8 weeks before the Date for Practical Completion.

Requirement: Prepare a schedule of tools, portable instruments and spare parts necessary for maintenance of the installation. For each item state the recommended quantity and the manufacturer's current price. Include the following in the prices:

- Checking receipt, marking and numbering in conformance with the spare parts schedule.
- Packaging and delivery to Site.
- Painting, greasing and packing to prevent deterioration during storage.
- Referencing equipment schedules in the operation and maintenance manuals.
- Suitable means of identifying, storing and securing the tools and instruments. Include instructions for use.

### 8.2 TRAINING

#### **General**

Standard: To SA TS 5342 (2021).

Duration: Instruction to be available for the whole of the commissioning and running-in periods.

Format: Conduct training at agreed times, at system or equipment location. Also provide seminar instruction to cover all major components.

Operation and maintenance manuals: Use items and procedures listed in the final draft operation and maintenance manuals as the basis for instruction. Review contents in detail with the Principal's staff.

Certification: Provide written certification of attendance and participation in training for each attendee. Provide register of certificates issued.

#### **Demonstrators**

General: Use only qualified manufacturer's representatives who are knowledgeable about the installations.

**Operation**

General: Explain and demonstrate to the Principal's staff the purpose, function and operation of the installations.

**Maintenance**

General: Explain and demonstrate to the Principal's staff the purpose, function and maintenance of the installations.

**Seasonal operation**

General: For equipment requiring seasonal operation, demonstrate during the appropriate season.

**8.3 CLEANING****Removal of material**

General: Dispose of building waste material off Site to the requirements of the relevant authorities.

**Final cleaning**

General: Before the Date for Practical Completion, clean throughout, including all exterior and interior surfaces except those totally and permanently concealed from view.

Labels: Remove all visible labels not required for maintenance.

**8.4 WARRANTIES****General**

Requirement: If a warranty is Documented, name the Principal as warrantee. Register with manufacturers as necessary. Retain copies delivered with components and equipment.

Approval of applicator or installer: If the warranty is conditional on the manufacturer's approval of the applicator or installer, submit the manufacturer's written approval of the installing company, and authorised personnel, with evidence of qualifications and experience in the specific use of the product, material or system.

Principal's responsibilities: Submit details of responsibilities of the Principal required to keep warranties in force.

**Warranty types**

Manufacturer's warranty: Warranty to cover manufacturing Defects and Defects with products and materials delivered to Site.

Manufacturer and applicator's/installer's interlocking warranty: Interlocking warranty to cover manufacturing Defects and Defects with products and materials delivered to Site, including their application or installation.

Supplier's warranty: Warranty to Defects in materials delivered to Site.

---

**9 TESTING AND COMMISSIONING**

---

**9.1 TESTING - GENERALLY****Inspection and Testing plan**

Requirement: Provide inspection and Testing plan consistent with the construction Program including details of Test stages and procedures.

**Notice**

Site Tests: Give notice of the time and place of Documented Tests.

Inspection: Give sufficient notice for inspection to be made of the commissioning, Testing and verification Tests on completion of commissioning.

**Attendance**

General: Provide attendance at Tests.

Suppliers: If necessary to carry out Documented Tests, arrange equipment suppliers to assist.

**Testing authorities**

Requirement: Have Tests carried out by an Accredited Testing Laboratory, accredited for the Documented Test method, except for Site Tests or Test methods that do not have an Accredited Testing Laboratory.

**Test equipment**

Accuracy: Use Testing equipment designed to Test and/or measure system performance within the Documented tolerances.

Calibration: Use only instruments that have current calibration certificates issued by an Accredited Testing Laboratory. Tag or label instruments with calibration date and calibration authority name. Provide copies of certification if requested.

Maximum period since last calibration: As recommended by the manufacturer but less than 12 months, except as Documented.

Recalibration: If dropped or damaged, recalibrate instruments.

Testing equipment: Provide Test equipment and tools to perform Documented Tests as follows:

- Special Testing equipment: If Documented, provide special equipment, tools and instruments required for Testing or calibration.
- Other Testing equipment: Provide standard Testing equipment.

### **Testing procedures**

Verification: Verify Test procedures by:

- Manual Testing.
- Monitoring performance and analysing results using the control system trend logs.
- A combination of the above methods.

Sampling: Sampling may be used subject to the following:

- Use a sampling strategy only for multiple identical pieces of non-life-safety or otherwise non-critical equipment.
- If at any point, more than one identical item has failed, stop Testing, determine the cause, rectify and Document changes made to remaining units, before continuing with functional Testing of the remaining units.

### **Type Tests**

Type Test reports: Required, as evidence of conformance of proprietary equipment.

### **Sound pressure level measurements**

Requirement: Conform to the following:

- Correction for background noise: To AS/NZS 2107 (2016) Table B1.
- External: To AS 1055 (2018).
- Internal: To AS/NZS 2107 (2016).
- Measurement positions: If a Test position is designated only by reference to a room or space, do not take measurements less than 1 m from the floor, ground or walls. For large equipment items including chillers, measure at 2 m and 7 m from the equipment item.
- Sound pressure level analysis: Measure the sound pressure level and the background sound pressure level over the full range of octave band centre frequencies from 31.5 Hz to 8 kHz at the designated positions.
- Sound pressure levels: Measure the A-weighted sound pressure levels and the A-weighted background sound pressure levels at the designated positions.

### **Test outcome**

Requirement: Test as Documented and achieve the following:

- Pass the Documented Pass/Fail Test, and/or
- Values that meet Documented requirements, and/or
- Verification of manufacturer's claimed performance.

### **Failure of multiple items**

Requirement: If 10% or 3, whichever is greater, of identical pieces (size does not constitute a difference) of equipment fail to perform as Documented for any reason, treat all identical units as having failed. Submit notice of failure and conform to the following:

- Within one week of notification, examine all other identical units and record the results. Submit a report of the findings within two weeks of the original failure notice.
- Within two weeks of the original failure notification, submit a signed and dated explanation of the problem, including the cause of failure, the proposed solution, full equipment details and any other information. Do not exceed the Documented requirements of the original installation with the proposed solution.

### **Rectification of failure under Test**

Requirement: If an item fails a Documented Test, rectify the cause of failure and repeat the Test.

Submissions: If submission of Test results is Documented, submit results of both successful and unsuccessful Tests.

#### **Test reports**

Requirement: Include the following:

- Documented performance criteria including, if Documented, tolerances.
- Observations and results of Tests and conformance or non-conformance with Documented requirements.

#### **Test validity period**

Requirement: As Documented or, if no validity period is Documented, no older than 5 years.

#### **Controls**

General: Calibrate, set and adjust control instruments, control systems and safety controls.

#### **Circuit protection**

General: Confirm that circuit protective devices are sized and adjusted to protect installed circuits.

#### **Certification**

General: On satisfactory completion of the installation, Testing and commissioning and before the Date for Practical Completion, certify that each installation is operating correctly.

#### **Integrated system Tests**

Requirement: Conduct integrated system Tests as Documented.

Tests: Provide the following:

- Test the integrated operation of the systems listed in each mode Documented.
- Restoration of the systems to their pre-Test condition on completion of the Tests above.

Failure: If any of the systems fails to perform as Documented, including return to normal operation, rectify the cause and repeat the integrated system Test.

#### **Deferred and seasonal Tests**

Deferred Tests: If Documented Testing cannot be completed at the scheduled or Documented time, the Superintendent may direct that they be deferred to a later time but as soon as possible after the scheduled or Documented time.

Seasonal Tests: If Documented Tests are dependent on specific weather conditions, they may be deferred to a time when weather conditions are close to the Documented Test conditions. Complete seasonal Testing as soon as possible but no later than one month before the end of the Defects Liability Period.

#### **Functional Tests**

Function: Carry out functional and operational Tests on each energised equipment item and circuit.

## **9.2 COMMISSIONING**

#### **Standard**

Requirement: Conform to SA TS 5342 (2021).

#### **Static completion**

Requirement: Systems, components and building elements are statically complete when:

- Their construction and installation is complete and as Documented, including completion of all systems, components and building elements on which they are dependent for commissioning.
- All pre-commissioning Tests have been successfully completed.
- They are safe and ready for commissioning.
- All cleaning that may adversely affect commissioning is complete.
- They have been inspected and all outstanding remedial work that may adversely affect commissioning is complete.
- All spaces required for access for commissioning are safe to use and cleared of obstructions that may adversely affect commissioning.

#### **Commissioning plan**

Requirement: Provide a commissioning plan to SA TS 5342 (2021) including the following:

- A summary of the Work covered by the commissioning plan.
- The parties responsible for this Work and any commissioning interrelationships.
- The basis of the design.

- General sequence of commissioning.
- Project specific commissioning methodologies for each system and building element to be commissioned.
- Pre-commissioning requirements.
- Project specific commissioning procedures for each commissioning activity including integrated system Tests, deferred and seasonal Tests.
- A project specific building tuning plan for all commissioned systems. Include building tuning procedures and tuning team members.
- Requirements for witnessing of Tests and Documented demonstrations of completion of commissioning.
- Commissioning program to **COMMISSIONING, Commissioning Program**.

#### **Commissioning Program**

Submissions: Submit a Program consistent with, and forming part of, the construction Program as follows:

- Set out the proposed program for completion, commissioning, Testing and instruction.
- Identify related works and timing of the works prerequisite to successful and timely completion of the Works.

Revisions: Submit revisions of the program as the project proceeds.

Plant operation period: Include time in the program for the Documented plant operation period before the Date for Practical Completion.

#### **Commissioning activities**

Requirement: Provide the following to SA TS 5342 (2021):

- Manage the commissioning process.
- Establish and manage the completion process.
- Review design Documents for commissionability. Submit a report including any recommended changes.
- Review Documented commissioning requirements. Submit a report including any recommended changes.
- Review construction Documents for commissionability. Submit a report including any recommended changes.
- Develop, review and update the commissioning plan and commissioning Program.
- Develop, review and update commissioning methodologies.
- Develop, review and update commissioning procedures.
- Report on interdependencies between trades that may affect commissioning.
- Develop, review and update procedures for initial start-up of systems.
- Develop, review and update integrated system Test procedures.
- Carry out pre-commissioning activities. Record results and submit pre-commissioning records.
- Conduct commissioning activities to the commissioning methodologies and procedures. Record and submit commissioning records.
- Facilitate and conduct integrated system Tests and demonstrations. Record and submit integrated system Test records.
- Conduct Documented demonstrations of completion of commissioning.
- Report on the progress of commissioning work.
- Report on conformance to the commissioning plan and program.
- Report on commissioning Defects and issues and progress on their resolution.
- Develop, review and update commissioning report.
- Develop, review and update training materials, conduct training sessions to **TRAINING**.
- Develop, review and update operation and maintenance manuals to **OPERATION AND MAINTENANCE MANUALS**.
- Manage and report deferred and seasonal Testing activities to **TESTING - GENERALLY**.
- Management and reporting of building tuning process.

- Periodically review performance data.

**Verification of commissioning**

Requirement: On completion of commissioning of the equipment or system, provide additional Tests to verify that it is fully commissioned and operating to Documented requirements.

**9.3 BUILDING TUNING****General**

Standard: To SA TS 5342 (2021).

Frequency: Three monthly or more frequently.

Duration: Until the end of the maintenance period. Provide last building tuning in the month before the end of the maintenance period.

Requirement: Provide the following:

- Review data from all recording systems against Documented requirements.
- Review of building occupant feedback.
- If discrepancies are identified from the above, take corrective action to rectify them.
- Report on the findings of the reviews, corrective action and effect of corrective action.
- Recommend other action to improve the effectiveness, reliability and efficiency of systems.

---

**10 PROJECT RECORDS****10.1 RECORD DRAWINGS****General**

Requirement: Prepare record drawings showing the following:

- Installed locations of building elements, services, plant and equipment.
- Off-the-grid dimensions and depth if applicable.
- Any provisions for the future.

**Recording, format and submission**

Requirement: Record changes made during the progress of the works on a set of drawings kept on Site for that specific purpose.

Drawing layout: Use the same borders and title block as the Contract drawings.

Quantity and format: Conform to **SUBMISSIONS**.

Endorsement: Sign and date all record drawings.

Accuracy: If errors in, or omissions from, the record drawings are found, amend the drawings and re-issue in the quantity and format Documented for **SUBMISSIONS**.

Date for submission: Not later than 2 weeks after the Date for Practical Completion.

**Services record drawings**

General: To **RECORD DRAWINGS, General** and **Recording, format and submission** and the following:

- Extensions and/or changes to existing: If a drawing shows extensions and/or alterations to existing installations, include sufficient detail of the existing installation to make the drawing comprehensible without reference to drawings of the original installation.
- Detention: If on-Site detention tanks or pondage are provided, include the volume required on the drawing and the permitted flow rate to the connected system.
- Domestic cold water or fire mains: Show the pressure available at the initial connection point and the pressure available at the most disadvantaged location on each major section of the Works.
- Stormwater: If storm water pipes are shown, include the pipe size and pipe grade together with the maximum acceptable flow and the actual design flow.

Diagrams: Provide diagrammatic drawings of each system including the following:

- Controls.
- Piping including all valves and valve identification tags.
- Principal items of equipment.
- Single line wiring diagrams.

- Acoustic and thermal insulation.
- Access provisions and space allowances.
- Fasteners.
- Fixtures.
- Switchgear and controlgear assembly circuit schedules including electrical service characteristics, controls and communications.
- Charts of valve tag numbers, with location and function of each valve, keyed to flow and control diagrams.

CAD base drawings: Refer consultant File Transfer Disclaimer

Subsurface services: Record information on underground or submerged services to the Documented quality level, conforming to AS 5488.1 (2022).

Subsurface services recording quality level: Level A

## 10.2 BASELINE DATA

### General

Requirement: Provide baseline data to permit routine service of fire protection systems and equipment to AS 1668.1 (2015), the NCC cited AS 1670.1 (2018), AS 1851 (2012) and AS/NZS 2293.1 (2018).

Include baseline data for the following:

- Active fire and smoke systems including automatic fire sprinkler systems, fire pumpsets, fire hydrant systems and water storage tanks for fire protection systems.
- Fire detection and alarm systems.
- EWIS, exit signs and emergency lighting.
- Standby generator sets and batteries.
- Lay flat fire hose, fire hose reels, portable and wheeled fire extinguishers and fire blankets.
- Passive fire and smoke systems, including the following:
  - . Fire and smoke elements (vertical and horizontal) - walls, floors, ceilings, access panels and hatches.
  - . Structural fire-resistant elements - beams, columns, girders and trusses.
  - . Fire-resisting doorsets - hinged, pivoted and horizontal sliding.
  - . Smoke doors - hinged and pivoted.
  - . Fire shutters.
  - . Fire-resisting glazing.
  - . Ducts.
  - . Dampers.
- Fire and smoke control features of mechanical services.
- Emergency planning in facilities.

Revisions: Amend baseline data to include changes made to the installation during construction and maintenance. Correct errors in baseline data.

Format: Provide baseline data in a format that facilitates the carrying out and recording of routine service tasks including drawings showing the extent and location of items to be serviced, schedules of items and unique identification of each item.

## 10.3 OPERATION AND MAINTENANCE MANUALS

### General

Standard: To SA TS 5342 (2021).

Authors and compilers: Personnel experienced in the maintenance and operation of equipment and systems installed, and with editorial ability.

Referenced Documents: If referenced Documents or worksections require submissions of manuals, include corresponding material in the operation and maintenance manuals.

Subdivision: By installation or system, depending on project size.

Revisions: Amend the operation and maintenance manuals to include changes made to the installation during construction and maintenance including changes to software and commissioning records.

### Contents of manual

Table of contents: Include a table of contents in each volume. Title to match cover.

Table of amendments: Include a table of amendments.

Directory: Names and contact details of principal consultant, subconsultants, Contractor, subcontractors and name of main contact.

Record drawings: Include complete set of record drawings, full size.

Drawings and technical data: Include as necessary for the efficient operation and maintenance of the installation.

Installation description: Include a general description of the installation.

Systems descriptions and performance: Include a technical description of the systems installed including the basis of design, the interrelation with other systems and the building and mode of operation, presented in a clear and concise format readily understandable by the Principal's staff. Identify function, normal operating characteristics, safety features and limiting conditions.

Baseline data: Include the baseline data to **BASELINE DATA**.

Commissioning records: Include commissioning records to SA TS 5342 (2021). Link commissioning records to item codes on the record drawings.

Training material: Include materials used to provide training, to **TRAINING**, in a form that can be used to train others.

Fire systems and equipment: Include Documentation to AS 1851 (2012), including the schedule of essential functionality and performance requirements.

Digital photographic and video records: Provide Documented digital photographic and video records, keyed to the drawings.

Equipment: Include schedules with the following details for installed equipment:

- Item code for use on record and diagrammatic drawings, and spare parts schedule.
- Equipment name plate data including serial number, if any.
- Name and contact details of the manufacturer and supplier.
- Catalogue list number(s).
- Location.
- Function.
- Performance figures and capacity data.
- Date of manufacture.
- Manufacturer's product data sheets including only relevant matter for the project. Mark each product data sheet to clearly identify specific products and component parts used in the installation, and data applicable to the installation.
- Additional information and commentary to illustrate relations of component parts.

Certificates:

- Certificates from authorities.
- Product certification.
- Test certificates for each service installation and all equipment.
- Warranties.

Trends: 7 day record of all trends at commissioning.

Operation procedures: Include for systems installed:

- Manufacturer's technical literature as appropriate.
- Safe starting up, running-in, operating and shutting down procedures. Include logical step-by-step instructions for each procedure.
- Control sequences and flow diagrams.
- Legend for colour-codes services.

- Schedules of fixed and variable equipment settings established during commissioning and maintenance.
- A list of special safety devices and their set points.
- Procedures for seasonal changeovers.
- Warnings to operators.
- Procedures for identifying and rectifying common faults.
- Recommendations for efficient plant operation.
- If the installation includes cooling towers, recommendations for water efficiency.
- Building tuning plan and procedure to **COMMISSIONING, Commissioning plan.**

Building occupants' guide: Include a concise guide written and illustrated for building occupants with no technical background. Include the following:

- Security provisions.
- Safety and access.
- Environmental features, including energy and water efficiency and waste management.
- Occupant relevant information on design and operation.
- Information for occupants on environmental systems that rely partially or wholly on local controls for heating, lighting, cooling, and ventilation.
- Contact details for faults, maintenance and emergencies.

Maintenance procedures:

- Detailed recommendations for periodic maintenance and procedures, including schedule of maintenance work with frequency and manufacturer's recommended Tests.
- Manufacturer's technical literature as appropriate. Register with manufacturer as necessary. Retain copies delivered with equipment.
- Safe trouble-shooting, disassembly, repair and reassembly, cleaning, alignment and adjustment, balancing and checking procedures. Provide logical step-by-step instructions for each procedure.
- Schedule of spares, recommended to be held on Site, for those items subject to wear or deterioration and that may involve the Principal in extended deliveries when replacements are required. Include complete nomenclature and model numbers, and local sources of supply.
- Schedule of normal consumable items, local sources of supply, and expected replacement intervals up to a running time of 40 000 hours. Include lubrication schedules for equipment.
- Instructions for use of tools and Testing equipment.
- Troubleshooting procedures.
- Emergency procedures, including telephone numbers for emergency services, and procedures for fault finding.
- Safety data sheets (SDS).
- Instructions and schedules conforming to AS 1851 (2012), AS/NZS 3666.2 (2011), AS/NZS 3666.3 (2011) and AS/NZS 3666.4 (2011).

Maintenance records:

- Prototype routine service records conforming to AS 1851 (2012) prepared to include project specific details.
- Prototype periodic maintenance records and report to AS/NZS 3666.2 (2011), AS/NZS 3666.3 (2011) and AS/NZS 3666.4 (2011) as appropriate, prepared to include project specific details.
- Hard copies: Binders to match the manuals, containing loose leaf logbook pages designed for recording completion activities including operational and maintenance procedures, materials used, Test results, comments for future maintenance actions and notes covering the condition of the installation. Include completed logbook pages recording the operational and maintenance activities performed up to the Date for Practical Completion.
- Number of pages: The greater of 100 pages or enough pages for the maintenance period and a further 12 months.

Emergency information: For each type of emergency, including fire, flood, gas leak, water leak, power failure, water failure, system or subsystem failure, chemical release or spill, include the following:

- Emergency instructions.
- Emergency procedures including:
  - . Instructions for stopping or isolating.
  - . Shutdown procedures and sequences.
  - . Instructions for actions outside the property.
  - . Special operating instructions relevant to the emergency.
  - . Contact details relevant to the emergency.

**Emergency information manual**

Form of emergency information: Provide one of the following:

- An index and coloured tabs identifying emergency information for each type of emergency within the operation and maintenance manual.
- A separate Emergency manual containing copies of emergency information from the main operation and maintenance manual.

**Format – electronic copies**

Scope: Provide the same material as Documented for hardcopy in electronic format.

Quantity and format: Conform to **SUBMISSIONS, Electronic submissions**.

Printing: Except for drawings required in **RECORD DRAWINGS** provide material that can be legibly printed on A4 size paper.

**Format – hard copies**

General: A4 size loose leaf, in commercial quality, 4 ring binders with hard covers, each indexed, divided and titled. Include the following features:

- Cover: Identify each binder with typed or printed title *OPERATION AND MAINTENANCE MANUAL*, to spine. Identify title of project, volume number, volume subject matter, and date of issue.
- Dividers: Durable divider for each separate element, with typed description of system and major equipment components. Clearly print short titles under laminated plastic tabs.
- Drawings: Fold drawings to A4 size with title visible, insert in plastic sleeves (one per drawing) and accommodate them in the binders.
- Pagination: Number pages.
- Ring size: 50 mm maximum, with compressor bars.
- Text: Manufacturer's printed data, including associated diagrams, or typewritten, single-sided on bond paper, in clear concise English.

Number of copies: 3.

**Date for submission**

Draft submission: The earlier of the following:

- 4 weeks before the Date for Practical Completion.
- Commencement of training.

Final submission: Within 2 weeks after the Date of Practical Completion.

---

**11 MAINTENANCE**

---

**11.1 PERIODIC MAINTENANCE****General**

Requirement: Provide Documented maintenance so that the condition and performance of the maintained work throughout and at the end of the maintenance period is equal to or better than that at the beginning of the maintenance period including with respect to the following:

- Performance, service delivery.
- Service life and reliability.
- Compliance with statutory requirements.
- Compliance with building rating requirements.
- Energy and water efficiency.
- Environmental impact.

- Health and safety.
- Risk management.

Inclusions: Include the following:

- Periodic and statutory maintenance, cleaning and replacement of consumables.
- Emergency repairs.
- Condition reporting.

Duration: From the time systems and equipment are put into service to the end of the maintenance period.

Maintenance period: The greater of the Defects Liability Period and the period Documented.

Faults: Rectify promptly.

Emergencies: Attend emergency calls promptly.

Annual maintenance: Carry out recommended annual maintenance procedures within the four weeks before the end of the maintenance period.

### **Maintenance Program**

General: Submit details of maintenance procedures and program, relating to installed plant and equipment, 6 weeks before the Date for Practical Completion. Indicate dates of service visits. State contact telephone numbers of service operators and describe arrangements for emergency calls.

### **Maintenance records**

General: Record in binders provided with the operation and maintenance manuals.

Referenced Documents: If referenced Documents or technical worksections require that logbooks or records be submitted, include this material in the maintenance records.

Certificates: Include Test and approval certificates.

Service visits: Record comments on the functioning of the systems, work carried out, items requiring corrective action, adjustments made and name of service operator. On completion of the visit, obtain the signature of the Principal's designated representative on the record of the work undertaken.

### **Site control**

General: Report to the Principal's designated representative on arriving at and before leaving the Site.

## **11.2 STATUTORY INSPECTIONS AND MAINTENANCE**

### **General**

Duration: From the time systems and equipment are put into service to the end of the maintenance period.

Requirement: Provide inspections and maintenance of safety measures required by the following:

- AS 1851 (2012).
- Other statutory requirements applicable to the WUC.

Records: Provide mandatory records.

Certification: Certify that mandatory inspections and maintenance have been carried out and that the respective items conform to statutory requirements.

Annual inspection: Perform an annual inspection and maintenance immediately before the end of the maintenance period.

## **12 SELECTIONS**

### **12.1 PERFORMANCE**

#### **Noise level schedule**

|                                         | <b>Whole Project</b>       |
|-----------------------------------------|----------------------------|
| Upper limit of noise caused by services | Refer mechanical Documents |

**12.2 SUBMISSIONS AND INSPECTIONS****Notices schedule**

| Item                           | Minimum notice  |
|--------------------------------|-----------------|
| On-Site or Off-Site Inspection | 5 Business Days |

**12.3 TESTS****Tests schedule**

| Test           | Requirements                                             |
|----------------|----------------------------------------------------------|
| Façade Testing | Refer to Façade Engineer Performance Based Design Brief. |

**12.4 MAINTENANCE****Maintenance requirements schedule**

Refer to Landscape Architect, Hydraulic, Mechanical, and Electrical Documents.

|                                      |
|--------------------------------------|
| <b>0172 ENVIRONMENTAL MANAGEMENT</b> |
|--------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide environmental management, as Documented.

**Management and control plans**

Implementation: To Documented management plans.

**Principle HSE requirements**

Requirements: Environmental management to conform with Stanwell Document: Principal Contractor HSE Requirements.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Clearances: A formal certificate, approval or condition issued by a statutory authority allowing work in a particular area.
- Contamination of land: The presence of a substance in, on or under the land that is designated hazardous material and/or is at a concentration above that which is normally found in that locality, and presents a risk of harm to human health or to the environment.
- Environment: The physical factors of the surroundings of human beings including the land, waters, atmosphere, climate, sound, odours, tastes; and the biological factors of animals; and plants and the social factor of aesthetics.
- Environmental audit: A review of environmental management practices, in particular the evaluation of a site for environmental liability.
- Environmental impact assessment (EIA): A method for predicting environmental impacts of a proposed development including strategies for minimising identified impacts.
- Environmental management plan (EMP): A project or site specific plan describing the management of the environmental issues and considerations for the activity being undertaken. This applies to the design, construction and operation of the buildings, external works and infrastructure.
- Organic waste: Includes all food wastes, vegetative wastes from land clearing and pruning operations, biosolids produced from the treatment of liquid wastes, garden wastes and forestry waste (bark and saw dust) and paper and cardboard products.
- Pollution Incident: An Incident or set of circumstances during or as a consequence of which there is, or is likely to be, a leak, spill or other escape of a substance as a result of which pollution has occurred, is occurring or is likely to occur.
- Weed: An invasive plant that degrades natural areas, reduces the sustainability or affects the health of people and animals.

**1.4 SUBMISSIONS**

Control plans

**Requirement: Submit the following:**

- **Environmental management plan.**

Time for submissions: before any Work has commenced on Site.

## 1.5 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Non-conforming items, such as ground contamination.
- Completed removal or rectification of non-conforming items.
- Unexpected finds.

## 2 ENVIRONMENTAL ADMINISTRATION, MONITORING AND REPORTING ---

### 2.1 PROCEDURAL AND PERSONNEL

#### Emergency response

Emergency response personnel: Provide staff names and contact details.

#### Complaints

Reporting: Within 1 working day of receiving a complaint about an environmental impact, including pollution incidents, prepare a written report detailing the complaint and remedial action taken.

Register: Keep a register of all complaints and action taken.

#### Unexpected finds

Requirement: Give notice and close off affected Site area with barrier tapes and warning signs to prevent access.

Unexpected finds: Include asbestos and other hazardous or volatile contaminants, archaeological finds and items of heritage value.

### 2.2 MONITORING

#### Internal monitoring

Approval authority: the Superintendent

Documentation: Provide details of the following:

- Environmental monitoring: Implementation and recording procedures.
- Control measures to be implemented: Non-conformance control and corrective action procedures.

Records: Maintain records of results of environmental monitoring, including the effectiveness of any remedial action taken.

Internal monitoring personnel: Provide staff names and contact details.

Machinery and equipment: Provide details of proposed plant.

Review timing: Review the EMP or control measures at the following stages:

- When there is a change in the project, including the scope.
- Following significant environmental impacts or pollution incidents.
- When improved performance is required to reduce a specific environmental impact.
- At completion of environmental audits.
- At the end of the project.

### 2.3 REPORTING

#### General

Requirement: Prepare environmental management reports that record the progress of the following:

- Performance against statutory requirements.
- Performance against the EMP, environmental objective and policy, and ecologically sustainable development outcomes and targets.
- Summary of monitoring, inspection and audits.
- Summary of reports required to meet the statutory requirements.
- Summary of environmental impacts, pollution incidents, non-conformance and complaints.
- Summary of corrective actions where required.
- Unexpected finds.

Reporting frequency: End of each month

### 3 ENVIRONMENTAL MANAGEMENT

#### 3.1 ENVIRONMENTAL MANAGEMENT PLAN

##### Control plan

EMP: Prepare an environmental management plan with the following details:

- Project description, including Site location, construction activities and project schedule.
- EMP context, describing how the EMP fits into the overall project planning process.
- EMP objective and environmental policy.
- Risk assessment.
- Assignment of responsibility for environmental controls, including hierarchy of management.
- Conditions of approvals, licences and permits to meet statutory requirements.
- Reporting requirements.
- Environmental training plan and procedures: Including a program to familiarise staff with the EMP and/or management controls, environmentally sensitive areas and responsibilities.
- Environmental audit program and corrective action procedures.
- Emergency response procedures, including response time.
- Control plans, as required.
- Details of operational control measures to reduce risk of environmental impacts on the following:
  - . Heritage.
  - . Visual values.
  - . Endangered species.
  - . Habitat.
- Locations of, and environmental controls for, environmentally sensitive areas.
- Details of environmental protection for each activity.
- Communication procedures.
- Other items necessary to protect the surrounding environment.

Activities staging: Address the phases of activity, as appropriate:

- Before construction and Site establishment.
- During construction.
- After construction, including rehabilitation activities and Site and landscaping maintenance.

#### 3.2 SOIL EROSION AND SEDIMENT CONTROL

##### Control plan

Plan: Prepare a soil erosion and sediment control plan to protect natural overland flow paths and stormwater drainage systems from sedimentation by minimising soil erosion and sediment transportation. Include the following details:

- Staging of operations and sequence of Works.
- Diversion of upstream water around the Site.
- Provision of temporary drains and catch drains.
- Erosion control measures.
- Sediment control measures.
- Stormwater control measures, including diversion, dispersal and/or retention to control and dissipate stormwater throughout the Site without damage.
- Maintenance procedures for control measures.
- Progressive restoration of disturbed areas.

Extent: All Site areas, access and haulage tracks, borrow pits, stockpile and storage areas and compound areas.

##### Control measures

Requirement: Protect natural overland flow paths and stormwater drainage systems from sedimentation by minimising soil erosion and sediment transportation. Refer also to Civil Engineering Documents.

### 3.3 WASTE MANAGEMENT

#### Control plan

Plan: Prepare a waste management control plan to minimise solid waste generated by construction activities and to dispose of the solid waste. Include the following details:

- Major waste streams.
- For each waste stream describe the following:
  - . Location.
  - . Disposal method.
  - . Transport method.
  - . Storage method.

### 3.4 GROUND CONTAMINATION

#### Control plan

Plan: Prepare a ground contamination control plan for land suspected of being contaminated or containing presence of acid sulfate soil, in conformance with the state or territory Environmental Protection Authority (EPA) and the relevant state or territory planning guidelines. Include the following details:

- Preliminary investigation.
- Detailed investigation.
- Site Remedial Action Plan (RAP).
- Stockpile sites.
- Site auditing and reporting procedures.
- Record of maintenance procedures, including record of remediation work, certificates issued and restrictions placed on the Site.

### 3.5 WEED MANAGEMENT

#### Control plan

Plan: Prepare a weed management control plan with the following details:

- Weed species.
- Weed infestation zones within the work Site and the investigation period.
- Treatment of infestation, including treatment type and application rate.
- Cleaning methods for vehicles and machinery, including cleaning bay location and treatment date.
- Subcontractors: Names and contact details for proposed subcontractors and evidence of experience in treatment of weed infestations and name and qualifications of chemical treatment handlers.

### 3.6 AIR QUALITY

#### Control plan

Plan: Prepare an air quality control plan to protect adjoining owners, residents and the public from emissions of dust and exhaust gases. Include the following details:

- Dust control.
- Exhaust gases emission control.

### 3.7 WATER QUALITY

#### Control plan

Plan: Prepare a water quality control plan to keep earthworks free of water and to reduce impacts on groundwater and surface water. Include the following details:

- Dewatering system.
- Washout prevention.
- Cross connection prevention to public sewerage system.
- Water disposal.

#### Control measures

Requirement: Keep earthworks free of water and reduce impacts on groundwater and surface water. Refer also to Civil Engineering Documents.

### **3.8 NOISE CONTROL AND VIBRATION**

#### **Control plan**

Plan: Prepare a noise and vibration control plan to protect adjoining owners, residents and the public from noise and vibration impacts. Include the following details:

- Noise control and vibration: To the recommendations of AS 2436 (2010).
- Maximum noise level at the Site boundary.
- Noise control.
- Vibration assessment.
- Vibration control.

### **3.9 FLORA AND FAUNA**

#### **Control plan**

Plan: Prepare a flora and fauna control plan to protect native flora and fauna on Site and prevent introduction of pest species. Include the following details:

- Native flora and fauna assessment.
- Pest species assessment.
- Tree pruning.
- Habitat provision.

### **3.10 CULTURAL AND ABORIGINAL AND/OR TORRES STRAIT ISLANDER HERITAGE**

#### **Control plan**

Plan: Prepare a cultural heritage management plan to protect and prevent damage or loss of items of cultural heritage or Aboriginal and/or Torres Strait Islander sites and artefacts. Include the following details:

- Cultural heritage assessment.
- Aboriginal and/or Torres Strait Islander heritage assessment.
- Unexpected finds.
- Training.

|                                               |
|-----------------------------------------------|
| <b>0181 ADHESIVES, SEALANTS AND FASTENERS</b> |
|-----------------------------------------------|

## **1 GENERAL**

### **1.1 RESPONSIBILITIES**

#### **General**

Requirement: Provide adhesives, sealants and fasteners, as Documented.

#### **Performance**

Requirement: Conform to the following:

- Fitness for purpose: Suitable for particular use, capable of transmitting imposed loads, and sufficient to maintain the rigidity of the assembly and integrity of the joint.
- Discolouration: Will not cause discolouration of the finished surface.
- Compatibility: Compatible with the products to which they are applied.
- Sealant replacement: Capable of safe removal without compromising the application of the replacement sealant for future refurbishment.
- Movement: If an adhered or sealed joint is subject to movement, accommodate the projected movement under the conditions of service.

### **1.2 CROSS REFERENCES**

#### **General**

Requirement: Conform to the following:

- 0171 General requirements.

### **1.3 SUBMISSIONS**

#### **Products and materials**

Manufacturer's data: Submit the manufacturer's product data sheets for each type of adhesive and sealant to be used.

Type Tests: Submit Test results demonstrating that the adhesive, sealant or fastener is compatible with the materials to be adhered, sealed or fixed and is suitable for the project conditions.

#### **Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

#### **Warranties**

Requirement: Submit warranties to **COMPLETION**, **Warranties**.

## **2 PRODUCTS**

### **2.1 GENERAL**

#### **Samples**

Visible joint sealants: Provide samples for each sealant showing the colour once cured.

#### **Storage and handling**

Adhesives and sealants: Maintain temperatures of between +5°C to +25°C in the storage area and within the manufacturer's recommended range. Do not use materials that have been stored longer than the manufacturer's maximum recommended shelf life.

### **2.2 ADHESIVES**

#### **Standards**

Gypsum plaster adhesive: To AS 2753 (2018).

#### **Contact adhesive**

Compatibility: Select adhesives that are compatible with the substrates to be bonded.

#### **High strength adhesive tape**

Description: A foam of cross linked polyethylene or closed cell acrylic coated both sides with a high performance acrylic adhesive system, encased in release liners of paper or polyester.

Type: Select tape to suit substrate as follows:

- High energy surfaces, including most bare metals such as stainless steel and aluminium: Firm high strength foam tapes.
- Medium energy surfaces, including many plastics, paints and unfinished metals: Conformable high strength foam tapes.
- Low energy surfaces, including many plastics, most paints and powder coatings, and unfinished metals: Conformable high strength foam tapes.

Thickness: Select tape so that the height deviation to adjoining surfaces does not exceed half the tape thickness under the required application pressure.

#### **Total VOC limits**

Requirement: Conform to the following maximum limits:

- General purpose adhesives: 50 g/L.
- Structural glazing adhesives, and timber flooring and laminate adhesives: 100 g/L.

### **2.3 SEALANTS**

#### **Standards**

General: To ISO 11600 (2002).

#### **External masonry joints**

General: Provide sealant and bond breaking materials that are non-staining to masonry. Do not use bituminous materials with absorbent masonry units.

Bond breaking backing:

- Bond breaking materials: Non-adhesive to sealant, or faced with a non-adhering material.
- Foamed materials: Closed cell or impregnated, not water-absorbing.

#### **Lightweight building element joints**

Joints subject to rapid changes of movement: Provide sealants that accommodate the movement of the contact materials.

#### **Floor control joints**

General: Provide trafficable sealants.

Bond breaking backing:

- Bond breaking materials: Non-adhesive to sealant, or faced with a non-adhering material.
- Foamed materials: Closed cell or impregnated, not water-absorbing.

#### **Total VOC limits**

Requirement: Conform to the following maximum limits:

- General purpose sealants: 50 g/L.
- Acoustic sealants, architectural sealants and waterproofing sealants: 250 g/L.
- Timber flooring and laminate sealants: 100 g/L.

### **2.4 FASTENERS**

#### **General**

Masonry anchors: Proprietary expansion or bonded type anchors, as Documented.

Post-installed and cast-in fasteners in concrete: To AS 5216 (2021).

Plain washers: To AS 1237.1 (2002). Provide washers to the heads and nuts of bolts, and the nuts of coach bolts.

Plugs: Proprietary purpose-made plastic.

Stainless steel fasteners: To ASTM A276/A276M (2025).

Steel nails: To AS 2334 (1980). Provide nails with length conforming to the following:

- Generally: At least 2.5 times the thickness of the member being secured.
- Sheet or panel products, including plywood, blockboard, fibreboard and hardboard: If less than 10 mm thick, at least 4 times the thickness of the sheet or panel product being secured.

Unified hexagon bolts, screws and nuts: To AS/NZS 2465 (1999).

#### **Bolts**

Coach bolts: To AS/NZS 1390 (1997).

Hexagon bolts - Grades A and B: To AS 1110.1 (2015).

Hexagon bolts - Grade C: To AS 1111.1 (2015).

### Nuts

Hexagon chamfered thin nuts - Grades A and B: To AS 1112.4 (2015).

Hexagon nuts - Grade C: To AS 1112.3 (2015).

Hexagon nuts - Style 1 Grades A and B: To AS 1112.1 (2015).

Hexagon nuts - Style 2 Grades A and B: To AS 1112.2 (2015).

### Screws

Coach screws: To AS/NZS 1393 (1996).

Hexagon screws - Grades A and B: To AS 1110.2 (2015).

Hexagon screws - Grade C: To AS 1111.2 (2015).

Hexagon socket screws: To AS 1420 (2008).

Self-drilling screws: To AS 3566.1 (2002).

Self-tapping screws:

- Cross-recessed countersunk (flat - common head style): To AS/NZS 4407 (2015).
- Cross-recessed pan: To AS/NZS 4406 (2015).
- Cross-recessed raised countersunk (oval): To AS/NZS 4408 (2015).
- Hexagon: To AS/NZS 4402 (2015).
- Hexagon flange: To AS/NZS 4410 (2015).
- Hexagon washer: To AS/NZS 4409 (2015).
- Slotted countersunk (flat - common head style): To AS/NZS 4404 (2015).
- Slotted pan: To AS/NZS 4403 (2015).
- Slotted raised countersunk (oval - common head style): To AS/NZS 4405 (2015).

### Blind rivets

Description: Expanding end type with snap mandrel.

Type:

- External application: Closed end.
- Internal application: Open end.

End material:

- Metallic-coated or prepainted steel: Aluminium base alloy.
- Stainless steel sheet: Stainless steel.
- Copper sheet: Copper.

Size:

- Sheet metal to sheet metal: 3 mm.
- Sheet metal to supports, brackets and rolled steel angles: 4.8 mm.

### Corrosion resistance

Atmospheric corrosivity category: To 0171 General requirements.

Steel products: Conform to the **Corrosion resistance table** or provide proprietary products with metallic and/or organic coatings of equivalent corrosion-resistance.

#### Corrosion resistance table

| Atmospheric corrosivity category to AS 4312 (2019) | Threaded fasteners and anchors           |                                               | Powder actuated fasteners |
|----------------------------------------------------|------------------------------------------|-----------------------------------------------|---------------------------|
|                                                    | Material                                 | Minimum local metallic coating thickness (µm) | Material                  |
| C1 and C2                                          | Electroplated zinc or hot dip galvanized | 30                                            | Stainless steel Type 316  |
| C3                                                 | Hot dip galvanized                       | 45                                            | Stainless steel Type 316  |

| Atmospheric corrosivity category to AS 4312 (2019)                                   | Threaded fasteners and anchors |                                               | Powder actuated fasteners |
|--------------------------------------------------------------------------------------|--------------------------------|-----------------------------------------------|---------------------------|
|                                                                                      | Material                       | Minimum local metallic coating thickness (µm) | Material                  |
| C4                                                                                   | Stainless steel Type 316       | -                                             | Stainless steel Type 316  |
| Note: For categories C5, CX and T to the AS/NZS 2312 series, seek specialist advice. |                                |                                               |                           |

**Finishes**

Electroplating:

- Metric thread: To AS 1897 (2016).
- Imperial thread: To AS 4397 (2007).

Galvanizing:

- Threaded fasteners: To AS/NZS 1214 (2016).
- Other fasteners: To AS/NZS 4680 (2025).

Mild steel fasteners: Provide galvanized mild steel fasteners for the following locations:

- Embedded in masonry.
- In external timbers.
- Exposed to or in air spaces behind the external leaf of masonry walls.
- In contact with chemically treated timber other than CCA treated timber.

Fasteners in CCA treated timber: Provide epoxy coated or stainless steel fasteners.

**3 EXECUTION****3.1 ADHESIVES****General**

Requirement: Apply to the manufacturer's recommendations.

**Substrate preparation**

Requirement: Conform to the following:

- Remove any deposit or finish that may impair adhesion.
- Framed or discontinuous substrates: Provide support members in full lengths without splicing.
- Solid or continuous substrates: Remove excessive projections.
- Previously painted substrates: Remove cracked or flaking paint and lightly sand the surface.

Surfaces where adhesive tape will be applied: Conform to the following:

- Non-porous surfaces: Clean with surface cleaning solvents such as isopropyl alcohol/water, wash down and allow to dry.
- Porous surfaces: Prime the surface with a contact adhesive compatible with the adhesive tape.

**Contact adhesive**

Requirement: Make sure application of the adhesive solvent will not discolour the finished surface and dispersal of the adhesive solvent is not impaired.

Substrate: Do not use contact adhesive if one of the substrates is polystyrene foam. If contact adhesive is to be applied to a PVC substrate, make sure there is no plasticiser migration.

Two-way method: Immediately after application, press firmly to transfer adhesive and then pull both surfaces apart. Allow to tack off and then reposition and press firmly together. Tap areas in contact with a hammer and padded block.

One-way method: Immediately after application, bring substrates together and maintain maximum surface contact for 24 hours by clamps, nails or screws as appropriate. If highly stressed, use permanent mechanical fasteners.

**High strength adhesive tape**

Application pressure: Make sure the tape undergoes at least 100 kPa pressure.

Application temperature: Do not apply at temperatures below 10°C.

Completion: Do not apply loads to the assembly for at least 72 hours.

### 3.2 SEALANTS

#### General

Requirement: Apply to the manufacturer's recommendations.

#### Joint preparation

Cleaning: Cut flush joint surface protrusions and rectify if required. Mechanically clean joint surfaces free of any deposit or finish that may impair adhesion of the sealant. Immediately before sealant application, remove loose particles from the joint, using oil-free compressed air.

Bond breaking: Install bond breaking backing material.

Protection: Provide 50 mm wide protection tape on each side of the joint.

Primer: Apply the recommended primer to the surfaces in contact with sealant materials.

#### Conditions

Requirement: Do not apply at temperatures below 5°C or above 40°C, and in conditions outside the manufacturer's recommendations.

#### Application

General: Apply the sealant to dry joint surfaces using a pneumatic applicator gun. Make sure the sealant completely fills the joint to the required depth, provides good contact with the full depth of the sides of the joint and traps no air in the joint. Do not apply the sealant outside the recommended working time for the material or the primer.

Joint proportions (width:depth):

- Joints not more than 12 mm wide: 1:1.
- Joints more than 12 mm wide: 2:1.

Joint finish: Finish with a smooth, slightly concave surface using a tool designed for the purpose.

Remove excess sealant from adjoining surfaces using cleaning material recommended by the sealant manufacturer.

#### Protection

General: Protect the joint from inclement weather during the setting or curing period of the material.

### 3.3 FASTENERS

#### General

Requirement: Install to the manufacturer's recommendations.

Fasteners in timber substrates: To AS 1720.1 (2010) Section 4.

Self-drilling screws in timber and steel substrates: To AS 3566.1 (2002).

Post-installed and cast-in fasteners in concrete substrates: To AS 5216 (2021) Section 2.

### 3.4 COMPLETION

#### Cleaning

Joint sealing: Remove protection tape and remove any stains or marks from adjacent surfaces.

#### Rectification

Sealants: Cut out and remove damaged portion of joint sealant and reinstall so repaired area is indistinguishable from undamaged portion.

#### Warranties

Type: Interlocked supply and installation

## 4 SELECTIONS

---

### 4.1 GENERAL

Unless noted otherwise in Documents, selection are to be made by the Contractor and are to be:

1. Fit for purpose
2. Conforming with the above criteria
3. Approved by way of a sample where visible

|                                    |
|------------------------------------|
| <b>0183 METALS AND PREFINISHES</b> |
|------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirements: Provide metal and prefinishes, as Documented. Refer to Appendix A – T-Sheet, and Appendix B – Façade and Wall Schedule.

**Performance**

Requirement: Provide metals in sections of strength and stiffness suited to their required function, finish and method of fabrication.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 SUBMISSIONS****Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples of the following:

- Stainless steel: One sample of every Documented surface finish.
- Anodising: One sample of every colour and finishing option.

**2.2 METALS****Aluminium and aluminium alloys**

Drawn pipe: To AS/NZS 1867 (1997).

Drawn rod, bar and strip: To AS/NZS 1865 (1997).

Extrusions: To AS 1866 (1997).

Plate and sheets: To AS/NZS 1734 (1997).

**Copper and copper alloys**

Casting: To AS 1565 (2023).

Plate, sheet and strip: To AS 1566 (1997).

Rods, bars and sections: To AS/NZS 1567 (2023).

Composition and designations: To AS 2738 (2023).

**Stainless steel**

Bars: To ASTM A276/A276M (2025).

Plate, sheet and strip: To ASTM A240/A240M (2025).

Welded pipe (plumbing applications): To AS 1769 (1975).

Welded pipe (round, square, rectangular): To ASTM A554 (2021).

**Steel**

Sheet: To AS/NZS 1595 (1998).

Structural bars and sections: To AS/NZS 3679.1 (2016).

Structural hollow sections: To AS/NZS 1163 (2016).

**Steel for prefinishes**

Cold-rolled bar: Bright bars to AS 1443 (2004).

Cold-rolled sheet: To AS/NZS 1595 (1998).

Electric resistance welded tube: To AS 1450 (2007).

### 3 EXECUTION

---

#### 3.1 GENERAL

##### **Metal separation**

Incompatible sheet metals: Prevent direct contact between incompatible metals. Provide separation by one of the following:

- Apply an anti-corrosion, low moisture transmission coating such as alkyd zinc phosphate primer or aluminium pigmented bituminous paint to contact surfaces.
- Insert a concealed, non-conductive separation layer such as polyethylene film, adhesive tape, neoprene, nylon or bituminous felt.

Incompatible fastenings: Do not use.

Incompatible service pipes: Install lagging or grommets. Do not use absorbent, fibrous or paper products.

##### **Brazing**

Lap-joints: Make sure brazed lap-joints have sufficient lap to provide a mechanically sound joint.

Butt joints: Do not use butt jointing for joints subject to load. If butt joints are used, do not rely on the filler metal fillet only.

Filler metal: To AS/NZS ISO 17672 (2025).

##### **Soldering**

Lap-joints: Provide a mechanically sound soldered joint with sufficient lap for roofing, guttering, metalwork.

Pipes: Make a leakproof soldered joint using joiners for copper pipes.

Solder: To AS 1834.1 (1991).

##### **Welding**

Aluminium: To AS/NZS 1665 (2004).

Stainless steel: To AS/NZS 1554.6 (2012).

Steel: To AS/NZS 1554.1 (2014).

##### **Finishing**

Visible joints: Finish visible joints made by welding, brazing or soldering using methods appropriate to the class of Work (including grinding or buffing) before further treatment such as painting, galvanizing or electroplating. Make sure self-finished metals are without surface colour variations after jointing.

##### **Preparation**

General: Before applying decorative or protective prefinishes to metal components, complete welding, cutting, drilling and other fabrication, and prepare the surface using a suitable method.

Standard: To the AS 1627 series.

Priming steel surfaces: If Site painting is Documented to otherwise uncoated mild steel or similar surfaces, prime as follows:

- After fabrication and before delivery to the Works.
- After installation, repair damaged priming and complete the coverage to unprimed surfaces.

#### 3.2 FERROUS STEEL FINISHES

##### **Metallic-coated steel**

General: Steel coated with zinc or aluminium-zinc alloy as follows:

- Electrogalvanized (zinc) coating on ferrous hollow and open sections: To AS 4750 (2003).
- Ferrous open sections by an in-line process: To AS/NZS 4791 (2006).
- Ferrous hollow sections by a continuous or specialised process: To AS/NZS 4792 (2006).
- Steel sheet and strip: To AS 1397 (2021).
- Steel wire: To AS/NZS 4534 (2006).

### 3.3 STAINLESS STEEL FINISHES

#### General

Requirement: Provide a surface finish to match the sample submitted.

#### Pre-assembly

Mechanically polished and brushed finishes: Apply grit faced belts or fibre brushes that achieve uni-directional finishes with buffing, as required to provide the following:

- Finish designation: No. 4

Bead blasted finish: Provide a uniform non-directional low reflective surface by bead blasting. Do not use sand, iron or carbon steel shot. Blast both sides of austenitic stainless steel to equalise induced stress.

#### Post-assembly pre-treatment

Heat discolouration: Remove by pickling to ASTM A380/A380M (2025).

Welds: Grind excess material, brush, and polish to match the pre-assembly finish.

#### Post-assembly finish

Electropolish finish: Provide an electro-chemical process to stainless steel Type 316.

Brushed electropolish finish: Conform to the following:

- Pre-assembly finish: No. 4 polished.
- Post-assembly finish: Provide an electro-chemical process to achieve a surface roughness  $R_a$ , no greater than 0.50  $\mu\text{m}$ .

Mirror finish: Conform to the following:

- Pre-assembly finish: 2B cold-rolled finish.
- Post-assembly finish: Apply a polishing and buffing process to achieve a No. 8 mirror finish.

#### Completion

Cleaning: Clean and rinse to an acid free condition and allow to dry. Do not use carbon steel abrasives or materials containing chloride.

Protection: Secure packaging or strippable plastic sheet.

### 3.4 NON-FERROUS METAL FINISHES

#### Mechanical finishes

Bright finished copper alloy surfaces: For indoor applications, apply a clear lacquer protective coating.

### 3.5 ELECTROPLATED FINISHES

#### Electroplated coatings

Chromium on metals: To AS 1192 (2004).

- Service condition number: At least 2.

Nickel on metals: To AS 1192 (2004).

- Service condition number: At least 2.

Zinc on iron or steel: To AS/NZS 1789 (2023).

### 3.6 ANODISED FINISHES

#### General

Standard: To AS 1231 (2000).

Thickness grade: To the recommendations of AS 1231 (2000) Appendix H.

### 3.7 SPRAYED METAL FINISHES

#### Thermal metal spraying

Standard: To AS/NZS 2312.3 (2025).

Minimum thicknesses: To the recommendations of AS/NZS 2312.3 (2025) Table 2.

Surface preparation: To AS/NZS 2312.3 (2025) Section 5.

Application process: Arc spray.

Sealing: Apply the sealer as soon as possible after metal spraying. Continue application of the sealer until absorption into the sprayed metal coating is complete.

### 3.8 PREPAINTED FINISHES

#### **Air-drying enamel**

Application: Spray or brush.

Finish: Full gloss.

General use:

- Primer: Two-pack epoxy primer to AS/NZS 3750.13 (1997).
- Topcoats: 2 coats to AS 3730.6 (2006).

Oil resistant use:

- Primer: Two-pack epoxy primer to AS/NZS 3750.13 (1997).
- Topcoats: 2 coats to AS/NZS 3750.22 (2008).

#### **Equipment paint system**

Description: Brush or spray application using paint as follows:

- Full gloss enamel finish coats, oil and petrol resistant: To AS/NZS 3750.22 (2008), two coats.
- Prime coat to metal surfaces generally: To AS/NZS 3750.19 (2008) or AS/NZS 3750.20 (2008).
- Prime coat to zinc-coated steel: To AS 3730.15 (2006) or AS/NZS 3750.16 (1998).
- Undercoat: To AS/NZS 3750.21 (2008).

#### **Prepainted metal products**

Standard: To AS/NZS 2728 (2013).

Product type: To AS/NZS 2728 (2013): Not lower than the type appropriate to the Documented atmospheric corrosivity category.

#### **Stoving enamel**

Application: Spray or dip.

#### **Two-pack liquid coating**

Application: Spray.

Finish: Full gloss.

Primer: Two pack epoxy primer to AS/NZS 3750.13 (1997).

Topcoat:

- Internal use: Proprietary polyurethane or epoxy acrylic system.
- External use: Proprietary polyurethane system.

### 3.9 COMPLETION

#### **Damage**

Damaged prefinishes: Remove and replace items, including damage caused by unauthorised Site cutting or drilling.

#### **Repair**

Anodising: Use sprayers or pens for minor scratches and mitre cuts as required.

Metallic-coated sheet: If repair is required to metallic-coated sheet or electrogalvanizing on inline galvanized steel products, clean the affected area and apply a two-pack organic primer to AS/NZS 3750.9 (2009).

#### **Cleaning**

General: On completion, clean all surfaces. Do not use abrasive cleaners.

|                                |
|--------------------------------|
| <b>0184 TERMITE MANAGEMENT</b> |
|--------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide termite management systems, as Documented.

**Performance**

Requirement: Building protection from termite attack.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARD****General**

Termite management systems: To AS 3660.1 (2014).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection the definitions given in AS 3660.1 (2014) apply.

**1.5 SUBMISSIONS****Certification**

Installation: On completion, submit certificate to AS 3660.1 (2014) clause A3.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

**Products and materials**

Manufacturer's data: Submit manufacturer's data including the following:

- Product data sheet.
- Recommendations for installation or application.

Type Tests: Submit Test results of the following:

- Termite management systems: To **PRODUCTS, GENERAL, Tests**.

**Records**

Chemical termite management systems: Submit report to **COMPLETION, Chemical termite management systems**.

Termite management system: Submit report to **MAINTENANCE, Inspection**.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

**Tests**

Site Tests: Submit results, as follows:

- Chemical termite management systems: To **TESTING, Site tests**.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Completed earthworks or substrate preparation before system application or installation.
- Completed termite management system before concealing.
- Termite management system at the end of the Defects Liability Period.

## 2 PRODUCTS

---

### 2.1 GENERAL

#### Tests

Standard: To AS 3660.3 (2014).

### 2.2 PHYSICAL SYSTEMS

#### Termite caps, collars and sheeting

General: To AS 3660.1 (2014) Section 5.

Collars: To AS 3660.1 (2014) clauses 4.3.2.4.2 and 5.3.6.

#### Granular materials

Standard: To AS 3660.1 (2014) Section 6.

### 2.3 CHEMICAL SYSTEMS

#### General

Standard: To AS 3660.1 (2014) Section 7.

System assessment: To AS 3660.3 (2014) Section 5.

## 3 EXECUTION

---

### 3.1 GENERAL

#### Concrete slabs

Standard: To AS 3660.1 (2014) Section 4.

### 3.2 PHYSICAL SYSTEMS

#### Termite caps, collars and sheeting

General: To AS 3660.1 (2014) Section 5.

Collars: To AS 3660.1 (2014) clauses 4.3.2.4.2 and 5.3.6.

#### Granular materials

Standard: To AS 3660.1 (2014) Section 6.

### 3.3 CHEMICAL SYSTEMS

#### General

Standard: To AS 3660.1 (2014) Section 7.

### 3.4 TESTING

#### Site tests

Chemical systems: To AS 3660.1 (2014) Appendix E.

### 3.5 COMPLETION

#### Chemical termite management systems

Requirement: Prepare a report including the following:

- Termiticide brand name and manufacturer.
- Date and time of application.
- Location and extent of application.
- Each location:
  - . Moisture content of soil before application.
  - . Volume of undiluted termiticide used and dilution rate.
  - . Method of application.
  - . Rate of application.
- Water source for application.

#### Termite management system notice

Signage: Permanently fix a durable notice in a prominent location to BCA (2022) B1D4(i)(ii).

**Cleaning**

Requirement: Clean progressively and remove from the Site waste building materials that could attract termites.

**Operation and maintenance manuals**

Requirement: For systems requiring post-construction monitoring, prepare a maintenance manual that includes the following:

- Inspection frequency.
- Instructions for inspection of termite activity and treatment effectiveness.
- Contact details of installers and manufacturer's authorised supplier of replacement components.
- Recommendations for reapplication, if required.

**Warranties**

Type: Interlocking supply and installation.

**3.6 MAINTENANCE****Inspection**

Requirement: At the end of the Defects Liability Period, inspect the termite management system to AS 3660.2 (2017) clause 3.3.2.2. Prepare a report on the efficacy and status to AS 3660.2 (2017) clause 3.4.

|                                                     |
|-----------------------------------------------------|
| <b>0185 TIMBER PRODUCTS, FINISHES AND TREATMENT</b> |
|-----------------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide timber products with finishes and treatments, as Documented. Refer to Appendix A – T-Sheet, Appendix B – Façade and Wall Schedule, and Structural Engineer's Documents.

**Performance**

Requirements:

- Appropriate for durability and fire-resistance.
- Appropriate surface finish.
- Appropriate certification for the finishing applications.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0671 *Painting.*

**1.3 STANDARDS****General**

Sawn and milled products:

- Hardwood: To AS 2796.1 (1999).
- Softwood: To AS 4785.1 (2002).

Reconstituted wood based panels:

- Particleboard: To AS 1859.1 (2017).
- Particleboard flooring: To AS/NZS 1860.1 (2017).
- Dry process fibreboard: To AS/NZS 1859.2 (2017).
- Decorative overlaid wood panels: To AS/NZS 1859.3 (2017).
- Wet process fibreboard: To AS/NZS 1859.4 (2018).

Plywood:

- Structural: To AS/NZS 2269.0 (2012).
- Interior: To AS/NZS 2270 (2006).
- Exterior: To AS/NZS 2271 (2004).
- Marine: To AS/NZS 2272 (2006).

Glued laminated timber: To AS/NZS 1328.1 (1998).

Laminated veneer lumber: To AS/NZS 4357.0 (2022).

Timber grading methods:

- Stress graded: To the AS/NZS 1748 series.
- Visually graded F-grade: To AS 2082 (2007) or AS 2858 (2023).

**1.4 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviations apply:

- LVL: Laminated veneer lumber.

**Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 4491 (1997) and the following apply:

- Dry process fibreboard: Panel material with a nominal thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and pressure, the bond of which is derived from a synthetic adhesive added to the fibres and the panels are manufactured with a forming moisture content less than 20%.
- Particleboard: Panel material manufactured under pressure and heat from particles of wood (wood flakes, strands, chips, shavings, sawdust and similar) and/or lignocellulosic material in particle form (flax shives, hemp hurds, bagasse fragments, rice hulls, wheat straw and similar) with the addition of an adhesive.
- Wet process fibreboard: Panel material with a nominated thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from the felting of the fibres and the panels are manufactured with a forming moisture content greater than 20%.

## 1.5 SUBMISSIONS

### Products and materials

Chain of custody of forest products: Submit the following as evidence of conformity to

#### **CERTIFICATION, Timber source certification:**

- Third party certification of supplier's chain of custody management system.
- Formal claim of chain of custody by supplier.

Preservative treatment of timber: Submit a certificate from an independent Testing authority to AS/NZS 1604.1 (2021) clause 1.5.3.6. Include details of treatment and a copy of the charge sheet.

Tests: Submit moisture content Test results.

### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties.**

## 2 PRODUCTS

---

### 2.1 GENERAL

#### Storage and handling

General: Deliver timber products to Site in unbroken wrapping or containers and store so that the moisture content is not adversely affected.

#### Product identification

Preservative-treated timber: Marking to AS/NZS 1604.1 (2021) clause 1.5.3 and including the following:

- A unique identifier for the treatment plant.
- A unique identifier for the preservative.
- Hazard class.

### 2.2 CERTIFICATION

#### Timber source certification

Requirement: Use timber products originating from sustainably managed forests.

Application: Items requiring source certification:

- Timber framed doors and windows
- Timber wall panelling
- Timber door jambs and sills
- Timber veneers

Certification: FSC (Forestry Stewardship Council) preferred.

#### Engineered timber product certification and identification

Application: Items requiring source certification:

- Finger jointed pine trims, battens and skirtings.

Certification: EWPA Product Certification Scheme preferred

Branding: Brand timber products under the authority of a certification scheme applicable to the product. Locate the brand on faces or edges that will be concealed in the Works.

## 2.3 FIRE-RESISTANCE

### General

Structural timber: To AS/NZS 1720.4 (2019) or alternative conforming to NCC (2022) A5G3.

## 2.4 DURABILITY

### General

Requirement: Provide timbers with natural durability appropriate to the conditions of use, or preservative-treated timber of equivalent durability.

Natural durability class: To AS 5604 (2022).

Naturally termite-resistant timbers: To AS 3660.1 (2014) Appendix C.

Timber quality: Free of core wood (material within 50 mm of the tree's centre) and free of splits, checks, loose knots and cavities. Free of sapwood (lighter coloured wood found on the outer layer of the tree).

Lyctid susceptible timbers: To AS 5604 (2022). Do not provide untreated timbers containing lyctid susceptible sapwood.

Untreated sapwood: Do not use in applications requiring treated timber or natural durability.

### Preservative treatment

Wood-based products: To AS/NZS 1604.1 (2021) or preservative-treated products conforming to NCC (2022) A5G3.

Verification requirements: To AS/NZS 1604.2 (2021).

Test methods: To AS/NZS 1604.3 (2021).

### Moisture content

Moisture content: Seasoned timber <15% moisture content.

Test: Methods as follows:

- Timber and glued laminated timber products: To AS/NZS 1080.1 (2012).
- Plywood and LVL: To AS/NZS 2098.1 (2006).
- Reconstituted wood-based products: To AS/NZS 4266.1 (2017).

Protection: Protect timber and timber products stored on Site from moisture and weather. For milled, prefinished, prefabricated and similar elements that are to be protected in the final structure, provide temporary weather protection until the permanent covering is in place.

## 2.5 FINISHING

### Production finish

Glued laminated timber: To AS/NZS 1328.1 (1998).

Hardwood: To AS 2796.1 (1999) Table B1.

Plywood: To AS/NZS 2269.0 (2012), AS/NZS 2270 (2006), AS/NZS 2271 (2004) and AS/NZS 2272 (2006).

Softwood: To AS 4785.1 (2002) Table B1.

Surface finish: Dressed for all visible surfaces.

Edge detail: Arrised edges unless noted otherwise. Abutting faces to have square edges.

### Surface coating

Painting and staining: To 0671 *Painting*.

Application: To the manufacturer's specification; refer also to Spec E-Appendix E – Paint Specification.

---

## 3 EXECUTION

### 3.1 JOINTS

#### General

Joints and connections: Use hot dip galvanized or stainless steel fasteners, composite bolts, nails or nailed metal connectors.

Timber-to-timber interfaces: To the manufacturer's recommendations and the following:

- Provide a seal coating of preservative treatment.

- Make sure the inside of bolt holes and the end grains of the timber are coated.

Water retention: Avoid details that may trap water including housing or birdsmouth joints.

Fasteners: To prevent chemical treatments reacting with fasteners, install to manufacturer's recommendations.

### **3.2 SHRINKAGE RESTRAINT**

#### **General**

Requirement: Use seasoned timber, particularly where timber elements are integrated with steel and/or concrete.

Moisture content: Maintain a timber moisture content near the anticipated in-service equilibrium moisture content. Use finishes and end-grain sealants to minimise moisture content changes.

Fasteners: Align fasteners along member axis.

Connections: Use connections that allow for movement without adversely affecting the performance of the connection.

Unseasoned timber: Provide as follows:

- Drill bolt holes 2 mm or 10% larger than the bolt diameter.
- Use species with similar shrinkage values to reduce movement and shrinkage.
- Provide adequate clearance between unseasoned timber framing, and interfacing structures and materials to allow for movement.

### **3.3 FINISHING**

#### **Ploughing**

General: Back plough boards liable to warp (e.g. if exposed externally on one face). Make the width, depth and distribution of ploughs appropriate to the dimensions of the board and degree of exposure.

#### **Painting**

Edges: Chamfer edges of work to receive paint or similar coatings.

Priming: For woodwork to be painted, prime hidden surfaces before assembly.

### **3.4 COMPLETION**

#### **Warranties**

Type: Manufacturer and applicator / installer interlocking warranty: Interlocking warranty to cover manufacturing Defects and Defects with products and materials delivered to Site, including their application and / or installation.

Period: As offered by the manufacturer and the installer.

## **4 SELECTIONS**

---

### **4.1 PRODUCT SCHEDULES**

Refer to Architectural Drawings and Appendix A – T – Sheet.

|                          |
|--------------------------|
| <b>0191 SUNDRY ITEMS</b> |
|--------------------------|

## **1 GENERAL**

### **1.1 RESPONSIBILITIES**

#### **General**

Requirement: Provide sundry items, as Documented. Refer to Appendix A – T-Sheet. This section is also to cover:

- Reclaimed scrap elements (for inclusion in Landscape works typically)
- Water feature components, including proprietary and non-proprietary components
- Security Turnstile; refer to Door Schedule
- All other elements nominated in Appendix A - T-Sheet not covered in other parts of this specification (eg. Toilet Roll Holders, Snake Barrier, TGSIs, etc).

#### **Performance**

Requirements: Installation as follows:

- Undamaged and free of surface Defects or distortions; for reclaimed scrap, limited changes to existing condition.
- Correctly located and aligned, plumb, level and straight.
- Fixed firmly in position.
- Connected to the nominated service(s).
- Where applicable, in accordance with manufacturer's instructions.

### **1.2 CROSS REFERENCES**

#### **General**

Requirement: Conform to the following:

- *0171 General requirements.*

### **1.3 SUBMISSIONS**

#### **Certification**

Sealant compatibility: Submit statements from all parties to the installation certifying the compatibility of sealants with items and substrates.

#### **Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

Products and materials

#### **Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

#### **Shop drawings**

General: Submit shop drawings, to a scale that best describes the detail, showing the following:

- Details of fabrication and components.
- Details of fabrication involving other trades or components.
- Information necessary for Site assembly.
- Proposals for the break-up of large items as required for delivery to the Site.
- Proposed method of joining the modules of large items.
- Fastening locations and types.

Requirement: Refer to Appendix A – T-Sheet.

#### **Tests**

Requirement: Submit completion Test results.

#### **Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

## 1.4 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Set-out of item locations before fixing.
- Completion of installation.

## 2 PRODUCTS ---

### 2.1 GENERAL

#### Storage and handling

General: Deliver, unload and store components and accessories in unbroken manufacturer's packaging.

#### Samples

Samples required: refer to Appendix A – T-Sheet.

Labelling: Label each sample, giving the brand and product name, manufacturer's code reference, date of manufacture and intended building location.

## 3 EXECUTION ---

### 3.1 PREPARATION

#### Substrates

General: Prepare the substrate to receive the item.

#### Protection

General: Protect existing Work from damage during the installation and rectify any damage. Provide temporary coverings if required.

### 3.2 INSTALLATION

#### Accessories and trim

Requirement: Provide accessories and trim necessary to complete the installation.

### 3.3 TESTING

#### Completion Tests

Requirement: Refer to Appendix A – T-Sheet.

### 3.4 COMMISSIONING

#### General

Requirement: Provide commissioning to *0171 General requirements* for the following:

Refer to Appendix A – T-Sheet.

### 3.5 COMPLETION

#### Cleaning

Requirement: Remove packaging. Clean the completed assembly and surrounds. Wipe down appliances and fittings with a damp, soft, clean cloth.

#### Operation and maintenance manuals

Requirement: Prepare a maintenance manual and, if required, an operation manual with the technical specification and manufacturer's recommendations for the item to be installed.

#### Warranties

Type: Manufacturer warranty where applicable; installer warranty where applicable.

Period: As provided by supplier and/or installer.

## 4 SELECTIONS ---

Refer to Appendix A – T-Sheet

|                                            |
|--------------------------------------------|
| <b>0193 BUILDING ACCESS SAFETY SYSTEMS</b> |
|--------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide building access safety systems, as Documented. Refer to Appendix G – Building Access System.

**Performance**

Roofing and cladding: Maintain waterproofing integrity without damage or distortion. Maintain the structural integrity of the supporting elements.

**1.2 DESIGN****General**

Designer: Preliminary design provided by Kattsafe; refer to Appendix G – Building Access System.

**Requirements**

General: To DESIGN in *0171 General requirements*.

Responsibility: Contractor to finalise design to the approval of the Superintendent.

Performance requirements: equal to or better than Appendix G – Building Access System.

Authority requirements: to comply with WHS Requirements.

Access: Provide a system for three workers at any one time, to access the following:

- Full extent of gutters.
- Roof areas within 2.5 m of fall hazards not otherwise protected by parapets or guardrails.

**1.3 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements*.

**1.4 STANDARDS****General**

Fixed platforms, walkways, stairways and ladders for use by operating, inspection, maintenance and servicing personnel: To AS 1657 (2018).

Personal equipment for working at height: To AS/NZS 1891.1 (2020), AS/NZS 1891.2 (2001), AS 1891.3 (2020), AS/NZS 1891.4 (2009) and AS 1891.5 (2020).

Rope access system: To AS/NZS ISO 22846.1 (2020) and AS/NZS ISO 22846.2 (2020).

**1.5 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviation applies:

- PPE: Personal protective equipment.

**Definitions**

General: For the purposes of this worksection, the definitions given in AS 1657 (2018), AS/NZS 1891.1 (2020), AS/NZS 5532 (2013) and AS/NZS ISO 22846.1 (2020) apply.

**1.6 SUBMISSIONS****Certification**

General: Submit certification of installed system to **COMPLETION, Certification**.

**Design Documentation**

General: To *0171 General requirements* and the following:

- Calculations: Submit calculations by a professional engineer experienced in building access safety systems.

- Certification: Submit certification by a professional engineer experienced in building access safety systems design as evidence of conformance to Documented requirements.
- Drawings: Submit the following drawings:
  - . Layout of roof access stairways and ladders, platforms and walkways.
  - . Details of stairways and ladders, walkways, including elevated walkways and required guardrailing.
  - . Layout of anchors, static lines and system components in plan and elevation.
  - . Proposed methods of fixing to each substrate type in the building.

**Marking and labelling**

Requirement: Samples and schedules of proposed marking and labels for each system component.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Products and materials**

Manufacturer's data: Submit manufacturer's data including the following:

- Product data sheets.
- Installation and maintenance recommendations.

Type Tests: Submit results for the following:

- Proprietary stairways and fixed ladders: To **FIXED STAIRWAYS AND LADDERS, Tests.**
- Proprietary platforms, walkways and guardrailing: To **FIXED PLATFORMS, WALKWAYS AND GUARDRAILING, Tests.**
- Personal equipment for working at height: To **FALL PROTECTION SYSTEMS, Tests.**
- Rope access systems: To **FALL PROTECTION SYSTEMS, Tests.**
- Single point anchors: To **FALL PROTECTION SYSTEMS, Tests.**

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples.**

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers as recommended by the manufacturer.

Evidence of experience: QBCC licence and CV.

**Tests**

Site Tests: Submit results of proof load Tests of all friction and glued-in anchors.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Shop fabricated or assembled items ready for delivery to the Site.
- Commencement of shop or Site welding.
- All equipment attachments with concealed fasteners, before they are covered.
- Site erected assemblies on completion of erection, before applying finishes.
- Steel surfaces prepared for, and immediately before, Site applied finishes.

Installation inspector: Registered height safety inspector or professional engineer.

---

**2 PRODUCTS****2.1 GENERAL****Product identification**

General: Marked to show the following:

- Manufacturer's identification.
- Installer's contact details.

- Intended location.
- Load rating and direction.
- Current inspection/service date.
- Batch number or serial number of the components.

**Signage**

General: Include all mandatory and Documented signage.

**2.2 FIXED STAIRWAYS AND LADDERS****General**

Product: Fixed stairways and fixed ladders complete with all required handrails, guardrailing, cages and restricted access devices.

Standard: To AS 1657 (2018).

**Tests**

Stair assembly: To AS 1657 (2018) Appendix D.

Fixed ladders: To AS 1657 (2018) Appendix F.

Test reports: To AS 1657 (2018) Appendix E.

**Ladder fall protection systems**

Product: Vertical systems including cables, rails, guides and fall arrestor trolleys.

Standard: Selection of system to AS 1657 (2018) Table H1.

Design of fall protection devices: To AS 1891.3 (2020).

**2.3 FIXED PLATFORMS, WALKWAYS AND GUARDRAILING****General**

Product: Fixed platforms and walkways complete with all required guardrailing and toe boards.

Balustrades: Provide a proprietary or bespoke fabricated system, as Documented.

Standard: To AS 1657 (2018).

Material: Aluminium, corrosion-resistant steel or fibre-reinforced polymer.

**Tests**

Guardrailing: To AS 1657 (2018) Appendix B.

Infill: To AS 1657 (2018) Appendix C.

Test reports: To AS 1657 (2018) Appendix E.

**2.4 FALL PROTECTION SYSTEMS****Access safety system**

System: Refer to Appendix G – Building Access System.

**Anchors**

Single point anchors: To AS/NZS 5532 (2013).

**PPE**

Harness: Supply two full body harnesses to AS/NZS 1891.1 (2020) with shock absorbing lanyards to AS 1891.5 (2020).

Storage: PPE storage holdall supplied by the manufacturer.

**Tests**

Personal equipment for working at height: Tested as follows:

- Harnesses: To AS/NZS 1891.1 (2020) Section 4 and Appendix A to F.
- Horizontal lifeline and rail systems: To AS/NZS 1891.2 (2001) Appendix A to E.
- Lanyard assemblies and pole straps: To AS 1891.5 (2020) clause 3.4.

Rope access systems: Fall-arrest devices to AS 1891.3 (2020).

### 3 EXECUTION

---

#### 3.1 INSTALLATION

##### General

Installation: To the manufacturer's recommendations and the following:

- Fixed platforms, walkways, stairways and ladders: To AS 1657 (2018).
- Personal equipment for working at height: To the AS/NZS 1891 series.
- Rope access systems: To AS/NZS ISO 22846.2 (2020).
- Incompatible materials: Separate using concealed layers of suitable materials in appropriate thicknesses.

##### Subcontractor

Installer: Registered installer, approved by the manufacturer.

##### Fixing to structure

General: Provide fabricated predrilled or purpose-made brackets and bases and attach to the building structure with fasteners compatible with the substrate.

Proprietary items: Install to the manufacturer's recommendations.

##### Labels and signage

General: To AS/NZS 1891.4 (2009) clause 2.2.9.

#### 3.2 TESTING

##### Testing of installed Work

General: On completion of the installation, carry out Tests to confirm the system's competence in accordance with AS/NZS 5532 (2013), AS/NZS 1891.2 (2001) and AS/NZS 1891.4 (2009), and issue certification confirming compliance with all installation requirements.

##### Proof load Test for anchors

Testing: Following completion of the installation, proof load Test all friction and glued-in anchors to 50% of the design ultimate strength capacity in conformance with AS/NZS 1891.4 (2009) clause 3.1.2(g).

Report: Record the following:

- Details of organisation and inspector performing the Test.
- Date Test was performed.
- Load cell details and calibration date.
- A reference of each item Tested including a description of the item location.
- The item ultimate load.
- The Test load applied.
- Duration of Test.
- Results of Test.
- Photo of each item showing peak load applied.

#### 3.3 TRAINING

##### General

Responsibilities: Coordinate the training of owner's facilities management personnel in conformance with *0171 General requirements*.

Training records: Video record all training sessions. Catalogue and include recordings with the operation and maintenance manuals.

#### 3.4 COMPLETION

##### Certification

Completion certificate: Provide inspection, Testing and certification by an Accredited Installer and/or Accredited Height Safety Inspector:

- Upon completion of the installation at the Date for Practical Completion.
- Upon the expiry of the Defects Liability Period or 12 months after completion of the installation, whichever is the lesser, and valid for a further 12 month period.

**Reinstatement**

Extent: Repair or replace damage to the roofing and rainwater system. If the Work cannot be repaired satisfactorily, replace the whole area affected.

Touch up: If it is necessary to touch up minor damage to prepainted metal roofing, do not overspray onto undamaged surfaces.

**Cleaning**

Roofing and rainwater drainage system: Remove debris, metal swarf, solder, sealants and unused materials.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the following:

- Instructions and recommended procedures for operating and routinely maintaining the equipment.
- Technical information relating to all access and anchorage equipment provided including design, Testing, installation, and conformance of the system to relevant industry standards and manufacturer's recommendations.
- Special provisions for use (e.g. Training, additional equipment, rescue provisions).
- A statement that devices are for personnel use and note specific equipment that is not for personnel use.
- Include certification and equipment logs in conformance with AS 1657 (2018) Section 8 and Appendices, and AS/NZS 1891.4 (2009) Section 9.

**Warranties**

Type: Interlocking supplier and installer warranties to cover materials, systems and installation.

Period: As offered by the supplier and installer.

**3.5 MAINTENANCE****General**

Preventive and mandatory system maintenance: By an Accredited Height Safety Inspector/Certifier, in conformance with AS/NZS 1891.4 (2009) Section 9 and manufacturer's maintenance/recertification recommendations.

Checklist for all inspections: To AS/NZS 1891.2 Supp 1 (2001) Table 8, and AS/NZS 1891.4 (2009) Section 9 and Appendices C and D.

The competent person: To AS/NZS 1891.4 (2009) clause 1.4.8.

**Regular scheduled periodic inspections**

Standard: To AS/NZS 1891.4 (2009) clause 9.3 and the following:

- Provide inspection, Testing and certification by an Accredited Installer and/or Accredited Height Safety Inspector.
- Record the date of the next system inspection and period of validity and display the certificate at the access points of the work area or on the individual system components where provision is made.

**Inspection after a fall or other event**

Standard: To AS/NZS 1891.4 (2009) clause 9.5.

**Ongoing maintenance**

Certificate: Submit the completion certificates and notify the proprietor of the requirement for continued interval Testing.

---

**4 SELECTIONS**

**Refer to Appendix G – Building Access System**

|                        |
|------------------------|
| <b>0201 DEMOLITION</b> |
|------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Carry out demolition, as Documented.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****General**

Demolition: To AS 2601 (2025).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Demolition: The complete or partial removal of a building or structure using planned and controlled methods.
- Dilapidation record: The photographic or video and written record of the condition of the portion of the existing building retained, adjacent buildings and other relevant structures or facilities, before the start of demolition work.
- Dismantle: The reduction of an item to its components in a manner to allow re-assembly.
- Recover: The disconnection and removal of an item in a manner to allow re-installation.

**1.5 SUBMISSIONS****Authority approvals**

Evidence of compliance: Before starting demolition, submit the following:

- Evidence that requirements of authorities relating to the WUC have been obtained.
- A permit to demolish from the appropriate authority.
- A hoardings or scaffold permit from the appropriate authority, if hoardings or scaffolding is proposed to be used.
- Certification that each person having access to the construction Site has completed Site-specific WHS Policies and Procedures induction training.
- Evidence that precautions necessary for the protection of persons and property have been taken and suitable protective and safety devices have been provided to the approval of the relevant authority.
- Certification from the relevant authority confirming treatment for any vermin infestation has been carried out.
- Fees and/or costs have been paid.

**Execution details**

Requirement: Submit the following:

- Hazardous substances management plan: To **HAZARDOUS SUBSTANCES MANAGEMENT, Hazardous substances management plan**.
- Demolition work plan: To **DEMOLITION WORK PLAN, General**.
- Safe Work Method Statement.

Recycling: Submit certification as evidence of disposal of recycled materials.

**Records**

Dilapidation record:

- Before demolition: Submit to each owner of each adjoining or adjacent property, a copy of the part of the record relating to that property and obtain their written agreement to the contents.
- Rectification work: Submit written acceptance of rectification works from the owner of each adjoining or adjacent property affected.

### Tests

Requirement: Submit compliance Test results for building services components to be re-used.

## 1.6 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Adjoining and adjacent structures before starting demolition.
- Services before disconnection or diversion.
- Trees Documented to be retained, before starting demolition.
- Contents of building before starting demolition.
- Structure after stripping and removal of roof coverings and external cladding.
- Underground structures after demolition above them.
- Remaining excavations after removal of underground Work.
- Site after removal of demolished materials.
- Services after reconnection or diversion.
- Adjoining and adjacent structures at completion of demolition.

## 2 PRODUCTS

### 2.1 DEMOLISHED MATERIALS

#### Demolished material classes table

| Class                                         | Requirement                                                                                                                                                                                | Ownership            |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Recovered items for re-use in the Works       | Recover without damage items identified in the <b>Recovered items for re-use in the Works schedule</b>                                                                                     | Principal/proprietor |
| Dismantle for relocation as part of the Works | Dismantle without damage and store items identified in the <b>Dismantle for relocation schedule</b>                                                                                        | Principal/proprietor |
| Demolish for removal                          | Remove from Site demolished materials identified in the <b>Demolish for removal schedule</b> . Do not burn or bury on Site<br>Transit: Prevent spillage of demolished materials in transit | Contractor           |

## 3 EXECUTION

### 3.1 SITE INVESTIGATION

#### Investigation reports

Reports: Contractor to liaise directly with Principal for available reports.

### 3.2 HAZARDOUS SUBSTANCES MANAGEMENT

#### Hazardous substances management plan

Requirement: Prepare a hazardous substances management plan to AS 2601 (2025) clause 3.2.12.2. Include the following:

- Asbestos-containing materials.

- Flammable or explosive liquids or gases.
- Toxic, infective or contaminated materials.
- Radiation or radioactive materials.
- Noxious or explosive chemicals.
- Tanks or other containers that have been used for storage of explosive, toxic, infective or contaminated substances.
- Laboratory analysis of hazardous substances.

#### **Removal of hazardous substances**

Requirement: To AS 2601 (2025) clause 5.3.2 and the hazardous substances management plan.

### **3.3 DEMOLITION WORK PLAN**

#### **General**

Requirement: Before demolition, prepare a demolition work plan based on the information provided in **SITE INVESTIGATION**. Conform to AS 2601 (2025) Sections 3 and 4. Include the checklist items appropriate to the project from AS 2601 (2025) Appendix A and the following:

- Stockpile locations: Details of the proposed locations of on-Site stockpiles for demolished materials for recycling in the Works. Coordinate with the locations for storage of other waste streams. Prevent mixing and pollution.
- Off-Site disposal locations: Details of the proposed locations for the disposal of material required to be removed from the Site and evidence of conformance with the requirements of relevant authorities.
- Recycling: Details of the proposed recycling facilities. If concrete crushing is proposed on Site, include details of plant and environmental controls.
- Method of protection and support for adjoining or adjacent structures.
- Locations and details of service deviations and terminations.
- Sequence of WUC.
- If the demolition program results in components temporarily cantilevered, provide a certificate from a professional engineer.
- Proposals for the safe use of mobile plant on suspended structural members including provisions for the protection of lower floors in the event of structural failure.
- Structural engineering report and demolition methodology, as appropriate, if the structure is suspected to contain unbonded prestress tendons.
- Wheel loads of tipping or loading vehicles.

### **3.4 SUPPORT**

#### **Temporary support**

General: If temporary support is required, certification for its design and installation is required from a professional engineer engaged by the Contractor.

Existing buildings: Until permanent support is provided, provide temporary support for sections of existing buildings or parts of buildings being retained and which normally rely on support from Work to be demolished.

Suspended slabs: If mobile plant is required for use on suspended structural members, conform to professional engineer's requirements and the demolition work plan.

Ground support: Support excavations for demolition of underground structures.

Adjoining or adjacent structures: Provide supports to adjoining or adjacent structures if necessary, sufficient to prevent damage resulting from the Works.

Lateral and vertical supports: At least equal in capacity to that originally provided by the structural element or structure to be demolished.

#### **Permanent supports**

General: If permanent supports for adjacent structures are necessary and are not Documented, give notice and obtain instructions.

### 3.5 PROTECTION

#### Encroachment

General: Prevent the encroachment of demolished materials onto adjoining property, including public spaces.

#### Dust protection

General: Provide dustproof screens, bulkheads and covers to protect existing finishes and the immediate environment from dust and debris.

#### Security

General: If fences or gates are opened for alterations or additions, provide security against unauthorised entry to the adjoining site.

#### Temporary screens

General: Fill the whole of designated temporary openings or other spaces using dustproof and weatherproof temporary screens, fixed securely to the existing structure. Install to shed water to avoid damage to retained existing elements, and adjacent structures and contents.

Type: Timber framed screens sheathed with 12 mm plywood and painted. Seal the junctions between the screens and the openings.

#### Temporary access

General: If required, provide a substantial temporary doorset fitted with a rim deadlock and remove on completion of demolition.

#### Exposed surfaces

General: If necessary, protect and weatherproof the surfaces of adjoining structures exposed by demolition.

Treatment of exposed surface: seal with a protective treatment suitable for the material, and prepare for new applied finish if applicable.

#### Existing services

Location: Before starting demolition, locate and mark existing underground services by potholing or other non-destructive digging, in the areas that will be affected by the demolition operations.

Utility services: Contact BEFORE YOU DIG AUSTRALIA to identify location of underground utility services pipes and cables.

Essential services: Shut off, cap or control services not required for the demolition work, at or outside the building line before commencing demolition. Conform to the requirements of the relevant existing utility authority, as appropriate.

Underground utility services to be retained: Do not excavate by machine within 1 m of existing underground services.

#### Fixed items

Individual protection: Protect the following items in their existing positions:

Memorial plaque in north garden area.

#### Recovered items

General: If items are Documented for recovery and re-use, minimise damage during removal and recover all associated components required for their re-use.

### 3.6 DEMOLITION - BUILDING WORKS

#### General

Requirement: To the relevant parts of AS 2601 (2025) Section 6, the Safe Work Method Statement and the demolition work plan.

#### Encroachment

General: If encroachments from adjacent structures are encountered and are not Documented, give notice and obtain instructions.

#### Concrete slabs and pavements

Partial demolition or penetrations: Using a diamond saw, neatly cut back or trim to new alignment with a clean true face. Do not overcut at corners. If required, provide protection to exposed reinforcement along the newly sawn concrete slab edge, as Documented.

Storage: Do not store demolished materials on suspended slabs.

**Material below grade**

Extent: Demolish the following:

Redundant footings, pits and services.

Remaining voids: Backfill.

**Explosives**

General: Do not use explosives.

**3.7 DEMOLITION - BUILDING SERVICES****General**

Requirement: Conform to the demolition work plan. Decommission, isolate, demolish and remove from the Site all equipment and associated components that have become redundant as a result of the demolition.

Breaking down: Disassemble or cut up equipment if necessary to allow removal.

**Demolition of refrigeration systems**

Standard: To AS/NZS 5149.4 (2016).

Refrigeration system to be demolished: existing office chillers.

**Components for re-use**

General: Before returning to service, clean components and Test for conformance to Australian standards, as required.

**3.8 COMPLETION****Notice of completion**

General: Give at least 5 Business Days' notice of completion of demolition so that adjoining or adjacent structures may be inspected following completion of demolition.

**Reinstatement**

Assessment of damage: Use the dilapidation record to assess the damage and rectification work arising from the demolition work.

Rectification: Repair damage arising from the demolition work. Obtain written acceptance from the owner of each adjoining or adjacent property of the completeness and standard of the rectification work.

**Removal of temporary supports**

General: Obtain written instructions from a professional engineer at the completion of demolition before removing temporary supports.

**4 SELECTIONS****4.1 DEMOLITION****Recovered items for re-use in the Works schedule**

| Item           | Location for re-use |
|----------------|---------------------|
| Opening Plaque | Entry landscape     |
|                |                     |
|                |                     |

**Dismantle for relocation schedule**

| Item             | Location for storage | Location for re-assembly |
|------------------|----------------------|--------------------------|
| 2no Bus Shelters | By Principal         | Southern walkway         |
|                  |                      |                          |
|                  |                      |                          |

**Demolish for removal schedule**

| Item                                              |
|---------------------------------------------------|
| Existing Project's Office and associate services. |

---

| Item                                                                    |
|-------------------------------------------------------------------------|
| Northern shade structure and associated services, appliances and Works. |
| 1no Bus Shelter                                                         |
| Refer also to drawings A-040002-Site Plan – Existing + Demolition       |

|                              |
|------------------------------|
| <b>0221 SITE PREPARATION</b> |
|------------------------------|

## **1 GENERAL**

### **1.1 RESPONSIBILITIES**

#### **General**

Requirement: Provide Site preparation, as Documented.

#### **Performance**

Areas for protection: existing planting areas not Documented as demolished. Refer also to Landscape Architect's Documents.

### **1.2 CROSS REFERENCES**

#### **General**

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0172 *Environmental management.*

### **1.3 INTERPRETATION**

#### **Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Authority: Any organisation with statutory authority relating to the project, including clearances.
- Clearances: A formal certificate, approval or condition issued by a statutory authority allowing work in a particular area.
- Plant establishment period: The period between the Date of Practical Completion and the end of the Defects Liability Period.

### **1.4 SUBMISSIONS**

#### **Certification**

Vermis: Submit pest exterminator's certification as evidence that the completed Site Works are free from vermin.

#### **Execution details**

Requirement: Submit details of methods and equipment proposed for the following:

- Clearing and grubbing.
- Tree removal and transplanting.
- Protecting the ground within and adjacent to tree driplines from compaction by proposed earthworks machinery.

### **1.5 INSPECTION**

#### **Notice**

Inspection: Give notice so that inspection may be made of the following:

- Enclosures around trees requiring protection.
- Trees requiring removal.
- Trees for transplanting to determine final orientation.

## **2 EXECUTION**

### **2.1 COMMUNITY LIAISON**

#### **Notification**

General: Notify residents about construction activities that will affect access to or disrupt the use of their properties.

Notice: Minimum 5 Business Days, unless the work is of an urgent nature with safety implications.

Notification content:

- Description of the work.
- The reason for the work.
- The expected duration.
- Changes to traffic arrangements and property access.
- The 24-hour contact number of the representative responsible.

## 2.2 EXISTING SERVICES

### General

Requirement: Before starting earthworks, locate and mark existing underground services in the areas affected by the earthworks operations including clearing, excavating and trenching.

Utility services: Contact BEFORE YOU DIG AUSTRALIA to identify location of underground utility services pipes and cables.

Construction Plant: Conform to the utility service provider requirements for the operation of Construction Plant within the zone of influence of existing services. Maintain the required cover and do not exceed the allowable load limit.

Excavation: Do not machine excavate within 1000 mm of existing services.

Existing service lines: If required, divert services detected during excavation clear of the building and reconnect to the utility service provider requirements.

## 2.3 SITE CLEARING

### Extent

Requirement: Clear only areas occupied by Works such as structures, paving, excavation, regrading and landscaping or other areas Documented for clearing.

Contractor's Site areas: If not included within the areas Documented above, clear only to the extent necessary for the performance of the WUC.

### Clearing and grubbing

Clearing: Remove everything on or above the Site surface, including rubbish, scrap, grass, vegetable matter and organic debris, scrub, trees, timber, stumps, boulders and rubble.

Grubbing: Grub out stumps and roots over 75 mm diameter to a minimum depth as follows:

- Below subgrade under buildings, embankments or paving: 500 mm.
- Below finished surface in unpaved areas: 300 mm.

Backfilling: Fill holes remaining after grubbing with sand material to prevent ponding of water.

Compact the material to the relative density of the existing adjacent ground material.

Redundant/decommissioned works: Remove works no longer required, including slabs, foundations, paving, drains, and access chambers and covers within the Works zone.

### Batters

Temporary protection: If the change in level between crest and toe is more than 1500 mm, protect from erosion with geofabric, hessian and tar or heavy duty black polyethylene sheet cover. Securely fix down at crest and toe.

### Surplus material

Topsoil and excavated material: Remove unwanted stripped soil and other material from the Site as the WUC proceeds, including any material dropped on footpaths or roadways.

## 2.4 STORMWATER AND SEDIMENT CONTROL

### General

Erosion and sediment control measures: To 0172 *Environmental management*.

### Waterways and drains

Waterways: If required, temporarily divert ditches, field drains and other waterways affected by excavation and reinstate on completion.

Stormwater drains: Divert drains detected during excavation clear of the building and reconnect as Documented. Conform to the Network Utility Operator's requirements.

## 2.5 EXISTING WORKS TO REMAIN

### Marking

Requirement: Identify existing Works to remain with 1000 mm high, 50 x 50 mm timber stakes connected by yellow plastic tape to prevent accidental damage.

## 2.6 TREE REMOVAL

### Designation

Marking: Identify trees and shrubs for removal by tagging 1000 mm above ground level.

Extent: As Documented. Refer also to Landscape Architect's Documents.

## 2.7 TREE PROTECTION

### General

Warning signs: Display warnings that trees and plantings require protection during the Contract in a prominent position at each entrance to the Site. Remove on completion.

Lettering: Road sign type sans serif letters, 100 mm high to AS 4970 (2025) Appendix C.

Protection measures: Provide before starting the earthworks.

### Trees to remain

Extent: Trees not marked for removal.

### Tree protection

Tree protection zone (TPZ): To AS 4970 (2025) Section 3.

Tree protection measures: To AS 4970 (2025) Section 4.

Monitoring and certification: To AS 4970 (2025) Section 5.

### Work near trees

Materials placement: Conform to the following:

- Keep the area within the dripline of trees free of sheds and paths, construction material and debris.
- Do not place bulk materials and harmful materials within the dripline of trees.
- Do not place spoil from excavations against tree trunks.
- Prevent wind-blown materials such as cement from harming trees and plants.

Damage: Prevent damage to tree bark. Do not attach stays, guys and similar material to trees.

Work under trees: Do not remove topsoil from or add topsoil to the area within the dripline of the trees.

Excavation: If excavation is required near trees, give notice. Minimise period and extent of excavation within the dripline.

Hand methods: If excavation is required within the dripline, use hand methods so that root systems remain intact and undamaged.

Roots: Do not cut tree roots exceeding 50 mm diameter. If required to cut tree roots, use cutting methods that do not excessively disturb the remaining root system. Immediately after cutting, water the tree and apply a liquid rooting hormone to stimulate the growth of new roots.

Backfilling: Backfill excavations around tree roots. Place the backfill in layers of 300 mm maximum depth and compact to a dry density similar to that of the surrounding soil. Do not backfill around tree trunks to a height greater than 200 mm above the original ground surface. Immediately after backfilling, thoroughly water the root zone surrounding the tree.

Backfill material:

- Mix proportions by volume (topsoil: well-rotted composts): 3:1.
- Neutral pH value.
- Free from weed growth and harmful materials.

Compaction protection: Protect ground adjacent to the tree dripline.

Compacted ground: Do not compact the ground or use skid-steer vehicles under the tree dripline. If compaction occurs, give notice.

Watering: Water trees as necessary, including where roots are exposed at ambient temperatures more than 35°C.

Mulching: Spread 100 mm thick organic mulch conforming to AS 4454 (2012), to the whole of the area within the dripline of all existing trees to remain.

## 2.8 TEMPORARY LANDSCAPE FENCING

### Fence dimensions

Height: 1200 mm.

Maximum post spacing: 5000 mm.

### Component sizes

Corner and gate posts: Hardwood or preservative-treated softwood, 250 mm diameter.

Intermediate posts: Star picket.

Gate: Provide a suitable hinged gate with a gate latch.

Wire: Top, intermediate and bottom rows of 3.2 mm plain galvanized steel wire. Thread the top wire through pieces of plastic tube and through corner posts.

### Removal

Completion: Remove the fence at the end of the plant establishment period.

## 2.9 TREE MAINTENANCE

### General

Requirement: As Documented.

Notice: Give notice before starting tree maintenance.

Pruning: To AS 4373 (2007) using a fully qualified and experienced arborist. Carry out all required WUC in a safe manner.

### Execution

Requirement: Rectify any damage to existing trees to remain.

Operations: Remove deadwood and decayed wood or damaged limbs. Make final cuts as close to the branch collar as possible without cutting into the branch collar. If trees show signs of deterioration after the work is completed, ameliorate the soil by soil aeration, irrigation or incorporation of organic material. Continue this program until the end of the plant establishment period.

Root pruning: Do not excessively disturb the remaining root system. Cut off damaged roots cleanly inside the exposed or damaged area. Cover exposed root area with soil immediately after pruning, do not leave roots exposed.

Wetting and new root stimulation: Form a water collecting basin and apply a rooting hormone and wetting agent to the rootball.

Precautions: Avoid damage to trees being treated and to nearby trees and surroundings. Do not use trees as anchors for winching operations or bracing. Provide bracing as necessary before cutting to prevent uncontrolled breakages and damage to surroundings.

Failure: If repair work is impracticable or is attempted and is rejected, remove the tree and root system and make restitution.

Restitution by replacement tree: Replace with tree of the same species and similar size.

## 2.10 COMPLETION

### Temporary Works

Remove at completion: temporary tree enclosures, site hoardings and site plant.

### Site restoration

Requirement: Reinstate undeveloped ground surfaces to the condition existing at the commencement of the Contract.

### Clean up

Progressive cleaning: Keep the Works clean and tidy. Regularly remove waste and surplus material arising from execution of the WUC from the Site.

Waste disposal: To 0172 *Environmental management*.

### Vermin management

Requirement: Employ a suitably qualified pest exterminator to remove vermin found during Site preparation.

|                       |
|-----------------------|
| <b>0222 EARTHWORK</b> |
|-----------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide earthworks to the levels, dimensions and tolerances, as Documented. Refer also to Civil Engineer and Landscape Architect's Documents.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0172 *Environmental management.*

### 1.3 STANDARDS

#### General

Earthworks: Conform to the recommendations of those parts of AS 3798 (2007) that are referenced in this worksection.

Description and classification of soils: To AS 1726 (2017).

### 1.4 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the definitions given in AS 3798 (2007) and the following apply:

- Bad ground: Ground unsuitable for the Works, including fill liable to subsidence, ground containing cavities, faults or fissures, ground contaminated by harmful substances and ground that is or becomes soft, wet or unstable.
- Rock: Monolithic material with volume greater than 0.3 m<sup>3</sup> that cannot be removed until broken up by rippers or percussion tools.
- Site topsoil: Natural soil, excavated from the Site, which contains organic matter, supports plant life, conforms to the fine-to-medium texture classification to AS 4419 (2018) and is free from the following:
  - . Stones more than 25 mm diameter.
  - . Clay lumps more than 50 mm diameter.
  - . Weeds and tree roots.
  - . Sticks and rubbish.
  - . Material toxic to plants.
- Subgrade: The trimmed or prepared earth material on which the pavement, footing or slab is constructed. Generally taken to relate to the upper line of the earth material.
- Zone of influence: A foundation zone bounded by planes extending downward and outward from the bottom edge of a footing, slab or pavement and defining the extent of foundation material having influence on the stability or support of the footings, slab or pavement.

### 1.5 TOLERANCES

#### General

Finish: Finish the surface to the required level, grade and shape within the following tolerances:

- Under building slabs and load bearing elements: +0, -25 mm.
- Pavement subgrades: +0, -40 mm.
- Batters: No steeper than the slope shown on the drawings. Make sure flatter slopes do not impact on boundaries or required clearances to buildings, pavements or landscaping.
- Other ground surfaces: ±50 mm, provided the area remains free draining and matches adjacent construction where required. Provide smoothness as normally produced by a scraper blade.

## 1.6 SUBMISSIONS

### Design Documentation

Calculations: Submit calculations by a professional engineer showing the stability and safety of proposed excavations and temporary supports, including supports required for adjacent structures.

### Execution details

Report: Submit a time-based schedule detailing the methods and equipment proposed for the earthworks, including the following:

- Dewatering and groundwater control and disposal of surface water.
- Excavation methods, stages, clearances, batters and temporary supports.
- Stockpiles and borrow pits.
- Placing and compaction methods and stages.

Geotechnical Site investigations: Provide a geotechnical report supporting the excavation methods proposed.

Disposal location: Submit details of the location and evidence of compliance with the appropriate authority requirements for the disposal of material requiring removal from Site.

Temporary shoring: Submit a proposal for any temporary shoring required, including the progressive removal.

Proof rolling: Submit details of proposed method and equipment.

Records of measurement: Submit a certified copy of the agreed records of measurement.

Site records: Submit the following to AS 3798 (2007) clause 3.4 and Appendix B:

- Geotechnical Site visit record.
- Earthworks summary report or daily geotechnical reports.

### Products and materials

Imported fill: Submit certification from a geotechnical inspection and Testing authority (GITA) or Test results of the imported fill as evidence of conformity with the Contract, including the source.

### Tests

Requirement: Submit Test results of the following:

- Compaction control: To **TESTING, Site tests**.

## 1.7 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Items to be measured as listed in **RECORDS OF MEASUREMENT**.
- Areas to be cleared and/or stripped of topsoil.
- Areas stripped of topsoil.
- Excavation completed to Contract levels or founding material.
- Proof rolling of the subgrade.
- Proof rolled subgrade before placing fill.
- Filling and compaction completed to Contract levels.
- Stockpiled topsoil before spreading.

---

## 2 PRODUCTS

### 2.1 FILL MATERIALS

#### General

Suitable material: To AS 3798 (2007) clause 4.4 including inorganic, non-perishable material suitably graded and capable of compaction to the Documented density.

Unsuitable materials: Do not use fill defined as unsuitable by AS 3798 (2007) clause 4.3.

Sulfur content: Do not provide material with sulfur content exceeding 0.5% within 500 mm of concrete and cement bound elements (for example masonry) unless the elements are protected by impermeable membranes or equivalent means.

Re-use of excavated material: Only re-use suitable material.

**Stockpiles**

General: Segregate earth and rock material and stockpile for re-use in backfilling operations.

Location: Do not stockpile excavated material against tree trunks, buildings or fences. Do not obstruct the free flow of water along drainage channels.

**2.2 BORROW OR IMPORTED FILL****General**

Requirement: Use only when suitable material obtained from Site excavations are insufficient for completing the Documented earthworks.

- Suitable material: To AS 3798 (2007) clause 4.4.

Material conforming to the following: by Civil Engineer.

Borrow pits:

- Locate more than 3000 mm from any fence line, boundary, edge of excavation or embankment.
- Strip and stockpile topsoil.
- Provide erosion protection during borrowing operations and make sure drainage is maintained.
- On completion of borrowing operations, grade abrupt changes of slope, respread topsoil, and apply and maintain hydroseeded grassing.

**2.3 GEOTEXTILE****General**

Material: UV-stabilised, permeable, polymeric, woven or non-woven textile material used in contact with soil/rock material.

Identification and marking: To AS 3705 (2012).

---

**3 EXECUTION**

---

**3.1 SITE PREPARATION****Erosion and sedimentation control**

Requirement: To 0172 *Environmental management*.

**3.2 GEOTECHNICAL****As found Site conditions**

General: If the following are encountered, give notice and obtain instructions before carrying out any further WUC in the affected area:

- Bad ground.
- Discrepancy in expected conditions.
- Rock.
- Springs, seepages.
- Topsoil more than 100 mm deep.

**Inspection and Testing**

Level of inspection and Testing: by Civil Engineer.

Frequency of Testing: To AS 3798 (2007) Table 8.1.

**3.3 RECORDS OF MEASUREMENT****Excavation and backfilling**

Agreed quantities: If a Schedule of Rates applies, provisional quantities are Documented or there are variations to the Contract levels or dimensions of excavations, do not backfill or place permanent Works in the excavation until the following have been agreed and recorded:

- Depths of excavations in relation to the datum.
- Final plan dimensions of excavations.
- Quantities of excavations in rock.

Method of measurement: By a registered surveyor.

**Rock**

Level and class: If rock is measured for payment purposes, either as extra over excavation of material other than rock or for adjustment of provisional measurements, do not remove the rock until the commencing levels and classes of rock have been determined.

**3.4 REMOVAL OF TOPSOIL****General**

Extent: Areas of cut or fill and areas to be occupied by structures, pavements and embankments.

Maximum depth: 200 mm.

Disposal: Remove topsoil unsuitable for re-use from the Site to AS 3798 (2007) clause 6.1.8.

**Topsoil stockpiles**

General: Stockpile Site topsoil intended for re-use.

Stockpile maximum height: 1.5 m.

Identification: Mark and label stockpiles of different soil types.

Vegetation: Do not burn off or remove plant growth that occurs during storage.

Protection: Conform to the following:

- Provide drainage and erosion protection.
- Do not allow traffic on stockpiles.
- If a stockpile is to remain for more than four weeks, sow with temporary grass.
- Protect the topsoil stockpiles from contamination by other excavated material, weeds and building debris.

**3.5 EXCAVATION****Extent**

Site surface: Excavate the Site to the levels and profiles required for the Documented structures, pavements, filling and landscaping. Make allowance for compaction, settlement or heaving.

Footings, pits, wells and shafts: Excavate to the required sizes and depths. Confirm the foundation conditions meet the design bearing capacity.

**Bearing surfaces**

Requirement: Provide even plane bearing surfaces for load bearing elements including footings. Step to accommodate level changes, as Documented. If supporting masonry, make the steps appropriate to the courses.

**Rock**

General: Do not use explosives.

**Existing footings**

Requirement: If excavation is required within the zone of influence of an existing footing, provide supports to the footing sufficient to prevent damage arising from the Works. Use methods including temporary shoring or underpinning.

**Existing services**

Location: Before starting earthworks, locate and mark existing underground services in the areas that will be affected by the earthworks operations including clearing, excavating and trenching.

Utility services: Contact BEFORE YOU DIG AUSTRALIA to identify location of underground utility services pipes and cables.

Excavation: Do not excavate by machine within 1000 mm of existing services.

**Proof rolling**

Extent: Proof roll excavations for pavements, filling and non-spanning slabs on ground to determine the presence of bad ground.

Proof rolling method and equipment: To AS 3798 (2007) clause 5.5.

Requirement: If excessive settlement, rebound or heaving is encountered, provide Test pits or trenching to determine the extent of bad ground.

Subgrade replacement: Excavate the full extent and depth of bad ground. Remove and replace with selected fill. Place and compact to **PLACING FILL** to achieve the required capacity and levels.

**Disposal of excess excavated material**

General: If not required or unsuitable for fill, remove from Site.

Standard: To AS 3798 (2007) clause 6.1.8.

### 3.6 REINSTATEMENT

#### Deterioration of bearing surfaces

Requirement: If the bearing surface deteriorates, excavate to a sound surface before placing the load bearing element.

#### Subgrades affected by moisture

Requirement: If, due to high moisture content, the subgrade cannot support construction equipment or the overlying pavement cannot be compacted, perform one or more of the following:

- Allow the subgrade to dry until it provides support for equipment and allows compaction.
- Scarify the subgrade to a depth of 150 mm, work as necessary to accelerate drying, and recompact when the moisture content is satisfactory.
- Excavate the wet material and move to spoil stockpile. Backfill excavated areas.

#### Over excavation

Requirement: If excavation exceeds the required depths, reinstate to the correct depths, levels and bearing capacity.

Zone of influence: Within the zone of influence of footings, beams or other structural elements, use concrete of strength equal to the structural element, minimum 15 MPa. Make sure that remedial concrete does not create differential bearing conditions.

Below slabs or pavements: Rectify the over excavation as follows:

- Generally: Provide selected fill compacted to the Documented density.
- Less than 100 mm: Do not backfill. Increase the thickness of the layer above.

Subsoil drains: Backfill over excavation of subsoil drains using coarse filter material conforming to the NCC cited AS/NZS 3500.3 (2021) clause 2.13.1.

### 3.7 SUPPORTING EXCAVATIONS

#### Removal of supports

General: Remove temporary supports progressively as backfilling proceeds.

#### Voids

General: If voids occur outside sheeting or sheet piling, fill and compact voids to a dry density similar to that of the surrounding material.

### 3.8 ADJACENT STRUCTURES

#### Temporary supports

General: If required, provide supports to adjacent structures, sufficient to prevent damage arising from the Works.

Lateral supports: Provide lateral support with shoring.

Vertical supports: If required, provide vertical support with piling or underpinning or both.

#### Permanent supports

General: If permanent supports for adjacent structures are required and are not Documented, give notice and obtain instructions.

#### Encroachments

General: If encroachments from adjacent structures are encountered and are not Documented, give notice and obtain instructions.

#### Zone of influence

Angle from horizontal: by Structural Engineer.

### 3.9 ROCK BOLTING

#### General

Requirement: For temporary or permanent support of rock faces, provide proprietary high strength steel bars or cables anchored into holes drilled in the rock and tensioned against plates bearing on the rock face. Schedule the installation to conform to systematic bolting or calculated relief, as Documented.

Standard: To AS 4678 (2002).

**Protection**

General: Protect permanent rock bolts by grouting the drilled hole with cement grout after tensioning the rock bolt. Protect the bearing plate and the exposed portion of rock bolt and anchorage with a protective coating or by embedment in concrete.

**3.10 GEOTEXTILE****General**

Preparation: Trim the ground to a smooth surface free from cavities and projecting rocks.

Installation: Lay the fabric flat, not stretched tight, and secure with anchor pins. Overlap joints 300 mm minimum.

**3.11 PREPARATION FOR FILLING****Preparation**

Stripping: Prepare the ground surface before placing fill (including topsoil fill), ground slabs or load bearing elements to AS 3798 (2007) clause 6.1.5. Remove loose material, debris, organic matter and material that inhibits or prevents satisfactory placement of fill layers.

Foundation preparation: To AS 3798 (2007) clause 6.1.7.

Compaction: Compact the ground exposed after stripping or excavation, to a minimum depth of 150 mm, to the minimum relative compaction in AS 3798 (2007) Table 5.1.

Ground treatment or improvement methods:

- Scarify method: Loosen exposed excavation by scarifying to a minimum of 150 mm, moisture-condition and compact to AS 3798 (2007) Section 5.
- Impact roller and impact compaction: Use an approved method.

Slope preparation: If fill is placed on a surface steeper than 4:1 (horizontal:vertical), bench the surface to form a key for the fill. As each layer of fill is placed, cut the existing ground surface progressively to form a series of horizontal steps more than 1 m in width and more than 100 mm deep. Recompact the excavated material as part of the filling. Shape to provide free drainage.

**Under earth mounds**

General: Cultivate the ground to a depth of 200 mm before mound formation.

**Under slabs, paving and embankments**

General: If required, loosen the ground to a depth of more than 200 mm and adjust the moisture content before compaction to a density consistent with subsequent filling.

**Rock ledges**

General: Remove overhanging rock ledges.

**3.12 PLACING FILL****General**

Extent: Place fill to the Documented dimensions, levels, grades and cross-sections so that the surface is always self-draining.

Layers: Place fill in near-horizontal layers of uniform thickness no greater than 150 mm after compaction, deposited systematically across the fill area.

Edges: At junctions of fill and existing surfaces, do not feather the edges.

Mix: Place fill in a uniform mixture.

Previous fill: Before placing subsequent fill layers, make sure that previously accepted layers still conform to requirements, including moisture content.

Protection: Protect the Works from damage due to compaction operations. If required, limit the size of compaction equipment or compact by hand.

Protective covering to membranes: Do not disturb or damage during backfilling.

**Placing at structures**

Fill adjacent structures and trenches: To AS 3798 (2007) clause 6.2.6.

Requirement: Place and compact fill in layers simultaneously on both sides of structures, culverts and pipelines to avoid differential loading. Commence compacting each layer at the structure and proceed away from structure.

Over the top of structures: Carefully place first layers of fill.

Retaining walls: Do not place fill against concrete retaining walls until the concrete has been in place for 28 days unless the structure is supported by struts.

### **Compaction**

General: Compact the subgrade and each layer of fill to the required depth and density, as a systematic construction operation. Shape surface to provide drainage and prevent ponding.

Maximum rock and lump size in layer after compaction: To AS 3798 (2007) clause 6.2.2.

Fill batter faces: Either compact separately or overfill and cut back. Form roughened surfaces to the faces.

Minimum relative compaction: To AS 3798 (2007) Table 5.1.

## **3.13 PLACING TOPSOIL**

### **Stockpiled topsoil**

Cultivation: Rip subgrade to a depth of 100 mm or to the depth of rippable subgrade if less. Cultivate around services and tree roots by hand. Trim to allow for the required topsoil depth.

Herbicide: Apply before placing topsoil.

Placing: Spread and grade evenly.

Compaction: Lightly compact topsoil so that the finished surface is smooth, free from lumps of soil, at the required level, ready for cultivation and planting.

Edges: Finish topsoil flush with abutting kerbs, mowing strips and paved surfaces. Feather edges into adjoining undisturbed ground.

### **Disposal of excess topsoil**

On site: Dispose of surplus topsoil remaining on Site by spreading evenly over the areas already placed.

Off site: Remove excess topsoil from the Site and dispose of legally.

## **3.14 FILL MOISTURE CONTROL**

### **General**

Moisture content: Determine in conformance with AS 1289.5.1.1 (2017) or AS 1289.5.2.1 (2017), as appropriate. Adjust the moisture content of fill to  $\pm 2\%$  of the optimum moisture content during compaction as required to achieve the Documented density.

## **3.15 TESTING**

### **Site tests**

Compaction control Tests: To AS 1289.5.4.1 (2007) or AS 1289.5.7.1 (2006).

Test frequency: To AS 3798 (2007) Table 8.1.

## **3.16 COMPLETION**

### **Geotechnical report**

Inspection and Testing report to AS 3798 (2007): by Civil Engineer.

### **Grading**

External areas: Grade to give falls away from buildings, minimum 1:100.

Subfloor areas: Grade the ground surface under suspended floors to drain ground or surface water away from buildings without ponding.

### **Site restoration**

Requirement: If Variation of existing ground surfaces is not required as part of the WUC, restore surfaces to the condition existing at the commencement of the Contract.

|                               |
|-------------------------------|
| <b>0223 SERVICE TRENCHING</b> |
|-------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide trenching for underground services, as Documented. Refer also Electrical, Hydraulic and Mechanical engineer's Documents.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****Trenching**

Earthworks: To AS 3798 (2007).

Electrical services: To AS/NZS 3000 (2018).

Hydraulic services: To the NCC cited AS/NZS 3500.1 (2021), the NCC cited AS/NZS 3500.2 (2021) and the NCC cited AS/NZS 3500.3 (2021).

Communication services: To AS/CA S009 (2020).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS 3798 (2007) and the following apply:

- Bad ground: Ground unsuitable for the Works, including fill liable to subsidence, ground containing cavities, faults or fissures, ground contaminated by harmful substances and ground that is or becomes soft, wet or unstable.
- Rock: Monolithic material with volume greater than 0.3 m<sup>3</sup> that cannot be removed until broken up by rippers or percussion tools.
- Zone of influence: A foundation zone bounded by planes extending downward and outward from the bottom edge of a footing, slab or pavement and defining the extent of foundation material having influence on the stability or support of the footings, slab or pavement.

**1.5 TOLERANCES****Excavation width**

Requirement: ±50 mm, unless constrained by adjacent structures.

**Surface levels**

Earthworks: Finish the surface to the required level, grade and shape within the following tolerances:

- Under building slabs and load bearing elements: +0, -25 mm.
- Pavement subgrades: +0, -40 mm.
- Batters: No steeper than the slope shown on the drawings. Make sure flatter slopes do not impact on boundaries or required clearances to buildings, pavements or landscaping.
- Other ground surfaces: ±50 mm, provided the area remains free draining and matches adjacent construction where required. Provide smoothness as normally produced by a scraper blade.

Pavement base and subbase: Finish the surface to the required level, grade and shape within the following tolerances:

- Subbase: +10 mm, -25 mm.
- Base: +10 mm, -5 mm.

Finished pavement or paving surface: Conform to the Documented level within the following tolerances:

- Asphalt: ±10 mm.

- Concrete: +10 mm, -0 mm.
- Paving:
  - . Finished level:  $\pm 8$  mm.
  - . Height deviation between adjacent units (lippage): 2 mm.
- Granular surfaces:  $\pm 10$  mm.
- Lippage between restored surface and adjacent existing surface: 5 mm.

## 1.6 SUBMISSIONS

### Execution details

Excavation method: Submit details of proposed equipment and method of excavation, including the following:

- Service location and type: A plan of the trench works showing the location and type of service.
- Open excavation: Proposed duration.
- Shuttering and/or bracing of trench sides: If required for safety and stability, provide proposals.
- Geotechnical data: Geotechnical report supporting the trenching and/or boring procedures proposed.
- Boring: Proposals for the following:
  - . Limits on length.
  - . Existence of other services and method of protection.
  - . Pressure grouting to voids.
  - . The effect of pressure grouting on other services, ground heave and proposals for minimising such effects.
  - . Access to properties outside the Site.
  - . Council permits.
  - . Service interruptions including a plan for minimising unintended interruptions.
- Hazards: Identify WHS hazards that may be encountered with deep trenches including toxic gases and liquids.

Off-site disposal location: Submit details of the proposed disposal location and evidence of conformance with the relevant authorities for the disposal of material required to be removed from the Site.

### Records

Requirement: Upon completion, submit to the relevant authority record drawings showing the location of the installed services.

### Tests

Trench backfill: Submit results of the following:

- Compaction control Tests.

## 1.7 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Items to be measured as listed in **GROUND CONDITIONS, Records of measurement**.
- Service trenches excavated before laying the service.
- Services laid in trenches and ready for backfilling.
- Completed surface restoration.

---

## 2 PRODUCTS

### 2.1 FILL MATERIALS

#### General

Suitable material: To AS 3798 (2007) clause 4.4 including inorganic, non-perishable material suitably graded and capable of compaction to the Documented density.

Unsuitable materials: Do not use fill defined as unsuitable by AS 3798 (2007) clause 4.3.

Sulfur content: Do not provide filling with sulfur content exceeding 0.5% within 500 mm of concrete and cement bound elements (for example masonry) unless such elements are protected by impermeable membranes or equivalent means.

Re-use of excavated material: Only re-use suitable material.

Material in reactive clay areas: In sites classified M, M-D, H1, H1-D, H2, H2-D, E or E-D to AS 2870 (2011), re-use excavated site material at a moisture content within  $\pm 1\%$  of that of the adjoining in situ clay.

## **2.2 SURFACE RESTORATION MATERIALS**

### **General**

Re-use: If possible, re-use existing surface materials removed during excavation. Make sure replacement materials conform to the Documented requirements.

### **Subbase and base**

Requirement: Provide crushed rock material configured in layers and depths to match existing and adjacent work, as follows:

- Base: 20 mm nominal size.
- Subbase: 40 mm nominal size.

### **Pathways and paved surfaces generally**

Requirement: Provide materials consistent with those of the existing surface before service trenching works commence.

### **Concrete surfaces**

Material requirements: To AS 1379 (2007).

Concrete strength grade: N25.

Slump: Maximum 100 mm.

### **Asphalt surfaces**

Aggregate: To AS 2758.5 (2020) or to AS 2758.2 (2021) for sprayed bituminous surfaces.

Asphalt: To AS 2150 (2020).

Binder:

- Bitumen emulsion: To AS 1160 (1996).
- Conventional bitumen: Class 170 to AS 2008 (2013).
- Cutback bitumen: To AS 2157 (1997).

### **Pavers**

Concrete and clay pavers: To AS/NZS 4455.2 (2010).

Bedding and joint filling sand: Well-graded, free of deleterious material such as soluble salts that may cause efflorescence.

Stone paving and setts: Provide sound stone paving and setts, of uniform quality. Reject any with Defects liable to affect strength and durability.

Bedding mortar mix (cement:sand): Select from the range 1:3 to 1:4 to obtain satisfactory adhesion. Provide minimum water.

---

## **3 EXECUTION**

### **3.1 EXISTING SERVICES**

#### **Location**

Requirement: Before commencing service trenching, locate and mark existing underground services in the areas that will be affected by the service trenching operations.

Utility services: Contact BEFORE YOU DIG AUSTRALIA to identify location of underground utility services pipes and cables.

#### **Excavation**

General: Do not excavate by machine within 1 m of existing underground services.

### 3.2 EXISTING SURFACES

#### Concrete and asphalt pavements

Method: Sawcut trench set-out lines for the full depths of the bound pavement layers except where the set-out line is located along expansion joints.

Removal of concrete and asphalt: Break out concrete or asphalt pavement material between the trench set-out lines, remove and dispose of off site.

#### Paving

Removal: Take up paving units both full and cut by hand, between the trench set-out lines. Neatly stack on wooden pallets.

Concrete edging: Break out, remove and dispose of off site.

Concrete subbase: If present, sawcut along the trench set-out lines to allow the concrete subbase to be removed and disposed of off site.

#### Grass

Removal method: Neatly cut grass turf between trench set-out lines into 300 mm squares.

Grass suitable for re-use: Take up and store the turf. Water during the storage period.

Unsuitable grass: Remove and dispose of off site.

#### Small plants, shrubs and trees

Remove for re-planting: Take up and store. Wrap the rootball in a hessian or plastic bag with drain holes and water during the storage period.

Unsuitable vegetation: Remove and dispose of off site.

### 3.3 GROUND CONDITIONS

#### As found Site conditions

Unexpected conditions: If any of the following are encountered, give notice immediately and obtain instructions before carrying out any further Work in the affected area:

- Bad ground.
- Discrepancies from expected ground conditions.
- Rock.
- Springs, seepages.
- Topsoil > 100 mm deep.

#### Records of measurement

Excavation and backfilling: If a Schedule of Rates applies, provisional quantities are specified or there are variations to the Contract levels or dimensions of excavations, do not backfill or place permanent Works in the excavation until the following have been agreed and recorded:

- Depths of excavations in relation to the datum.
- Final plan dimensions of excavations.
- Quantities of excavations in rock.

Rock: If rock is measured for payment purposes, whether as extra over excavation of material other than rock or for adjustment of provisional measurements, do not remove the rock until the commencing levels and classes of rock have been determined.

### 3.4 EXCAVATION

#### General

Requirement: Excavate for underground services in conformance with the following:

- To required lines and levels, with uniform grades.
- Straight between access chambers, inspection points and junctions.
- With stable sides.
- Tree protection: To AS 4970 (2025).

#### Adjacent structures

Existing footings: If excavation is required within the zone of influence of an existing footing, provide supports to the footing sufficient to prevent damage arising from the Works. Use methods including temporary shoring or underpinning.

Temporary supports: If required, provide supports to adjacent structures, sufficient to prevent damage arising from the Works, as follows:

- Lateral supports: Provide lateral support using shoring.
- Vertical supports: Provide vertical support using piling or underpinning or both.

Permanent supports: If permanent supports for adjacent structures are required and are not Documented, give notice and obtain instructions.

Encroachments: If encroachments from adjacent structures are encountered and are not Documented, give notice and obtain instructions.

#### **Trench widths**

General: Keep trench widths to the minimum, consistent with the laying and bedding of the relevant service and construction of access chambers and pits.

#### **Trench depths**

General: As required by the relevant service and its bedding method.

#### **Obstructions**

General: Clear trenches of sharp projections. Cut back roots encountered in trenches to at least 600 mm clear of services. Remove other obstructions including stumps and boulders that may interfere with services or bedding.

#### **Dewatering**

General: Keep trenches free of water. Place bedding material, services and backfilling on firm ground, free of surface water.

Pumping: Provide pump-out from adjacent sumps or install well points.

Adjacent subsidence: Provide recharge points to isolate the dewatering zone.

#### **Excess excavation**

General: If trench excavation exceeds the correct depth, reinstate to the correct depth and bearing capacity using compacted bedding material or sand stabilised with 1 part of cement to 20 parts of sand by volume.

#### **Stockpiles**

Topsoil removal: Stockpile topsoil intended for re-use to a maximum height of 1500 mm.

Excavated material for backfill: If required, segregate the earth and rock material and stockpile, for re-use in backfilling operations.

Locations: Do not stockpile excavated material against tree trunks, buildings or fences. Do not obstruct the free flow of water along drainage channels.

Disposal: If stockpiling is not permitted, dispose of excavated material off site.

#### **Unsuitable material**

Disposal: Remove unsuitable material from the bottom of the trench or at foundation level and dispose of off site. Replace with trench backfill material.

#### **Boring**

Subcontractor: If boring is required instead of trenches, engage a suitably qualified subcontractor to do the work.

### **3.5 TRENCH BACKFILL**

#### **General**

Place fill: To AS 3798 (2007) clauses 6.2.2 and 6.2.6.

Timing: Backfill service trenches as soon as possible after laying and bedding the service, if possible on the same working day.

Removal of supports: Remove temporary supports progressively as backfilling proceeds.

#### **Marking services**

Marking tape: Provide marking tape above services, with appropriate labelling, to AS/NZS 2648.1 (1995) and as follows:

- Non-metallic services: Provide tape capable of being detected by in-ground scanning devices.
- Location: Locate tape approximately half the depth of the service being marked, to a maximum depth of 200 mm below the finished ground level.

Boring: If boring techniques are used to install the service, provide permanent on site labelling at the start and end of the service run and record on the record drawings.

**Bedding, haunch, side and overlay zones**

Installation and material: To the particular utility authority or utility service requirements. Secure pipes against flotation.

Bedding of services: Surround pipes or conduits on all sides with a minimum of 75 mm compacted bedding sand and as Documented.

Overlay zone thickness: Maximum 300 mm immediately over the utility service.

**Trees**

Backfill at trees: Backfill minimum 300 mm thick, around tree roots with a topsoil mixture. Place and compact in layers of 150 mm minimum depth to a dry density equal to that of the surrounding soil.

Original surface level: Do not place backfill above the original ground surface around tree trunks or over the root zone.

Watering: Thoroughly water immediately after backfilling the tree root zone.

**Compaction**

Backfill moisture control: To AS 3798 (2007) clause 6.2.3.

Layers: Compact all material in layers not exceeding 150 mm compacted thickness. Compact each layer to the Documented relative compaction before starting the next layer.

Compaction: To AS 3798 (2007) Section 5 and clause 6.2.4.

Precautions: Use compaction methods that do not cause damage or misalignment to utility services.

**3.6 TESTING****General**

Test frequency: To AS 3798 (2007) Table 8.1.

**Compaction control**

Requirement: To AS 1289.5.4.1 (2007) or AS 1289.5.7.1 (2006).

**3.7 SURFACE RESTORATION****Subbase and base**

Compaction: Uniformly compact each layer of the subbase and base courses over the full area and depth within the trench to a relative compaction of 100% when Tested in conformance with AS 1289.5.4.1 (2007).

Compacted layer thickness:

- Maximum: 200 mm.
- Minimum: 100 mm.

Compaction Test frequency: Minimum 1 Test every 2 layers, for each length of 40 linear metres.

**Concrete surfaces**

Construction: Conform to the following:

- Prime coat the cut edges of the existing surfaces with cement slurry. Lay and compact concrete so that the edges are flush with the adjoining existing surfaces.
- Surface finish and pattern: Match existing adjoining work.
- Minimum thickness: 75 mm or the adjacent pavement thickness, whichever is thicker.
- Reinforcement and dowels: If required, provide steel reinforcement with dowels into the adjacent concrete.
- Expansion joints: 15 mm thick preformed bituminous fibreboard jointing material placed in line with joints in existing concrete.
- Control joints:
  - . Form control joints in line with the control joints in existing concrete.
  - . Around service poles: Terminate the concrete paving 200 mm from the pole and fill the resulting space with cold mix asphalt.

Weather: Do not place concrete in ambient temperatures above 30°C or below 10°C, without adequate precautions. Protect surface from rain damage.

Compaction: Compact as follows:

- Thickness 100 mm or less: Compact by placing, screeding and finishing processes. If required, use a hand-held vibrating screed at the surface. Do not use immersion vibrators.

- Thickness more than 100 mm and downturns: Use an immersion vibrator.

Curing: Cure by keeping continuously wet for 7 days.

**Asphalt surfaces**

Placement: To AS 2150 (2020).

Operations: Spread the asphalt mix in layers covering the full width of the trench.

Thickness: Match the adjoining asphalt surface.

Finish: Compact to a smooth even surface.

Sprayed bituminous surfaces: To AS 3727.1 (2016) Section 8.

**Pavers**

Bedding: Replicate the bedding used for the original paved surface. Use bedding methods and materials that are appropriate to the paver, the substrate and the conditions of service, and which leave the paver firmly and solidly bedded in the bedding material.

Laying: Relay to match the pattern and surface levels of the existing paving.

Damaged paving unsuitable for relaying: Replace with new paving of the same material, type, size and colour as the existing.

**Landscaped areas**

In topsoil areas: Complete the backfilling with topsoil for at least the top 100 mm.

Grass: Relay stockpiled turf. If existing turf is no longer viable, re-sow grass over the trench and other disturbed areas.

Planted areas: Overfill to allow for settlement.

### **3.8 COMPLETION**

**General**

Requirement: Upon completion, record the location of all installed services on the record drawings.

|                               |
|-------------------------------|
| <b>0224 STORMWATER - SITE</b> |
|-------------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide stormwater drainage, as Documented. Refer also Civil Engineer and Landscape Architect's Documents.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.

### 1.3 STANDARDS

#### Stormwater drainage

Standard: To the NCC cited AS/NZS 3500.3 (2021).

### 1.4 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the following definitions apply:

- Pipe surround: Includes pipe overlay, pipe side support, side zone and haunch zone.

### 1.5 TOLERANCES

#### Pipeline tolerances table

|                                                                                                                                            | Permissible angular deviation<br>from the Documented<br>alignment | Permissible displacement<br>from the Documented<br>positions |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------|
| Horizontal                                                                                                                                 | 1:300                                                             | 15 mm                                                        |
| Vertical                                                                                                                                   | 1:500                                                             | 5 mm                                                         |
| Note: These tolerances are conditional on maintaining falls to outlets and no part of a pipeline having less than the Documented gradient. |                                                                   |                                                              |

### 1.6 SUBMISSIONS

#### Certification

Requirement: Submit certificate stating that network is leak free upon completion.

#### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

#### Records

Record drawings: To **RECORD DRAWINGS** in 0171 General requirements.

#### Samples

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

#### Shop drawings

Requirement: To **SHOP DRAWINGS** in 0171 General requirements. Submit detail drawings at minimum 1:100 scale, showing the following:

- Pipework and equipment layout and sections showing the Work to be installed on the level that the services are installed.
- Long sections of below ground drainage.
- Access openings, cover plates, valve boxes and access pits.
- Location, type, grade and finish of piping, fittings and pipe supports.
- On-site detention pondage areas.

- Provision of blue metal backfill to seepage drain system.
- Provision of erosion control measures.
- Provision of road barriers and lighting.
- Provision of temporary sanitary accommodation for construction workers.
- Provision of trafficable cover plates in the public domain.
- Relevant survey levels.
- Site and floor set-out points.

**Tests**

Requirement: Submit Test results of the following:

- Pre-completion Testing to **TESTING**.

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Excavated surfaces before placing bedding material.
- Underground services or services to be concealed, before being covered.
- Pipe joints before covering.
- Placing of concrete.
- Completion of stormwater drainage.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples of the following:

- Each type of imported pipe bedding material.
- Each type of filter material.

**2.2 MATERIALS****Concrete and mortar**

Requirement: To the NCC cited AS/NZS 3500.3 (2021) clause 2.9 and the following:

- Concrete:
  - . Standard: To AS 1379 (2007).
  - . Grade: Minimum N25.
- Cement:
  - . Standard: To AS 3972 (2010).
  - . Type: GP, GL or GB.
- Steel reinforcement: To AS/NZS 4671 (2019).

**Joints**

General: To the NCC cited AS/NZS 3500.3 (2021) clause 2.7.

Solvent cement and priming fluid: To AS 3879 (2011).

Rubber ring joints/elastomeric seals: To AS 1646 (2007).

**Pipes and fittings**

Materials and products: To the NCC cited AS/NZS 3500.3 (2021) Section 2 and the following:

- Best environmental practice (BEP) PVC: To AS/NZS 5395 (2024).
- Precast concrete: To AS 3850.3 (2021).
- Vitrified clay or ceramic: To AS 1741 (1991).

**Bedding material**

Bed and haunch zones: Conform to the NCC cited AS/NZS 3500.3 (2021) clause 6.3.5.1.

**Filter material**

Requirement: Materials consisting of natural clean washed sands and gravels and screened crushed rock, to the NCC cited AS/NZS 3500.3 (2021) clause 2.13.1.

Filter socks: Permeable polyester socks, capable of retaining particles 0.25 mm and greater.

**2.3 GEOTEXTILES****General**

Material: UV-stabilised, permeable, polymeric, woven or non-woven textile material used in contact with soil/rock material.

Identification and marking: To AS 3705 (2012).

**Subsoil drainage**

Filter: To the NCC cited AS/NZS 3500.3 (2021) clause 2.13.2.

**2.4 PREFABRICATED PITS****General**

Precast or prefabricated pits: To the NCC cited AS/NZS 3500.3 (2021) clauses 2.12.8 and 7.5, and AS 3850.3 (2021).

**Access covers and grates**

Cover finish and load classification: Provide access cover and grate with the Documented finish, slip resistance and load class.

---

**3 EXECUTION**

---

**3.1 PIPING****Installation**

Laying: Lay lengths separately with the barrel bearing evenly on the prepared bedding.

Sockets: Lay with socket ends pointing upstream.

Cleaning: Clean pipe interior of dirt, debris, mortar and other foreign matter.

Protection: Provide temporary caps over the ends of incomplete sections to prevent the entry of foreign matter.

**3.2 STORMWATER DRAINS****Location**

General: Provide stormwater drains to connect surface drains, subsoil drains and drainage pits to the outlet point or point of connection. Make sure location of piping will not interfere with other services and building elements not yet installed or built. Subject to the preceding and Documented layouts, follow the most direct route.

**Laying**

Installation: Lay in straight lines between changes in direction or grade. If other pipes are adjacent, set each pipe true to line and complete each joint before laying the next pipe. If Work is not continuous, cap open ends to prevent entry of foreign matter.

**Identification**

Requirement: Lay a detectable strip or plastic tape in the trench after pipe laying, Testing and initial backfilling.

**Pipe underlay**

General: Bed piping on a continuous underlay of bedding material, minimum 75 mm compacted thickness. Grade the underlay evenly to the gradient of the pipeline.

Chases: If required, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

**Pipe surrounds**

General: Place the material in the pipe surround in layers, maximum 200 mm loose thickness. Compact without damaging or displacing the piping.

**Trench backfill**

General: Backfill the remainder of the trench to the level of the bottom of the subgrade with fill material placed in maximum 200 mm loose layers and compacted as follows:

- Under pavements: 95% of the standard maximum dry density.
- Other locations: 90% of the standard maximum dry density.

**Lifting holes**

Sealing: Seal lifting holes in all pipes with plastic preformed plugs or 3:1 (sand:cement) mortar, before the commencement of backfilling.

**Anchor blocks**

Restraint: Provide reinforced concrete anchor blocks at junctions and changes of grade or direction conforming to the NCC cited AS/NZS 3500.3 (2021) clause 7.9 and as Documented.

**3.3 SUBSOIL DRAINS****General**

Requirement: Provide subsoil drains to intercept groundwater seepage and prevent water build-up behind walls and under pavements. Connect subsoil drains to surface drains or to the stormwater drainage system as applicable.

Piping: As Documented.

Trench width: Minimum 450 mm.

Trench floor: Grade the trench floor evenly to the gradient of the pipeline. If the trench floor is rock, correct any irregularities with compacted bedding material.

Pipe cover: To the NCC cited AS/NZS 3500.3 (2021) clause 6.2.5

**Jointing**

General: At junctions of subsoil pipes, provide tees, couplings or adaptors to AS 2439.1 (2007).

**Pipe underlay**

Bedding: Bed piping on a continuous underlay of bedding material, minimum 75 mm compacted thickness. Grade the underlay evenly to the gradient of the pipeline.

Chases: If required, form chases to prevent projections such as sockets and flanges from bearing on the trench bottom or underlay.

**Pipe surrounds**

General: Place the material in the pipe surround in layers, maximum 200 mm loose thickness. Compact without damaging or displacing the piping.

Depth of overlay:

- To the underside of the bases of overlying structures such as pavements, slabs and channels.
- To within 150 mm of the finished surface of unpaved or landscaped areas.

**Geotextiles**

Protection: Provide heavy duty protective covering. Store clear of the ground and out of direct sunlight. During installation, do not expose the filter fabric to sunlight for more than 14 days.

Laying: Place geotextile, as Documented.

**Filter socks**

General: Securely fit or join the sock at each joint.

**3.4 PITS****Installation**

General: Prepare foundation, install pit and connect pipes, to manufacturer's recommendations.

Location: At junctions, changes of gradient and changes of direction of stormwater drains, as Documented.

**Finish to in situ exposed surfaces**

General: Provide a smooth, seamless finish, using steel trowelled render or concrete cast in steel forms.

Corners: Cove or splay internal corners.

**Metal access covers and grates**

Cover levels: Top of cover or grate, including frame:

- In paved areas: Flush with the paving surface.
- In landscaped areas: 25 mm above finished surface.
- Gratings taking surface water runoff: Locate to receive runoff without ponding.

### 3.5 TESTING

#### Site tests

General: Before backfilling or concealing, carry out an air or water pressure Test to the NCC cited AS/NZS 3500.3 (2021) Section 9.

Leaks: If leaks are found, rectify and retest.

### 3.6 COMPLETION

#### Cleaning

General: Clean and flush the whole installation.

#### Operation and maintenance manuals

Requirement: Conform to **OPERATION AND MAINTENANCE MANUALS** in *0171 General requirements*.

|                               |
|-------------------------------|
| <b>0274 CONCRETE PAVEMENT</b> |
|-------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide concrete pavement, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Civil Engineer's Documents, Structural Engineer's Documents, and Landscape Architect's Documents.

**Performance**

Requirement:

- Free draining and evenly graded between level points.
- Consistent in colour and finish.
- Even and smooth riding surfaces.

Conformance: Conform to the local authority requirements for levels, grades and minimum thickness, reinforcement and concrete strength for pavements within the kerb-and-gutter property boundaries.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements.*
- *0222 Earthwork.*
- *0271 Pavement base and subbase.*

**1.3 STANDARDS****Concrete**

Specification and supply: To AS 1379 (2007).

Materials and construction: To AS 3600 (2018).

Residential pavements: To AS 3727.1 (2016).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Ambient temperature: The air temperature at the time of mixing and placing of concrete.
- Concrete class - normal: Concrete that is specified primarily by a standard compressive strength grade up to 50 MPa and otherwise in conformance with AS 1379 (2007) clause 1.5.3.
- Concrete class - special: Concrete that is specified to have certain properties or characteristics different from, or additional to, those of normal-class concrete and otherwise in conformance with AS 1379 (2007) clause 1.5.4.
- Sample: A portion of material taken for assessment, representative of the larger quantity of material from which it was taken.
- Weather - cold: Ambient shade temperature less than 10°C.
- Weather - hot: Ambient shade temperature greater than 30°C.

**1.5 TOLERANCES****General**

Surface abutting gutters:  $\pm 5$  mm from the level of the gutter edge.

Rigid pavement surface:

- From design level: +10 mm, -0 mm.
- From a 3 m straightedge placed anywhere on surface: 5 mm.

Horizontal position of outer concrete edge:  $\pm 20$  mm from Documented position.

Joint locations in plan: 10 mm from Documented position.

## 1.6 SUBMISSIONS

### Certification

Test certificates and records: Submit Test certificates and also retain results on Site.

Compliance certificate: If product Testing is not proposed, submit a manufacturer's certificate together with results of recent Tests undertaken by the manufacturer, showing compliance with Test criteria.

### Execution details

Concrete: Submit proposals for mixing, placing, finishing and curing concrete including the following:

- Addition of water on Site.
- Changes to the concrete mix.
- Changes to Documented joint locations.
- Curing and protection methods.
- Cutting or displacing reinforcement, or cutting or coring hardened concrete.
- Handling, placement, compaction and finishing methods and equipment, including pumping.
- Sawn joints: Submit details of proposed methods, timing and sequence of sawing joints.
- Sequence and times for concrete placement and construction joint locations. Include any proposed sequential placement of slab segments.
- Site storage, mixing and transport methods and equipment, if applicable.
- Temperature control methods to suit hot or cold atmospheric conditions during concrete placement.

Reinforcement: Submit the following:

- General: Details of any proposed changes to Documented reinforcement.
- Damaged galvanizing: Details of proposed repair to AS/NZS 4680 (2025) clause 4.3.
- Mechanical bar splices: Details and Test certificates for each size and type of bar to be spliced.
- Provision for concrete placement: Details of spacing or cover to reinforcement that does not conform to AS 3600 (2018).
- Splicing: Details of any proposed changes to Documented requirements.
- Welding: Details of any proposed welding of reinforcement to AS/NZS 1554.3 (2014).

Joint sealants: Submit proposals for installation methods and sealant performance.

Crack assessment: If unplanned cracks occur in the finished pavement, submit proposals for investigation.

Surface repair method: If required, submit details of the proposed method before commencing repairs.

### Products and materials

Aggregates: Nominate the source for all aggregates.

Reinforcement: Submit Test certificates or the manufacturer's certificate of compliance, as evidence of product conformance with AS/NZS 4671 (2019).

Pre-mixed supply delivery dockets: For each batch, submit a docket listing the information required by AS 1379 (2007) and the following:

- Special-class performance concrete: Documented performance and type of cement binder.
- Special-class prescription concrete: Details of mix, additives and type of cement binder.
- Method of placement and climate conditions during pour.
- Name of concrete delivery supervisor.
- Project assessment carried out each day.
- The concrete element or part of the Works for which the concrete was ordered and where it was placed.

Liquid curing compounds: Submit certified Test results, including the application rate and the efficiency index to AS 3799 (1998) Appendix B.

Curing by covering: Submit details of the proposed covering material.

Repair materials: Submit proposals for epoxy resin/grout and elastomeric sealant.

Trial mix design report: Six weeks before commencing production, submit a report for each mix design containing the information required in AS 1012.2 (2014), the individual and combined aggregate particle size distribution, and the records and reports for the Tests.

**Subcontractors**

Pre-mixed supply: Submit names and contact details of proposed pre-mixed concrete suppliers and alternative source of supply in the event of breakdown of pre-mixed or Site mixed supply.

**Tests**

Requirement: Submit Test results of the following:

- Concrete:
  - . Compressive strength.
  - . Drying shrinkage.
  - . Flexural strength.
  - . Slump.
- Luminance contrast of completed tactile ground surface indicator installations.
- Slip resistance of completed installations.

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Base or subgrade before covering.
- Membrane or film underlay installed on the base or subgrade.
- Concrete formwork, reinforcement and dowels in position.
- Commencement of concrete placement and compaction.
- Finishing and curing of concrete.
- Evaluation of surface finish.

---

**2 PRODUCTS****2.1 REINFORCEMENT****Steel reinforcement**

Standard: To AS/NZS 4671 (2019).

Surface condition: Provide surfaces conforming to the following:

- Free of loose or flaking mill scale and rust.
- Clean from oil, grease, mud or other material that may reduce the bond between the reinforcement and concrete.

Storage: Store reinforcement above the surface of the ground and protect from damage and from deterioration by exposure.

**Protective coatings**

Standard: To AS 3600 (2018) clause 17.2.1.2.

Requirement: For concrete containing protective coated reinforcement, provide the same coating type to all reinforcement and embedded ferrous metal items, including tie wires, stools, spacers, stirrups, plates and ferrules, and protect other embedded metals with a suitable coating.

Epoxy coating: Provide high build, high solids chemically resistant coating to AS/NZS 3750.14 (1997).

- Thickness: 200 µm minimum.

Galvanizing: To AS/NZS 4680 (2025) and as follows:

- Sequence: If fabricating after galvanizing, repair damaged galvanizing and coat cut ends.
- Zinc-coating (minimum): 600 g/m<sup>2</sup>.

**Fibre reinforcement**

Steel fibres: To AS 3600 (2018) clause 16.7.1.

Synthetic fibres: To EN 14889-2 (2006).

Storage: Store in a dry environment. Do not stack.

**Dowels**

General: Provide each dowel in one piece, straight, cut accurately to size with ends square and free from burrs.

Standard: To AS/NZS 3679.1 (2016).

Finish: Hot dip galvanized.

#### **Tie wire**

General: Annealed steel 1.25 mm diameter (minimum).

External and corrosive applications: Galvanized.

#### **Supports**

Standard: To AS/NZS 2425 (2015).

## **2.2 CONCRETE MIX**

#### **Standard**

Concrete mix and supply: To AS 3600 (2018) clause 17.1 and AS 1379 (2007).

#### **Properties**

Slump: Maximum 100 mm.

Drying shrinkage: Maximum 450  $\mu\epsilon$  after 21 days of air drying.

## **2.3 AGGREGATE**

#### **Characteristics**

Standard: To AS 2758.1 (2014).

Durability: Tested to AS 1141.22 (2019):

- Wet strength not less than 80 kN.
- 10% Fines Wet/Dry variation not to exceed 35%.

Recycled concrete aggregate (RCA): If blending coarse RCA with natural aggregates, make sure substitution rates are below 30%.

## **2.4 CEMENT**

#### **General**

Standard: To AS 3972 (2010).

Age: Less than 6 months old.

Moisture: Protect from moisture until used. Do not use caked or lumpy cement.

Storage: Store cement bags in a dry, under cover and above-ground environment.

#### **Supplementary cementitious materials**

Fly ash: To AS/NZS 3582.1 (2016).

Slag: To AS 3582.2 (2016).

Amorphous silica: To AS/NZS 3582.3 (2016).

Manufactured pozzolans: To AS 3582.4 (2022).

## **2.5 WATER**

#### **General**

Mixing water: To AS 1379 (2007) clause 2.4.

Requirement: Clean potable water, free from any material that may be harmful to the concrete or reinforcement including oil, acid, alkali, organic or vegetable matter.

Limits of impurities in mixing water: To AS 1379 (2007) Table 2.2 and the following:

- Chloride ion: Maximum 300 parts per million to AS 1478.1 (2000) Appendix C.
- Sulfate ion: Maximum 400 parts per million to AS 1289.4.2.1 (2020).

## **2.6 ADMIXTURES**

#### **General**

Standard: Chemical admixtures to AS 1478.1 (2000), used to the manufacturer's recommendations.

Quality: Free from calcium chloride, calcium formate, triethanolamine or any other accelerator. Do not use admixtures or combinations of admixtures without prior written approval.

Dosage: Vary the dosage of chemical admixture to account for factors such as air temperature, setting time and cement content to the manufacturer's recommendations.

**Types of admixtures**

Air entraining agent: Adjust mix for workability allowing up to 5% air entrainment.

Cool season retarder: From April to September, use a lignin or lignin-based set-retarding admixture containing maximum 6% reducing sugars, Type WRRe conforming to AS 1478.1 (2000).

Warm season retarder: From October to March, use a lignin or lignin-based set-retarding admixture, Type Re or Type WRRe, for controlling slump within the limits in **CONCRETE MIX, Properties**.

**2.7 CURING COMPOUNDS****General**

Curing compounds: To AS 1160 (1996) and AS 3799 (1998), Type 2.

Sheet material covering: To ASTM C171 (2020), white opaque or clear polyethylene film, or white burlap-polyethylene sheet, or equivalent material.

**2.8 OTHER MATERIALS****Tactile ground surface indicators**

Standard: To AS/NZS 1428.4.1 (2009).

---

**3 EXECUTION**

---

**3.1 GENERAL****Traffic control**

Traffic restriction: Do not allow traffic or Construction Plant other than those associated with Testing, sawcutting, cleaning or joint sealing on pavement for minimum 10 days after placing or until the concrete has reached a compressive strength of at least 20 MPa, and joints have been sealed.

**3.2 SUBGRADE****Preparation**

Conformance: Prepare subgrade to *0222 Earthwork*.

Extent: Prepare a uniform subgrade for the full pavement formation, extending at least to the back of kerbs or at least 300 mm beyond each side of the carriageway if kerbs are not proposed.

Adjacent structures: Compact without causing damage to adjacent structures.

Reinstatement: Make sure of uniformity for backfilling of any utility trenches.

**3.3 SUBBASE****Thickness**

Subbase thickness: by Structural Engineer.

**Width**

Subbase width: Extend the subbase at its full depth to at least the back of kerbs or other edge stops before their installation.

No integral kerbs: Extend granular unbound subbase at least 300 mm beyond each side of the carriageway.

**Tolerance**

Subbase finished surface level: +0 mm, -10 mm.

**Placement**

Bound and unbound subbase materials and placement: To *0271 Pavement base and subbase*.

**Friction reduction**

Requirement: Provide 200 µm thick polyethylene sheeting with 200 mm taped minimum laps and/or a 20 mm thick layer of sand (silt and clay material less than 5%) directly beneath the concrete pavement.

**3.4 INSTALLATION****Junctions with existing pavements**

Trimming: If new pavement is to be joined to an existing pavement, trim the edge of the existing pavement to create a neat vertical edge for its full depth before placing new pavement material.

**Fixed formwork**

Type:

- Steel forms.
- Seasoned, dressed timber planks, free of warps, bends or kinks.

Depth: Equal to the edge thickness of the slab and in one piece.

Tolerances on position:

- Level of top of form: -0 mm, +10 mm from pavement surface design level.
- Horizontal tolerance: 10 mm (maximum departure from a plane surface).
- Verticality: 3 mm departure from vertical.

Staking: Stake forms in position using at least 3 steel stakes per form, not more than 1.5 m apart. Lock joints between form sections to prevent movement.

Release agent: Before placing reinforcement, apply a release agent compatible with the contact surfaces, to the interior of the formwork, except if the concrete is to receive an applied finish for which there is no compatible release agent.

Re-use: Clean and recoat the forms each time before placing concrete.

Keyways: Form the keyways of keyed construction joints using steel or timber form strips accurately located at the mid-depth of the slab and securely fixed flush against the formwork face.

**Reinforcement**

Tolerances in fabrication and fixing: To AS 3600 (2018).

Locate reinforcement: Place reinforcement in the top half of the pavement.

Minimum cover to reinforcement: 30 mm.

Splicing mesh: Overlap a minimum of 2 crosswires.

Supports: Provide reinforcement supports as follows:

- Able to withstand construction and traffic loads and maintain the concrete cover, as Documented.
- With a protective coating if they are ferrous metal extending to the surface of the concrete.
- Use plastic or concrete supports with galvanized or zinc-coated reinforcement.
- Spacing:
  - . Bars:  $\leq 60$  diameters.
  - . Mesh:  $\leq 600$  mm.
- Supports over membranes: Prevent damage to waterproofing membranes or vapour barriers. If appropriate, place a metal or plastic plate under each support.

Projecting reinforcement: If starter or other bars extend beyond reinforcement mats or cages, through formwork or from cast concrete, provide a plastic protective cap to each bar until it is cast into later Work.

Tying: Secure the reinforcement against displacement at intersections with either wire ties, or clips. Bend the ends of wire ties away from nearby faces of formwork or unformed faces to prevent the ties projecting into the concrete cover.

Mats: For bar reinforcement in the form of a mat, secure each bar at alternate intersections.

**Cores, fasteners and embedded items**

Position: Fix cores and embedded items to prevent movement during concrete placing. In locating cores, fasteners and embedded items, displace but do not cut reinforcement, and maintain cover to reinforcement.

Isolation: Isolate embedded items to prevent water tracking to concrete providing minimum cover to reinforcement.

**3.5 CONCRETE SUPPLY****Elapsed delivery time**

General: Make sure that the elapsed time between the wetting of the mix and the discharge of the mix at the Site is in conformance with the **Elapsed delivery time table**. Do not discharge at ambient temperature below 10°C or above 30°C unless approved heating or cooling measures are taken to deliver concrete within the range 5°C to 35°C.

**Elapsed delivery time table**

| Concrete temperature at time of discharge (°C) | Maximum elapsed time (minutes) |
|------------------------------------------------|--------------------------------|
| 5 – 24                                         | 120                            |
| 24 – 27                                        | 90                             |
| 27 – 30                                        | 60                             |
| 30 – 35                                        | 45                             |

**Pre-mixed supply**

Transport method: Select a method that will prevent segregation, loss of material and contamination of the environment, and that will not adversely affect placing or compaction.

**Site mixed supply**

Emergencies: If mixing by hand, provide details.

Plant: Mix concrete in a plant located on the construction Site.

**3.6 TESTING****General**

Reports and records of Test results: To the relevant parts of the AS 1012 series. Keep results on Site.

**Standards**

Sampling, identification, Testing and recording: To the AS 1012 series.

Type and frequency: To AS 1379 (2007).

**Concrete Testing methods**

Specimens: Sample the concrete on Site, at the point of discharge from the agitator.

Slump: To AS 1012.3.1 (2014).

Compressive strength: To AS 1012.8.1 (2014) and AS 1012.9 (2014).

Drying shrinkage: To AS 1012.8.4 (2015) and AS 1012.13 (2015).

Flexural strength: To AS 1012.8.2 (2014) and AS 1012.11 (2000).

Acceptance criterion for strength: The average strength of any set of 3 consecutive project samples must be equal to or greater than the specified minimum value.

Sampling frequency: Provide a minimum of one sample from each 50 m<sup>3</sup> of concrete.

**Slip resistance Tests**

Slip resistance of completed installation: To AS 4663 (2013).

**Tactile ground surface indicators**

Luminance contrast of completed installation: To AS/NZS 1428.4.1 (2009) Appendix E.

**3.7 CONCRETE PLACING AND COMPACTION****Placing**

General: Place concrete uniformly over the width of the slab or lane and so that the face is generally vertical and normal to the direction of placement. Hand spread concrete using shovels, not rakes.

Fibre-reinforced concrete: For pumped concrete, use a 100 to 150 mm mesh screen on the pump hopper to catch fibre balls.

Ponding: Remove any water ponding on the base or subbase before starting placement.

Placing sequence: Commence from one corner (usually the lowest point) and proceed continuously out from that point.

Weather: Do not place concrete in ambient temperatures above 30°C or below 10°C, without adequate precautions.

**Compaction**

Thickness 100 mm or less: Compact by placing, screeding and finishing processes. If required use a hand-held vibrating screed at the surface. Do not use immersion vibrators.

Thickness more than 100 mm and downturns: Use an immersion vibrator.

**Placing records**

Logbook: Keep on Site and make available for inspection a logbook, recording each placement of concrete, including the following:

- Date.
- Specified grade and source of concrete.
- Slump measurements.
- The portion of WUC.
- Volume placed.

**Rain**

Protection: During placement and before setting, protect surface from damage.

**Placing in cold weather**

Cement: Do not use high alumina cement.

Temperature limits: Maintain the following:

- Freshly mixed concrete:  $\geq 5^{\circ}\text{C}$ .
- Formwork and reinforcement before and during placing:  $\geq 5^{\circ}\text{C}$ .
- Water: Maximum  $60^{\circ}\text{C}$  when placed in the mixer.

High early strength cement: If deteriorating weather conditions are predicted, use high early strength cement.

Temperature control: Heat the concrete materials, other than cement, to the minimum temperature necessary so that the temperature of the placed concrete is  $\geq 5^{\circ}\text{C}$ .

Admixtures: Do not use calcium chloride, salts, chemicals or other material in the mix to lower the freezing point of the concrete.

Frozen materials: Do not allow frozen materials or materials containing ice to enter the mixer, and keep formwork, materials, and equipment coming in contact with the concrete free of frost and ice.

Freezing: Prevent concrete from freezing.

**Placing in hot weather**

Requirement: Prevent premature stiffening of the fresh mix and reduce water absorption and evaporation losses.

Severe weather: If ambient shade temperature more than  $38^{\circ}\text{C}$ , do not mix concrete.

Temperature control: Select one or more of the following methods to make sure the temperature of the concrete mix does not exceed  $35^{\circ}\text{C}$ :

- Cool the concrete using liquid nitrogen injection before placing.
- Cover horizontal transport containers.
- Forms and reinforcement before and during placing:  $\leq 35^{\circ}\text{C}$ .
- Spray the coarse aggregate using cold water before mixing.
- Use chilled mixing water.

Evaporation control barriers: Erect barriers to protect freshly placed concrete from drying winds.

Evaporation rate limit:  $\leq 0.50 \text{ kg/m}^2/\text{h}$ .

**3.8 CONCRETE FINISH****General**

Commencement: Immediately after placement, spreading and compaction of the concrete, start initial finishing procedures to achieve the Documented finish.

Final finishing: Do not commence final finishing until all bleed water has evaporated from the surface after initial finishing procedures.

**Unformed surfaces**

General: Strike off, screed and level slab surfaces to finished levels, to the tolerance class and finish Documented.

**Formed surfaces**

Damage: Do not strip formwork prematurely if damage to the concrete may be caused.

Curing: If formwork is stripped before the minimum curing period for the concrete has elapsed, continue curing the exposed faces as soon as the stripping is completed and within an hour of exposure.

**Finishing methods - primary finish**

Machine float finish:

- After levelling, consolidate the surface using a machine float.
- Cut and fill and refloat immediately to a uniform, smooth, granular texture.
- Hand float in locations inaccessible to the machine float.

Wood float finish: After machine floating, use wood or plastic hand floats to produce the final consolidated finish free of float marks and uniform in texture and appearance.

Broom finish: After machine floating and steel trowelling use a broom or hessian belt drawn across the surface to produce a coarse even-textured transverse-scored surface.

Scored or scratched finish: After screeding, use a stiff brush or rake drawn across the surface before final set to produce a coarse scored texture.

#### **Finishing methods - supplementary finish**

Abrasive blast: After steel trowelling, abrasive blast the cured surface to provide texture or to form patterns without exposing the coarse aggregate, using hard, sharp graded abrasive particles.

Coloured applied finish: After machine floating, apply a proprietary liquid or dry shake material to the manufacturer's recommendations and trowel to achieve the required appearance.

Stamped and coloured pattern finish: Provide finishing system.

#### **Surface repairs**

Method: If surface repairs are required, detail proposals.

### **3.9 CURING**

#### **General**

Requirements: Take into account the average ambient temperature at Site over the relevant period affecting the curing and adopt procedures to make sure of the following:

- Curing: Cure continuously from completion of finishing, when the concrete has set sufficiently not to be damaged by the curing process.
- Minimum curing period: Total cumulative number of days or fractions of days, during which the air temperature in contact with the concrete is above 10°C, conforms to AS 3600 (2018) clause 17.1.5. Cure for at least 7 days.
- End of curing period: Prevent rapid drying out at the end of the curing period.
- Protection: Maintain at a reasonably constant temperature with minimum moisture loss, during the curing period.

Curing method: by Structural Engineer.

#### **Curing compounds**

Liquid membrane-forming compounds: Provide a uniform continuous flexible coating without visible breaks or pinholes, which remains unbroken for at least the required curing period after application. Respray Defective areas within 30 minutes. Respray within 3 hours after heavy rain.

Visually important surfaces: Apply curing compounds to produce uniform colour on adjacent surfaces.

#### **Water curing**

Method: Select a method of ponding or continuous sprinkling that does not damage the concrete surface during the required curing period.

#### **Wet hessian curing**

Method: Place wet hessian sheets/bags over concrete surface. Keep hessian wet during the required curing period by regularly sprinkling with water. Protect from wind and traffic.

#### **Impermeable sheet curing**

Method: Place impermeable sheets, to ASTM C171 (2020), over concrete surface. Anchor down and tape joints in material to retain concrete moisture. Keep the concrete surface covered for the required curing period.

#### **Cold weather curing**

Temperature: Maintain concrete surface temperatures above 5°C for the duration of the curing period.

#### **Hot weather curing**

Requirement: If the concrete temperature exceeds 25°C, or the ambient shade temperature exceeds 30°C, protect from drying winds and sun by using an evaporative retarder until curing has commenced.

### 3.10 JOINTS

#### General

Requirement: Construct expansion, contraction and construction joints straight and plumb. Make transverse joints normal to longitudinal joints. Extend transverse expansion and contraction joints continuously from edge to edge of the pavement through interconnected slabs.

Joint layout: Install joints as Documented.

Joint spacings: min 3m

Joint widths: 3 to 5mm. Refer to SEALANT requirements.

#### Non-dowelled contraction joints

Installation: Construct transverse and longitudinal contraction joints by early power sawing at an appropriate time, tooling or by placing an insert in the fresh concrete.

#### Dowelled joints

Requirement: Place dowels as Documented, orthogonal to the joint direction and parallel to the pavement surface, accurate alignment is critical.

Dowel assembly: Use a dowel-assembly support frame firmly secured to the subbase during concrete placement. Prevent the dowel assembly support frame from passing through the joint. Do not insert dowels during the placement of concrete.

Debond dowel: Provide proprietary sleeves or debonding coatings to 0.5 length +25 mm and embed the dowels as follows:

- Proprietary sleeve: Cast the proprietary sleeves in the slab placed first. Insert the dowels into the proprietary sleeves before the subsequent pour.
- Coated dowel: Embed the unpainted half of the dowels in the slab placed first.

Dowelled expansion joints: Cap dowels at one end with a compressible material.

Movement: Do not distort or displace beyond the alignment tolerances under Testing or during construction. Do not remove and replace dowels in preformed holes.

Dowel tolerances:

- Alignment: 1:150.
- Location:  $\pm$  half the diameter of the dowel.

#### Construction joints

Installation: Place header board on the subbase or subgrade at right angles to the pavement centreline.

Planned location: Terminate each day's placing operation at a transverse construction joint located to coincide with a planned contraction or expansion joint.

Unplanned joints: If placement is interrupted for 30 minutes or longer, form a tied transverse construction joint within the middle third of the distance between planned joints but no closer than 1.5 m to the nearest planned joint. If necessary remove placed concrete back to the required location.

#### Expansion joints

Joint filling: Fill with jointing materials as Documented. Finish visible jointing material neatly flush with adjoining surfaces.

Jointing materials: Provide jointing materials compatible with each other and non-staining to concrete in visible locations.

Foamed materials (in compressible fillers): Closed cell or impregnated, not water absorbing.

#### Isolation joints

Requirement: Provide formed full depth joints, made from compressible filler material, around structures and features that project through, into or against the pavement, and elsewhere as required.

#### Formed joints

Full depth joints: Form the edge of the concrete placed first to provide a smooth, vertical face. After stripping and cleaning, fix the joint filler with a suitable waterproof adhesive to the face of the slab. Place the adjoining concrete after the adhesive has set.

Weakened plane joint: Cut a crack-inducing groove by using a suitable tool into the plastic concrete during finishing of the concrete surface. Compact and refinish the plastic concrete around the groove after forming the joint.

Rebated groove joints: Form the rebate by securely fixing removable steel or timber form strips to the form or forms on the slab that is placed first, so that the top of the strip is flush with the top of the form. After stripping and cleaning, fix the joint filler in the rebate after placing the adjoining concrete.

#### **Sawn joints**

Weakened plane joint: Saw the hardened concrete to depth at least 0.25 to 0.33 of the pavement thickness and to a uniform width in the range of 3 to 5 mm as follows:

- Timing: Commence sawing, regardless of time or weather conditions, as soon as the concrete has hardened sufficiently to permit cutting with only minor ravelling of the edges of the sawcut. Complete sawing no later than 24 hours after concrete placement.
- Sequence: If possible, saw every third transverse joint initially, then saw the intermediate joints. Start where concrete placement commenced.
- Cracking: If the concrete has already cracked near the location chosen for a joint, do not saw a joint in that location. If a crack develops ahead of the sawcut, discontinue sawing and submit proposals for extra sawn joints.
- Standby machines: Provide one standby sawing machine for each machine planned to be used.
- Cleaning and protection: Immediately after each joint is sawn, flush the sawcut and adjacent concrete surface using water, until the waste from sawing is removed from the joint.

Rebated groove joints: Saw straight, parallel-sided grooves for joint seals on top of and centred on the sawn weakened plane joints.

- Timing: Commence sawing after the curing period has ended, immediately before joint sealing. Saw during daylight hours.

#### **Preparing joints**

Stripping time: At least 12 hours.

Clean: Immediately before installation of the sealer, make sure the joint space is dry, clean and free from loose material. Remove laitance, curing compound and protrusions of hardened concrete from the sides and upper edges of the joint.

#### **Joint sealing**

Sealant type: Provide silicone sealant in conformance with the manufacturer's recommendations.

Backing rod: Compressible closed cell polyethylene foam with a bond breaking surface.

### **3.11 SURFACE SEALERS**

#### **General**

Sealer: Refer to Appendix A – T-Sheet

Application: Apply surface sealer after the curing period to the manufacturer's recommendations.

Curing sealer compound: If using the sealer as a curing compound, apply directly after finishing.

### **3.12 TACTILE GROUND SURFACE INDICATORS**

#### **Preparation**

Requirement: Conform to the manufacturer's requirements and make sure surfaces are clean and free of dust and contaminants.

#### **Installation**

General: Install on a dry and flat surface. Conform to the manufacturer's recommendations.

### **3.13 COMPLETION**

#### **Rectification**

Reinstating adjacent surfaces: Reinstate surfaces next to new pavements and associated elements. If an existing road pavement has been disturbed, trim back to a straight, neat and undisturbed edge, parallel to the new concrete for the full depth of the slab.

Traffic control provisions: Make sure completed Work is safe for traffic, reinstate linemarkings if necessary.

Concrete pavement: If pavement does not conform to the tolerances, submit rectification proposal.

Unplanned cracking:

- Acceptable limit: Maximum 0.4 mm crack width.
- If cracking is more than 1 mm, provide an assessment of the possible cause and provide a proposal for rectification.

**Material removal**

Excavated material: Remove from Site.

**4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                                |
|--------------------------------|
| <b>0334 BLOCK CONSTRUCTION</b> |
|--------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide block construction, as Documented. Refer to Architectural Drawings, Appendix B – Façade and Wall Schedule, and Structural Engineer's Documents.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARD****General**

Materials and construction: To AS 3700 (2018).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions in AS 3700 (2018), AS/NZS 4455.1 (2008) and the following apply:

- Facework: Masonry intended to be exposed in a wall.
- Special mortar: Mortar with compositions not covered in AS 3700 (2018) Table 11.1.

**1.5 TOLERANCES****General**

Standard: To AS 3700 (2018) clause 12.5 and Table 12.1.

**1.6 SUBMISSIONS****Products and materials**

Type Tests: Submit Test results of the following:

- Characteristic unconfined compressive strength of masonry unit: To **MATERIALS, Masonry units.**

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

**Tests**

Site Tests: Submit results for special mortar and masonry as follows:

- Characteristic compressive strength.
- Characteristic flexural tensile strength.
- Scratch index for mortar joints.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Set-out.
- Structural steelwork, including bolts and shelf angles, in position.
- Unit type, colour and texture.
- Bottoms of cavities, after cleaning out.
- Bottoms of core holes, before grouting.
- Reinforcement type and diameter.

- Positioning of reinforcement before grouting.

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide samples, as follows:

- Each type of face unit, showing the range of variation available, including colour, texture, surface irregularities, Defective arrises, and shape.
  - . Number of each type: 6.
- Each type of sealant, showing the finished colour.
  - . Quantity: Minimum two of each colour.

### 2.2 DURABILITY

#### General

Exposure environment: Moderate

Exposure locations: To AS 3700 (2018) clause 5.4.

### 2.3 MATERIALS

#### Masonry units

Selections: As Documented. Refer to Appendix B – Façade and Wall Schedule, and Structural Engineer's Documents.

Standard: To AS/NZS 4455.1 (2008) and AS/NZS 4455.3 (2008).

Salt attack resistance grade: To AS 3700 (2018) Table 5.1.

Characteristic unconfined compressive strength: To AS/NZS 4456.4 (2003).

#### Mortar materials

Mortar class: To AS 3700 (2018) Table 5.1.

Cement: To AS 3972 (2010).

White cement: With not more than 1% iron salts content.

Lime: To AS 1672.1 (1997).

Sand: Fine aggregate with a low clay content, free from efflorescing salts and deleterious matter, selected for colour and grading.

Water: Clean and free from any deleterious matter.

Admixtures: To AS 3700 (2018) clause 11.4.2.4.

Pigment: To EN 12878 (2014), and as follows:

- Integral pigment mix proportion: Not more than 10% by weight of cement.

#### Masonry cement mortar mix proportions table (cement:lime:sand), by volume

| Mortar class to AS 3700 (2018) | Clay  | Concrete | Calcium silicate | Water thickener |
|--------------------------------|-------|----------|------------------|-----------------|
| M3                             | 1:0:4 | 1:0:4    | N/A              | Yes             |
| M4                             | 1:0:3 | N/A      | N/A              | Yes             |

#### Cement (GP/GB) mortar mix proportions table (cement:lime:sand), by volume

| Mortar class to AS 3700 (2018) | Clay       | Concrete   | Calcium silicate | Water thickener |
|--------------------------------|------------|------------|------------------|-----------------|
| M2                             | 1:2:9      | N/A        | N/A              | No              |
| M3                             | 1:1:6      | 1:1:6      | N/A              | Optional        |
| M3                             | 1:0:5      | 1:0:5      | 1:0:5            | Yes             |
| M4                             | 1:0.5:4.5  | 1:0.5:4.5  | N/A              | Optional        |
| M4                             | 1:0:4      | 1:0:4      | 1:0:4            | Yes             |
| M4                             | 1:0-0.25:3 | 1:0-0.25:3 | N/A              | Optional        |

**Grout**

Standard: To AS 3700 (2018) clause 11.7.

Maximum aggregate size: 12 mm.

Minimum cement content: 300 kg/m<sup>3</sup>.

Characteristic compressive strength: Minimum 12 MPa.

Nominal slump: 200 mm.

**2.4 BUILT-IN COMPONENTS****General**

Durability class of built-in components: To AS 3700 (2018) Table 5.1.

**Reinforcement**

Standard: To AS/NZS 4671 (2019).

Corrosion protection: To AS 3700 (2018) clause 5.9.

Minimum cover: To AS 3700 (2018) Table 5.1.

**Connectors and accessories**

Standard: To AS 2699.2 (2020).

Corrosion protection: To AS 2699.2 (2020).

---

**3 EXECUTION****3.1 GENERAL****Mortar mixing**

General: Measure volumes accurately to the Documented proportions. Machine mix for at least six minutes. If the initial set of the cement has taken place, discard the mortar. Do not retemper.

**Storage and handling**

Masonry units: Store above the surface of the ground and cover to prevent entry of rainwater and contaminants. Locate away from surface and ground water runoff.

Mortar materials: Protect from contamination and as follows:

- Sand: Store away from surface and ground water runoff and allow for free drainage of rainwater.
- Cement and lime: Store bags in a dry, under cover and above ground environment.

**Bond**

Type: Stretcher bond.

**Mortar joints**

General: Set out masonry with joints of uniform width and minimum cutting of masonry units.

Solid and cored units: Lay on a full bed of mortar. Fill perpends solid. Cut mortar flush.

Hollow units: Face-shell bedded. Fill perpends solid. Cut mortar flush.

Joint thickness: 10 mm.

Finish: Conform to the following:

- Externally: Tool to give a dense water-shedding finish.
- Internally: If wall is to be plastered, do not rake more than 10 mm to give a key.

**Rate of construction**

General: Regulate the rate of construction to eliminate joint deformation, slumping or instability.

**Rods**

Set-out: Construct masonry to the following rods:

- 190 mm high units: 3 courses to 600 mm.

**Temporary support**

General: If the final stability of the masonry is dependent on construction of (structural) elements after the masonry is completed, provide proposals for temporary support or bracing.

### 3.2 FACEWORK

#### Cleaning

General: Clean progressively as the WUC proceeds to remove mortar smears, stains and discolouration. Do not erode joints if using pressure spraying.

#### Colour mixing

Distribution: In facework, distribute the colour range of units evenly to prevent colour concentrations and banding.

#### Below ground

Facework: Commence facework at least one full course, below the adjacent finished surface level.

### 3.3 CONTROL JOINTS

#### General

Location and spacing: Provide control joints to AS 3700 (2018) clause 4.8.

#### Control joint filling

Filler material: Provide compatible sealant and bond breaking backing materials that are non-staining to masonry. Do not use bituminous materials with absorbent masonry units.

- Bond breaking materials: Non-adhesive to sealant, or faced with a non-adhering material.
- Foamed materials: Closed cell or impregnated, not water-absorbing.

Installation: Clean the joints thoroughly and insert an easily compressible backing material before filling with a gun-applied flexible sealant.

Sealant joint depth to width ratio (depth:width): 1:2.

Minimum sealant depth: 6 mm.

### 3.4 BED JOINT REINFORCEMENT

#### Location

Stack bonded masonry: Conform to AS 3700 (2018) clause 4.12 and the following:

- Spaced vertically at centres not exceeding six times the thickness of the stack bonded leaf.
- In the first bed joint above or below an unrestrained horizontal edge of the masonry.
- One bed joint minimum, within 300 mm above or below a horizontal line of lateral support.

#### Installation

General: Lap 450 mm at splices. Fold and bend at corners so that the longitudinal wires are continuous. Stop 50 mm short of control joints. Extend 450 mm beyond each side of openings.

#### Reinforcement

Material: Galvanized welded wire mesh.

Width: Equal to the width of the leaf, less 15 mm cover from each exposed surface of the mortar joint.

### 3.5 REINFORCED AND GROUTED BLOCKWORK

#### Reinforcement

Cover: Maintain cover to vertical and horizontal steel reinforcement using plastic clips or wheels, as appropriate.

Vertical reinforcement: Tie vertical steel reinforcement to the starter bars through cleanout holes in each reinforced hollow masonry unit and fix in position at the top of the wall with plastic clips.

Horizontal: Lay horizontal steel reinforcement in contact with rebated webs. Hold in position using plastic clips if vertical steel is subsequently positioned to wall construction.

#### Cleaning core holes

General: Provide purpose-made cleanout blocks or machine cut a cleaning hole at the base of each grouted core.

Location: Locate on the side of the wall that is to be rendered or otherwise concealed.

Cleaning: Rod cores to dislodge mortar fins protruding from the blocks and mortar droppings from reinforcement. Remove through the cleanout blocks.

#### Grouting

Commencement: Do not commence until grout spaces have been cleaned out and the mortar joints have attained sufficient strength to resist blow-outs.

Height of lift: Limit the height of individual lifts in any pour to make sure that the grout can be thoroughly compacted to fill all voids.

Compaction: Compact by vibration or by rodding.

Topping up: On the completion of the last lift, top up the grout after 10 minutes and within 30 minutes, and vibrate or rod to mix with the previous pour.

### **3.6 TESTING**

#### **Special mortar**

Durability: Scratch index Test to AS 3700 (2018) Appendix E.

Compressive strength: To AS 3700 (2018) Appendix C.

Flexural strength: To AS 3700 (2018) Appendix D.

### **3.7 COMPLETION**

#### **Warranties**

Type: Interlocking material and installation warranty.

Period: As provided by manufacturer and installer.

## **4 SELECTIONS**

---

**Refer to Appendix B – Façade and Wall Schedule**

**Refer to Structural Engineer's Documents.**

|                                 |
|---------------------------------|
| <b>0342 LIGHT STEEL FRAMING</b> |
|---------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide light steel floor, external wall, internal wall, and ceiling framing, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Appendix B – Façade and Wall Schedule, and Structural Engineer's Documents.

**Performance**

Requirements:

- Suitable for installation of flooring, lining, cladding and roofing, as required.
- Conforms to the Documented performance criteria.
- Conforms to NASH-1 (2005) or NASH-2 (2014).

**1.2 DESIGN****General**

Designer: RPEQ registered engineer.

**Requirements**

General: To DESIGN in *0171 General requirements*.

**1.3 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements*.

**1.4 STANDARDS****General**

Design, materials and protection: To AS/NZS 4600 (2018).

Residential and low-rise steel framing: To the National Association of Steel Housing standards NASH-1 (2005) or NASH-2 (2014).

**1.5 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in the NASH-1 (2005), NASH-2 (2014) apply.

**1.6 TOLERANCES****General**

Manufacturing, assembly and installation tolerances: To NASH-1 (2005) Appendix D and NASH-2 (2014) Appendix A.

**1.7 SUBMISSIONS****Certification**

Erected frame: Submit certification that the erected frame conforms to the Documented project requirements.

**Design Documentation**

General: If the structural Documentation defines performance criteria, submit as follows:

- To AS/NZS 4600 (2018) or NASH-1 (2005): Independent design, Documentation and certification by a professional engineer.
- To NASH-2 (2014): Certification of conformance to the requirements of NASH-2 (2014) by a professional engineer.

Reactions: Submit the location and magnitude of reactions that are to be accommodated by the support structure.

Floor and wall frame member sizes: Submit a schedule of proposed member sizes, certified as meeting Documented project requirements, and AS/NZS 4600 (2018), NASH-1 (2005) or NASH-2 (2014) requirements for span, spacing and loading.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, requirements for the Documented configurations and loading.

Wall frames: Include the following:

- Plan: Wall layout.
- Elevation: Arrangement of members, and size and section type of each member.
- Method of assembly, connection, holding down and bracing.

Floor frames: Include the following:

- Plan: Level of installation, arrangement of members, and size and section type of each member, including prefabricated floor joists.
- Method of assembly, connection, holding down and bracing.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.8 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Damp-proof course installed before installation of steel framing.
- Steel framing erected on Site before lining or cladding.

---

**2 PRODUCTS****2.1 GENERAL****Storage and handling**

Requirement: Transport all components to Site and store, if required, so that components and their coating are not damaged or distorted.

Frames and trusses: If required, store on a flat even surface and do not load with other items.

Exposure: Minimise exposure of components to the weather during storage, handling and after erection.

**2.2 COMPONENTS****Damp-proof course**

Membrane: To the membrane requirements of AS 2870 (2011) or AS/NZS 2904 (1995).

**Light steel framing**

General: Cold-formed sections from steel, metallic-coated to AS 1397 (2021).

Corrosion protection: To NASH-2 (2014) Section 8.

**Prefabricated elements**

Standard: To NASH-1 (2005) or NASH-2 (2014).

---

**3 EXECUTION****3.1 GENERAL****Frame fabrication**

Length: Cut members accurately to length so they fit firmly against abutting members.

Service holes: If not pre-punched, form holes by drilling or punching, without compromising the structural integrity of the frame, located centrally in the web of the section, conforming to the requirements of NASH-2 (2014).

Swarf: Immediately remove swarf and other debris from the light steel framing.

**Connections**

Prefabricated framing: Fix framing elements to the fabricator's requirements.

Framing built in situ: Fix framing elements, as Documented.

Fasteners: Compatible with steel frame to prevent galvanic corrosion of dissimilar metals.

#### **Welding**

Burning: Avoid procedures that result in greater than localised burning of the sheets or framing members.

#### **Prefabricated frames**

General: Protect frames from damage or distortion during erection.

#### **Unseasoned or CCA treated timber**

General: Do not fix in contact with framing without fully painting the timber and/or the steel.

#### **Earthing**

Requirement: To AS/NZS 3000 (2018). Provide temporary earthing during erection until the permanent earthing is installed.

#### **Protection**

General: Restore coatings that have been damaged. Thoroughly clean affected areas back to base metal and coat with a zinc-rich organic primer.

Metal separation: Install lagging to separate non-ferrous service pipes and accessories from the framing.

Grommets: Provide grommets to isolate piping and wiring from the light steel framing.

Site cut holes: Provide plastic bushes or grommets to Site cut holes.

### **3.2 FLOOR FRAMING**

#### **General**

Protection: If floor framing is for ground floor construction, make sure that it is protected from moisture.

Construction loads: If construction loading exceeds design loading, provide additional supports to avoid overstressing the members.

### **3.3 WALL FRAMING**

#### **Wall studs**

General: Provide studs in single lengths without splices. Place a stud and a stiffened top plate under each structural load point from the roof or ceiling (except at openings). Provide multiple studs at points of concentrated load.

Maximum stud spacing: 600 mm.

#### **Heads to openings**

Requirement: Provide lintels appropriate for the load and span.

#### **Additional support**

General: Provide additional support in the form of noggings, trimmers and studs for support and fixing of lining, cladding, hardware, accessories, fixtures and fittings.

#### **Vermin barriers**

Sub floor snake barrier: Fit snake barrier under floor as Documented in Architectural Drawings. Any gaps are to be sealed.

#### **Damp-proof course**

Requirement: Provide damp-proof courses under the bottom plate of stud walls built off slabs or masonry dwarf walls as follows:

- External walls (not masonry veneer): Turn up a minimum of 75 mm on the inside and tack to studs. Project 10 mm beyond the external slab edge or dwarf wall and turn down at 45°.
- Walls of bathrooms, shower rooms and laundries: Turn up a minimum of 150 mm on the wet side and tack to studs.

Installation: Lay in long lengths. Lap full width at corners and intersections and at least 150 mm at joints.

Junctions: Preserve continuity of damp-proofing at junctions of damp-proof courses, sarking and waterproof membranes.

#### **Flashings**

Location: Provide flashings to external openings sufficient to prevent the entry of moisture. Form trays at the ends of sill flashings.

### 3.4 ROOF AND CEILING FRAMING

#### **Beam framing**

General: Construct framing for flat or pitched roofs where the ceiling follows the roof line, consisting of prefabricated roof beams, rafters or purlins supporting both ceiling and roof covering.

#### **Additional support**

Requirement: Provide additional framing members at the following locations:

- Hanging light fittings.
- Ceiling fans.
- Access panels.
- Any other hanging services or fixtures and fittings.

Water tank or heater in roof space: Provide a support platform to the NCC cited AS/NZS 3500.4 (2021) clause 5.5.1.

#### **Battens**

Requirement: Supply and fix battens suitable for the span, spacing and proposed roofing material.

#### **Anti-ponding boards**

Standard: To AS 2050 (2018) clause 3.1.2.

### 3.5 ROOF TRIM

#### **Fascia, valley and barge boards**

Requirement: Fix fascia, valley gutter boards and barge boards to the manufacturer's recommendations.

### 3.6 COMPLETION

#### **Cleaning**

General: On completion of framing, remove debris from any gaps between members and make sure void between bottom chord of roof trusses and top of any non-supporting internal wall is clear.

#### **Warranties**

Type: Interlocking warranties between the manufacturer and the installer.

Period: As offered by the manufacturer and the installer.

|                                        |
|----------------------------------------|
| <b>0345P DULUX PROTECTIVE COATINGS</b> |
|----------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide Dulux protective paint coatings for the protection of steel products and structural steelwork against interior and exterior atmospheric corrosion, as Documented. Refer to Appendix A – T-Sheet for paint type and colour selections. Refer to Appendix E(c) for Dulux system specifications.

**Performance**

Requirement: Control atmospheric corrosion to structural steelwork and steel products until the first scheduled maintenance.

**1.2 COMPANY CONTACTS****DuluxGroup/Dulux technical contacts**

Architects and Specifiers' Hotline: 13 23 77.

Website: [www.dulux.com.au/specifier](http://www.dulux.com.au/specifier)

**1.3 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.4 STANDARDS****General**

Surface preparation and coating: To AS/NZS 5131 (2016) Section 9 and the recommendations of AS 2312.1 (2014).

**Site Testing of protective coatings**

Test methods: To the AS 3894 series.

**1.5 MANUFACTURER'S DOCUMENTS****Technical manuals**

Duspec PDS, SDS, paint system selection: [www.duspec.com.au](http://www.duspec.com.au)

**1.6 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviations apply:

- ACA: Australasian Corrosion Association.
- DFT: Dry film thickness.
- ITP: Inspection and Test plan.
- NACE: National Association of Corrosion Engineers (USA).
- PDS: Product data sheet.
- SDS: Safety data sheet.
- SSPC: The Society for Protective Coatings (USA).
- $\mu\text{m}$ : Micron ( $10^{-6}$  m).

**Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 2310 (2002) and the following apply:

- Coating subcontractor: The protective coatings application subcontractor conducting the on- or off-site coating application works.
- Coating manufacturer: Dulux Protective Coatings.

- Inspection and Test plans (ITP): A series of formal inspection and Test plans, prepared by the coating subcontractor to reflect the specific inspection and Testing that will be carried out on the surface preparation, coating application and the record keeping tasks to be undertaken.
- Safety data sheet (SDS): Prepared in conformance with Safe Work Australia's requirements and distributed by the coating manufacturer to provide information on the safe handling, storage, personal protective equipment requirements, use and disposal of a coating product. Previously called a material safety data sheet (MSDS).

## 1.7 QUALITY ASSURANCE

### General

Standard: Applicator Quality Assurance system to AS/NZS ISO 9001 (2016).

Applicator's quality assurance officer: Nominate a qualified NACE Certified Coating Inspector or a ACA Certified Coatings Technician under direction of a NACE inspector.

Records: Maintain records:

- Access: Have records available for inspection.

Verification: Nominate an independent NACE Certified Coating Inspector to carry out quality audits.

Defects: Provide written inspector reports.

## 1.8 SUBMISSIONS

### Execution details

Detailing features: If design and fabrication features of the items to be coated may lead to difficulties, identify these and submit details for improvement.

Repair of damaged coating: If the protective coating is damaged, submit a coating repair proposal, based on the coating manufacturer's recommendations for reinstating the corrosion protection function of the system.

Reinstatement: If final coat varies from the submitted sample, submit proposals for reinstatement of the visible final coating system.

### Products and materials

Multi-component coatings: If partial mixing of packs is proposed, submit details.

### Quality

ITPs: Submit for each proposed coating system.

Quality supervisor: Submit the name and record of experience of the person responsible for the implementation of the ITPs.

### Records

Requirement: Submit records to EXECUTION, **GENERAL, Records**.

### Samples

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

### Subcontractors

General: Submit names and contact details of proposed suppliers and applicators.

Evidence of experience: QBCC Licence and CV.

Requirement: Submit proof of currency of the applicator's environmental operating licence.

### Substrate acceptance

Requirement: Submit evidence of applicator's acceptance of the coating substrate before starting application.

### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 1.9 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Items after fabrication, before commencing surface cleaning and preparation.

- Surfaces after preparation, before application of first coating.
- Coating stages:
  - . After application of primer or seal coats.
  - . After application of each subsequent coat.

---

## 2 PRODUCTS

### 2.1 GENERAL

#### Product substitution

Other products: Conform to **SUBSTITUTIONS** in 0171 *General requirements*.

#### Samples

Requirement: Provide samples of each coating system, showing the painting and coating colour, and other physical properties.

Sample size: Minimum 400 x 400 mm.

Retention: Retain samples for comparison during application.

#### Storage and handling

General: Store in a cool shady place.

Care: Handle, store, mix and apply all protective coatings in conformance with Dulux recommendations.

Original containers: Deliver coating products to Site in manufacturer's labelled and sealed containers.

Ambient temperature range for storage: 15°C to 25°C.

Sunlight: Protect coating materials from direct sunlight before mixing or adding the converter (catalyst).

Use-by-date: Use products with limited shelf life before their use-by-date, unless written authorisation from the coating manufacturer's technical services section is provided.

#### Paint material

Requirement: To AS/NZS 5131 (2016) clause 9.9.3.

#### Proprietary products

Substitution: Dulux paint products and specified coatings systems have been selected for this project and unauthorised product substitution will jeopardise or void the Warranties.

Product data sheets (PDS): Keep on Site copies of all relevant Dulux technical data sheets.

Safety data sheets (SDS): Keep on Site copies of all relevant Dulux SDSs.

Recording: To AS/NZS 5131 (2016) clause 9.9.5.

---

## 3 EXECUTION

### 3.1 GENERAL

#### Product warnings

Requirement: Conform to the requirements and recommendations of the relevant Dulux SDS.

#### Qualifications

Requirement: All Work is to be completed by suitably qualified professionals holding TAFE or other recognised qualifications.

#### Surroundings

Protection: Prevent the release of abrasives, overspray or paint waste debris into the air, ground or to any watercourse. Prevent damage to other assets, services or equipment.

Reinstatement: Repair and/or clean affected surrounding areas.

#### Working area

General: Perform all painting under cover and/or protected from rain, condensation, dew, excessive wind, overspray or wind-blown dust.

Period: Continue protection where any of these conditions exist until the coating is no longer affected.

#### Records

Requirement: Prepare and maintain records of all surface preparation and coating application works, as follows:

- Standards: To AS 3894.10 (2002), AS 3894.11 (2002), AS 3894.12 (2002), AS 3894.13 (2002) and AS 3894.14 (2002).
- Reference the relevant parts of the ITP and record conformance.

### 3.2 SURFACE PREPARATION

#### General

Requirement: Conform to AS/NZS 5131 (2016) clauses 9.3, 9.4 and 9.5.

Treatment grade to AS/NZS 5131 (2016): refer to Appendix E(c).

Surface cleansing: Wash and degrease all surfaces to be coated, to AS 1627.1 (2003), with a free-rinsing, alkaline detergent, such as Gibson F310B or Gamlen CA No. 1 used in conformance with the manufacturer's written instructions and all safety warnings.

Bolts: Provide washers at heads and nuts at replacement bolts.

#### Zinc primed surfaces

Requirement: Remove grease, oil and other solvent-soluble contaminants to AS 1627.1 (2003). Allow to dry and immediately proceed with priming.

Zinc primed surfaces: If present, remove zinc salts from zinc primers.

#### Treatment of welds

Requirement: Clean welds to remove roughness, using power tools to AS 1627.2 (2002). Remove filings by vacuuming or compressed air.

Temporary welds: Grind flush any temporary welds.

Porous, skip or stitch welds: Not permitted.

Site welding: If possible, avoid Site welding. If on-site welding is required, prepare and treat the weld to AS/NZS 5131 (2016) clause 9.12.2.

#### Shop priming

Requirement: Dust off and apply a coat of primer, according to the technical specification.

#### Site coating

General: High pressure wash down all surfaces with clean water. Lightly sand down primer/intermediate coats, which have been shop applied, before Site application of next coat.

### 3.3 PREPARATION ASSESSMENT

#### General

Conformance: Assess all surfaces of each steel member for conformance with the Documented preparation requirements.

#### Abrasive blast cleaning

Assessment: To AS 1627.4 (2005) and AS 1627.9 (2002).

Minimum acceptance class: Sa 2.5.

#### Mechanical cleaning

Assessment: To AS 1627.9 (2002) and ISO 8501-2 (1994).

Minimum acceptance class: St 2.

#### Surface profile

General: To AS 3894.5 (2002) Method A.

Acceptable profile range: 40 to 75 µm.

#### Surface dust from abrasion

General: To AS 3894.6 (2002) Method C.

Rating criteria for rejection of surface treatment: refer to Appendix E(c) – Paint Specification.

#### Chloride level Testing

Test: To AS 3894.6 (2002) Method A.

Maximum allowable chloride levels: 50 mg/m<sup>2</sup> for critical applications (heavy condensation, fresh water ponding or immersion) or to manufacturer's recommendations.

Conformance: If the maximum allowable chloride level is exceeded, rewash the affected surface area until the chloride level is within the acceptable limits using clean water or chloride neutralising solutions. Jet-washing or steam cleaning is also acceptable before re-testing and re-abrasive blasting.

Timing of Testing: Early in the blasting work so that removal procedures can be started before the blasting is completed.

### 3.4 MIXING

#### General

Requirement: To AS/NZS 5131 (2016) clause 9.9.6.

Powered agitators: Mix package sizes larger than 4 litres using powered agitators driven by air motors.

Multi-component coatings: Combine as whole pack units before application.

Thinners: If addition of thinners is proposed, conform to the Dulux PDS for the Documented product.

Colour consistency: If colour consistency is required, pre-mix tinted products, before the addition of the curing agent or converter and before coating application.

### 3.5 COATING APPLICATION

#### General

Requirement: Conform to the Dulux PDS, the Dulux specification and AS/NZS 5131 (2016) clause 9.9.

Painting and coating colour: Verify all project finish colours with the retained samples.

#### Final surface preparation or coating application

Limits: Do not apply coating if any of the environmental/climatic/substrate conditions listed in AS/NZS 5131 (2016) clause 9.9.10 exist or if the following conditions are present:

- Ambient air temperature below 5°C or above 40°C.
- Substrate temperature below 5°C or above 35°C.
- The specified surface cleanliness will deteriorate before the full prime coat application can be completed.
- Surface preparation standard has not been achieved.
- Time between final surface preparation and the commencement of coating has exceeded 4 hours.
- Visual tarnishing or black spots develop on the surface of the steel.

Exception: Preliminary blast or other surface preparations may be performed in conditions that are outside the limits, provided the final surface preparation and all coating applications are undertaken under the limit conditions.

Pre-coating: Before the spray application of each coating, stripe coat by brush method all edges, welds, seams, rivets, bolts, boltholes (including slots) and difficult to spray areas. Prime the underlying surfaces of replacement bolts, washers and nuts before installation.

Procedure: Conform to the coating order of each protective paint coating system, as Documented.

Subsequent coats: Make sure that before any subsequent coating layer is applied, the surface condition of the preceding coat is complete and correct in all respects, including its DFT achievement, cleanliness and freedom from Defects. These are detailed on the Dulux Protective Coating specification. Depending on the applicators chosen method additional coats may be required to achieve the nominated minimum DFT.

Conformance: To AS 2312.1 (2014) for the specified film thickness of individual coats.

Correction: Correct any Defect in a coating layer before the subsequent coating layer is applied.

#### Wet film thickness (WFT)

Method of measurement: To AS 3894.3 (2002) Appendix C using an approved wet film gauge continuously during application.

#### Dry film thickness (DFT)

Method of measurement: To AS 3894.3 (2002) Section 10.

Extent: Measure all surfaces at the completion of each prime, intermediate and finish coats, including areas of the element difficult to paint, masked by structure, or where double or light coating is likely.

Number of measurements: To AS 3894.3 (2002) Section 7.

Coatings with DFT 150 µm or less: If Testing, deduct the effect of the measured surface profile from all DFT readings.

Single readings: Conform to the following:

- The average of 5 point readings for each 10 m<sup>2</sup> area of coating surface to be within the Documented coating thickness range.

- No single point reading in any 10 m<sup>2</sup> to be less than 80% of the specified minimum coating thickness. If the average of three readings is used to produce a point reading, an individual reading may be less than 80% of the minimum coating thickness.
- Check any single reading that is greater than 150% of the Documented maximum DFT with three additional readings within 50 mm of the original reading. If the average of these three readings is not greater than 150% of the specified DFT, take the average reading as the point reading. If greater than 150%, reject the DFT for that area. If no maximum limit for DFT is Documented, consult manufacturer.

#### **Rectification and Defects**

Rectification: Re-work areas rejected, using the same surface preparation, coatings and sequence as for the original work.

Defects (including under-thickness and over-thickness): Mark with dustless chalk, adhesive inspection labels or masking tape. Do not use crayon, paint or spirit based ink pens.

Defects for rejection: dry spray, splatter, inclusions, lumps or excessive orange peel in visible locations.

### **3.6 PROTECTION**

#### **Contamination**

Surfaces: Prevent contamination of coated surfaces, which are not yet dry, from blasting dust, abrasive or surface preparation debris and any other foreign matter.

#### **Post application care**

General: Protect the coating against physical, chemical, or atmospheric damage until all components are fully cured.

Care: Stack and handle all coated items using fabric slings or padded chains. Use soft packaging, carpet strips or other deformable materials between all coated items.

Water ponding: Stack coated items to prevent water ponding.

### **3.7 COATING REPAIR**

#### **Repair of coating damage**

Preparation: Feather back by hand or machine sand all leading edges of intact coating adjacent to the repair, to remove any sharp edge.

Surface contamination: Remove by dusting or blowing down before applying the first coat of paint.

Sequence: Apply the repair coating in the same sequence and manner as the original coating.

Areas damaged without exposing the primer: Wash with a proprietary detergent solution, rinse with clean water and abrade so that edges of sound paint are feathered. Coat the area with the appropriate intermediate and finishing coat materials.

Areas damaged exposing the primer or steel surface: Blast clean to the original standard. Prepare at least 50 mm into the sound coating and to a further feathering zone of approximately 50 mm. Recoat with the Documented system to restore the film thickness and integrity over the whole prepared surface including the feathered zone.

Aesthetic reinstatement: If required, repaint to a physical or discernible boundary line.

Defects: If corrosion pitting or areas of significant metal loss and Defects are exposed by the blasting process, advise for inspection and have areas passed as being fit for service before proceeding with the coating system.

Timing: Apply the Dulux Protective Coating system within 4 hours of blast cleaning or in any case before visual tarnishing of the steel occurs.

Cleaning: Provide, at no additional cost, surface treatment as follows:

- Surfaces left longer than 4 hours: Re-blast cleaning before coating.
- Surfaces that develop visual tarnishing (red rust or black spots) at any time before coating: Wash down with clean water then blast clean before coating. There are commercially available chloride reducing solutions that may assist.

### **3.8 COMPLETION**

#### **General**

Joints: On completion, seal all joints and mating surfaces with a compatible polyurethane sealant.

**Warranties**

Requirement: Cover materials and workmanship in the terms of the warranty in the form of interlocking warranties from the supplier and the applicator.

- Form: Against failure of materials and execution under normal environment and use conditions.
- Period: As offered by the supplier and the applicator.

**4 SELECTIONS**

---

**REFER TO APPENDIX A – T-SHEET AND APPENDIX E – PAINT SCHEDULE**

|                                  |
|----------------------------------|
| <b>0382 LIGHT TIMBER FRAMING</b> |
|----------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide light timber framing, as Documented. Refer also to Structural Engineer's Documents for timber truss specification.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0181 *Adhesives, sealants and fasteners.*
- 0185 *Timber products, finishes and treatment.*

**1.3 STANDARDS****General**

Framing: To AS 1684.2 (2021), AS 1684.3 (2021) or the NCC cited AS 1684.4 (2010), as appropriate.

Design: To AS 1720.1 (2010).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in the AS 1684 series apply.

**1.5 TOLERANCES****Floors**

Maximum deviation from a 3 m straightedge laid in any direction on the floor framing: 5 mm.

**Wall tolerances table**

| Property                                                                                                                                                                                                 | Permitted deviation |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Generally: Verticality                                                                                                                                                                                   | 1:500               |
| Generally: Flatness <sup>a</sup>                                                                                                                                                                         | 3 mm                |
| Features <sup>b</sup> : Verticality                                                                                                                                                                      | 1:1000              |
| Features <sup>b</sup> : Horizontality                                                                                                                                                                    | 1:1000              |
| a. Flatness: Measured under a 2 m straightedge laid in any direction on a surface.<br>b. Features: Conspicuous horizontal or vertical lines including external corners, parapets, reveals, heads, sills. |                     |

**1.6 SUBMISSIONS****Certification**

Requirement: Submit certification by an appropriately qualified person of the design, Documentation and erected Work to the AS 1684 series and/or by a professional engineer to AS 1720.1 (2010).

Include the following:

- Reactions: Provide location and magnitude of reactions to be accommodated by the support structure. If part of the structure is manufactured by a prefabricator (e.g. roof trusses), provide location and magnitude of reactions and tie down forces.
- Floor, wall and roof frame member sizes: A schedule of proposed member sizes, certified as meeting stated project requirements for spans, spacings, loadings and deflections.
- Species and stress grade.

**Products and materials**

Supply: Submit supplier's evidence of conformity, which may be included on an invoice or delivery docket, verifying the timber conforms to the Documented requirements.

Inspection: Submit the inspection authority's certificate verifying the erected timber frame conforms to the Documented requirements.

Moisture content: Submit records of moisture content to AS/NZS 1080.1 (2012).

CCA treated timber: If proposed to be used, submit details.

### Shop drawings

General: Submit shop drawings, to a scale that best describes the detail, certified by an appropriately qualified person stating that the design has been carried out to the requirements of the AS 1684 series and/or by a professional engineer to AS 1720.1 (2010), for the Documented configurations and loadings.

Prefabricated roof trusses: Include the following:

- Marking plans.
- Truss plan layout.
- Elevations, showing the arrangement of members, allowing for the accommodation of in-roof services, and the size and section type of each member.
- Camber of all elements.
- Method of assembly, connection and lifting.
- Location and details of tie down and bracing.

Prefabricated wall frames: Include the following:

- Wall plan, showing all wall layouts.
- Elevations, showing the arrangement of members, and the size and section type of each member.
- Method of assembly, connection and lifting.
- Location and details of tie down and bracing.

### Subcontractors

Prefabricated items: Submit the name and contact details of proposed manufacturers, suppliers and installers.

## 1.7 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Prefabricated units before installation.
- Fabricated items before priming or water-repellent treatment.
- Connections/joints after final tightening of fasteners.
- Timberwork after erection but before it is covered.

## 2 PRODUCTS

### 2.1 GENERAL

#### Certification

Requirement: Forest certification, chain of custody and product labelling to *0185 Timber products, finishes and treatment*.

#### Identification

Method: Identify timber using branding, certification or both.

Branding: Brand structural timber, under the authority of a recognised product certification scheme to *0185 Timber products, finishes and treatment* as applicable to the product. Locate the brand mark on faces or edges that will be concealed in the Works. Include the following data for timbers not covered by branding provisions in Australian standards or regulations for which branding is required:

- Stress grade.
- Method of grading.
- If seasoned, the word, SEASONED or DRY, or an abbreviation of seasoned, such as SEAS or S.
- The certification mark of the product certification scheme.
- The applicable standard.

Trusses: Permanently label each truss to show:

- Manufacturer.
- Tag or number with reference to location.
- Support and tie down points.
- Labelling in coordination with installation Documentation.

### **Preservative treatment**

Requirement: To *0185 Timber products, finishes and treatment*, including for termite treatments.

## **2.2 TIMBER**

### **Framing**

Requirement: To AS 1684.2 (2021), AS 1684.3 (2021) or the NCC cited AS 1684.4 (2010), as appropriate.

### **Fascias and barge boards**

Hardwood: To AS 2796.1 (1999).

Seasoned cypress pine: To AS 1810 (1995).

Softwood: To AS 4785.1 (2002).

### **Trusses**

Design: To AS 1720.1 (2010).

Nailplated roof trusses: To AS 1720.5 (2015).

Overhangs: Free from spring or splits.

## **2.3 GLUED LAMINATED TIMBER AND LAMINATED VENEER LUMBER**

### **Glued laminated timber**

Standard: To AS/NZS 1328.1 (1998).

### **Laminated veneer lumber**

Standard: To AS/NZS 4357.0 (2022).

Formaldehyde emission class to AS/NZS 4357.0 (2022): E<sub>1</sub>(1.0 mg/L)

## **2.4 STRUCTURAL PLYWOOD**

### **General**

Standard: To AS/NZS 2269.0 (2012).

Bond: Type A to AS/NZS 2754.1 (2016).

Formaldehyde emission class to AS/NZS 2269.0 (2012): E<sub>1</sub> (1.0 ml/L)

### **Veneer**

Veneer quality to visible surfaces: AC (minimum) to AS/NZS 2269.0 (2012).

## **2.5 COMPONENTS**

### **Fasteners**

Requirement: Conform to *0181 Adhesives, sealants and fasteners*.

CCA treated timber: If in contact with CCA treated timber, provide hot dip galvanized bolts with plastic sheaths, or bituminous or epoxy coatings to manufacturer's recommendations.

### **Damp-proof course**

Material: To AS/NZS 2904 (1995) or suitable alternative material conforming to NCC (2022) A5G3.

### **Flashings**

Material: To AS/NZS 2904 (1995) or suitable alternative material conforming to NCC (2022) A5G3.

## **2.6 FINGER JOINTED STRUCTURAL TIMBER**

### **General**

Performance: To AS/NZS 8008 (2022).

Adhesive bond performance: To AS/NZS 8008 (2022).

Production: To AS 5068 (2006).

Material requirements: As Documented.

## 2.7 RECONSTITUTED WOOD PRODUCTS

### Wet process fibreboard (including hardboard)

Standard: To AS/NZS 1859.4 (2018).

Bending strength: To AS/NZS 1859.4 (2018) Section 7.

Material requirements: As Documented.

## 3 EXECUTION

### 3.1 GENERAL

#### Storage and handling

Handling: Do not distort or damage timber or timber products. Do not mark or stain the surface of architecturally expressed structural elements. Use identified loading and lifting points.

Storage: To the manufacturer's specifications and the following:

- Maintain integrity of timber and treatments.
- Store above ground. Provide supports at maximum 1.2 m centres and of sufficient height to allow air circulation around the timber elements.
- Store architecturally expressed structural elements and elements for internal use under cover.

Moisture content of seasoned timber: Provide protection throughout handling and storage to maintain a moisture content within the targets for seasoned timber (15% maximum) and ideally near the equilibrium moisture content anticipated in service.

#### Installation

Framing: To the AS 1684 series.

Fastener installation: To *0181 Adhesives, sealants and fasteners*. Do not split or damage the timber.

### 3.2 FLOOR FRAMING

#### Bearers and joists

Levelling: Level bearers and joists by checking or by packing for the full width of the member with dense corrosion-resistant material that is secured in place.

Maximum thickness of packing: 3 mm.

Spring: Lay bearers and joists to allow for straightening under loading.

#### Joints

Requirement: Locate joints only over supports:

- Minimum bearing of bearers: 50 mm.
- Minimum bearing of joists: 30 mm.

#### Fixing and restraint

Fixing: Secure bearers and joists to supports to provide restraint against lateral movement.

Deep joists: To AS 1684.2 (2021) clause 4.2.2.3 or AS 1684.3 (2021) clause 4.2.2.3, as appropriate.

Trimmers or blocking dimensions:

- Depth: Joist depth less 25 mm.
- Minimum thickness: 25 mm.

Engineered timber joists: Provide lateral restraint to the manufacturer's recommendations.

### 3.3 WALL FRAMING

#### Additional support

Requirement: Provide additional support in the form of noggings, trimmers and studs for fixing lining, cladding, hardware, accessories, fixtures and fittings, as required.

Spacing of noggings: Maximum 1350 mm centres.

#### Vermin barriers

Brick veneer barrier: Nail 10 mm galvanized steel wire mesh to the underside of the bottom plate of external stud walls, extending across the cavity for building into brickwork.

**Damp-proof course**

Requirement: Provide damp-proof courses under the bottom plate of stud walls built off slabs or masonry dwarf walls as follows:

- External walls (not masonry veneer): Turn up at least 75 mm on the inside and tack. Project 10 mm beyond the external slab edge or dwarf wall and turn down at 45°.
- Walls of bathrooms, shower rooms and laundries: Turn up at least 150 mm on the wet side and tack to studs.

Installation: Lay in long lengths. Lap full width at corners and intersections, and at least 150 mm at joints.

Junctions: Preserve continuity at junctions of damp-proof courses, sarking and waterproof membranes.

**Flashings**

Location: Provide flashings to external openings to prevent the entry of moisture. Form trays at the ends of sill flashings.

Masonry veneer construction: Extend flashing across cavities and build into brickwork.

**3.4 ROOF AND CEILING FRAMING****Wall plates**

Fixing: Fix timber wall plates to masonry, with straps, bolts or both.

**Fixing plates**

Requirement: Provide timber fixing plates to transfer the design loads where timber joists, rafters or purlins bear on or into steel members, as Documented. Bolt to the steel member at maximum 500 mm centres and at a maximum of 100 mm from the end of the fixing plate.

**Beam framing**

Ridge straps: Butt ends of rafters together at ridge. Strap each pair together with 900 mm long steel strap passing over the ridge, triple nailed to each rafter.

**Additional support**

Requirement: Provide additional frame members at the following locations:

- Hanging light fittings.
- Ceiling fans.
- Access panels.
- Any other hanging services or fixtures and fittings.

Water tank or heater in roof space: Provide a support platform to the NCC cited AS/NZS 3500.4 (2021) clause 5.5.1.

**Anti-ponding boards**

Standard: To AS 2050 (2018) clause 3.1.2.

**Trusses**

Nailplated prefabricated roof trusses: To AS 4440 (2004).

Support: Support trusses on bottom chord at two points only, unless designed for additional support.

Plumb: The lesser of H/50 or 50 mm, where H is the height of the truss at the point where plumb is being measured.

Vertical movement: Provide minimum vertical clearance of 10 mm plus ceiling batten depth over internal non-load bearing walls. Use bracing methods that accommodate design vertical movements.

**3.5 ROOF TRIM****Fascia, valley and barge boards**

Requirement: Fix fascia, valley gutter boards and barge boards.

**3.6 COMPLETION****Protection**

Protection from weather: Provide temporary protection for members until permanent covering is in place.

**Tightening**

Requirement: Retighten bolts, screws and other fasteners so that all joints and anchorages are secure at the Date of Practical Completion.

**Cleaning**

General: On completion of framing remove debris from any gaps between members and make sure

|                                            |
|--------------------------------------------|
| <b>0423 ROOFING - PROFILED SHEET METAL</b> |
|--------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide a profiled sheet metal roofing system and associated work, as Documented. Refer to Appendix A – T-Sheet and Architectural Drawings.

**Ambient climatic conditions**

Design rainfall intensity (mm/h) to the NCC cited AS/NZS 3500.3 (2021): by Hydraulic Engineer.

**Corrosion resistance**

Material: To the manufacturer's recommendations for distance from marine influence.

Distance from marine influence: by Stratco/Colorbond.

**Roof access**

Requirement: To *0193 Building access safety systems*.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements*.
- *0193 Building access safety systems*.

**1.3 STANDARDS****General**

Standard: To AS 1562.1 (2018).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS 1562.1 (2018) apply.

**1.5 TOLERANCES****Sheet metal roofing**

Supporting members: To AS 1562.1 (2018) clause 4.2.3.

**1.6 SUBMISSIONS****Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

**Products and materials**

Type Tests: Submit Test results for the following:

- Profiled sheet metal roofing: To **PRODUCTS, PROFILED SHEET METAL ROOFING, Tests**.

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Roof supports before covering up or concealing.
- Glazing products before they are installed.

- The parts of the roofing, sarking, vapour barrier, insulation and roof plumbing installation before covering up or concealing.

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide samples of the following, showing the range of variation available:

- Trim and accessories with a colour finish.
- Custom profiled flashings and cappings.
- Sheet metal finishes.
- Sealants.

#### Storage and handling

Storage: To the manufacturer's recommendations and as follows:

- Keep clean, dry and unexposed to weather.
- Store away from uncured concrete and masonry, on a level base and not in contact with other materials that cause staining, denting or other surface damage.
- Stack flat and off the ground on at least 3 evenly placed bearers.

Handling: Handle metal roofing materials as follows:

- Use gloves when handling precoated metal roofing material.
- Use soft soled shoes when fixing or working on roofs.
- Protect edges and surfaces from damage. Do not drag sheets across each other or over other materials.

Storage area conditions: Allocate a safe and trade free area.

#### Welded safety mesh

Standard: To AS/NZS 4389 (2015).

### 2.2 PROFILED SHEET METAL ROOFING

#### Standards

Design and materials: To AS 1562.1 (2018).

#### Fastenings

Requirement: Starter clips, fixing clips and fasteners to the roofing system supplier's recommendations.

Prefinished exposed fasteners: Finish with an oven baked polymer coating to match the roofing material.

Fastenings to timber battens: Fasteners long enough to penetrate the thickness of the batten without piercing the underside.

#### Profiled fillers

Type: Purpose-made closed cell polyethylene foam profiled to match the roofing profile.

Location: Provide profiled fillers under flashings to the following:

- Ridges.
- Eaves.
- Lapped joints in roof sheeting.

#### Insulation spacers

Description: Proprietary spacer system to prevent excessive compression of insulation between roof sheeting and framing.

#### Components

Sealant: 100% neutral cure non-acid based silicone rubber to match roofing.

Accessories: Provide accessories with the same finish as roofing sheets.

#### Tests

Resistance to concentrated loads: To AS 1562.1 (2018) clause 5.4.

Resistance to wind pressures:

- Non-cyclonic wind regions: To AS 1562.1 (2018) clause 5.5.
- Cyclonic wind regions: To AS 1562.1 (2018) clause 5.6.

### 2.3 ROOF PLUMBING

#### General

Description: Flashings, cappings, gutters, rainheads, outlets, external downpipes and accessories necessary to complete the roofing system.

Flashing and capping: Notched to match profile of roofing.

Matching fascia/barge capping: If the selected eaves gutter is a proprietary high front pattern forming part of a combined system of gutter, fascia and barge, provide matching proprietary fascias and barge cappings to roof verges and edges.

#### Standards

Roof drainage: To the NCC cited AS/NZS 3500.3 (2021).

Metal rainwater goods: To AS/NZS 2179.1 (2014).

Flashings and cappings: To AS/NZS 2904 (1995).

### 2.4 ROOF VENTILATORS

#### General

Description: A proprietary roof ventilator system including framing, fastenings, trim, seals, accessories and flashings. Refer also to Mechanical Engineer's Documents.

## 3 EXECUTION

---

### 3.1 GENERAL

#### Preparation

Substrates or framing: Before fixing roofing, check the alignment of substrates or framing and adjust if required.

Flexible underlay: Check that the underlay or insulation is restrained.

Roofing: Make sure the roofing is clean and free of dust and loose particles.

### 3.2 INSTALLATION

#### Protection

General: Keep the roofing and rainwater system free of debris and loose material during construction.

Protection: Protect surfaces and finishes, including the retention of protective film during installation.

#### Thermal movement

Requirement: Allow for thermal movement in the roof installation and the structure, including movement in joints and fastenings.

#### Metal separation

Requirement: Prevent direct contact between incompatible metals, and between green hardwood or chemically treated timber and aluminium or coated steel, by :

- Inserting a separation layer.

### 3.3 PROFILED SHEET METAL ROOFING

#### Roof sheet installation

Standard: To AS 1562.1 (2018).

Set-out point: to be agreed on Site with the Superintendent.

Fastener type, size, corrosion resistance class, and spacing: To the sheet metal roofing manufacturer's recommendations.

Swarf: Remove swarf and other debris as soon as it is deposited.

Accessories: Provide accessories as required to complete the roofing installation.

Maximum expansion joint spacing (m): 4.2m; refer to drawings.

#### Pan type sheets

Removal: Install sheets so that individual sheets can be removed without damage.

**Curved corrugated sheet**

General: Form by rolling from material recommended for curving or bullnosing. Minimise crimping or creasing across the face of the sheet. Trim off crimped or creased edges and ends.

**Ridges and eaves**

Sheet ends: Treat as follows:

- Project sheets 50 mm into gutters.
- Close off ribs at bottom of sheets using mechanical means or with purpose-made fillers or end caps.
- Turn pans of sheets up at tops and down into gutters by mechanical means.
- Pre-cut notched eaves flashing and birdproofing if required.
- Close off ridges with purpose-made ridge fillers of closed cell polyethylene foam.

**Ridge and barge**

Capping: Finish off along ridge and verge lines with purpose-made ridge capping or barge rolls.

**Sprung curved ridge**

General: Lay the roofing sheets in single lengths from eaves to eaves by naturally curving the sheets over the ridge.

Ridge: Seal side laps at the ridge and extend the sealant to the point where the roof pitch equals the recommended pitch of the roofing profile.

**End laps**

General: If end laps are unavoidable, and the sheet profile is not suitable for interlocking or contact end laps, construct a stepped type lap.

**3.4 ROOF PLUMBING****Jointing sheet metal rainwater goods**

Butt joints: Make joints over a backing strip of the same material.

Soldered joints: Do not solder aluminium or aluminium/zinc-coated steel.

Sealing: Seal fasteners and mechanically fixed joints. Fill the holes of blind rivets with silicone sealant.

**Flashings**

Installation: Flash roof junctions, upstands, abutments and projections through the roof. Preform to required shapes if possible. Notch, scribe, flute or dress down as necessary to follow the profile of adjacent surfaces. Mitre angles and lap joints 150 mm in running lengths. Provide matching expansion joints for every two lengths of flashing, at a maximum of 12 m centres.

Upstands: Flash projections above or through the roof with two-part flashings, consisting of a base flashing and a cover flashing, with at least 100 mm vertical overlap. Provide for independent movement between the roof and the projection.

Large penetrations in low pitch roofs: Extend the flashing over the roofing to the ridge to prevent ponding behind the penetrating element.

Wall abutments: If a roof abuts a wall, provide overflashing as follows:

- In masonry walls, planked cladding or concrete: Step in courses to the roof slope. Interleave with damp-proof course, if any.
- Raking in masonry: Build into the full width of the outer leaf. Turn up and across the cavity and fix to or build into the inner leaf at least 75 mm above the roofing line.
- Raking in concrete: Turn 25 mm into joints or grooves, wedge at 200 mm centres with compatible material and point up.

Fixing to pipes: Solder or seal with neutral cure silicone rubber and secure with either of the following:

- Clamping ring.
- Proprietary flexible clamping shoe with attached metal surround flashing.

**Gutters**

Gutter and sump support: Provide framing and lining to support valley gutters, box gutters and sumps. Line the whole area under the gutters and sumps.

Support: Proprietary metallic coated adjustable strap and channel system

Lining: 17.5mm structural plywood.

Box gutter: Prefabricate box gutters to the required section and shape as follows:

- Form stop ends, downpipe nozzles, bends and returns.

- Dress downpipe nozzles into outlets.
- Hail guards: Install grating over the whole of the box gutter, over all box gutter sumps and over the edges of roofing sheeting entering box gutters.
- Overflows: Provide overflows to prevent back-flooding. Size to pass 100% of the design rainfall. Discharge overflows in visible locations and so water does not enter the building or cause damage to the building.
- Sumps: Minimum 150 mm deep and the full width of the box gutter.

Valley gutters: Refer to Hydraulic Engineer's Performance Based Design Brief for approved alternative solution. .

#### **External downpipes**

General: Prefabricate downpipes to the required section and shape. Connect heads to gutter outlets and, if applicable, connect feet to rainwater drains.

Access cover: Provide a removable watertight access cover at the foot of each downpipe stack.

- Size: Not less than the diameter of the downpipe.

Downpipe support: Provide supports and fastenings for downpipes.

#### **Rainwater disposal**

System: Overland swale; refer to Civil Engineer's Documents.

### **3.5 ROOF VENTILATORS**

#### **Installation**

Fixing method: to manufacturer's requirements.

### **3.6 COMPLETION**

#### **Reinstatement**

Extent: Repair or replace damage to the roofing and rainwater system. If the Work cannot be repaired satisfactorily, replace the whole area affected.

Damage to prepainted finish: Replace sheets with scratches in the prepainted finish greater than 2 mm in width.

Fasteners: Make sure weathertight and external sheet facings are not distorted.

#### **Cleaning**

Roofing and rainwater drainage system: Remove debris, metal swarf, solder, sealants and unused materials.

Exposed metal surfaces: Clean surfaces of substances that interfere with uniform weathering or oxidation.

Roof plumbing: Clean out spoutings, gutters and rainwater pipes after completion of roof installation.

Protection: After completion, remove protective film using methods to the manufacturer's recommendations.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes recommendations from the roofing manufacturer or supplier for the maintenance of the roofing system including frequency of inspection and recommended methods of access, inspection, cleaning, repair and replacement.

#### **Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: as provided by manufacturer and installer.

## **4 SELECTIONS**

---

Refer to Appendix A – T-Sheet, Hydraulic Engineer and Civil Engineer's Documents.

|                                          |
|------------------------------------------|
| <b>0424 ROOFING - SEAMED SHEET METAL</b> |
|------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide a seamed sheet metal roofing system and associated work, as Documented. Refer to Appendix A – T-Sheet and Architectural Drawings.

**Ambient climatic conditions**

Design rainfall intensity (mm/h) to the NCC cited AS/NZS 3500.3 (2021): by Hydraulic Engineer.

**Corrosion resistance**

Material: To the manufacturer's recommendations for distance from marine influence.

Distance from marine influence: by Colorbond.

**Roof access**

Requirement: To 0193 Building access safety systems.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0185 Timber products, finishes and treatment.
- 0193 Building access safety systems.

**1.3 STANDARDS****General**

Standard: To AS 1562.1 (2018).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS 1562.1 (2018) apply.

**1.5 TOLERANCES****Sheet metal roofing**

Supporting members: To AS 1562.1 (2018) clause 4.2.3.

**1.6 SUBMISSIONS****Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Products and materials**

Type Tests: Submit Test results for the following:

- Seamed sheet metal roofing: To PRODUCTS, **SEAMED SHEET METAL ROOFING, Tests.**

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail. **Subcontractors**

Installer experience: Submit evidence of experience with non-ferrous roof installation.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.7 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Roof supports before covering up or concealing.

- Glazing products before they are installed.
- The parts of the roofing, sarking, vapour barrier, insulation and roof plumbing installation before covering up or concealing.

## 2 PRODUCTS

---

### 2.1 GENERAL

#### **Samples**

Requirement: Provide samples of the following, showing the range of variation available:

- Trim and accessories with a colour finish.
- Custom profiled flashings and cappings.
- Sealants.
- Pre-weathered finish to sheet metal.

#### **Storage and handling**

Storage: To the manufacturer's recommendations and as follows:

- Keep clean, dry and unexposed to weather.
- Store away from uncured concrete and masonry, on a level base and not in contact with other materials that cause staining, denting or other surface damage.
- Stack flat and off the ground on at least 3 evenly placed bearers.

Handling: Handle metal roofing materials as follows:

- Use gloves when handling precoated metal roofing material.
- Use soft soled shoes when fixing or working on roofs.
- Protect edges and surfaces from damage. Do not drag sheets across each other or over other materials.

Storage area conditions: Allocate a safe and trade free area.

#### **Welded safety mesh**

Standard: To AS/NZS 4389 (2015).

### 2.2 SEAMED SHEET METAL ROOFING

#### **General**

Description: Sheet metal roll formed into pan profiles, laid with seamed joints on flush finished, continuous plywood sheeting over an underlayer and separation layer.

#### **Plywood sheeting**

Standard: To AS/NZS 2269.0 (2012).

Surface grade: DD.

Bond: Type A.

Formaldehyde emission class: E1.

Thickness: 19 mm.

Identification: Sheets labelled under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*.

#### **Underlayer**

Description: Self-adhesive, rubberised asphalt/polyethylene waterproofing membrane.

#### **Separation layer**

Description: Fire-resisting mat of a nylon core of fused entangled filaments.

#### **Fastenings**

Requirement: Starter clips, fixing clips and fasteners to the roofing system supplier's recommendations.

#### **Components**

Solder (tin/lead): 40/60 soft solder.

Flux: Liquid or paste as appropriate for the roofing.

Sealant: 100% neutral cure non-acid based silicone rubber to match roofing.

**Tests**

Resistance to concentrated loads: To AS 1562.1 (2018) clause 5.4.

Resistance to wind pressures:

- Non-cyclonic wind regions: To AS 1562.1 (2018) clause 5.5.
- Cyclonic wind regions: To AS 1562.1 (2018) clause 5.6.

**2.3 ROOF PLUMBING****General**

Description: Flashings, cappings, and accessories necessary to complete the roofing system.

Flashing and capping: Notched to match profile of roofing.

Matching fascia/barge capping: If the selected eaves gutter is a proprietary high front pattern forming part of a combined system of gutter, fascia and barge, provide matching proprietary fascias and barge cappings to roof verges and edges.

**Standards**

Roof drainage: To the NCC cited AS/NZS 3500.3 (2021).

Metal rainwater goods: To AS/NZS 2179.1 (2014).

Flashings and cappings: To AS/NZS 2904 (1995).

---

**3 EXECUTION****3.1 GENERAL****Preparation**

Substrates or framing: Before fixing roofing, check the alignment of substrates or framing and adjust if required.

Flexible underlay: Check that the underlay or insulation is restrained.

Roofing: Make sure the roofing is clean and free of dust and loose particles.

**3.2 INSTALLATION****Protection**

General: Keep the roofing and rainwater system free of debris and loose material during construction.

Protection: Protect surfaces and finishes, including the retention of protective film during installation.

**Thermal movement**

Requirement: Allow for thermal movement in the roof installation and the structure, including movement in joints and fastenings.

**Metal separation**

Requirement: Prevent direct contact between incompatible metals, and between green hardwood or chemically treated timber and aluminium or coated steel, by one of the following methods:

- Applying an anti-corrosion, low moisture transmission coating to contact surfaces.
- Inserting a separation layer.

**3.3 SEAMED SHEET METAL ROOFING****Plywood sheeting**

Installation: Lay the length of the sheets at right angles to the supports.

End joints: Stagger the end joints and locate centrally over framing members.

Edge support: If panels are not tongue and grooved, provide noggings or trimmer joists to support the edges.

Support spacing: Maximum 300 mm centres.

Fixing method:

- To timber support: Screw or adhesive and nail.
- To steel support: Metallic-coated, self-drilling/tapping screws with the heads finishing below the surface.

Control joints: 12 mm gap at abutting building elements.

**Fabrication**

Requirement: Factory fabricate roofing trays.

Minimum bending radius: 1.75 mm.

**Fixing**

Method: Fix pans to the plywood sheeting with concealed clips screw fixed at 250 mm maximum centres.

**Seams**

Roof pitch 3° to 25°: Double standing seam.

Roof pitch > 25°: Roll cap seam.

Method: Mechanically form and welt seal in situ using a self-propelled seaming machine, to stand 25 mm high on completion. Dress seams flat at gutters, ridges and hips, and fold both pan and seam down into gutters and up to form stop ends at ridges and hips.

**Ridge and hip capping**

Installation: Lock welt to the upturn of the roofing.

**End laps**

General: To the roofing manufacturer's recommendations.

**3.4 ROOF PLUMBING****Jointing sheet metal rainwater goods**

Butt joints: Make joints over a backing strip of the same material.

Soldered joints: Do not solder aluminium or aluminium/zinc-coated steel.

Sealing: Seal fasteners and mechanically fixed joints. Fill the holes of blind rivets with silicone sealant.

**Flashings**

Installation: Flash roof junctions, upstands, abutments and projections through the roof. Preform to required shapes if possible. Notch, scribe, flute or dress down as necessary to follow the profile of adjacent surfaces. Mitre angles and lap joints 150 mm in running lengths. Provide matching expansion joints for every two lengths of flashing, at a maximum of 12 m centres.

Upstands: Flash projections above or through the roof with two-part flashings, consisting of a base flashing and a cover flashing, with at least 100 mm vertical overlap. Provide for independent movement between the roof and the projection.

Large penetrations in low pitch roofs: Extend the flashing over the roofing to the ridge to prevent ponding behind the penetrating element.

Wall abutments: If a roof abuts a wall, provide overflashing as follows:

- In masonry walls, planked cladding or concrete: Step in courses to the roof slope. Interleave with damp-proof course, if any.
- Raking in masonry: Build into the full width of the outer leaf. Turn up and across the cavity and fix to or build into the inner leaf at least 75 mm above the roofing line.
- Raking in concrete: Turn 25 mm into joints or grooves, wedge at 200 mm centres with compatible material and point up.

Fixing to pipes: Solder or seal with neutral cure silicone rubber and secure with either of the following:

- Clamping ring.
- Proprietary flexible clamping shoe with attached metal surround flashing.

Box gutter: Prefabricate box gutters to the required section and shape as follows:

- Form stop ends, downpipe nozzles, bends and returns.
- Dress downpipe nozzles into outlets.
- Hail guards: Install grating over the whole of the box gutter, over all box gutter sumps and over the edges of roofing sheeting entering box gutters.
- Overflows: Provide overflows to prevent back-flooding. Size to pass 100% of the design rainfall. Discharge overflows in visible locations and so water does not enter the building or cause damage to the building.
- Sumps: Minimum 150 mm deep and the full width of the box gutter.

Valley gutters: Profile to suit the valley boarding. Turn back both edges 180 x 6 mm radius. Nail or screw to the valley boarding at the top end to prevent the gutter creeping downwards.

Gratings: Install removable gratings over rainheads and sumps.

Leaf guard location: All gutter outlets.

**Rainwater disposal**

System: Landscape Catchment; refer Civil Engineer Documents.

**3.5 COMPLETION**

**Reinstatement**

Extent: Repair or replace damage to the roofing and rainwater system. If the Work cannot be repaired satisfactorily, replace the whole area affected.

Damage to prepainted finish: Replace sheets with scratches in the prepainted finish greater than 2 mm in width.

Fasteners: Make sure weathertight and external sheet facings are not distorted.

**Cleaning**

Roofing and rainwater drainage system: Remove debris, metal swarf, solder, sealants and unused materials.

Exposed metal surfaces: Clean surfaces of substances that interfere with uniform weathering or oxidation.

Roof plumbing: Clean out spoutings, gutters and rainwater pipes after completion of roof installation.

Protection: After completion, remove protective film using methods to the manufacturer's recommendations.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes recommendations from the roofing manufacturer or supplier for the maintenance of the roofing system including frequency of inspection and recommended methods of access, inspection, cleaning, repair and replacement.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As provided by the manufacturer and installer.

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet**

|                                                        |
|--------------------------------------------------------|
| <b>0436 CLADDING - PROFILED AND SEAMED SHEET METAL</b> |
|--------------------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide external wall cladding and associated work, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Appendix B – Façade and Wall Schedule, and Façade Engineer's Documents, including Façade Performance Specification.

Folded Metal Sheet Cladding: for this section Seamed Metal Sheet Cladding has been changed to Folded Metal Sheet Cladding to more accurately represent the project.

**Corrosion resistance**

Material: To the manufacturer's recommendations for distance from marine influence.

Distance from marine influence: by metal sheet Manufacturer.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0185 Timber products, finishes and treatment.
- 0342 Light steel framing.

**1.3 STANDARDS****General**

Standard: To AS 1562.1 (2018).

**1.4 INTERPRETATION****Abbreviation**

General: For the purposes of this worksection, the following abbreviation applies:

- FC: Fibre cement.
- RAB: Rigid air barrier

**Definitions**

General: For the purposes of this worksection, the definitions given in AS 1562.1 (2018) apply.

**1.5 TOLERANCES****Permitted deviations**

Supporting members for profiled and seamed metal sheet cladding: To AS 1562.1 (2018) clause 4.2.3.

Structural steelwork for wall cladding:  $\pm 5$  mm between bearing planes of adjacent supports.

**1.6 SUBMISSIONS****Certification**

Design: Refer to façade engineer's specification for components requiring a Form 15 to be provided by an RPEQ registered engineer.

Installation: Provide installation certification to the satisfaction of the assessing authority.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

**Products and materials**

Type Tests: Submit Test results for the following:

- Façade: To PRODUCTS, **GENERAL**, **Tests**.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

### **Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC licence and CV.

Seamed sheet metal cladding: Submit evidence of experience with non-ferrous cladding installation.

### **Substrate acceptance**

Requirement: Submit evidence of installer's acceptance of the wall substrate or framing before starting installation.

### **Shop drawings**

General: Submit shop drawings to a scale that best describes the detail. **Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

## **1.7 INSPECTION**

### **Notice**

Inspection: Give notice so that inspection may be made of the following:

- Framing, pliable membranes and insulation before covering up or concealing.

## **2 PRODUCTS**

---

### **2.1 GENERAL**

#### **Samples**

Requirement: Provide samples of the following, showing the range of variation available:

- Cladding material.
- Trim and accessories with a colour finish.
- Custom profiled flashings and cappings.
- Profiled sheet metal cladding:
  - . Profiled sheet metal cladding finishes.
  - . Sealants.
- Seamed sheet metal cladding:
  - . Seamed sheet metal cladding finishes.
  - . Sealants.
- . Thermal Break

#### **Storage and handling**

Requirement: Store and handle materials to the manufacturer's recommendations and the following:

- Protect materials including edges and surfaces from damage.
- Keep clean, dry and unexposed to weather.
- Do not drag sheets or panels across each other or over other materials.
- Sheeting: Stack flat and off the ground on at least 3 evenly placed bearers.
- Store metal materials away from uncured concrete and masonry on a level base.
- Do not store metal materials in contact with other materials that may cause staining, denting or other surface damage.
- Use gloves when handling metal cladding material.

#### **Tests**

Resistance to wind pressures:

- Non-cyclonic wind regions: To AS 1562.1 (2018) clause 5.5.
- Cyclonic wind regions: To AS 1562.1 (2018) clause 5.6.

### **2.2 FOLDED SHEET METAL**

#### **General**

Requirement: Sheet metal press formed into Documented profiles, laid with folded joints to battens on a breathable waterproof membrane. Refer also to Façade Engineer's Documents.

### **Underlayer**

Requirement: Breathable waterproof membrane to FC substrate, installed as a vented cavity or rainscreen system. Minimum 35 mm ventilation gap between substrate and framing.

## **2.3 COMPONENTS**

### **Fastenings**

General: Type, size, corrosion resistance class and spacing to the cladding manufacturer's recommendations.

### **Folded sheet metal**

General: as Documented in the Architectural Drawings and the Façade Engineer's Documents.

### **Subframe**

General: cold formed metal battens to façade engineer's specification.

### **Rigid air barrier**

General: including bonded membrane to façade engineer's specification.

### **Thermal break**

General: fixed directly to metal framing; to have a thermal value of R0.2.

### **Flashings**

Requirement: To AS/NZS 2904 (1995).

## **3 EXECUTION**

---

### **3.1 GENERAL**

#### **Preparation**

Substrates or framing: Before fixing cladding, check the alignment of substrates or framing and adjust if required.

Flexible underlay: Check that the underlay or insulation is restrained.

Cladding: Make sure the cladding is clean and free of dust and loose particles.

#### **Installation**

Standard: To AS 1562.1 (2018).

Requirement: Install cladding as follows:

- Fix firmly against framing to the manufacturer's recommendations.
- Plumb, level, straight and to Documented tolerances.
- Fixed or anchored to the building structure in conformance with the wind action loading recommendations.
- Isolated from any building loads, including loads caused by structural deflection or shortening.
- Allow for thermal movement.

Cladding layout: Cut/fabricate and install cladding to suit the layout as Documented.

Protection: Protect surfaces and finishes, and retain protective film during installation.

#### **Prototypes**

Requirement: Provide a prototype of each panel type, including at least one example of each component in the system to verify selections submitted as samples, to demonstrate aesthetic effects, to set quality standards for materials and execution.

Inclusions:

- Typical components, attachments to building structure and methods of installation.
- Window opening with cladding panel, trim and returns.
- Sealant filled joint.

Testing: Critical weatherproofing junctions are to be flood or hose Tested as determined by the façade engineer.

#### **Accessories and trim**

Requirement: Provide accessories, flashings and trim required to complete the installation.

#### **Metal separation**

Requirement: Prevent direct contact between incompatible metals, and between green hardwood or chemically treated timber and aluminium or coated steel, by either of the following methods:

- Apply an anti-corrosion, low moisture transmission coating to contact surfaces.
- Insert a separation layer.

Incompatible metal fastenings: Do not use.

#### **Defective and damaged parts**

Defective components: Do not install component parts that are Defective, including warped, bowed, dented, chipped, scratched, abraded or broken members.

### **3.2 FOLDED SHEET METAL CLADDING**

#### **Installation**

Ground clearance: Maintain Documented clearance.

Cutting sheets: Wherever possible, factory cut to length. Do not use an abrasion disc.

Laser cutting: It has been verified through Stratco and Laser 3D that laser cutting is a viable method for Zincalume sheet up to 1.2mm BMT and this does not affect the material warranty of the product or compromise the integrity of the AM125 coating. Any laser cutting provider should be ISO 9001 certified and use fibre optic laser systems, not gas-based systems.

Accessories: Provide material with the same finish as cladding sheets.

Swarf: Remove swarf and other debris as soon as it is deposited.

#### **Fabrication**

Requirement: Factory fabricated panels.

Minimum bending radius: 1.75 mm.

#### **Fixing**

Requirement: As Documented.

### **3.3 COMPLETION**

#### **Fasteners**

Requirement: Adjust for weathertightness without distortion of external panel face.

#### **Reinstatement**

Requirement: Replace damaged cladding, including cladding with scratches in the prepainted finish.

#### **Cleaning**

Requirement: Remove debris, metal swarf, solder, sealants and unused materials.

Exposed metal surfaces: Remove substances that interfere with uniform weathering or oxidation.

Protection: Remove protective film using methods required by the manufacturer after completion.

Panels: Clean surfaces with soft, clean cloths and clean water to the manufacturer's recommendations.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published use, care and maintenance requirements.

#### **Warranties**

Type: Material manufacturer, fabricator and installer interlocking warranty.

Period: As offered by material manufacturer.

## **4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                                      |
|--------------------------------------|
| <b>0451 WINDOWS AND GLAZED DOORS</b> |
|--------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide windows and glazed doors, as Documented. Refer to Architectural Drawings, and Appendix B – Façade and Wall Schedule. Refer also to Façade Engineer's Performance Based Design Brief for performance and certification requirements. Refer also to ESD consultant's Section J report for energy efficiency performance requirements. Refer also to Acoustic Engineer's Report for acoustic performance requirements.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****General**

Selection and installation: To AS 2047 (2014).

Building classification: Class 5

**Glazing**

Glass type and thickness: To AS 1288 (2021), as nominated in the façade engineer's Documents. .

Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000).

**1.4 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviations apply:

- AGWA: Australian Glass and Window Association.
- WERS: Window Energy Rating Scheme.

**Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 4668 (2000), AS/NZS ISO 22496 (2024) and the following apply:

- Hardware: To AS 4145.1 (2008) Section 2.
- Louvres - continuous: Louvres that run continuously past, and are supported by, concealed framing or brackets.
- Louvres - horizontal: Louvres that span horizontally between frame stiles, mullions or vertical supports.
- Louvres - vertical: Louvres that span vertically between frame heads and sills, or horizontal supports.
- Total system SHGC: Solar heat gain coefficient as defined by the NCC and Tested in conformance with NFRC 200 (2023).
- Total system U-Value: Thermal transmittance as defined by the NCC and Tested in conformance with NFRC 100 (2023).
- Weathering: Inclined upper external surface, such as of a coping, sill, or top of a buttress or chimney, designed to shed rainwater quickly and throw it clear of the facing material below.

**1.5 SUBMISSIONS****Certification**

Windows and glazed doors: Submit evidence of conformity to AS 2047 (2014). Refer also to façade engineer's Documents.

Sealant compatibility: Submit statements from all parties to the installation certifying the compatibility of sealants and glazing systems to all substrates.

Toughened glass: Submit certification from the manufacturer for all toughened glass as evidence of heat soaking to **GLAZING, Heat soaking**.

#### **Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

#### **Products and materials**

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

Type Tests: Submit Test results of the following:

- Acoustic performance of windows and doors: To PRODUCTS, **GENERAL, Acoustic performance**.
- Protection of openable windows: To PRODUCTS, **GENERAL, Protection of openable windows**.

#### **Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

#### **Shop drawings**

General: Submit shop drawings, to a scale that best describes the detail, showing the following:

- Full size sections of members.
- Hardware, fittings and accessories including fastener details.
- Junctions and trim to adjoining surfaces.
- Layout (sectional plan and elevation) of the window assembly.
- Methods of assembly.
- Methods of installation, including fixing, caulking and flashing.
- Provision for vertical and horizontal expansion.
- Method of glazing, including the following:
  - . Rebate depth.
  - . Edge restraint.
  - . Clearances and tolerances.
  - . Glazing gaskets and sealant beads.

#### **Subcontractors**

General: Submit names and contact details of proposed manufacturers and installers.

Evidence of experience: QBCC licence and CV.

#### **Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

## **1.6 INSPECTION**

#### **Notice**

Inspection: Give notice so that inspection may be made of the following:

- Openings prepared to receive windows.
- Fabricated window assemblies at the factory ready for delivery to the Site.
- Fabricated window assemblies delivered to the Site, before installation.
- Commencement of window installation.

---

## **2 PRODUCTS**

### **2.1 GENERAL**

#### **Samples**

Requirement: Provide samples labelled with the series code reference and date of manufacture.

Window and door framing: Provide samples of the following:

- Prefinished production materials showing the limits of the range of variation in the Documented colour.
- Joints made by proposed techniques.
- Sections for frames, sashes, louvres and slats.

Glazing: Provide samples of glazing materials, each at least 200 x 200 mm, showing the visual properties and range of variation, if any, for each of the following:

- Tinted or coloured glass or plastics glazing.
- Surface modified or surface coated glass.
- Patterned or obscured glass or plastics glazing.
- Ceramic-coated glass.
- Wired glass.
- Mirror glass.

Hardware and accessories: Provide samples of the following:

- Window manufacturer's standard hardware and accessories including locks, latches, handles, catches, sash operators, anchor brackets and attachments, masonry anchors and weatherseals (pile or extruded).
- Generic hardware: Provide samples of generic hardware not Documented as proprietary items.

### **Storage and handling**

Storage: Store in a clean, dry area unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

Handling: Handle frames to the manufacturer's recommendations and as follows:

- Stack upright, off the ground and against a flat, vertical surface.
- Carry in the vertical position with sashes locked.
- Do not rack frames out of square.
- Do not remove any bands and corner bracing until after installation.

### **Acoustic performance**

Windows and doors: Rating to the NCC cited AS/NZS ISO 717.1 (2004), as Documented.

### **Protection of openable windows**

Fall prevention: To BCA (2022) D3D29.

### **Marking**

Window assemblies: To AS 2047 (2014) Section 8.

## **2.2 FRAMES**

### **Aluminium frames**

Standard: To AS 2047 (2014) clause 3.1.

Construction: Assembled from aluminium sections, including accessories such as pile strips, fixing ties or brackets and cavity flashings, with provision for fixing Documented hardware and seals.

Subsill: If the frame includes a subsill, provide a self-draining section.

### **Timber frames**

Standard: To AS 2047 (2014) clause 3.2.

Hardwood: To AS 2796.1 (1999):

- Grade: Select.

Construction: Assembled from timber sections, with provision for fixing Documented hardware including rebates for seals.

## **2.3 GLAZING**

### **Performance**

Glass: Free from Defects that detract from appearance or interfere with performance under normal conditions of use.

Plastics glazing: Free from surface abrasions and resistant to yellowing or other colour change. Capable of maintaining physical properties including strength and impact resistance for its design life.

### **Heat soaking**

Requirement: Heat soak glass to AS 1288 (2021) clause 3.8.

Standard: To EN 14179-1 (2016).

Marking: To EN 14179-1 (2016) or certified by the manufacturer to AS 1288 (2021) clause 3.8.2.

**Safety glazing materials**

Standard: To AS 2208 (2023).

Type: Grade A.

Marking: To AS 1288 (2021) clause 5.23.

**Heat-strengthened glass**

Standard: To ASTM C1048 (2025).

**Insulating glass units (IGUs)**

Requirement: Provide insulating glass units, as Documented.

Manufacture, Testing and installation: To AS 4666 (2012).

**2.4 GLAZING MATERIALS****General**

Requirement: Putty, glazing compounds, sealants, gaskets, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks and compression wedges appropriate for the conditions of application and required performance.

**Jointing materials**

Requirement: Jointing and pointing materials that are compatible with each other and the contact surfaces, and non-staining to finished surfaces to manufacturer's recommendations. Do not provide bituminous materials on absorbent surfaces.

**Elastomeric sealants**

Sealing compounds (polyurethane, polysulfide, acrylic): To ASTM C920 (2018) or ISO 11600 (2002).

Sealing compounds (silicone): To ASTM C920 (2018) or ISO 11600 (2002).

Sealing compounds (butyl): To ASTM C1311 (2022).

**Primer**

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials. Note: a change in the specified timber joinery supplier may result in a paint specification Variation; this will be considered a substitution that requires a submission and assessment.

**Control joints**

Depth of elastomeric sealant: Half the joint width or 6 mm, whichever is the greater.

Foamed materials (in compressible fillers and backing rods): Closed cell or impregnated types that do not absorb water.

Bond breaking: Provide backing rods, and other back-up materials for sealants, that do not adhere to the sealant.

**2.5 ALUMINIUM FRAME FINISHES****Powder coatings**

Service condition category to AS 3715 (2025): refer to Appendix E – Paint Specification

Coating performance: refer to Appendix E – Paint Specification

Coating type: refer to Appendix A – T-Sheet

**Anodised**

Standard: To AS 1231 (2000).

Thickness:

- Internal: 15 µm.
- External: 20 µm.

**2.6 ANCILLARY COMPONENTS AND FITTINGS****Trim**

General: Provide trim, shadow angles and architraves, as Documented in the Architectural Drawings.

**Extruded gaskets and seals**

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

**Flashings**

General: Corrosion-resistant, compatible with the other materials in the installation, and coated with a non-staining compound if necessary.

Standard: To AS/NZS 2904 (1995).

**Nylon brush seals**

General: Dense nylon bristles locked into holding strips and fixed in a groove in the edge of the door or in purpose-made anodised aluminium holders fixed to the door or frame to the manufacturer's recommendations.

**Pile weatherstrips**

General: Provide weatherstrips, as Documented.

Standard: To AAMA 701/702 (2023).

Material: Pile and backing or equivalent polypropylene, low friction silicone treated, ultraviolet stabilised, fixed to the frame to the manufacturer's recommendations.

Finned type: A pile weatherseal with a central polypropylene fin bonded into the centre of the backing rod and raised above the pile level.

**Weather bars**

General: Provide corrosion-resistant weather bars or threshold plates for hinged external doors, located under the centres of closed doors or to manufacturer's recommendations.

**2.7 HARDWARE****Hardware Documented generically**

General: Provide hardware of sufficient strength and quality to perform its function, appropriate to the intended conditions of use, compatible with associated hardware, and fabricated with fixed parts firmly joined.

**Window locks and latches**

Standard: To AS 4145.2 (2008).

Window catches: Provide 2 catches per sash to manually latched awning or hopper sashes over 1000 mm wide.

**Sash balances**

Requirement: Match the spring strength of the balances to the sash weight they support.

**Sash operators**

Requirement: Provide sash operators, as Documented.

---

**3 EXECUTION**

---

**3.1 GLAZING PROCESSING****General**

Processing: Perform required processes on glazing, including cutting, obscuring, silvering and bending. Form necessary holes, including for fastenings, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arised.

**3.2 INSTALLATION****General**

Requirement: Install windows and glazed doors as follows:

- Plumb, level, straight and true within building tolerances.
- Fixed or anchored to the building structure in conformance with the wind action loading requirements.
- Isolated from any building loads, including loads caused by structural deflection or shortening.
- Allow for thermal movement.

**Glazing**

Requirement: Install the glass as follows:

- Permanently fix in place each piece of glass to withstand the normal loadings and ambient conditions at its location without distortion or damage to glazing materials.

- No transfer of building movements to the glazing.
- Watertight and airtight for external glazing.

Temporary marking: Use a method that does not damage the glazing. Remove marking only after certification and acceptance of the installation.

Toughened glass: Do not cut, drill, edgework or permanently mark after toughening. Use installation methods that prevent the glass making direct contact with metals or other non-resilient materials.

Frameless installations: Join the vertical edges of adjacent glass panels with silicone jointing compound.

Heat-absorbing glass: In locations exposed to direct sunlight, provide wheel cut edges free from damage or blemishes, with minimum feather.

### **Preglazing**

Window assemblies and glazed doors: Supply inclusive of glazing, shop preglazed.

### **Weatherproofing**

Flashing and weatherings: Install flashings, weather bars, threshold plates, drips, storm moulds, joint sealant and pointing to prevent water penetrating the building between the window frame and the building structure under the prevailing service conditions, including normal structural movement of the building.

### **Fixing**

General: Conform to the manufacturer's recommendations. Make sure fasteners are concealed.

Packing: Pack behind fixing points with durable full width packing.

Prepared masonry openings: If fixing of timber windows to prepared anchorages needs to be fixed from the frame face, sink the fastener heads below the surface and fill the depression flush with a material compatible with the surface finish.

### **Joints**

General: Form tight fitted joints so that fasteners and fixing methods, such as pins, screws and adhesives do not create pressure indentations visible on exposed surfaces.

Sealants:

- If priming is recommended, prime surfaces in contact with jointing materials.
- If frames are powder coated, apply a neutral cure sealant.

### **Operation**

General: Make sure moving parts operate freely and smoothly, without binding or sticking, at correct tensions or operating forces and are lubricated.

### **Protection**

Removal: Remove temporary protection measures from the following:

- Contact mating surfaces before joining up.
- Exposed surfaces before completion of the Works.

### **Seals**

General: Provide the fastenings, rebates, grooves, and clearances required for installation and operation of the seals. Allow seals unwound from coils to settle before use. Install proprietary seals to manufacturer's recommendations and adjust correctly.

### **Trim**

General: Provide mouldings, architraves, reveal linings, and other internal trim using materials and finishes matching the window frames. Install to make neat and clean junctions between frames and the adjoining building surfaces.

## **3.3 HARDWARE**

### **Fasteners**

Materials: Use materials compatible with the item being fixed and of sufficient strength, size and quality to perform their function.

- Concealed fasteners: Provide a corrosion-resistant finish.
- Exposed fasteners: Match exposed fasteners to the material being fixed.

Support: Provide appropriate back support (for example lock stiles, blocking, wall noggings and backing plates) for hardware fasteners.

- Hollow metal sections: Provide backing plates drilled and tapped for screw fixing, or provide rivet nuts with machine thread screws. Do not use self-tapping screws or pop rivets.

**Proprietary window systems**

Requirement: Provide the standard hardware and internal fixing points for personnel safety harness attachment, if required by and conforming to the governing regulations.

**Operation**

General: Make sure working parts are accurately fitted to smooth close bearings, without binding or sticking, free from rattle or excessive play, lubricated if appropriate.

Opening force performance: To AS 1428.1 (2021).

**Supply**

Delivery: Deliver window hardware items, ready for installation, in individual complete sets for each window set, as follows:

- Clearly labelled with the intended location.
- In a separate dust and moisture proof package.
- Including the necessary templates, fastenings and fixing instructions.

**3.4 COMPLETION****Hardware**

Adjustment: Leave the hardware with working parts in working order, and clean, undamaged, properly adjusted, and lubricated if appropriate.

**Repair of finish**

Polyester or fluoropolymer coatings: Contact supplier for approval to apply touch up products, otherwise replace damaged material.

**Cleaning**

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive, acidic or alkaline materials.

Extent: All frames and glass surfaces internally and externally.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the following:

- Window and glazed door manufacturer's published recommendations for operation, care and maintenance.
- Hardware manufacturer's published recommendations for use, care and maintenance.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As provided by manufacturer and installer.

---

**4 SELECTIONS**

---

**Refer to Appendix B – Façade and Wall Schedule**

**Refer to Façade Engineer's Performance Based Design Brief**

**Refer to ESD Consultant's Section J Report**

**Refer to Acoustic Engineer's Report.**

|                                     |
|-------------------------------------|
| <b>0453 DOORS AND ACCESS PANELS</b> |
|-------------------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide doors, frames, doorsets, and smoke doorsets as Documented. Refer to the Architectural Drawings, Appendix A – T-Sheet, and Façade Engineer's Documents.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.
- 0185 Timber products, finishes and treatment.
- 0455 Door hardware.

### 1.3 STANDARDS

#### General

Timber and composite doors: To AS 2688 (2017).

### 1.4 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the definitions given in AS 2688 (2017), AS/NZS ISO 22496 (2024) and the following apply:

- Smoke doorset: A doorset that restricts the passage of smoke.

### 1.5 SUBMISSIONS

#### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

#### Products and materials

Type Tests: Submit Test results for the following:

- Acoustic performance of doorsets: To PRODUCTS, **DOORSETS, Acoustic performance.**
- Smoke doorsets: To PRODUCTS, **DOORSETS, Smoke doorsets.**

#### Samples

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

#### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties.**

### 1.6 INSPECTION

#### Notice

Inspection: Give notice so that inspection may be made of the following:

- Door frames installed before fixing trim.

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide 2 samples as follows:

- Colour samples from prefinished production material, including anodised or powder coated extrusions and sheet, showing the limits of the range of variation, if any, for each component Documented.
- Door manufacturer's standard hardware items.
- Finishes to prepared surfaces, including timber stains or veneers.

- Joints using proposed techniques.
- Proposed sections for frames.

## 2.2 FRAMES

### Aluminium frames

Construction: Assembled from aluminium sections, including accessories such as buffers, pile strips, strike plates, fixing ties or brackets and cavity flashings, with provision for fixing Documented hardware and seals.

Threshold: If the frame includes a threshold member, provide a self-draining section with slip-resistant surface.

### Steel frames

Construction: Continuously welded from metallic-coated steel sheet sections, including accessories such as buffers, strike plates, spreaders, mortar guards, switch boxes, fixing ties or brackets, and cavity flashing with provision for fixing Documented hardware, seals and electronic security assemblies, and prefinished with a protective coating.

Base metal thickness (minimum):

- General: 1.1 mm.

Metallic-coating class to AS 1397 (2021) interior: ZF100.

Finish: Grind the welds smooth, cold galvanize the welded joints and shop prime.

Hardware and accessories: Refer to Appendix C – Door Hardware Schedule.

### Timber frames

Hardwood: To AS 2796.1 (1999):

- Grade: Select.

Softwood: To AS 4785.1 (2002):

- Grade: Select.

Joints:

- Morticed head and through tenons.
- Trenched head:
  - . Bare faced tenons on jambs.
  - . Full let-in jambs.

Construction: Assembled from timber sections, with provision for fixing Documented hardware including rebates for door seals.

## 2.3 DOORS

### General

Doors: Proprietary products manufactured for interior or exterior applications and for the finish required.

### Materials

Standards: Conform to the following:

- Decorative laminated sheets: To AS/NZS 2924.1 (2024).
- Wet process fibreboard (including hardboard): To AS/NZS 1859.4 (2018).
- Dry process fibreboard (including medium density fibreboard): To AS/NZS 1859.2 (2017).
- Particleboard: To AS 1859.1 (2017).
- Plywood and blockboard for interior use: To AS/NZS 2270 (2006).
- Plywood and blockboard for exterior use: To AS/NZS 2271 (2004).
- Seasoned cypress pine: To AS 1810 (1995).
- Timber – hardwood: To AS 2796.1 (1999).
- Timber – softwood: To AS 4785.1 (2002).

### Identification

Panel doors: Provide panels branded under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*, as applicable to the product. Locate the brand on faces or edges that will be concealed in the Works.

**Joinery doors**

General: Provide joinery doors, as Documented.

**Flush panel doors**

General: Provide flush panel doors of balanced construction, as Documented.

Medium density fibreboard doors: Single thickness of moisture resistant general purpose medium density fibreboard with the same surface finish to both sides, for internal use.

**Construction**

General: To AS 2688 (2017).

Adhesives:

- Internal: To AS/NZS 2270 (2006).
- External: To AS/NZS 2271 (2004).

Door thickness:

- General: as Documented.

Cut-outs: If openings are required in flush panel doors (e.g. for louvres or glazing), do not make cut-outs closer than the width of the stiles at the edges of the doors.

Edge strips: Minimum thickness 10 mm. Increase overall thickness to greater than 15 mm to accommodate the full depth of the rebate in rebated doors. Apply to the external edges of door after the facings are bonded to the door framing/core and finish flush with outside surface of the facings.

Edge strip location: fix all round

**Double doors**

Square edged doors: Bevel as necessary to prevent binding between the leaves.

Double doors other than double acting doors: Provide rebated meeting stiles. Form rebates to suit standard rebated hardware.

**2.4 DOORSETS****Marking and labelling**

Fire-resisting doorsets: To AS 1905.1 (2015) Section 6.

Doors and doorsets: To AS 2688 (2017) clause 2.5.

**Acoustic performance**

Doorsets: Rating to the NCC cited AS/NZS ISO 717.1 (2004), as Documented.

**Revolving doorsets**

Standard: To AS 5007 (2007).

Control systems: To *0455 Door hardware*.

**Smoke doorsets**

Construction: Solid core doors not less than 35 mm thick.

Standard: To AS 6905 (2007) and BCA (2022) Spec 12. Tested to AS 1530.7 (2007).

**2.5 ANCILLARY MATERIALS****Trim**

General: Provide trim, shadow gaps and architraves, as Documented.

**Door seals**

Acoustic applications: Tested to AS 1191 (2002) or EN ISO 10140-2 (2021) and rated to the NCC cited AS/NZS ISO 717.1 (2004).

Weather and energy saving seals: To AS 4420.1 (2016) Sections 5 and 6, and AS 2047 (2014).

**Extruded gaskets and seals**

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

**Flashings**

General: Corrosion-resistant, compatible with the other materials in the installation, and coated with a non-staining compound if necessary.

Standard: To AS/NZS 2904 (1995).

**Jointing materials**

General: Compatible with each other and with the contact surfaces and non-staining to finished surfaces. Do not provide bituminous materials on absorbent surfaces.

**Nylon brush seals**

General: Dense nylon bristles locked into holding strips and fixed in a groove in the edge of the door or in purpose-made anodised aluminium holders fixed to the door or frame to the manufacturer's recommendations.

Standard: To AAMA 701/702 (2023).

---

**3 EXECUTION**

---

**3.1 FRAMES****General**

Frames: Install the frames as follows:

- Plumb, level, straight and true.
- Fixed or anchored to the building structure.
- Isolated from any building loads, including loads caused by structural deflection or shortening.

**Frame fastenings**

Brackets: Metallic-coated steel:

- Width: Minimum 25 mm.
- Thickness: Minimum 1.5 mm.

Depth of fastening for building into masonry:

- Brackets: Minimum 200 mm.
- Expansion anchors: Minimum 50 mm.
- Plugs: Minimum 50 mm.
- Rods: Minimum 60 mm.

Jamb fastening centres: Maximum 600 mm.

**Joints**

General: Form tight fitted joints so that fasteners and fixing methods such as pins, screws and adhesives do not create pressure indentations visible on exposed surfaces.

**Aluminium frames**

Building into masonry: Screw galvanized steel brackets twice to jambs and build in.

Fixing to masonry openings: Use proprietary expansion anchors and screw twice through jambs at each fixing location.

Fixing to stud frame openings: Screw once to studs at each fixing location.

**Steel frames**

Building into masonry: Attach galvanized steel rods to jambs, build in and grout up.

Fixing to masonry openings: Build in hairpin anchors and install locking bars, or use proprietary expansion anchors and screw twice through jambs at each fixing location.

Fixing to stud frame openings: Attach galvanized steel brackets to jambs and screw twice to studs at each fixing location.

**Timber frames**

Building into masonry: Screw galvanized steel brackets twice to jambs and build in.

Fixing to masonry openings: Use proprietary expansion anchors and screw twice through jambs at each fixing location.

Fixing to stud frame openings: Back screw twice to jambs at each fixing location.

Fixing to thresholds: Dowel external door frames to thresholds other than timber with 10 mm diameter brass dowels, 100 mm long.

Heads of fasteners: Conceal if possible, otherwise sink the head below the surface and fill the depression flush with a material compatible with the surface finish.

**Finishing**

Trim: Provide mouldings, architraves, reveal linings, and other internal trim using materials and finishes matching the door frames to make neat and clean junctions between the frame and the adjoining building surfaces.

**Seals**

General: Provide the fastenings, rebates, grooves, and clearances required for installation and operation of the seals. Allow seals unwound from coils to settle before use. Install proprietary seals to manufacturer's recommendations and adjust correctly.

**Weatherproofing**

Flashings and weatherings: Install flashings, weather bars, threshold plates, drips, storm moulds, caulking and pointing to prevent water from penetrating the building between the door frame and the building structure under the prevailing service conditions, including normal structural movement of the building.

**3.2 DOORS****Priming**

General: Prime timber door leaves on top and bottom edges before installation.

**Tolerances**

Installation: To AS 2688 (2017) Section 7.

**3.3 DOORSETS****General**

Installation: To AS 2688 (2017) Section 7.

**3.4 COMPLETION****Operation**

General: Make sure moving parts operate freely and smoothly, without binding or sticking, at correct tensions or operating forces and that they are lubricated if appropriate.

Opening force performance: To AS 1428.1 (2021).

**Protection**

Temporary coating: On or before the Date for Practical Completion, or before joining up to other surfaces, remove all traces of temporary coatings used as a means of protection.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published recommendations for service use.

**Warranties**

Type: Interlocking manufacturer and installation warranty.

Period: As provided by the manufacturer and installer.

---

**4 SELECTIONS**

---

**Refer to drawing A-730001 Door Types**

**Refer to Appendix C – Door Hardware Schedule**

|                           |
|---------------------------|
| <b>0455 DOOR HARDWARE</b> |
|---------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide door hardware, as Documented. Refer to Appendix C – Door Hardware Schedule, prepared by Asso Abloy.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0453 Doors and access panels.

**1.3 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the abbreviations given in AS 4145.1 (2008) Appendix D apply.

**Definitions**

General: For the purposes of this worksection, the general definitions given in AS 4145.1 (2008) Section 2 and Appendix E apply.

**1.4 SUBMISSIONS****Execution details**

Door-by-door schedule: Submit a door-by-door hardware schedule.

- Information sources: This worksection and the Contract drawings.

Key control system:

- Keyed hardware: Submit details to extend the existing key control security system for locks required to accept a group key.
- Electronic hardware: Submit details to extend the existing electronic control security system for locks required to accept a swipe card key.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Records**

Door hardware schedule: Submit a schedule, prepared by the door hardware supplier, showing changes to the Contract door hardware schedule resulting from the following:

- Approval of a hardware sample.
- Acceptance of an equivalent to a specified proprietary item.
- A variation to a door hardware requirement.

Key coding system: Submit the lock manufacturer's record of the key coding system showing each lock type, number and type of key supplied, key number for re-ordering, and name of supplier.

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples.**

**Subcontractors**

Automatic door operators: Submit names and contact details of proposed manufacturer and installer.

Pressure floor mat: Submit names and contact details of proposed manufacturer and installer.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Generic items: Provide samples of hardware items offered as meeting the description of items not specified as proprietary items.

#### Supply

Delivery: Deliver door hardware items, ready for installation, in individual complete sets for each door, as follows:

- Clearly labelled to show the intended location.
- In a separate dust and moisture proof package.
- Including the necessary templates, accessories, fastenings and fixing instructions.

Hardware specified generically: Hardware of the required strength and quality to perform its function, appropriate to the intended conditions of use, suitable for use with associated hardware, and fabricated with fixed parts firmly joined.

#### Storage and handling

General: Store and handle to protect materials from damage, to the manufacturer's recommendations.

### 2.2 LOCKS AND LATCHES

#### Standard

General: To AS 4145.2 (2008).

#### Padlocks

Standard: To AS 4145.4 (2002).

#### Lock and latch classification

Rating systems: To AS 4145.1 (2008) Section 3.

Performance requirements: To AS 4145.2 (2008) Section 3.

### 2.3 HINGES

#### Butt hinge materials

Timber doors in timber or steel frames:

- Material: refer to Appendix C – Door Hardware Schedule.

Aluminium framed doors or timber doors in aluminium frames:

- Material: refer to Appendix C – Door Hardware Schedule.

Doors fitted with closers: Provide low friction ball bearing hinges.

Fire-resisting doors: To AS 1905.1 (2015).

Power transfer hinges: Provide hinges that allow wires to be transferred unbroken between the door and frame.

Lift-off doors: If sanitary compartments require lift-off doors, provide lift-off hinges and allow sufficient clearance at the head to allow door removal.

#### Timber solid core doors

Number of hinges: Determine the number of hinges required based on the nominated door leaf size and weight only. For other door leaf sizes or for doors with applied finishes, use the weight of the door to determine the number of hinges required. For a door leaf over 80 kg, use pivot hinges.

Size of hinges: Determine the size of the hinge based on the door leaf thickness:

- 35 to 43 mm: 100 x 75 mm butt hinges with a minimum thickness of 2.5 mm.
- 44 to 55 mm: 100 x 100 mm butt hinges with a minimum thickness of 2.5 mm.
- > 55 mm: As Documented.

Hinge pin: Supply fixed pins to hinges of doors opening out or designated as a security doors. For all other doors, provide loose pins.

Wide throw: If necessary, provide wide throw hinges to achieve the required door swings in the presence of obstacles such as nibs, deep reveals and architraves.

**Hinges for timber doors table (unless noted otherwise in Appendix C – Door Hardware Schedule)**

| Nominal door leaf size (H x W x T) (mm) | Maximum door leaf weight (kg) | Number of hinges |
|-----------------------------------------|-------------------------------|------------------|
| 2040 x 400 x 35                         | 19                            | 2                |
| 2040 x 600 x 35                         | 29                            | 2                |
| 2040 x 720 x 35                         | 35                            | 3                |
| 2040 x 820 x 35                         | 39                            | 3                |
| 2040 x 920 x 35                         | 44                            | 3                |
| 2040 x 1020 x 35                        | 49                            | 4                |
| 2040 x 720 x 40                         | 37                            | 3                |
| 2040 x 820 x 40                         | 42                            | 3                |
| 2040 x 920 x 40                         | 48                            | 3                |
| 2040 x 1020 x 40                        | 52                            | 4                |
| 2040 x 720 x 50                         | 45                            | 3                |
| 2040 x 820 x 50                         | 50                            | 3                |
| 2040 x 920 x 50                         | 57                            | 3                |
| 2040 x 1020 x 50                        | 68                            | 4                |
| 2400 x 720 x 40                         | 50                            | 4                |
| 2400 x 820 x 40                         | 52                            | 4                |
| 2400 x 920 x 40                         | 55                            | 4                |
| 2400 x 1020 x 40                        | 60                            | 4                |
| 2400 x 1220 x 50                        | 72                            | 5                |
| 2040 x 920 x 70                         | 88                            | Pivot hinges     |

**Aluminium doors**

Application: Aluminium hinges for aluminium doors, or for doors of other materials in aluminium frames 40 kg or less.

**Hinges for aluminium doors table (unless noted otherwise in Appendix C – Door Hardware Schedule)**

| Nominal hinge size (L x W x T) (mm)                                                                                                                         | Maximum door leaf weight (kg) | Knuckles (minimum) | Screws/hinge leaf (minimum) |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|--------------------|-----------------------------|
| 100 x 70 x 3                                                                                                                                                | 30                            | 3                  | 3                           |
| 100 x 80 x 3.5                                                                                                                                              | 50                            | 5                  | 4                           |
| 130 x 50 x 3.4                                                                                                                                              | 75                            | Interfold          | 3                           |
| Length (L) is the dimension along the knuckles, not including hinge tips, if any, and width (W) is the dimension across both hinge leaves when opened flat. |                               |                    |                             |

## 2.4 SLIDING DOOR HARDWARE

### General

Requirement: Provide sliding door tracks and guides, as Documented. For Acoustic Sliding door refer to Appendix F – Acoustic Sliding Door Specification.

## 2.5 ANCILLARIES

### Bolts

General: Barrel bolts, flush bolts and tower bolts with keepers, including lock plates, staples, ferrules or floor sockets.

**Door seals**

Acoustic applications: Tested to AS 1191 (2002) or EN ISO 10140-2 (2021) and rated to the NCC cited AS/NZS ISO 717.1 (2004).

Weather and energy saving seals: To AS 4420.1 (2016) Sections 5 and 6, and AS 2047 (2014).

**Extruded gaskets and seals**

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

**Nylon brush seals**

General: Dense nylon bristles locked into holding strips and fixed in a groove in the edge of the door or in purpose-made anodised aluminium holders fixed to the door or frame to the manufacturer's recommendations.

**Pile weatherstrips**

General: Polypropylene or equivalent pile and backing, low friction silicone treated, ultraviolet stabilised, fixed to the door or frame to the manufacturer's recommendations.

Standard: To AAMA 701/702 (2023).

**Rebated doors**

General: For mortice locks or latches to rebated doors, provide purpose-made rebated pattern items.

**Strike plates**

General: Use strike plates supplied with the locks or latches. Do not provide universal strike plates.

**Weather bars**

General: Provide corrosion-resistant weather bars or threshold plates under hinged external doors, located under the centres of closed doors or to manufacturer's recommendations.

**2.6 DOOR CONTROLLERS****Standard**

General: To AS 4145.5 (2011).

**Performance**

Requirement: Door controllers, pivots, floor or overhead door closers, and automatic door operators, suitable for the door type, size, weight, sliding action and swings required and the operating conditions, including wind and air conditioning pressure.

**Closers**

Hinged or pivot doors used as fire-resisting doors: Closers Tested and certified for use as components of fire-resisting door assemblies to AS 1905.1 (2015).

**2.7 ELECTRONIC CONTROL DEVICES****General**

Requirement: Electric strikes, electric locks, drop bolts and/or similar devices to suit door construction and hardware.

Electromagnetic hold-open devices: To AS 1905.1 (2015) and the NCC cited AS 1670.1 (2018).

Glass doors: Tumbler, drop bolts or magnetic holders.

Double leaf doors (solid frame): Electric strike or lock on the inactive leaf, connected to the door frame by concealed flexible wiring.

**Activation**

Activation device: Keypads, card readers or other activation devices located next to entry points.

External: Weatherproof (IP56) hoods or housings for external units.

Mounting height: 900 to 1100 mm from floor level and not less than 500 mm from internal corners. Refer also to Architectural Drawings.

**2.8 PANIC EXIT DEVICES****General**

Standard: To EN 1125 (2008).

Requirements:

- Field sizable.
- Keyed dogging.

## 2.9 KEYING

### Keying requirements

Standard: To AS 4145.1 (2008) for keying security.

Requirement: Provide door hardware and keys, as Documented.

### Temporary construction keys and cylinders

Requirement: Provide one of the following:

- Loan cylinder: Install for construction locks and replace at the Date for Practical Completion.
- Construction keyed master key cylinder: Keep up-to-date records of keys issued including Recipient's name, company and contact details, date issued and date returned.

### Delivery of keys

Great grandmaster, grandmaster and master keys: Arrange for delivery direct to the Principal.

Locks keyed to differ and locks keyed alike: Check the quantity against key records, and deliver keys to the Superintendent at the Date of Practical Completion.

### Group keying

Keying system: As Documented.

Existing system extension: Obtain the details of existing group or master key systems of the system to be extended.

Future extensions: Provide master and grandmaster group keying systems capable of accommodating future extensions.

Proprietary keying control security system: Provide for cylinder or pin-tumbler locks that accept a group key (e.g. master key, maison key).

Stamping: Stamp keys and lock cylinders to show the key codes and/or door number as scheduled.

### Identification

Labelling: Supply each key with a purpose-made plastic or stamped metal label legibly marked to identify the key, attached to the key by a metal ring.

### Key material

Lever locks: Malleable cast iron or mild steel.

Pin tumbler locks: Nickel alloy, not brass.

### Number of keys table

| Key code | Key type                          | Minimum number of keys        |
|----------|-----------------------------------|-------------------------------|
| GGMK     | Great grandmaster keys            | 2                             |
| GMK      | Grandmaster keys                  | 2                             |
| MK       | Master keys                       | 2 per code group              |
| KD       | Locks keyed to differ             | 2 per lock                    |
| KA       | Locks keyed alike:                |                               |
|          | - 2 locks in code group           | 4                             |
|          | - 3 to 10 locks in code group     | 6                             |
|          | - 11 to 40 locks in code group    | 10                            |
|          | - 41 and over locks in code group | 1 per 4 locks or part thereof |

## 3 EXECUTION

### 3.1 INSTALLATION

#### General

Handing: Before supply, verify on Site, the correct handing of hardware items.

Operation: Make sure working parts are accurately fitted to smooth close bearings, without binding or sticking, free from rattle or excessive play, lubricated if appropriate.

**Mounting height**

Locks and latches: Centreline of the door knob or lever spindle above finished floor: 920mm AFFL.

**Locks**

Cylinders: Fix vertically and with consistent key alignment.

**Door stops**

Fixing location: Fix on the floor, skirting or wall, as appropriate, to prevent the door or door furniture striking the wall or other surface.

**Fasteners**

Materials: Provide materials compatible with the item being fixed, and of sufficient strength, size and quality to perform their function, and as follows:

- Concealed fasteners: Provide a corrosion-resistant finish to concealed fasteners.
- Exposed fasteners: Match exposed fasteners to the material being fixed.

Security: Locate exposed fasteners to lock furniture on the inside faces of external doors and on the inside faces of internal doors to lockable rooms.

Support:

- Hardware fasteners: Provide appropriate back support, such as lock stiles, blocking, wall noggings and backing plates.
- Hollow metal sections: Provide backing plates drilled and tapped for screw fasteners, or provide rivet nuts with machine thread screws. Do not use self-tapping screws or pop rivets.

**Floor springs**

General: Form a recess in the floor slab for the floor spring box, securely fix and grout the box in place so that the cover plate is flush with the finished floor.

**Hinges**

Setout:

- For two hinge doors:
  - . Top hinge: Top of hinge 150 mm down from top of door.
  - . Bottom hinge: Bottom of hinge 250 mm up from bottom of door.
- For three hinge doors:
  - . Top hinge: Top of hinge 150 mm down from top of door.
  - . Bottom hinge: Bottom of hinge 250 mm up from bottom of door.
  - . Centre hinge: Centre between top and bottom hinges.

Metal frames: Fix hinges using metal thread screws. Do not weld hinges to frames.

Timber doorsets: Install butt hinges in housings equal in depth to the thickness of the hinge leaf (except for hinges designed for mounting without housing), and fix with countersunk screws.

**Seals**

General: Provide the fastenings, rebates, grooves, and clearances required for installation and operation of the seals. Allow seals unwound from coils to settle before use. Install proprietary seals to manufacturer's recommendations and adjust correctly.

**3.2 COMPLETION****Adjustment**

General: Leave the hardware properly adjusted with working parts in working order, and clean, undamaged, properly adjusted, and lubricated if appropriate.

Opening force performance: To AS 1428.1 (2021).

Automatic door operators: Maintain and adjust the system throughout the Defects Liability Period.

**Keys**

Contractor's keys: Immediately before the Date of Practical Completion, replace or reset cylinders to which the Contractor has had key access during construction to exclude the Contractor's keys.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published recommendations for use, care and maintenance of the hardware provided.

Automatic door operators: Include the installer's proposal for continuing maintenance after completion on an annual renewal basis.

**Warranties**

Type: As provided by the manufacturers and installers.

Period: As provided by the manufacturers and installers.

**4 SELECTIONS**

---

**Refer to Appendix C – Door Hardware Schedule and Architectural Drawings.**

|                            |
|----------------------------|
| <b>0456 LOUVRE WINDOWS</b> |
|----------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide louvre windows, as Documented. Refer to Architectural Drawings, Appendix B – Façade and Wall Schedule, and Façade Engineer's Documents.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.

### 1.3 STANDARDS

#### General

Selection and installation: To AS 2047 (2014).

Building classification: class 5

#### Glazing

Glass type and thickness: To AS 1288 (2021), if no glass type or thickness is nominated. Refer to Appendix B - Façade and Wall Schedule.

Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000).

### 1.4 INTERPRETATION

#### Abbreviations

General: For the purposes of this worksection, the following abbreviations apply:

- AGWA: Australian Glass and Window Association.
- WERS: Window Energy Rating Scheme.

#### Definitions

General: For the purposes of this worksection, the definitions given in AS/NZS 4668 (2000) and the following apply:

- 
- Louvres - horizontal: Louvres that span horizontally between frame stiles, mullions or vertical supports.
- 
- Total system SHGC: Solar heat gain coefficient as defined by the NCC and Tested in conformance with NFRC 200 (2023).
- Total system U-Value: Thermal transmittance as defined by the NCC and Tested in conformance with NFRC 100 (2023).

### 1.5 SUBMISSIONS

#### Certification

Performance Solutions: Refer to Façade Engineer's Documents for submissions required for Performance Solution Report.

Conformance: Submit evidence that the louvre windows conform to AS 2047 (2014).

#### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

#### Products and materials

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

Type Tests: Submit Test results of the following:

- Acoustic performance: To PRODUCTS, **GENERAL, Acoustic performance**.
- Protection of openable windows: To PRODUCTS, **GENERAL, Protection of openable windows**.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Frame member profiles.
- Hardware, fittings and accessories including fastener details.
- Junctions and trim to adjoining surfaces.
- Layout (sectional plan and elevation) of the window assembly.
- Methods of assembly.
- Methods of installation including fixing, joint sealing and flashing.
- Provision for vertical and horizontal expansion.
- Automatic control components.

**Subcontractors**

General: Submit names and contact details of proposed manufacturers and installers.

Evidence of experience: QBCC licence and CV.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Openings prepared to receive louvre windows.
- Fabricated louvre window assemblies at the factory ready for delivery to the Site.
- Fabricated louvre window assemblies delivered to the Site, before installation.
- Commencement of louvre window installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples of louvre window system components as follows:

- Manufacturer's standard hardware and accessories, including louvre holders and operators, locks, latches, handles, catches, anchor brackets and attachments, masonry anchors and weatherseals (pile or extruded).
- Colour samples of prefinished production materials, showing the limits of the range of variation in the Documented colour.
- Frame member profiles and louvre materials.
- Frame member joining techniques.

Glazing: Submit samples of glazing materials, each at least 200 x 200 mm, showing the visual properties and range of variation, if any, for each of the following:

- Tinted or coloured glass.
- Patterned or obscured glass or plastics glazing.

Labelling: Label each sample with the series code reference and date of manufacture.

**Storage and handling**

Storage: Store in a clean, dry area unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

**Acoustic performance**

Louvre windows: Rating to the NCC cited AS/NZS ISO 717.1 (2004), as Documented.

**Marking**

Louvre window assemblies: To AS 2047 (2014) Section 8.

**Protection of openable windows**

Fall prevention: To BCA (2022) D3D29.

Testing: To AS 5203 (2016).

**2.2 LOUVRE WINDOW ASSEMBLIES****General**

Requirement: Louvre blades mounted in a frame or subframe, able to withstand the ultimate limit state wind pressures for that location without failure or permanent distortion of members, and without louvre blade flutter.

**Power operated louvres**

Requirement: Louvres with automated operating mechanisms to suit louvre size, integrated with lighting control system. Refer to Electrical Engineer's Documents.

**2.3 GLAZING****Performance**

Glass: Free from Defects that detract from appearance or interfere with performance under normal conditions of use.

**Safety glass**

Standard: To AS 2208 (2023).

Type: Grade A.

Marking: To AS 1288 (2021) clause 5.23.

**2.4 GLAZING MATERIALS****General**

Requirement: Putty, glazing compounds, sealants, gaskets, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks and compression wedges appropriate for the conditions of application and required performance.

**Jointing materials**

Requirement: Provide jointing and pointing materials that are compatible with each other and the contact surfaces, and non-staining to finished surfaces to manufacturer's recommendations. Do not provide bituminous materials on absorbent surfaces.

**Elastomeric sealants**

Sealing compounds (polyurethane, polysulfide, acrylic): To ASTM C920 (2018) or ISO 11600 (2002).

Sealing compounds (silicone): To ASTM C920 (2018) or ISO 11600 (2002).

Sealing compounds (butyl): To ASTM C1311 (2022).

**Primer**

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials.

**2.5 ALUMINIUM FRAME FINISHES****Powder coatings**

Refer to Appendix A – T-Sheet and Appendix E – Paint Schedule.

**2.6 ANCILLARY COMPONENTS AND FITTINGS****Fasteners**

Size and type: To suit louvre window unit size and wind loading conditions.

**Extruded gaskets and seals**

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

**Flashings**

General: Corrosion-resistant, compatible with the other materials in the installation, and coated with a non-staining compound if necessary.

Standard: To AS/NZS 2904 (1995).

**3 EXECUTION****3.1 LOUVRE WINDOW ASSEMBLIES****General**

Requirement: Install louvre windows frames, as follows:

- Plumb, level, straight and true within building tolerances.
- Fixed or anchored to the building structure in conformance with the wind action loading requirements.
- Isolated from any building loads, including loads caused by structural deflection or shortening.
- Allow for thermal movement.

**Glazing**

Requirement: Install the glass as follows:

- Permanently fix in place each piece of glass to withstand the normal loadings and ambient conditions at its location without distortion or damage to glass and glazing materials.
- No transfer of building movements to the glass.
- Watertight and airtight for external glass.

Temporary marking: Use a method that does not damage the glass. Remove marking on completion.

Toughened glass: Do not cut, drill, edge-work or permanently mark after toughening. Use installation methods that prevent the glass making direct contact with metals or other non-resilient materials.

Heat absorbing glass: In locations exposed to direct sunlight, provide wheel cut edges free from damage or blemishes, and with minimum feather.

**Weatherproofing**

Flashing and weatherings: Install flashings, weather bars, drips, storm moulds, joint sealant and pointing to prevent water penetrating the building between the window frame and the building structure under the prevailing service conditions, including normal structural movement of the building.

**Fixing**

General: Do not penetrate metal flashings with fasteners.

Packing: Pack behind fixing points with durable full width packing.

Fasteners: Conceal fasteners.

Fastener spacing (nominal): 600 mm and maximum 150 mm from reveal ends.

**Joints**

General: Form tight fitted joints so that fasteners and fixing methods, such as pins, screws and adhesives do not create pressure indentations visible on exposed surfaces.

Sealants:

- If priming is recommended, prime surfaces in contact with jointing materials.
- If frames are powder coated, apply a neutral cure sealant.

**Operation**

General: Make sure moving parts operate freely and smoothly, without binding or sticking, at correct tensions or operating forces and are lubricated.

**Protection**

Removal: Remove temporary protection measures from the following:

- Contact mating surfaces before joining up.
- Exposed surfaces before completion of the Works.

**Trim**

General: Provide mouldings, architraves, reveal linings, and other internal trim using materials and finishes matching the window frames. Install to make neat and clean junctions between frames and the adjoining building surfaces.

### 3.2 COMPLETION

#### **Repair of finish**

Polyester or fluoropolymer coatings: Contact supplier for approval to apply touch up products, otherwise replace damaged material.

#### **Cleaning**

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive or alkaline materials.

Extent: All frames and glass surfaces internally and externally.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published instructions for operation, care and maintenance.

#### **Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offer by manufacturer and installer.

## 4 SELECTIONS

---

**Refer to Appendix B – Façade and Wall Schedule.**

|                     |
|---------------------|
| <b>0461 GLAZING</b> |
|---------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide glazing, as Documented. Refer to Architectural Drawings, and Appendix B – Façade and Wall Schedule. Refer also to Façade Engineer's Performance Based Design Brief for performance and certification requirements. Refer also to ESD consultant's Section J report for energy efficiency performance requirements. Refer also to Acoustic Engineer's Report for acoustic performance requirements.

#### Performance

Thermal qualities: U-Value and Solar heat gain coefficient (SHGC) as Documented.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.

### 1.3 STANDARDS

#### Glazing

Glass type and thickness: To AS 1288 (2021), if no glass type or thickness is nominated.

Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000).

Roof glazing: To AS 1288 (2021) Section 6.

### 1.4 INTERPRETATION

#### Abbreviations

General: For the purposes of this worksection, the following abbreviations apply:

- AGWA: Australian Glass and Window Association.

#### Definitions

General: For the purposes of this worksection, the definitions given in AS/NZS 4668 (2000) apply.

### 1.5 SUBMISSIONS

#### Certification

Design: Submit an engineer's certificate confirming conformance to AS 1288 (2021).

Opacified glass: Submit a report, from the manufacturer certifying that the proposed method of opacifying the glass will not be detrimental to the glass or affect the glass product warranty.

Toughened glass: Submit certification from the manufacturer for all toughened glass as evidence of heat soaking to **GLAZING, Heat soaking**.

Installation: Submit certification from the fabricator that the method of glazing, the selection of sealant systems and conditions next to the glass conform to the following:

- Compatible with the edge seal of insulating glass units (IGUs) and self-cleaning glass.
- Will not be detrimental to the long-term structural performance, weathering capabilities and visual qualities of the glass.

Glazier's data: Submit the glazing subcontractor's statement certifying the following:

- A satisfactory thermal safety assessment.
- The assembled frame provides the required glazing clearances and tolerances, and maximum and minimum joint configurations, based on the bow, warp and kink characteristics of the required glass types, and is ready for glazing.

#### Execution details

Site glazing: If Site glazing is intended, submit proposals.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

**Products and materials**

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

**Samples**

Requirement: Submit samples to **PRODUCTS**, **GENERAL**, **Samples**.

**Shop drawings**

Requirement: Submit shop drawings showing the following:

- Method of glazing.
- Rebate depth.
- Edge restraint.
- Clearances and tolerances.
- Glazing gaskets and sealant beads.

**Warranties**

Requirement: Submit warranties to **COMPLETION**, **Warranties**.

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Glazing products before they are installed.

---

**2 PRODUCTS**

---

**2.1 GENERAL****Samples**

Requirement: Provide samples of glazing materials, each at least 200 x 200 mm, showing the visual properties and range of variation, if any, for each of the following:

- Tinted or coloured glass or plastics glazing.
- Surface modified or surface coated glass.
- Patterned or obscured glass or plastics glazing.
- Ceramic-coated glass.
- Wired glass.
- Insulating glass units.
- Mirror glass.

**Storage and handling**

Storage: Store glass and glazing materials in a clean, dry area unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

Handling: Handle glass to the manufacturer's recommendations.

**2.2 GLAZING****Performance**

Glazing: Free from Defects that detract from appearance or interfere with performance under normal conditions of use.

Plastics glazing: Free from surface abrasions and resistant to yellowing or other colour change. Capable of maintaining physical properties including strength and impact resistance for its design life.

**Heat soaking**

Requirement: Heat soak glass to AS 1288 (2021) clause 3.8.

Standard: To EN 14179-1 (2016).

Marking: To EN 14179-1 (2016) or certified by the manufacturer to AS 1288 (2021) clause 3.8.2.

**Safety glazing materials**

Standard: To AS 2208 (2023).

Type: Grade A.

Marking: To AS 1288 (2021) clause 5.23.

**Heat-strengthened glass**

Standard: To ASTM C1048 (2025).

**Insulating glass units (IGUs)**

Requirement: Provide insulating glass units, as Documented.

Manufacture, Testing and installation: To AS 4666 (2012).

**2.3 GLAZING MATERIALS****General**

Requirement: Putty, glazing compounds, sealants, gaskets, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks, shims and compression wedges appropriate for the conditions of application and required performance.

**Primer**

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials.

**2.4 ANCILLARY COMPONENTS AND FITTINGS****Extruded gaskets and seals**

General: Provide seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

**Pile weatherstrips**

Standard: To AAMA 701/702 (2023).

Material: Polypropylene or equivalent pile and backing, low friction silicone treated, ultraviolet stabilised.

Finned type: A pile weatherseal with a central polypropylene fin bonded into the centre of the backing rod and raised above the pile level.

**3 EXECUTION**

---

**3.1 GLAZING PROCESSING****General**

Processing: Perform required processes on glazing, including cutting, obscuring, silvering and bending. Form necessary holes, including for fastenings, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arised.

**3.2 INSTALLATION****Glazing**

Requirement: Install the glazing as follows:

- Permanently fix in place each piece of glass to withstand the normal loadings and ambient conditions at its location without distortion or damage to glazing materials.
- No transfer of building movements to the glazing.
- Watertight and airtight for external glazing.

Temporary marking: Use a method that does not damage the glazing. Remove marking only after certification and acceptance of the installation.

Toughened glass: Do not cut, work, or permanently mark after toughening. Use installation methods that prevent the glass making direct contact with metals or other non-resilient materials.

Heat-absorbing glass: In locations exposed to direct sunlight, provide wheel cut edges free from damage or blemishes, with minimum feather.

### **Preglazing**

Window assemblies and glazed doors: Supply inclusive of glazing, shop preglazed.

Curtain walls: Supply inclusive of glazing, shop preglazed.

### **Site glazing**

Minimum dimensional requirements: Edge clearance, edge cover, front clearance and back clearance to AS 1288 (2021).

Vision panels: Install so that they are to be removable from: interior.

External timber framed glazing: Glaze with putty.

## **3.3 COMPLETION**

### **Replacement**

Requirement: After replacing damaged glass, leave the Work clean, polished, free from Defects, and in good condition.

### **Cleaning**

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive, acidic or alkaline materials.

Extent: All frames and glass surfaces internally and externally.

### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturers' published recommendations for in-service use.

### **Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offered by manufacturer and installer.

Glazing subcontractor's warranty: Provide an undertaking conditional only on compliance with the manufacturers' recommendations for maintenance, to repair or replace glass and glazing materials that become Defective or prove unsuitable for the nominated application; during the warranty period.

Glass manufacturer's warranty: Provide an undertaking, conditional only on compliance with the manufacturer's recommendation for installation and maintenance, to supply replacement glass units to the Site for replacement of Defective units defined as follows:

- IGU: Units in which the hermetic seal has failed as evidenced by intrusion of foreign matter, or internal condensation at temperature above 2°C.
- Coated glass units (including coated super insulating glass units): Units in which the metallic coating shows evidence of manufacturing Defects, including but not necessarily limited to cracking or peeling, as determined in conformance with ASTM C1048 (2025).

Toughened glass warranty: Provide a manufacturer's warranty that toughened glass supplied for use in curtain walls has been subjected to a heat soaking process that has converted at least 95% of the nickel sulfide content to the stable beta-phase.

## **4 SELECTIONS**

**Refer to Appendix B – Façade and Wall Schedule.**

**Refer to Façade Engineer's Documents.**

|                              |
|------------------------------|
| <b>0467 GLASS COMPONENTS</b> |
|------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide glass components, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****General**

Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 4668 (2000) apply.

**1.5 SUBMISSIONS****Certification**

Barrier design: Submit a professional engineer's certificate confirming conformance with AS/NZS 1170.1 (2002) clause 3.6, AS/NZS 1170.2 (2021) and AS 1288 (2021) Section 7.

Shower screens: Submit evidence of conformance to AS 1288 (2021) clause 5.8.

Sealant compatibility: Submit statements from all parties to the installation certifying the compatibility of sealants and glazing systems to all substrates.

Installation: Submit certification from the fabricator, that the method of glazing, the selection of sealant systems and conditions next to the glass comply with the following:

- Not detrimental to the long-term structural performance, weathering capabilities and visual qualities of the glass.
- Not cause delamination or other impairment to laminated glass during the service life.

Toughened glass: Submit certification from the manufacturer for all toughened glass as evidence of heat soaking.

**Products and materials**

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC Licence and CV.

**Warranties**

Requirement: Submit warranties to **COMPLETION**, **Warranties**.

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Glass components before they are installed.

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide samples, each at least 200 x 200 mm, showing specified visual properties and the range of variation, if any, for each of the following:

- Mirror glass.

#### Storage and handling

Storage: Store in a clean, dry area, unaffected by weather, to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding splatter.

Handling: Handle glass to the manufacturer's recommendations and as follows:

- Stack upright, off the ground and against a flat, vertical surface.
- Carry in the vertical position.

### 2.2 MIRRORS

#### Reflective surface

Type: Silver layer deposited on the glass or glazing plastic.

Protective coatings: Copper free coating, at least 5 µm thick, and 2 coats of mirror backing and edge sealing paint having a total dry film thickness of at least 50 µm.

Venetian silvered mirror (one-way vision glass): 15 mm wide silvered strips alternating with 3 mm wide clear strips.

#### Safety glass mirrors

Standard: To AS 2208 (2023).

Type: Grade A safety glass.

#### Solid backed annealed glass mirrors

Type: Annealed glass mirror with backing.

Backing: 9 mm waterproof plywood.

## 3 EXECUTION

### 3.1 GLASS PROCESSING

#### General

Processing: Perform required processes on glass, including cutting, obscuring, silvering and bending. Form necessary holes, including for fastenings, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arrised.

### 3.2 FIXING MIRRORS

#### General

Adhesive fixing: Clean surfaces to be bonded. Apply non-acidic silicone adhesive to the manufacturer's recommendations. Secure the rear of the mirror to the substrate with double-sided adhesive tape until the adhesive cures.

Screw fixing: Fix direct to wall plugs with dome-headed chromium-plated screws in each corner and at 900 mm maximum centres around perimeter. Provide polyethylene sleeves and washers to prevent contact between screw and glass. Do not over-tension the screws.

Frame fixing: Fix with proprietary aluminium frames to mirror perimeter, corners mitred. If unbacked, bed glass edges in a continuous resilient gasket. Attach the frame to the substrate with concealed screw fixings. Seal the frame to the substrate with paintable sealant that will not react with the mirror coating. Do not allow the sealant to contact the mirror back.

Bead fixing: Fix with rebated timber beads to mirror perimeter, corners mitred. If unbacked, bed glass edges in a continuous resilient gasket. Screw fix the beads to the substrate.

Clip fixing: Fix direct to wall plugs with chromium-plated fixed clip and spring clip fasteners at 900 mm maximum centres around perimeter. If unbacked, provide polyethylene or cork washers to prevent contact between clips and mirror back.

### **3.3 COMPLETION**

#### **Cleaning**

Method: Clean with soft clean cloths and clean water, finishing with a clean squeegee. Do not use abrasive, acidic or alkaline materials.

Extent: All frames and glass surfaces internally and externally.

#### **Warranties**

Type: Interlocking manufacturer and installer's warranty.

Period: As offered by the manufacturer and installer.

### **4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet.**

|                    |
|--------------------|
| <b>0511 LINING</b> |
|--------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide internal lining systems, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Appendix B – Façade and Wall Schedule.

**Performance**

Requirement: Provide lining system with a surface that is:

- Resistant to impacts expected in use.
- Resistant to moisture encountered under expected environmental conditions.
- Free of irregularities.
- A suitable substrate for the nominated final finish.
- Set out to suit the timber battens as Documented.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviation applies:

- FC: Fibre Cement.
- MDF: Medium density fibreboard.

**Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 4491 (1997) and the following apply:

- Decorative overlaid wood panels: Particleboard or fibreboard with a bonded decorative finishing surface such as thermosetting resin (low pressure melamine), PVC film, paper foils or wood veneer.
- Dry process fibreboard: Panel material with a nominal thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and pressure, the bond of which is derived from a synthetic adhesive added to the fibres and the panels are manufactured with a forming moisture content less than 20%.
- Fibre cement sheet linings: Treated cellulose fibre in a matrix of cement and sand autoclaved sheet, sealed on one side.
- High pressure decorative laminates (HPDL):
  - . Panels consisting of core layers impregnated with phenolic and/or aminoplastic resins and a surface layer(s) impregnated with aminoplastic resins (mainly melamine resins).
  - . Sheets consisting of a decorative face and layers of fibrous sheet material (e.g. Paper) impregnated with thermosetting resins and bonded together under heat and pressure of at least 5 MPa.
- Wet process fibreboard: Panel material with a nominated thickness of 1.5 mm or greater, manufactured from lignocellulosic fibres (derived from wood or other materials) with application of heat and/or pressure, the bond of which is derived from the felting of the fibres and the panels are manufactured with a forming moisture content greater than 20%.

**1.4 TOLERANCES****Permitted deviations**

Bearing surface of finished framing:

- Gypsum lining: To AS/NZS 2589 (2017) clause 4.2.2.

- Other lining: 4 mm from a 1.8 m straightedge.

## 1.5 SUBMISSIONS

### Fire performance

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties**.

### Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

### Shop drawings

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Decorative panels: Panel set-out, large scale panel fixing details, attachment devices and other components.

### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 1.6 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrate or framing before installation of linings.
- Finished surface of installation before applying:
  - . Sealer.
  - . Finish coatings or decorative papers.

---

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Prefinished panels: For each finish, provide a 300 x 600 mm sample panel with associated trim.

#### Storage and handling

Requirement: Store lining stacked in pallets horizontally on a smooth, level surface. Prevent distortion or moisture ingress.

Timber or fibreboard panels: Store off the ground in a well-ventilated area.

Handling: Do not drag sheets across each other or across other materials. Protect edges, corners and surface from damage.

#### Certification

Timber based products: Label panels under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*, as applicable to the product. Locate the label on faces or edges that will be concealed in the Works.

### 2.2 FIRE PERFORMANCE

#### Fire hazard properties

Group number: To AS 5637.1 (2015).

### 2.3 PLASTERBOARD

#### General

Standard: To AS/NZS 2588 (2018).

### 2.4 FIBRE CEMENT

#### General

Standard: To AS/NZS 2908.2 (2000).

Wall and ceiling linings: Type B Category 2.

Minimum thickness: 8.5 mm.

## 2.5 TONGUE AND GROOVE BOARDS

### Hardwood

Standard: To AS 2796.1 (1999).

## 2.6 DECORATIVE OVERLAID WOOD PANELS

### General

Standard: To AS/NZS 1859.3 (2017).

## 2.7 HIGH PRESSURE DECORATIVE LAMINATE (HPDL)

### General

Standard: To AS/NZS 2924.1 (2024).

## 2.8 ADHESIVES, SEALANTS AND FASTENERS

### Adhesives

Wallboards: Gunnable synthetic rubber/resin based mastic contact adhesive formulated for bonding flooring and wallboards to a variety of substrates.

### Sealants

Fire-resisting sealant: Non-hardening sealant, compatible with the materials to be sealed and having a fire-resistance rating equal to that of the building element it seals.

Acoustic sealant: Non-hardening sealant compatible with the materials to be sealed.

### Fasteners

Steel nails: Hot dip galvanized.

## 3 EXECUTION

---

### 3.1 CONSTRUCTION GENERALLY

#### Conditions

Requirement: Do not start lining Work until the building or installation area is enclosed and weathertight, and all wet trades have been completed.

#### Preparation

Substrate: Before fixing linings, make sure substrates are plumb, level, in true alignment and conform to the lining manufacturer's recommendations.

Timber, steel framing and battened masonry: To AS/NZS 2589 (2017) clause 4.2.

#### Pre-conditioning

General: Acclimatise timber panels in the in-service conditions for 2 to 3 weeks before installing.

#### Battens

General: Fix at each crossing with structural framing members, to solid walls or ceiling support. Provide wall plugs in solid substrates.

#### Ceiling linings

General: Do not install until the timber roof structure has been fully loaded for at least 14 days.

#### Accessories and trim

General: Provide accessories and trim necessary to complete the installation.

#### Adhesives

General: Provide adhesive types appropriate for the purpose and apply them so they transmit the loads imposed without causing discolouration of the finished surfaces.

#### Fire-resisting and acoustic rated installations

Sealing: Apply sealant to the manufacturer's recommendations and as follows:

- Around services pipes and penetrations.
- Electrical outlets and recessed lights: Line back and sides of fixture with plasterboard and seal around fixture junction with sealant.
- Around perimeter of lining panels: Provide continuous runs of sealant.

### 3.2 PLASTERBOARD

#### Installation

Gypsum plasterboard and fibre-reinforced gypsum lining: To AS/NZS 2589 (2017).

Level of finish and jointing: To AS/NZS 2589 (2017) clause 3.1.

#### Supports

General: Install timber battens or proprietary cold-formed galvanized steel furring channels as follows:

- If framing member spacing exceed the recommended spacing.
- If direct fixing of plasterboard is not possible, due to the arrangement or alignment of the framing or substrate.
- If the lining is the substrate for tiled finishes.
- If required for penetrations for services, including mechanical grilles and lighting fixtures.
- If required to support fixtures.

#### Multiple sheet layers

Application: Fire-resisting and acoustic rated walls.

Joints: Fill and flush up all joints and fasteners in each layer and caulk up perimeters and penetrations before installing following layers. Stagger all sheet joints by minimum 200 mm.

#### Joints

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

Butt joints: Make joints over framing members or provide back blocking.

External corner joints: Make joints over metallic-coated steel corner beads.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Control joints: Provide purpose-made metallic-coated control joint beads at not more than 12 m centres in walls and ceilings and to coincide with structural control joints.

Wet areas: Install additional supports, flashings, trim and sealants as required.

Joints in tiled areas: Do not apply a topping coat after bedding perforated paper tape in bedding compound.

### 3.3 FIBRE CEMENT

#### Installation

Joints and layout: Run sheets perpendicular to the framing members. In flush jointed applications, stagger end joints in a brick pattern and locate them on framing members, away from the corners of large openings. Provide supports at edges and joints.

#### Supports

General: Install timber battens or proprietary cold-formed galvanized steel furring channels as follows:

- If framing member spacing exceed the recommended spacing.
- If direct fixing of fibre cement is not possible, due to the arrangement or alignment of the framing or substrate.
- If the lining is the substrate for tiled finishes.
- If required for penetrations for services, including mechanical grilles and lighting fixtures.
- If required to support fixtures.

#### Fixing

Timber framed construction: Nail only or combine with adhesive to manufacturer's recommendations.

Steel framed construction: Screw only or combine with adhesive to manufacturer's recommendations.

Wall framing: Conform to the following:

- Do not fix to top and bottom plates or noggings.
- In tiled areas: Provide an extra row of noggings immediately above wall-to-floor flashings. Fix sheet at 150 mm centres to each stud and around the perimeter of the sheet.

Masonry wall construction: Conform to the following:

- Direct fixing: Adhesive fix to the masonry except where lining forms a substrate for tiled finish.
- Furring channels: Fix using screw and/or adhesive.

Ceilings: Fix using screw and/or adhesive to ceiling furring members. Do not fix sheets directly to the bottom chords of trusses.

- Ceiling battens: Fix at 600 mm maximum centres.

Wet areas: Do not fix using adhesive alone.

### **Multiple sheet layers**

Application: Fire-resisting and acoustic rated walls.

Joints: Fill and flush up all joints and fasteners in each layer and caulk up perimeters and penetrations before installing following layers. Stagger all sheet joints by minimum 200 mm.

### **Joints**

Joint width:

- Butt joints: 1 to 2 mm.
- Exposed joints: 10 mm maximum.

Joint backing for exposed joints: Black self-adhesive polyurethane tape.

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

External corner joints: Make joints over metallic-coated steel corner beads.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Control joints: Provide control joints to coincide with structural control joints and as follows:

- Walls:
  - . Timber framing:  $\leq 7.2$  m centres.
  - . Steel framing 0.55 to 0.75 mm BMT:  $\leq 9$  m centres.
  - . Steel framing 0.8 to 1.6 mm BMT:  $\leq 6$  m centres.
- Ceilings: To divide into bays not larger than 10.8 x 7.2 m.
- Soffit linings: To divide into bays not larger than 4.2 x 4.2 m or 5.6 x 3.6 m.
- Control joint beads: Purpose-made metallic-coated.
- Support: Provide framing parallel to the joint on each side. Do not fix the lining to abutting building surfaces.

Wet areas: Provide additional supports, flashings, trim and sealants to manufacturer's recommendations.

Joints in tiled areas:

- Perforated paper tape: Bed in bedding compound. Do not apply a topping coat.
- Control joints:
  - . Timber framing: Not more than 4.2 m centres and space to suit joints required in tiling.
  - . Steel framing: Not more than 4.8 m centres and space to suit joints required in tiling.
- Internal corners: Reinforce with preformed angles to manufacturer's recommendations.

## **3.4 TONGUE AND GROOVE BOARDS**

### **Installation**

General: Conform to the following:

- Horizontal installation: Provide single lengths of boards if possible.
- Vertical installation: Provide single lengths only.

Stained or clear finished boards: Select board to give a random pattern. At corners, return the same board to give a continuous grain pattern.

Fixing: Nail twice to each crossing, except for secret nailed profiles.

Secret nail fixing: Fix nail diagonally through the tongue only. Punch nails to maintain correct alignment of the next board.

Nailheads: Treat visible nailheads as follows:

- In stained or clear finishes: Drive flush.
- In opaque finishes: Punch below surface and fill flush with putty after the surface has been primed.

Corners and junctions: Allow for movement at all corners and junctions.

**Joints**

Requirement: Select board lengths to give minimum number of joints.

End grain joints: Install boards so that butt joints are in compression.

Internal corners: Scribe.

External corners: Mitre.

**3.5 TIMBER PANEL LINING****General**

Installation: Set out in even panels with joints coinciding with framing members. Fit and fix panels and trim plumb, level and in true alignment of face and grain.

Fixing:

- Plywood and hardboard: Wallboard adhesive or pin fixed to timber frame, screw fixed to steel frame. Punch pin heads just below surface.
- Laminated plastic: Wallboard adhesive.

**Plywood**

Expansion joints: Provide expansion joints as follows:

- Perimeter of linings and each panel joint: 2 to 3 mm gap.
- Interval: One of the following:
  - . Maximum 3.6 m centres: 6 to 9 mm gap.
  - . Maximum 4.8 m centres: 8 to 12 mm gap.

Areas with an expected high level of internal moisture: Provide a gap of 4 to 6 mm every 1.2 m.

**3.6 TRIM AND ACCESSORIES****General**

Requirement: Provide trim such as beads, mouldings and stops to make neat junctions between lining components, finishes and adjacent surfaces.

Proprietary items: Provide complete with installation accessories.

Timber : Fix using full length so that trim is secure and without movement. Where nail or screw fasteners are used, make sure fastener finishes sufficiently below face of trim so that stopping piece finishes flush with the face.

**3.7 COMPLETION****General**

Damaged or marked lining and components: Replace.

Exposed surfaces: Touch up shop applied finishes and restore damaged or marked areas.

Timber panels: If appearance is not uniform, replace panels.

Cleaning: Clean completed surfaces to remove irregularities and leave panels smooth and clean, to the manufacturer's recommendations. If required, sand with fine paper to remove irregularities and refinish panel surface.

- Debris and unused material: Remove from Site.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offered by the manufacturer and installer.

---

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet.**

**Refer to Appendix B – Façade and Wall Schedule.**

|                                           |
|-------------------------------------------|
| <b>0522 PARTITIONS - FRAMED AND LINED</b> |
|-------------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide partition systems, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Appendix B – Façade and Wall Schedule, Acoustic Engineer's Performance Specification, and Structural Engineer's Documents.

**Performance**

Strength and stability: To remain stable, and without rattle and signs of deflection or permanent deformation under normal conditions of use, including the slamming of doors.

Serviceability: To support imposed dead loads, seismic loads, wind loads, including designated eccentric loads and not to deflect in excess of the following, with H being the height of the partition:

- The lesser of H/240 or 30 mm for partitions subjected to wind loads and lined with flexible material.
- The lesser of H/360 or 20 mm for partitions subjected to wind loads and lined with brittle materials.
- H/500 for eccentric loads.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 TOLERANCES****Framed and lined partitions**

Finished framing: To AS/NZS 2589 (2017) clause 4.2.2.

**1.4 SUBMISSIONS****Fire performance**

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties**.

Fire-resistance level: Submit evidence of conformity to **FIRE PERFORMANCE, Fire-resistance of building elements**.

**Products and materials**

Manufacturer's data: Submit manufacturer's standard product literature for each system type.

Type Tests: Submit Test results to PRODUCTS, **GENERAL, Tests** for the following:

- Impact resistance.
- Pressure resistance.
- Weighted sound reduction index ( $R_w$ ).

**Records**

Timber framing moisture content: Submit records of moisture content values.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

**1.5 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Set-out before installation.
- Installation of plenum baffles.
- Partition framing before installation of linings and finishes.

- Framed and lined partitions ready to receive framed and glazed components.
- Completion of installation.

## **2 PRODUCTS**

---

### **2.1 GENERAL**

#### **Samples**

Partition components: Provide samples of the following:

- Skirting, ducted skirting and ducted skirting stop ends, returns and removable covers, at least 100 mm long.
- Doors and door hardware.
- Prefinished panels: For each finish, provide a 300 x 600 mm sample panel with associated trim.

Sealant: One sample of each type and colour.

#### **Storage and handling**

Requirement: Store lining stacked in pallets horizontally on a smooth, level surface. Prevent distortion or moisture ingress.

Plasterboard and fibre cement panels: Store off the ground in a well-ventilated area.

Handling: Do not drag sheets across each other or across other materials. Protect edges, corners and surface from damage.

#### **Tests**

Impact resistance: To the NCC cited ASTM E695 (2003).

Pressure resistance: To the NCC cited ASTM E72 (2015).

Weighted sound reduction index ( $R_w$ ): To the NCC cited AS/NZS ISO 717.1 (2004).

### **2.2 FIRE PERFORMANCE**

#### **Fire hazard properties**

Group number: To AS 5637.1 (2015).

#### **Fire-resistance of building elements**

Fire-resistance level: Tested to AS 1530.4 (2014).

### **2.3 FRAMING**

#### **Light steel framing**

Requirement: Proprietary framing system of metallic-coated folded steel strip lipped studs and channel section top and bottom tracks and noggings.

Sections and members: To AS/NZS 4600 (2018).

#### **Light timber framing**

Timber species: Radiata pine.

Seasoning: Required.

Stress grade: F5 to AS/NZS 1748.1 (2011).

### **2.4 LININGS**

#### **Plasterboard**

Standard: To AS/NZS 2588 (2018).

#### **Fibre cement**

Standard: To AS/NZS 2908.2 (2000).

Wall and ceiling linings: Type B category 2.

Minimum thickness: 4.5 mm.

#### **Accessories**

General: Accessories necessary to complete the installation including the following:

- Corner beads.
- Stop beads.
- Trim.

- Control joints.
- Sheet metal and MDF partition end caps.

**Adhesives**

General: Adhesives of types appropriate to their purpose and substrates, applied to transmit the loads imposed without causing discolouration of finished surfaces.

**Sealants**

General: Sealant types appropriate for the partition's Documented acoustic rating and fire-resistance level, and compatible with partition materials and building substrate.

**2.5 AUTOCLAVED AERATED CONCRETE (AAC) PANELS****General**

Requirement: Lightweight concrete partition panels manufactured from a proprietary mixture of sand, lime and cement with a gas-forming additive, and with internal welded steel reinforcing mesh, cured in an autoclave.

Standard: To AS 5146.1 (2015) and AS 5146.2 (2018).

Properties: As Documented.

**Accessories**

Requirement: Accessories to the manufacturer's recommendations for the AAC panel system including the following:

- Base angle.
- Deflection head track.
- Steel top hat.
- Mortar.
- Fire-resisting and acoustic rated sealant.
- Panel joint adhesive.
- AAC thin bed adhesive for panel joints.

---

**3 EXECUTION**

---

**3.1 GENERAL****Preparation**

Substrate: Prepare the substrate to receive the partitions.

Carpet: Fix bottom tracks over polyethylene film. Prevent carpet threads from pulling if drilling or installing fasteners.

**Protection**

General: Protect existing Work from damage during the installation and rectify any damage. Provide temporary coverings if required.

**Pre-conditioning**

General: Acclimatise wood-based system components in the in-service conditions for a minimum period of two weeks before assembly.

**Set-out**

General: Partition set out dimensioned on drawings. Studs and noggins set out to suit cover batten set out. Refer to drawings for acoustic partitions required above ceiling level.

**3.2 INSTALLATION****Partition erection**

General: Install partitions plumb, level, on their correct alignment and firmly fixed.

Building movements:

- Provide clearances or deflection heads so that partitions are not damaged by structural building movements including long-term slab deflection.
- If fire-resistance levels or acoustic ratings are required, provide a resilient foam or mastic seal with properties equal to those required for the partition.

Suspended slabs: Provide deflection heads.

**Structural floor control joints**

General: Do not run or fix partitions framing across control joints.

**Acoustic rated partitions**

General: Isolate the frames from floors, ceilings and vertical abutments with beads of non-hardening sealant compatible with the materials to be sealed.

**Trim**

General: As Documented.

**Sealing fire-resisting and acoustic rated partitions**

General: Apply sealant to the manufacturer's recommendations and as follows:

- Around services pipes and penetrations.
- Electrical outlets and recessed lights: Line back and sides of fixture with plasterboard and seal around fixture junction with sealant.
- Around perimeter of lining panels: Provide continuous runs of sealant.

**3.3 LIGHT STEEL FRAMES****Tracks**

General: Conform to the following:

- Fix bottom tracks to floor substrate.
- Fix top wall tracks to suspended ceiling grid.
- Fix deflection head tracks to the structural soffit above.

Fixing to masonry: Provide masonry anchors of expansion or chemical grout type. Do not use explosive-driven fasteners.

Fixing to metal deck roofs: Provide for vertical uplift movement, as Documented.

Fixing to suspended ceilings: Provide intermediate support and bracing at maximum 1500 mm centres and at all load concentrations, doorways and jamb studs.

Seismic movement: If required, do not butt wall tracks or deflection heads against each other. Provide 10 mm clearance between tracks.

Track fixing: Fix top and bottom tracks at 600 mm maximum centres generally, and 100 mm from ends. Splice plates at ends to maintain continuity and alignment.

**Stud framing**

General: Provide studs in single lengths without splices. Rotate intermediate studs into tracks for friction fixing. Screw fix jamb studs, corner studs and wall intersection studs to tracks.

Fixing: Fix noggings at 1350 mm maximum centres vertically and as required for cover battens, skirtings and wet area lining. Make sure that faces of noggings and studs are accurately aligned.

Lintels: Install a stiffened top plate lintel for spans of 1800 mm or greater.

**Curved partitions**

Tracks: Install proprietary flexible curved floor and head tracks.

Stud spacing: Conform to the sheeting manufacturer's recommendations for curved partitions.

**Jambs**

General: Install boxed double studs at jambs and heads to all openings.

**Additional frame support**

General: Provide frame support for fixing the following:

- Floor and wall mounted fixed joinery units, furniture and equipment.
- All wet area fittings and fixtures.
- All grabrails and handrails.

Timber nogging: Provide 240 x 40 mm timber nogging with proprietary stud fastenings for wall-hung sanitary fittings.

Stud stiffening: Provide stud stiffening to support wall-hung joinery units and equipment with:

- Full height close-fitting timber inserts.
- Boxed steel lipped studs.

**Stud service holes**

General: Use factory pre-cut flared holes, or provide Site cut holes punched or drilled on the centreline of the member and fit proprietary plastic bushes or grommets. Splice additional stiffening to studs if Site cut service holes exceed 1/3 the depth of the member.

**Metal separation**

General: Isolate non-ferrous service pipes and accessories from the metal framing.

**Earthing**

Permanent earthing: Required.

Temporary earthing: Provide temporary earthing during erection until the permanent earthing is installed.

**Cavity walls**

General: If bridging is nominated, provide to the manufacturer's recommendations.

**Staggered stud framed walls**

General: Provide studs staggered at 300 mm centres set in oversized top and bottom plates so that each face has stud fastenings at 600 mm centres.

**3.4 LIGHT TIMBER FRAMES****Moisture content**

General: Install framing that meets the following values tested to AS/NZS 1080.1 (2012):

- Air conditioned buildings: 8 to 10%.
- Intermittently heated buildings: 10 to 12.5%.
- Unheated buildings: 12 to 15%.

**Framing**

General: Construct wall frames to the NCC cited AS 1684.4 (2010) Section 6, as appropriate for internal walls.

Double faced walls: Provide gauged timbers in studs, noggings and plates.

**3.5 PLASTERBOARD****Installation**

Gypsum plasterboard and fibre-reinforced gypsum lining: To AS/NZS 2589 (2017).

**Multiple sheet layers**

Application: Fire-resisting and acoustic rated partitions.

Joints:

- Fill and flush up all joints and fasteners in each layer and caulk up perimeters and penetrations before installing following layers.
- Stagger all sheet joints: Minimum 200 mm.

**Joints and joint treatment**

General: Install joint accessories as Documented, in conformance with manufacturer's recommendations. Install plumb, level and true to line.

Flush joints: Use joint reinforcing tape bedded in joint compound with recessed edge sheets and finish flush.

Butt joints: Make joints over framing members or provide back blocking.

External corner joints: Bed purpose fabricated perforated metallic-coated steel corner beads in joint compound.

Ceiling junctions: Install purpose fabricated perforated metallic-coated steel trim to top of partition.

Sheet metal partition end caps: Provide purpose fabricated perforated metallic-coated steel end caps, sized for partition thickness and bedded in joint compound.

MDF end caps: Provide recessed edge sheets and finish flush using joint reinforcing tape and joint compound.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Control joints: Provide purpose-made perforated metallic-coated control joint beads at not more than 12 m centres in partitions and to coincide with structural control joints. Bed in joint compound.

Wet areas: Provide additional supports, flashings, trim and sealants to manufacturer's recommendations.

Joints in tiled areas: Bed reinforcing tape in joint compound. Do not apply a topping coat.

### 3.6 FIBRE CEMENT

#### Installation

General: Install as follows:

- Run sheets across the framing members.
- In flush jointed applications, stagger end joints in a brick pattern and locate joints on framing members, away from the corners of large openings.
- Provide supports at edges and joints.
- Do not fix to top and bottom plates or noggings.

Timber framing: Nail only or combined with adhesive.

Steel framing: Screw only or combined with adhesive.

Tiled and wet areas: Provide an extra row of noggings immediately above wall-to-floor flashings. Fix sheet at 150 mm centres to each stud and around the perimeter of the sheet. Do not fix using adhesive alone.

#### Multiple sheet layers

Application: Fire-resisting and acoustic rated partitions.

Joints:

- Fill and flush up all joints and fasteners in each layer and caulk up perimeters and penetrations before installing following layers.
- Stagger all sheet joints: Minimum 200 mm.

#### Joints and joint treatment

General: Install joint accessories as Documented, in conformance with manufacturer's recommendations. Install plumb, level and true to line.

Flush joints: Use joint reinforcing tape bedded in joint compound with recessed edge sheets and finish flush.

External corner joints: Bed purpose fabricated perforated metallic corner beads in joint compound.

Ceiling junctions: Install purpose fabricated perforated metallic-coated steel trim to top of partition.

Sheet metal partition end caps: Provide purpose fabricated perforated metallic-coated steel end caps, sized for partition thickness and bedded in joint compound.

MDF end caps: Provide recessed edge sheets and finish flush using joint reinforcing tape and joint compound.

Dry joints: Use square edged sheet and finish with a PVC-U joining section.

Control joints: Provide control joints to coincide with structural control joints and as follows:

- Walls:  $\leq 7.2$  m centres.
- Control joint beads: Purpose-made metallic-coated.
- Support: Provide framing parallel to the joint on each side. Do not fix the lining to abutting building surfaces.

Wet areas: Provide additional supports, flashings, trim and sealants to manufacturer's recommendations.

Joints in tiled areas: Bed reinforcing tape in joint compound. Do not apply a topping coat.

- Control joints: At maximum 4.2 m centres and spaced to suit joints required in tiling.
- Internal corners: Reinforce with preformed angles to manufacturer's recommendations.

### 3.7 COMPLETION

#### Rectification

General: Correct any Defects, remove any excess joint compound and leave the partition installation complete, clean and ready for the application of finishes.

**4 SELECTIONS**

---

**Refer to Appendix B – Façade and Wall Schedule.**

|                                 |
|---------------------------------|
| <b>0524 PARTITIONS - GLAZED</b> |
|---------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide glazed partition systems, as Documented. Refer to Architectural Drawings, Appendix B – Façade and Wall Schedule, and Acoustic Engineer's Performance Specification.

**Performance**

Strength and stability: To remain stable, and without rattle and signs of deflection or permanent deformation under normal conditions of use, including the slamming of doors.

Serviceability: To support imposed dead loads, seismic loads, wind loads, including designated eccentric loads and not to deflect in excess of the following, with H being the height of the partition:

- The lesser of H/240 or 30 mm for partitions subjected to wind loads and lined with flexible material.
- The lesser of H/360 or 20 mm for partitions subjected to wind loads and lined with brittle materials.
- H/500 for eccentric loads.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements.*

**1.3 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Partition - glazed: A partition system consisting of a suite of exposed sections forming door and window frames, ceiling channels, sills, glazing and accessories; and generally intended for use in conjunction with framed and lined partition systems.

**1.4 SUBMISSIONS****Certification**

Toughened glass: For each batch of glass, submit certification from the manufacturer as evidence of heat soaking.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

**Products and materials**

Manufacturer's data: Submit manufacturer's standard product literature for each system type.

Safety glazing materials: Submit evidence of conformity to AS 2208 (2023).

Type tests: Submit Test results to PRODUCTS, **GENERAL**, **Tests** for the following:

- Impact resistance.
- Pressure resistance.
- Weighted sound reduction index ( $R_w$ ).

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL**, **Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, including the following:

- Plans, sections and elevations of the installation.
- Full size sections of members and details of partition intersections and terminations.
- Dimensions, clearances, tolerances and provision for expansion.
- Junctions and trim to adjoining surfaces.

- Doors (note if supplied by others) and frames, including door seals, door hardware and door stops coordinated with Documented door thicknesses.
- Coordination with Documented door hardware.
- Glass types, thicknesses and glazing methods.
- Details of safety markings that make glass visible.
- Glass processing required for fixing hardware to frameless glass doors.
- Methods of fixing partitions.
- Details of acoustic treatments to joints.
- Method of providing reticulation of services, access to services and service outlets.
- Performance data of components and assemblies.
- Specification of materials and finishes.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC licence and CV.

**Substrate acceptance**

Requirement: Submit evidence of the installer's acceptance of the wall, floor and ceiling substrate before starting installation.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.5 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Set-out before installation.
- Installation of plenum baffles.
- Framed and lined partitions ready to receive the framed and glazed component.
- Completion of installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Glazing framing systems: Provide samples of the following:

- Prefinished production extrusions showing the limits of the range of variation in the selected finish and colour, at least 100 mm long.
- Joints made by proposed techniques.

Partition components: Provide samples of the following:

- Skirting, skirting duct, and skirting duct stop ends, returns and removable covers, at least 100 mm long.
- Doors and door hardware.

Glazing materials: Provide samples of glazing materials, each at least 200 x 200 mm, showing specified visual properties and the range of variation, if any, for each of the following:

- Tinted or coloured glass or plastics glazing.
- Surface modified or surface coated glass.
- Patterned or obscured glass or plastics glazing.
- Ceramic-coated glass.
- Mirrored glass.

Gaskets and seals: Provide samples of each type of gasket and seal, each at least 200 mm long.

Sealant: One sample of each type and colour.

**Storage and handling**

Glass and glazing materials: Store in a clean, dry area unaffected by weather and to the manufacturer's recommendations. Protect from building materials and loose debris such as wet plaster, mortar, paint and welding spatter.

Handling glass: To the manufacturer's recommendations and without damage.

**Tests**

Impact resistance:

- Flat glass: To AS 2208 (2023) Appendix D.
- Curved glass: To AS 2208 (2023) Appendix J.

Pressure resistance: To the NCC cited ASTM E72 (2015).

Weighted sound reduction index ( $R_w$ ): To the NCC cited AS/NZS ISO 717.1 (2004).

**2.2 GLAZED PARTITIONS****General**

Requirement: Proprietary non-load bearing glazed partition suite comprising main frames, door frames, sills, ceiling channels and other extrusions and accessories to form a complete and finished system, as Documented.

**Frames**

Aluminium extrusions: To AS 1866 (1997).

**Partition components**

Requirement: As Documented.

**Sealants**

General: Sealant types appropriate for the partition's Documented acoustic rating and compatible with partition materials and building substrate.

**2.3 GLASS****Standards**

Glass type and thickness: To AS 1288 (2021), if no glass type or thickness is nominated.

Materials and installation: To AS 1288 (2021).

Quality requirements for cut-to-size and processed glass: To AS/NZS 4667 (2000).

Terminology for work on glass: To AS/NZS 4668 (2000).

**Performance**

Glass: Free from Defects that detract from appearance or interfere with performance under normal conditions of use.

Plastics glazing: Free from surface abrasions and resistant to yellowing or other colour change. Capable of maintaining physical properties including strength and impact resistance for its design life.

**Heat soaking**

Requirement: Heat soak glass to AS 1288 (2021) clause 3.8.

Standard: To EN 14179-1 (2016).

Marking: To EN 14179-1 (2016) or certified by the manufacturer to AS 1288 (2021) clause 3.8.2.

**Safety glazing materials**

Standard: To AS 2208 (2023).

Type: Grade A.

Marking: To AS 1288 (2021) clause 5.23.

**Unacceptable blemishes in heat-treated flat glass (including tinted and coated glass)**

Standard: To AS/NZS 4667 (2000).

## 2.4 GLAZING MATERIALS

### General

Requirement: Putty, glazing compounds, sealants, glazing tapes, spacing strips, spacing tapes, spacers, setting blocks, shims and compression wedges appropriate for the conditions of application and required performance.

### Glazing tapes

Standards: To AAMA 800 (2016), Products coded 804.3, 806.3 or 807.3, as applicable.

### Jointing materials

General: Jointing and pointing materials that are compatible with each other and with the contact surfaces and non-staining to finished surfaces. Do not provide bituminous materials on absorbent surfaces.

### Primer

Compatibility: Apply the manufacturer's recommended primer to the surfaces in contact with sealant materials.

### Extruded gaskets and seals

Requirement: Provide gaskets and seals, as Documented.

Materials: Non-cellular (solid) elastomeric seals as follows:

- Rubber products: Neoprene, ethylene propylene diene monomer (EPDM) or silicone rubber.
- Flexible polyvinyl chloride (PVC): E type compounds, colourfastness grade B.

## 2.5 PLENUM BAFFLES

### General

Requirement: Plenum baffles that maintain the Documented fire-resistance level and acoustic performance of the partitions.

### Types

Bulk insulation: Layers of bulk insulation batts compressed between the top of the partition and the slab soffit.

Flexible sheet insulation: Mass loaded vinyl sheeting hung as a curtain from the slab soffit.

Plasterboard: Plasterboard sheets bonded together (if more than one layer).

---

## 3 EXECUTION

### 3.1 GENERAL

#### Preparation

Substrate: Prepare the substrate to receive the partitions.

Carpet: Fix bottom tracks over polyethylene film. Prevent carpet threads from pulling if drilling or installing fasteners.

#### Protection

General: Protect existing Work from damage during the installation and rectify any damage. Provide temporary coverings if required.

#### Set-out

General: Set out the partition grid on the centreline of framing members, and to coincide with the ceiling grid and other major building grid, as applicable.

### 3.2 INSTALLATION

#### General

Requirement: Conform to manufacturer's recommendations and assembly details.

#### Frame erection

Frames: Install main frames, sills, ceiling channels, door and window frames and other framing members as follows:

- Plumb, level, square, straight and true.
- Fixed or anchored to the building structure.
- Isolated from any building loads, including loads caused by structural deflection or shortening.

- Joints tightly fitted and neatly aligned.
- Door and window openings accurately sized.
- Use concealed fastenings.

**Sealant**

Acoustic sealant: If required to maintain rated acoustic performance, bed sill and ceiling channels in acoustic sealant.

**3.3 PARTITION GLAZING****Glass processing**

Processing: Perform required processes on glass, including cutting, obscuring, silvering and bending. Form necessary holes, including for fastenings, hardware, equipment, access openings and speaking holes. Process exposed glass edges to a finish not inferior to ground arised.

**Framed glazing**

Assembly: Provide proprietary glazing beads and resilient (PVC, butyl or similar) glazing tapes, gaskets and inserts, to hold the glass firmly without distortion and to withstand the Documented loadings.

**Frameless glazing**

Assembly: Join the vertical edges of adjacent glass panels with a clear structural silicone jointing compound.

Support: For frameless installations not fixed directly to the building structure, provide adequate connection of the top and bottom glazing channels or bead to resist lateral loads.

**3.4 PLENUM BAFFLES****Baffles**

General: Install plenum baffles tightly butted to building structure, service ducts, pipes and conduits and to the top of the partition or to the top of the suspended ceiling directly above the line of the partition. Seal joints, penetrations and intersections and maintain the required performance.

Bulk insulation: Install individual layers to fill space between building structure and the top of the partition or the top of the suspended ceiling.

Flexible sheet insulation: Fix to soffit through a continuous furring channel, hang to meet the top of the partition and extend horizontally 900 mm over the suspended ceiling.

**Fire-resisting partitions**

General: If a suspended ceiling of equivalent fire-resistance is not provided, provide plenum baffles of equivalent fire-resistance level to the underside of the structural soffit.

**Acoustic rated partitions**

General: If a suspended ceiling of equivalent sound insulation rating is not provided, provide acoustic rated plenum baffles. Make sure the ceiling and baffle provide a combined rating equivalent to the partition rating.

**3.5 COMPLETION****Cleaning**

General: Remove protective coverings, replace damaged glass and leave the Work clean, polished, free from Defects and in good condition.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the following:

- Full product information for each system, including product designations, components list, colours and finishes, and accessories.
- Information on all glass, including type, thickness and details of any colouration or treatment affecting the physical appearance of the installation.
- Information on all doors and hardware supplied as part of the system, including door type, size, finishes and hardware details.
- Maintenance recommendations.
- Copies of type Tests and compliance certificates for fire, acoustic or other system performance requirements.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offered by the manufacturer and installer.

**4 SELECTIONS**

---

**Refer to Appendix B – Façade and Wall Schedule**

|                             |
|-----------------------------|
| <b>0525 CUBICLE SYSTEMS</b> |
|-----------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide cubicles, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviations apply:

- ABS: Acrylonitrile butadiene styrene.

**Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Acrylic polymer resin panels: Thermoplastic or thermosetting acrylic resin panels derived from acrylic acid, methacrylic acid or other related compounds, with mineral fillers and pigments.
- High pressure decorative laminates (HPDL):
  - . Panels consisting of core layers impregnated with phenolic and/or aminoplastic resins and a surface layer(s) impregnated with aminoplastic resins (mainly melamine resins).
  - . Sheets consisting of a decorative face and layers of fibrous sheet material (e.g. Paper) impregnated with thermosetting resins and bonded together under heat and pressure of at least 5 MPa.

**1.4 TOLERANCES****General**

Deviation from true grid lines and planes: 1:1000 to a maximum of 3 mm.

Misalignment of adjoining surfaces at panel junctions: 1 mm.

Panel thickness:  $\pm 0.5$  mm.

Length and width: 0.1% of the dimension or 0.5 mm, whichever is the greater.

Flatness, twist, winding and bow: 1 mm deviation from a 2.4 m straightedge placed in any position.

Maximum deviation of edges from the intended true line:  $\pm 1$  mm.

**1.5 SUBMISSIONS****Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

**Products and materials**

Manufacturer's data: Submit manufacturer's standard product literature for each system type.

**Samples**

Requirement: Submit samples to **PRODUCTS**, **GENERAL**, **Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, including the following:

- Overall layout and dimensions.
- Materials, thicknesses and finishes of elements including doors, divisions, fronts, pedestals and top rail.
- Assembly components, including suspension beam and fixing hardware.
- Door hardware type and location.

- Relationship of assembly to adjacent building elements.
- Junctions and trim to adjoining surfaces.
- Methods of fixing partitions.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC Licence and CV.

**Substrate acceptance**

Requirement: Submit evidence of the installer's acceptance of the wall, floor and ceiling substrate before starting installation.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Set-out before installation.
- Completion of installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples of the following:

- Each selected panel and door finish, at least 300 x 300 mm, with associated selected edge strips and trim.
- Sample panel Tested for graffiti removal:
  - . Nominate the cleaning agent.
  - . Mark the sample with spray paint and permanent markers.
  - . Clean half of each marked area.
  - . Repeat with alternate cleaning agents until removal has been achieved.
- All hardware and metal components in the selected finish.

**Storage and handling**

Requirement: Store in a clean, dry area, unaffected by weather and to the manufacturer's recommendations. Protect from building materials and loose debris, such as wet plaster, mortar, paint and welding spatter.

**2.2 PRODUCT SYSTEMS****Acrylic polymer resin panels**

Requirement: Chip, crack and scratch resistant acrylic polymer resin panels.

**High pressure decorative laminate (HPDL) panels and doors**

Material: Compact high pressure decorative laminate panels with an integral surface finish and edges sealed by the manufacturer.

Standard: To AS/NZS 2924.1 (2024).

- Classification: Compact general purpose standard (CGS).

Panel edge: Factory prefinished square cut, ground smooth and arrised. Cleaned and oiled by the manufacturer.

**2.3 COMPONENTS****Suspension beam**

General: For suspended systems, provide a suspension beam consisting of a galvanized mild steel channel, located immediately above the ceiling framing along the line of the partition fronts.

**Hardware**

Fixing hardware: Bolts, dowels, brackets, standards, cappings and stabilising bars supplied to complete the cubicle assembly.

Door furniture: As Documented.

---

**3 EXECUTION**

---

**3.1 GENERAL****Pre-conditioning**

General: Acclimatise wood-based system components in the in-service conditions for a minimum period of two weeks before assembly.

**Control of movement**

Assembly: Accommodate thermal expansion of panels.

**3.2 PANELS****Manufactured cubicle system installation**

Assembly: Attach divisions and nibs to walls and fronts with purpose-made proprietary fastenings. Cut nibs and divisions that abut walls, as required, so that assembly is plumb. Seal edges as recommended by the manufacturer.

Floor mounted/overhead braced type: Fix fronts to the floor with proprietary fittings and at the top to a metal channel head rail, supplied as part of the system. Run head rail across the fronts and fix to the walls at each end. Form the channel into a box section over doorways by snapping in a mating channel insert.

Heads of openings: Fix stabilising head channels by screwing to the top of the partitions. Provide an infill strip to the channel across the opening.

Ceiling-hung type: Hang the fronts from a suspension beam with attachments incorporating a means of height adjustment, supplied as part of the system.

Freestanding type: Fix fronts to the floor with proprietary fittings.

Stone panels: Avoid on-site cutting and mechanical processing of panels, if possible.

**3.3 COMPONENTS****Shower seats**

Fixing: Fix to structural elements using one of the following methods:

- Anodised aluminium channel to exposed edge, secured to walls at each end.
- Product assembly detail.
- Proprietary wall bracket.

**Suspension beam**

Installation: Install suspension beam as follows:

- Build the ends into masonry structure or provide end fastenings to the structure, as necessary, to transfer the load.
- Drill the bottom flange of the channel for the partition fasteners.

**3.4 COMPLETION****Operation and maintenance manuals**

Requirement: Prepare a manual that includes the following:

- Full product information for each system, including product designations, components list, colours and finishes, and accessories.
- Information on all doors and hardware supplied as part of the system including door type, size, finishes, and hardware details.
- Maintenance recommendations.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: as offered by the manufacturer and installer.

#### 4 SELECTIONS

---

Refer to Appendix A – T-Sheet.

## 0532 SUSPENDED CEILINGS - FLUSH LINED

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide suspended ceilings, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Structural Engineer's Documents, and Acoustic Engineer's Report.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0342 Light steel framing.
- 0382 Light timber framing.
- 0511 Lining.
- 0671 Painting
- 0902 Electrical design and installation.

**1.3 STANDARDS****General**

Suspended ceilings: To AS/NZS 2785 (2020).

**1.4 TOLERANCES****Suspension system**

Flatness, twist, winding and bow: 1.5 mm deviation from a 1.5 m straightedge placed in any position.

Deflection: To AS/NZS 2785 (2020) Table 2.4.5.

Setting out and levelling: To AS/NZS 2785 (2020) Appendix D.

**Sheeted or flush ceiling suspension system**

Suspension system bearing surface for flush lined ceiling: To AS/NZS 2589 (2017) Table 4.2.2.

Deflection: To AS/NZS 2589 (2017) Table 3.5.1.2.

**1.5 SUBMISSIONS****Fire performance**

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties.**

Fire-resistance level: Submit evidence of conformity to **FIRE PERFORMANCE, Fire-resistance of building elements.**

**Products and materials**

Type Tests: Submit Test results to PRODUCTS, **GENERAL, Tests** for the following:

- Weighted suspended ceiling normalised level difference.
- Weighted sound absorption coefficient.
- Weighted sound reduction index.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC Licence and CV.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

## 1.6 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- The suspension system before the installation of lining.
- The ceiling assembly before the installation of fittings and Site painting, if applicable.
- The completed ceiling.

## 2 PRODUCTS ---

### 2.1 GENERAL

#### Samples

Requirement: Provide samples as follows:

- Suspension system: Sections proposed for the suspension system, including suspension rods, clips and wall angles.
- Accessories including access panels and wall trim.
- Ceiling material: Lining, with insulation, showing the proposed finish.

#### Storage and handling

Requirement: Store suspended ceiling components in a dry and secure area, and to the manufacturer's recommendations.

#### Tests

Weighted suspended ceiling normalised level difference: To the NCC cited AS/NZS ISO 717.1 (2004).

Weighted sound absorption coefficient: To AS ISO 11654 (2002), as Tested to AS ISO 354 (2006).

Weighted sound reduction index: To the NCC cited AS/NZS ISO 717.1 (2004).

### 2.2 FIRE PERFORMANCE

#### Fire hazard properties

Group number: To AS 5637.1 (2015).

#### Fire-resistance of building elements

Fire-resistance level: Tested to AS 1530.4 (2014).

### 2.3 SUSPENSION SYSTEM

#### Ceiling suspension system

Consistency: Provide ceiling systems as complete proprietary systems, fabricated by one manufacturer, as Documented.

#### Materials

Coated steel: To AS 1397 (2021).

Aluminium: To AS 1866 (1997).

Protective coatings for steel components: To AS/NZS 2785 (2020) Appendix F.

Protection against atmospheric corrosion: To AS 2312.1 (2014) and AS/NZS 2312.2 (2014).

### 2.4 LININGS

#### Plasterboard

Standard: To AS/NZS 2588 (2018).

Minimum thickness: 10 mm.

#### Fibre cement

Standard: To AS/NZS 2908.2 (2000).

Internal ceiling linings: Type B Category 2.

External ceiling linings: Type A Category 3.

Minimum thickness: 9 mm.

#### Sealants

Fire-resisting sealant: Non-hardening sealant compatible with the materials to be sealed and having a fire-resistance rating equal to that of the building element it seals.

Acoustic sealant: Non-hardening sealant compatible with the ceiling materials and rated to match the ceiling system's acoustic performance.

## **2.5 ACCESS PANELS**

### **General**

Requirement: Proprietary products comprising hinged, faced panels fixed in steel frames, including hardware and accessories such as hinges and lock and installation lugs.

### **Finish**

General: Match the access panels to the ceiling in appearance and performance.

## **2.6 TRIM**

### **General**

Trim: Provide trim consistent with the materials and finishes of the ceiling system.

### **Accessories**

General: Provide accessories as part of the proprietary ceiling system necessary to complete the installation.

## **3 EXECUTION**

---

### **3.1 GENERAL**

#### **Working environment**

General: Do not start work before the building is enclosed, wet work is complete and dry, and all work above the ceiling, including services, is complete.

#### **Protection**

General: Protect existing Work from damage during the installation.

#### **Partitions**

General: If partitions are attached to the underside of the ceiling systems, include the partition mass in the seismic mass of the ceiling.

#### **Stability**

General: Install the ceilings level and fix to prevent looseness or rattling of ceiling components under normal conditions.

#### **Structure-borne sound**

General: Provide a ceiling system that does not amplify structure-borne sound. Provide suitable proprietary products or systems for reducing contact vibrations between structure and ceiling.

#### **Control of movement**

Abutments: Install the ceiling to allow for differential movement at abutting surfaces.

Alignment: Align ceiling control joints with structural control joints. Do not bridge structural control joints.

#### **Prefinishes**

General: Repair damaged prefinishes by recoating.

### **3.2 SUSPENSION SYSTEM**

#### **Ceiling suspension system**

General: Fix suspension system to the structural trusses.

Support members: Install support members as follows:

- Coordinate framing with sheet and batten set out as Documented.
- Space as required by the loads on the system and the type of ceiling.
- Allow for the installation of services and accessories, including ductwork, light fittings and diffusers.
- Provide additional back support or suspension members for the fixing of services and accessories to prevent distortion, overloading or excessive vertical deflection.
- Allow for access for maintenance of services.

Failure: Provide a ceiling system that will not progressively fail with failure of any one suspension point.

Height adjustment: Provide height adjustment with a length adjustment device at each suspension point, permitting length variation of at least 50 mm.

Restriction: Do not attach the suspension system to the lip or flange of purlins.

#### **Services**

Support: Conform to the following:

- If the service has not been designed to accept the ceiling load, do not fix suspension members to services.
- Where services obstruct the ceiling supports, provide bridging and suspension on each side of the services.

#### **Bracing**

General: Provide bracing to prevent lateral movement and to resist the imposed horizontal seismic force.

#### **Bulkheads**

General: Integrate bulkheads with the ceiling structure and brace to prevent lateral movement. If the ceiling is terminated at a bulkhead, provide for seismic requirements.

#### **External suspended soffits**

General: Support external suspended soffits on rigid members capable of carrying the loads from imposed actions. Install members to minimise any eccentricity, and carry the positive and negative loads from wind actions through to the supporting structure. Refer also to structural engineer's Documents.

#### **Fasteners**

General: Provide concealed fasteners. If material supporting hangers is less than 3 mm thick, do not use screw fasteners.

### **3.3 PLASTERBOARD**

#### **Installation**

Gypsum plasterboard and fibre-reinforced gypsum plaster: To AS/NZS 2589 (2017).

Level of finish and jointing: To AS/NZS 2589 (2017) clause 3.1.

Suspended flush ceilings: Fix using screws or screws and adhesive to ceiling members or support frame.

#### **Joints**

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

Butt joints: Make joints over framing members or otherwise provide back blocking.

External corner joints: Make joints over metallic-coated steel corner beads.

Control and movement joints: Align lining control joints with structural movement joints and as follows:

- Ceilings:
  - . Internal: At maximum 12 m centres.
  - . External: At maximum 6 m centres.
- Control joint beads: Purpose-made metallic-coated.
- Seismic joint: Purpose-made flexible joint and cover.
- Location: Position joints to intersect light fixtures, vents or air diffusers, as required.

Wet areas: Install additional supports, trim and sealants, to manufacturer's recommendations.

### **3.4 FIBRE CEMENT**

#### **Installation**

General: Run sheets across the framing members. In flush jointed applications, stagger end joints in a brick pattern and locate them on framing members, away from the corners of large openings. Provide supports at edges and joints.

Suspended flush ceilings: Fix using screws or screws and adhesive to ceiling members or support frame.

#### **Joints**

Flush joints: Provide recessed edge sheets and finish flush using perforated paper reinforcing tape.

External corner joints: Make joints over metallic-coated steel corner beads.

Dry joints: Provide square edged sheet and finish with a PVC-U joining section.

Non-set joints: Provide square edge joint with metal or socket strip backing.

Control and movement joints: Align lining control joints with structural movement joints to the manufacturer's recommendations and as follows:

- Control joint beads: Purpose-made metallic-coated.
- Seismic joint: Purpose-made flexible joint and cover.
- Support: Provide framing parallel to the joint on each side. Do not fix the lining to abutting building surfaces.
- Location: Position joints to intersect light fixtures, vents or air diffusers, as required.

Wet areas: Install additional supports, trim and sealants, to manufacturer's recommendations.

### **3.5 ACCESS PANELS**

#### **Identification**

General: Provide each access panel with an identification mark.

#### **Non-demountable ceilings**

General: Provide access panels supported and anchored to permit ready removal and refixing.

#### **Reinforcement**

General: Reinforce the back of the access panel to prevent warping and facilitate handling.

### **3.6 TRIM**

#### **General**

Trim: Install trim at junctions with other building elements and surfaces, including walls, beams and penetrations, consistent with the materials and finishes of the ceiling system.

#### **Accessories**

General: Install accessories as part of the proprietary ceiling system necessary to complete the installation.

### **3.7 COMPLETION**

#### **General**

Exposed surfaces: Touch up shop applied finishes and restore damaged or marked areas.

Cleaning: Clean completed surfaces

Debris and unused material: Remove from Site.

#### **Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offered by the manufacturer and installer.

## **4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                     |
|---------------------|
| <b>0551 JOINERY</b> |
|---------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide joinery, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, Hydraulic Engineer's Documents, and Electrical Engineer's Documents.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0185 *Timber products, finishes and treatment.*
- 0632 *Stone and terrazzo tiling.*
- 0802 *Hydraulic design and install.*
- 0811s *Sanitary fixtures.*
- 0812s *Tapware.*

### 1.3 TOLERANCES

#### General

Requirement: Fabricate and install joinery items to substrates undamaged, plumb, level, straight and free of distortion.

#### Tolerances table

| Property                                                               | Tolerance |
|------------------------------------------------------------------------|-----------|
| Plumb and level                                                        | 1:800     |
| Offsets in flush adjoining surfaces                                    | 0.5 mm    |
| Offsets in revealed adjoining surfaces                                 | 2 mm      |
| Alignment of adjoining doors                                           | 0.5 mm    |
| Difference in scribe thickness for joinery items centred between walls | 2 mm      |
| Doors centred in openings                                              | 0         |
| Joints in finished surfaces                                            | 0         |

### 1.4 SUBMISSIONS

#### Certification

Requirement: Submit one of the following, as evidence of conformity to Documented requirements for grading, species and board size:

- Supplier's certificate, which may be included on an invoice, delivery docket or packet label.
- Report by an independent inspecting authority.

Timbers for which certification is required: as Documented.

Moisture content: Submit Documentation noting moisture content of timber products.

Environmental rating evidence or certification: as Documented.

- Timing: Before fabrication.

#### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

#### Products and materials

Manufacturer's data: Submit manufacturer's product data.

Proprietary items: Submit the manufacturer's standard drawings and details showing:

- Methods of construction.
- Assembly and fasteners, with dimensions and tolerances.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Overall dimensions.
- Materials, thicknesses and finishes of elements including doors, divisions, shelves and benches.
- Type of construction including mitre joints and junctions of members.
- Hardware type and location.
- Temporary bracing, if required.
- Procedures for shop and Site assembly and fixing.
- Locations of benchtop joints.
- Stone benchtop layout including joint arrangement and penetrations.
- Locations of sanitary fixtures, stoves, ovens, sinks and other items to be installed in the units.
- Relationship of fixture to adjacent building elements.
- Details of fabrication involving other trades or components.
- Proposals for the break-up of large items as required for delivery to the Site.
- Proposed method of joining the modules of large items.

Timing: Before fabrication.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 1.5 INSPECTION

**Notice**

Inspection: Give notice so that inspection may be made of the following:

- Shop fabricated or assembled items ready for delivery to the Site.
- Openings prepared to receive assemblies.
- Site erected assemblies on completion of erection, before covering up by cladding and encasing.
- Surfaces prepared for, and immediately before, Site applied finishes.
- Completion of installation.

---

## 2 PRODUCTS

### 2.1 GENERAL

**Samples**

Requirement: Provide samples as follows:

- Boards: Two of each type, complete with finish and edge stripping.
- Joints: Two of each type.
- Typical hardware item: Two samples, showing each finish.
- Stone cladding: Provide three variants, two samples of each variant showing maximum expected variation.
- Timber veneer: Provide three variants, two samples of each variant showing maximum expected variation.
- Fabric: Two swatches of each type.
- Stainless steel items: Two of each type.
- Sealants: One sample of each type and colour.
- Bench cupboard door: One sample, complete with hardware.

- Overhead cupboard door: One sample, complete with hardware.
- Drawer front: One sample, complete with hardware.
- Timber stair stringer, riser and going, balustrade and handrail: One finished sample.

Clear finished timber: Provide samples as follows:

- Initial submission:
  - . Veneered board: Three samples each 600 x 600 mm for each species.
  - . Solid timber: Three samples each 40 x 19 x 600 mm for each species.
- Control sample: The approved selection from the initial submission.
- Finished sample: Cut the control sample in half and apply the finish to half the remaining area.

### **Storage and handling**

Requirement: Deliver joinery units to Site in unbroken wrapping or containers and store so that its moisture content is not adversely affected. Do not store in areas of wet plaster. Store in an adequately ventilated space away from heat and direct sunlight. Keep storage time to a minimum by delivering items only when required for installation.

## **2.2 JOINERY MATERIALS AND COMPONENTS**

### **Certification**

Timber based products: Label panels under the authority of a recognised certification scheme to *0185 Timber products, finishes and treatment*, as applicable to the product. Locate the brand on faces or edges that will be concealed in the Works.

### **Visible Work**

Clear finished timber and veneer: Make sure all visible surfaces are free of branding, crayon or chalk marks and of blemishes caused by handling.

### **Joinery timber**

Hardwood for trim: To AS 2796.1 (1999).

Hardwood for furniture: To AS 2796.3 (1999).

Hardwood grade: SEL

Seasoned cypress pine: To AS 1810 (1995).

Softwood for trim: To AS 4785.1 (2002).

Softwood for furniture: To AS 4785.3 (2002).

Softwood grade: clear (FCL).

Finished sizes of milled timbers: Not less than the Documented dimensions unless qualified by a term such as nominal, out of or ex to which industry standards for finished sizes apply.

### **Plywood**

Interior use generally: To AS/NZS 2270 (2006).

Interior use, exposed to moisture: To AS/NZS 2271 (2004).

Visible surface with a clear finish: Veneer quality A.

Other visible surfaces: Veneer quality B.

Formaldehyde emission class to AS/NZS 2270 (2006) and AS/NZS 2271 (2004): E<sub>0</sub>

### **Dry process fibreboard (including medium density fibreboard)**

Standard: To AS/NZS 1859.2 (2017).

Formaldehyde emission class to AS/NZS 1859.2 (2017): E<sub>0</sub>

### **Decorative overlaid wood panels**

Standard: To AS/NZS 1859.3 (2017).

### **High pressure decorative laminate (HPDL)**

Standard: To AS/NZS 2924.1 (2024).

Classification: Refer Appendix A – T-Sheet..

Minimum thickness: Conform to the following:

- Horizontal surfaces fixed to a continuous substrate: 1.2 mm.
- Vertical surfaces fixed to a continuous substrate: 0.8 mm.
- Post formed laminate fixed to a continuous substrate: 0.8 mm.

- Vertical surfaces fixed intermittently, including to studs: 3.0 mm.
- Edge strips: 0.4 mm.

**Stone facings**

General: Provide stone slabs within the visual range of the samples.

**Toughened glass panels**

General: Refer Appendix A – T-Sheet..

**Splashbacks**

Glass: Toughened safety glass to AS 2208 (2023).

Glass type: Refer Appendix A – T-Sheet..

Stainless steel: Type 304, No. 4 finish.

**2.3 JOINERY ASSEMBLIES****General**

Standard: To AS 4386 (2018).

**Plinths**

Material: Refer Appendix A – T-Sheet..

Thickness: Refer Appendix A – T-Sheet..

Fabrication: Form up with front and back members and full height cross members at not more than 900 mm centres.

Finish: Refer Appendix A – T-Sheet..

Fasteners: Conceal with finish.

Installation: Scribe to floor and secure to wall to provide level platform for carcasses.

**Carcasses**

Material: Refer Appendix A – T-Sheet.

Thickness: as Documented.

Joints: include on shop drawings for approval.

Adjustable shelves: Support on proprietary pins in holes bored at equal centres vertically.

- Spacing: 50 mm.

Finish: Refer Appendix A – T-Sheet..

Fasteners: Conceal with finish.

Installation: Secure to walls at not more than 600 mm centres.

**Drawer fronts and doors**

Material: Refer Appendix A – T-Sheet..

Thickness: 16 mm unless noted otherwise.

Door size: Not exceeding 1.5 m<sup>2</sup> on face, with 2400 mm maximum height or 900 mm maximum width.

Drawer fronts: Rout for drawer bottoms.

Finish: Refer Appendix A – T-Sheet..

Colour: Refer Appendix A – T-Sheet..

**Drawer backs and sides**

Proprietary systems: Refer Appendix A – T-Sheet..

**Drawer bottoms**

Material: PVC film laminated hardboard.

**Drawer and door hardware**

Hinge types: Concealed metal hinges with the following features:

- Nickel-plated.
- Adjustable for height, side and depth location of door.
- Integrated soft and self-closing action.
- Hold open function.

Piano hinges: Chromium-plated steel, extending full height of doors.

Slides: Unless Documented otherwise, metal runners and plastic rollers with the following features:

- 30 kg loading capacity.
- Integrated soft and self-closing action.
- Closure retention.
- White thermoset powder coating or nickel-plated.

## 2.4 WORKING SURFACES

### Laminated benchtops

Material: Refer Appendix A – T-Sheet..

Finish: High pressure decorative laminate, Refer Appendix A – T-Sheet.:

Exposed edges, unless Documented otherwise:

- Extend laminate over shaped nosing, finishing more than 50 mm back on underside. Splay outside corners at 45°.

Installation: Scribe to walls. Fix to carcass at least twice per 600 mm length of benchtop.

Joint sealing: Fill joint with sealant matching finish and clamp with proprietary mechanical connectors.

### Stone benchtops

Material:

- Type: Refer Appendix A – T-Sheet..

### Benchtop backing

Material: include in shop drawings for approval.

## 3 EXECUTION

### 3.1 TOLERANCES

#### General

Requirement: Fabricate and install joinery items to substrates undamaged, plumb, level, straight and free of distortion.

#### Tolerances table

| Property                                                               | Tolerance |
|------------------------------------------------------------------------|-----------|
| Plumb and level                                                        | 1:800     |
| Offsets in flush adjoining surfaces                                    | 0.5 mm    |
| Offsets in revealed adjoining surfaces                                 | 2 mm      |
| Alignment of adjoining doors                                           | 0.5 mm    |
| Difference in scribe thickness for joinery items centred between walls | 2 mm      |
| Doors centred in openings                                              | 0         |
| Joints in finished surfaces                                            | 0         |

### 3.2 JOINERY

#### General

Joints: Provide materials in single lengths if possible. If joints are necessary, locate over supports.

Framing: Frame and trim where necessary for openings, including those required by other trades.

Concealed surfaces: Prime surfaces concealed by substrates.

Deficiencies: Examine joinery units for completeness and remedy deficiencies.

Substrate: Damp clean and vacuum substrate surfaces that will be permanently concealed.

Openings: Provide openings for the following:

- Cable feeds as Documented in meeting rooms.

#### Acclimatisation

General: Acclimatise the joinery items by stacking in the in-service conditions with air circulation to all surfaces after the following are complete:

- Air conditioning operational.

- Lighting operational.
- Site drainage and stormwater works are complete.
- Space fully enclosed and secure.
- Wet work complete and dry.

#### **Accessories and trim**

General: Provide accessories and trim necessary to complete the installation.

#### **Fasteners**

Visibility: Do not provide visible fasteners except in the following locations:

- Inside cupboards and drawer units.
- Inside open units, in which case provide proprietary caps to conceal fasteners.

Visible fasteners: Where fasteners are unavoidable on visible joinery faces, sink the heads below the surface and fill the sinking flush with a material compatible with the surface finish. In surfaces that are to have clear or tinted finish, provide matching wood plugs showing face (not end) grain. In surfaces that are to have melamine finish, provide proprietary screws and caps finished to match.

Fixing to substrate: Fix joinery units to substrates as follows:

- Floor mounted units: 600 mm centres maximum.
- Wall mounted units: To each nogging and/or stud stiffener.

Fasteners: Screws with washers into timber or steel framing, or masonry anchors.

#### **Adhesives**

General: Provide adhesives to transmit the loads imposed and for the rigidity of the assembly, without causing discolouration of finished surfaces.

#### **Finishing**

Junctions with structure: Scribe plinths, benchtops, splashbacks, ends of cupboards, kickboards and returns to follow the line of structure.

Joints: Scribe internal and mitre external joints.

Edge strips: Finish exposed edges of sheets with edge strips that match sheet faces.

Matching: For surfaces that are to have clear or tinted finish, arrange adjacent pieces to match the grain and colour.

Hygiene requirements: For food handling areas, amenities, and voids at the backs of units in all areas, seal carcass junctions to walls and floors.

#### **Benchtops**

Installation: Fix to carcass at least twice per 600 mm length of benchtop.

Joint sealing: Fill joints with sealant matching the finish colour and clamp with proprietary mechanical connectors.

Edge sealing: Seal to walls and carcasses with a sealant that matches the finish colour.

Stone, porcelain or concrete benchtops: Avoid on-site cutting and mechanical processing of panels, if possible.

#### **Glass splashbacks**

Adhesive: Fix with non-acidic silicone adhesive. Apply at the rate recommended by the manufacturer.

Installation: Clean the back of the glass panel and apply walnuts of adhesive together with double-sided adhesive tape for temporary support, and affix directly to the substrate.

#### **Labelling**

General: Permanently mark each unit of furniture with the manufacturer's name, on an interior surface.

### **3.3 TRIM**

#### **General**

Requirement: Provide trims as Documented in the Architectural Drawings.

#### **Fixing**

Stud walls: Glue and nail to plate or framing at 600 mm centres maximum.

### 3.4 COMPLETION

#### **Cleaning**

Requirement: Remove all dust, marks and rubbish from all surfaces and internal spaces. Clean and polish all self-finished surfaces such as anodised and powder coated metals, sanitary ware, glass, tiles and laminates.

Temporary coatings: On or before completion of the works, or before joining up to other surfaces, remove all traces of temporary protective coatings.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published recommendations for service use.

#### **Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: as offered by the manufacturer and installer.

## 4 SELECTIONS

---

Refer to Appendix A – T-Sheet.

|                                    |
|------------------------------------|
| <b>0552 METALWORK - FABRICATED</b> |
|------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide metal fixtures, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Structural Engineer's Documents.

**Performance**

Requirement:

- Undamaged, plumb, level and straight.
- Free of surface Defects or distortions.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0183 Metals and prefinishes.
- 0342 Light steel framing.
- 0345p DULUX protective coatings.
- 0382 Light timber framing.

**1.3 STANDARDS****General**

Structural design actions: To AS/NZS 1170.1 (2002).

Stairs and balustrades: To AS 1428.1 (2021).

**1.4 TOLERANCES****General**

Requirement:  $\pm 2$  mm from design dimensions.

**1.5 SUBMISSIONS****Execution details**

Welding procedures: Submit details of proposed welding procedures and risk mitigation before fabrication.

Welding dissimilar metals: Submit the following details:

- Type and thickness of materials to be welded.
- Proposed joint preparation and welding procedures.
- Proposed filler metal.
- Expected dilution (proportion of fused parent metal in the weld metal).

Dissimilar metals: provide proposed method for separation of dissimilar metals.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION**, **Operation and maintenance manuals**.

**Products and materials**

Proprietary items: Submit the manufacturer's standard drawings and details showing:

- Methods of construction.
- Assembly and fasteners, with dimensions and tolerances.

Stainless steel: For each batch of stainless steel supplied to the Works, submit a certificate of conformance or Test certificate, as Documented.

Certification: Submit certification for the following:

- Form 16 for balustrades and railings.

Stainless steel welding: Before fabrication commences, submit evidence of qualification of the welding procedure by Testing to AS/NZS 1554.6 (2012) clause 4.7 or evidence of prequalification to AS/NZS 1554.6 (2012) clause 4.12.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, showing the following information:

- Overall and detail dimensions.
- Details of fabrication and components.
- Details of fabrication involving other trades or components.
- Information necessary for Site assembly.
- Proposals for the break-up of large items as required for delivery to the Site.
- Proposed method of joining the modules of large items.
- Proposed separation of dissimilar metals.

**Subcontractors**

General: Submit names and contact details of proposed suppliers, fabricators and installers.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Arrival of materials on Site or in workshop.
- Shop fabricated or assembled items ready for delivery to the Site.
- Commencement of shop or Site welding.
- Site erected assemblies on completion of erection, before covering up by cladding and encasing.
- Steel surfaces prepared for, and immediately before, Site applied finishes.

---

**2 PRODUCTS**

---

**2.1 GENERAL****Samples**

Requirement: Provide samples of the following:

- Each type of joint.
- Each type of finish illustrating the range of variation.
- Sections for use in fabricated work.

**Storage and handling**

Requirement: Store and handle fabricated metalwork, as follows:

- Deliver to Site in unbroken wrapping or packing.
- Store on a level base, away from uncured concrete and masonry and areas of wet plaster.
- Do not store in contact with other materials that may cause staining, denting or other surface damage.
- Use gloves if handling precoated finishes.
- Keep storage time to minimum by delivering items only when required for installation.

**Marking**

General: Provide suitable and sufficient marks or other means for identifying each member of site-erected assemblies, and for their correct setting out, location, erection and connection. Mark bolted connections to show the bolting category. Do not mark stainless steel by notching.

## 2.2 MATERIALS

### Metals and components

Performance: Provide metals and components in quantity, lengths and cross-sections of strength and stiffness suited to their required function and as Documented.

### Stainless steel

Plate, sheet and strip: To ASTM A240/A240M (2025).

Bar: To ASTM A276/A276M (2025).

Tube: To ASTM A554 (2021).

Type: 316 grade.

### Aluminium

Plate sheet and strip: To AS/NZS 1734 (1997).

Bar, rod and wire: To AS/NZS 1865 (1997).

Tube: To AS/NZS 1867 (1997).

Aluminium alloys, composition and designation: To AS 2848.1 (1998).

### Steel

Steel plate: To AS/NZS 3678 (2016).

Hot-rolled bars and sections: AS/NZS 3679.1 (2016).

Welded sections: To AS/NZS 3679.2 (2016).

### Fasteners

Performance: Provide fasteners to resist galvanic corrosion in materials of structural and mechanical strengths and corrosion resistance at least equal to that of the lowest resistant metal in the connection.

Materials: Provide fasteners as follows:

- To copper and copper alloys: Copper or copper-alloy fasteners only.
- To aluminium and aluminium alloys: Aluminium alloy or non-magnetic stainless steel fasteners only.
- To stainless steel: Appropriate stainless steel fasteners only.

## 2.3 OTHER MATERIALS

### Grabrail

Standard: To AS 1428.1 (2021).

### Tactile ground surface indicators

Tactile ground surface indicators to stairs: as Documented; to AS/NZS 1428.4.1 (2009).

## 3 EXECUTION

---

### 3.1 GENERAL

#### Prototypes

Requirement: Provide prototypes as Documented.

Installation: Erect prototype items in their final locations, incorporating at least one example of each component and fastening to building elements.

### 3.2 CONSTRUCTION GENERALLY

#### Aluminium structures

Standard: To AS/NZS 1664.1 (1997) or AS/NZS 1664.2 (1997).

#### Metals

Incompatible metals: Separate using concealed layers of suitable materials in appropriate thicknesses.

#### Fabrication

Workshop: Fabricate and pre-assemble items in the workshop if practicable.

Edges and surfaces: Keep clean, neat and free from burrs and indentations. Remove sharp edges without excessive radiusing.

Tube bends: Form bends in tube without deforming the cross-section and the material thickness.

Colour finished Work: Match colours of sheets, extrusions and heads of fasteners.

Thermal movement: Accommodate thermal movement in joints and fastenings.

### **Joints**

General: Fit joints to an accuracy appropriate to the class of Work. Finish visible joints made by cutting, drilling, welding, brazing or soldering using grinding, buffing or other methods appropriate to the class of Work, before further treatment.

Self-finished metals: Free of surface colour variations, after jointing.

Joints: Fit accurately to a fine hairline.

### **Splicing**

General: Provide structural members in single lengths.

## **3.3 WELDING AND BRAZING**

### **Welding**

Quality: Provide finished welds that are free of surface and internal cracks, welding slag, and porosity.

Corners and edges: Grind smooth sharp, marred or roughened corners and edges.

Rough surfaces: Deburr and grind smooth.

Site welds: Avoid Site welding if possible. If required, locate Site welds in positions for down hand welding.

Butt weld quality level: Not inferior to the appropriate level recommended in AS/NZS 1554.1 (2014) Section 6, AS/NZS 1554.6 (2012) Section 6 or AS/NZS 1665 (2004) Section 6, as appropriate.

### **Brazing**

General: Make sure brazed joints have sufficient lap to provide a mechanically sound joint.

Butt joints: Do not use butt joints for joints subject to load. If butt joints are used, do not rely on the filler material only.

## **3.4 STAINLESS STEEL FABRICATION**

### **Welding stainless steel**

Qualification of welders: To AS 1796 (2022).

### **Riveting**

General: Use only to join stainless steel sheet or strip less than 1 mm thick. Drill (not punch) the rivet hole and drive the rivet cold. On completion, clean and passivate the riveted assembly.

### **Soldering**

General: Do not solder stainless steel.

## **3.5 CUSTOM-BUILT STEEL STAIRS**

### **General**

Design and construction: To BCA (2022) D3.

### **Fabrication**

Method: Welding.

Joints: Produce smooth unbroken surfaces at joints. Scribe the joints to all steel members. Make end-to-end joints over an internal sleeve.

Bends: Make changes of direction in rails by evenly curved pipe bends.

Free ends: Seal the free ends of pipes with fabricated or purpose-made end caps.

### **Fixing to structure**

General: Provide fabricated predrilled or purpose-made brackets and bases, and attach the steel member to the building structure with fasteners compatible with the substrate.

Proprietary items: Install to the manufacturer's recommendations.

### **Galvanizing**

General: If possible, complete fabrication before galvanizing; otherwise apply a zinc-rich primer to affected joint surfaces.

### **Other protective coatings**

General: Apply other protective coatings as Documented and to the manufacturer's recommendations.

### 3.6 BALUSTRADES

#### **Fabrication**

Method: Welding.

Joints: Produce smooth unbroken surfaces at joints. Scribe the joints between posts and rails. Make end-to-end joints over an internal sleeve.

Bends: Make changes of direction in rails by evenly curved pipe bends.

Free ends: Seal the free ends of pipes with fabricated or purpose-made end caps.

#### **Fixing to structure**

General: Provide fabricated predrilled or purpose-made brackets or post bases, and attach the piping to the building structure with fasteners compatible with the substrate.

Proprietary items: Install to the manufacturer's recommendations.

#### **Other protective coatings**

General: Apply other protective coatings as Documented and to the manufacturer's recommendations.

### 3.7 GRABRAILS

#### **General**

Requirement: Conceal fix to wall. If additional support is required, provide additional support to **WALL FRAMING, Additional support** in 0382 *Light timber framing*.

### 3.8 OTHER CUSTOM-MADE ITEMS

#### **General**

Items: Fabricate as Documented, pre-assemble and fix in place.

Fixing: Fix to the building structure as Documented using fastenings compatible with the materials.

### 3.9 COMPLETION

#### **Cleaning**

Temporary coatings: On or before completion of the works, or before joining up to other surfaces, remove all traces of coatings used as temporary protection.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published recommendations for service use.

#### **Warranties**

Type: Cover materials and workmanship in the terms of the warranty from the fabricator, against failure of materials and execution under normal environment and use conditions.

---

## 4 SELECTIONS

---

Refer to Appendix A – T-Sheet.

|                                             |
|---------------------------------------------|
| <b>0573 FIRE EXTINGUISHERS AND BLANKETS</b> |
|---------------------------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide portable fire extinguishers and fire blankets, as Documented, including recessed cabinets where Documented with compliant signage. Refer to Architectural Drawings and Appendix A – T-Sheet.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.
- 0581 Signage.

### 1.3 SUBMISSIONS

#### Products and materials

Requirement: Submit evidence of suitability for use, to NCC (2022) A5G1, for all fire protection products.

#### Records

Requirement: Submit routine service records to AS 1851 (2012).

#### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 2 PRODUCTS

### 2.1 EXTINGUISHERS

#### Portable fire extinguishers

General: To AS/NZS 1841.1 (2007).

Type:

- Water: To AS/NZS 1841.2 (2007).
- Wet chemical: To AS/NZS 1841.3 (2007).
- Foam: To AS/NZS 1841.4 (2007).
- Powder: To AS/NZS 1841.5 (2007).
- Carbon dioxide: To AS/NZS 1841.6 (2007).
- Non-rechargeable: To AS/NZS 1841.8 (2007).

Selection, location and distribution: To AS 2444 (2001).

### 2.2 BLANKETS

#### Fire blankets

General: To AS/NZS 3504 (2006).

Selection and location: To AS 2444 (2001).

## 3 EXECUTION

### 3.1 INSTALLATION

#### Firefighting equipment

Standard: To AS 2444 (2001).

Signage: Provide signs to **STATUTORY SIGNS** in 0581 Signage.

### 3.2 COMPLETION

#### **Baseline data**

Requirement: Provide baseline data to AS 1851 (2012).

#### **Baseline data**

Requirement: Provide baseline data to AS 1851 (2012).

#### **Warranties**

Type: submit the manufacturer's published product warranties.

### 3.3 MAINTENANCE

#### **Routine service**

Portable fire extinguishers: To AS 1851 (2012) Section 10.

Fire blankets: To AS 1851 (2012) Section 11.

## 4 SELECTIONS

---

Refer to Appendix A – T-Sheet.

|                              |
|------------------------------|
| <b>0574 WINDOW COVERINGS</b> |
|------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide window coverings, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Electrical Engineer's Documents.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 SUBMISSIONS****Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Products and materials**

Manufacturer's drawings: Submit manufacturer's standard drawings and details showing the following:

- Methods of construction.
- Assembly and fastenings, with dimensions and tolerances.

Type Tests: Submit Test results of pest resistance to **PRODUCTS, MATERIALS, Pest resistance.**

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples.**

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Window coverings: Show sizes, locations, and details of fabrication and installation.
- Tracks: Show installation and anchorage details, and locations of controls.
- Motorised tracks: Indicate power requirements, locations, dimensions, weights, mounting methods, required clearances and required access for motor, and relationship of motor to track.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.4 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Building locations or substrates prepared to receive window coverings before installation.

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide two samples showing material, colour, texture, pattern and other visible characteristics for each of the following:

- Curtains including sheers: 200 x 200 mm.
- Venetian blinds and vertical louvre blinds: 400 mm long slat.
- Shutters: 400 mm long slat and frame sample.
- Holland blinds: 200 x 200 mm.
- Fabric shades including panel gliders: 200 x 200 mm.

- Track systems: 400 mm long and including manufacturer's standard control system furniture items and accessories.

## 2.2 FIRE PERFORMANCE

### Fire hazard properties

Windows coverings: Tested to AS/NZS 1530.3 (1999).

## 2.3 MATERIALS

### Fabrics

Uncoated woven and knitted fabrics:

- Standard: To AS 2663.1 (1997).
- Performance classification (minimum): 2.

Coated woven and knitted fabrics:

- Standard: To AS 2663.2 (1999).
- Performance classification (minimum): 2.

Vertical and holland blind fabrics: To AS 2663.3 (1999).

### Anti-microbial treatment

Requirement: Non-metallic, colourless, odourless, positively charged polymer applied during manufacturing to form a molecularly bonded surface to resist bacteria and mould growth.

### Pest resistance

Requirement: Provide window coverings composed entirely of materials either inherently resistant to insect attack or treated against insect attack, by application of insecticide to the yarn during the dyeing or scouring process.

Insect resist agents for wool: Conform to the recommended application levels published by the Woolmark Company for Level 4 protection.

### Powder coating

Application to aluminium and aluminium alloy substrates: To AS 3715 (2025).

Application to metal substrates other than aluminium: To AS 4506 (2024).

## 2.4 BLINDS

### Vertically retracting fabric blinds

Requirement: Complete proprietary systems, as Documented, fabricated by one manufacturer.

## 3 EXECUTION

---

### 3.1 INSTALLATION

#### General

Installation: Install tracks in Documented locations using manufacturer's purpose fabricated mounting brackets, clips, track splicing and other hardware. Install window coverings to hang plumb and level, and true to line.

Fixing: Provide concealed mechanical fasteners suitable for mounting tracks to substrate. Match exposed mounting hardware to the finish and colour of adjacent track.

Adjustment: Adjust all operating hardware for smooth operation free from binding and to provide even, accurate alignment of window covering in open and closed positions.

Safety: Install child safety devices on all control cords. Install all control cords in conformance with *Competition and Consumer (Corded Internal Window Coverings) Safety Standard (2014)*.

#### Vertically retracting fabric blinds

Clearance: Provide 5 mm horizontal clearance at each side of blind for recessed applications.

### 3.2 COMPLETION

#### Operation and maintenance manuals

Requirement: Prepare a manual that includes the manufacturer's data as follows:

- Recommendations for service use, care and maintenance.
- List of manufacturers and suppliers of replacement parts.

**Warranties**

Type: Manufacturer's warranty against Defects in workmanship. Installer's warranty against Defects in installation.

Period: as offered by manufacturer and installer.

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet.**

|                     |
|---------------------|
| <b>0581 SIGNAGE</b> |
|---------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide signage systems, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet. Refer also to services engineer's Documents for service signage requirements. Confirm labelling requirements to Stanwell labelling requirements.

#### Performance

Requirement: Provide signage as follows:

- Appropriately secured.
- Located within a clear line of vision.
- With characters and symbols contrasting with the background.
- With clean, well-defined edges or arrises, and free from blemishes.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.
- 0921 Low voltage power systems.

### 1.3 STANDARDS

#### Signs

Safety signs - design and use: To AS 1319 (1994).

Signs and graphics for disability access: To AS 1428.1 (2021) and AS 1428.2 (1992).

Tactile wayfinding signs: To AS 1428.4.2 (2018).

### 1.4 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the following definitions apply:

- Changeable letter systems: Sign systems consisting of display boards or holders into which removable individual letters, numbers, symbols or other characters can be inserted.
- Changeable plate systems: Sign systems consisting of fixed plate holders to which removable interchangeable sign plates can be attached or inserted.
- Digital signs: Sign systems consisting of electronic display panels that can be programmed to display images, videos or web pages.
- House signage: Internal and external project specific signs.
- Illuminated signs: Signs consisting of cabinets enclosing a light source, illuminating translucent face panels bearing the specified signage.
- Statutory signs: Signs prescribed by the NCC and statutory authorities.
- Variable room identification systems: Changeable plate systems incorporating fixed room numbers and removable name strips.

### 1.5 SUBMISSIONS

#### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

#### Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

#### Shop drawings

General: Submit shop drawings showing the following, if appropriate:

- Layout, construction and fixing method for custom designed (non-standard) signage.

- Large scale (full size if practicable) lettering layouts for individual letter signs.
- Computer generated graphic images.
- Full size spacing templates for individually mounted characters.
- Location templates for anchorages to permanent construction, including the type of anchorage.
- Wiring diagrams for illuminated signs.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Custom-built signage and graphic items fabricated and ready to be delivered to the Site.
- Signage and graphic items delivered to Site before installation.
- Building locations or substrates prepared to receive signage and graphic items before they are installed.
- Completion of installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples for each colour and finish of exposed signage materials and accessories, showing the extreme and mean of each colour and texture range.

Fabricated items: as Documented.

**2.2 MATERIALS****Standards**

Aluminium:

- Plate for engraving: Alloy and temper designation 6063-0 to AS 2848.1 (1998).
- Casting: To AS 1874 (2000).
- Finishes:
  - . Anodising: To AS 1231 (2000).
  - . Powder coating: To AS 3715 (2025) and AAMA 2604 (2022).

Stainless steel:

- External: Type 316. Mirror electropolish surface finish.
- Internal: Type 304. No. 4 brushed (general purpose polished) surface finish.

Plastics:

- PVC-U sheet: Semi-rigid sheet.
- Rigid cellular polystyrene: To AS 1366.3 (1992), class VH for cut-out shapes.

Brass and bronze: Plate, sheet and strip: To AS 1566 (1997).

- Finish: Patinated.

Glass type and thickness: To AS 1288 (2021).

Photoluminescent exit signs: To BCA (2022) E4D8(a)(ii).

**Adhesive**

General: Proprietary solvent based contact adhesive compatible with the substrate and signage material.

**2.3 SYSTEMS****Digital signage systems**

Requirement: Provide an integrated digital signage system comprising compatible display panels, media player/s, software and connecting network.

### 3 EXECUTION

#### 3.1 WORKMANSHIP

##### Production

General: Form signage and graphic items accurately with clean, well-defined edges or arises, free from blemishes.

Engraving to two-layer plastic laminate: Engrave lettering to expose the lower laminate.

Engraved and filled: Lettering precision cut and filled colouring material. Clean faces of all filling material.

Casting: Produce shapes free of pits, scale, blow holes or other Defects, hand or machine-finished if necessary.

Laser cut lettering: Individual vinyl letters with self-adhesive backing.

Printed lettering: Lettering and graphic images screen/digitally printed on:

- Film with self-adhesive backing.
- Acrylic sheet.
- Aluminium plate.
- Stainless steel plate.

Large format digital printing: Lettering and graphic images screen printed film with self-adhesive backing.

Signwriting: Lettering and graphic images hand painted direct to the background by a tradesman with recognised qualifications and demonstrated skills.

Fabricated: Three dimensional, formed as follows:

- Laser cutting from solid material and hand finished as necessary.
- Moulding: Individual plastic hollow three dimensional characters and shapes formed by:
  - . Injection moulding.
  - . Vacuum forming.
- Built-up individual shapes by fabricating the faces and edges from separate pieces neatly and securely joined.

#### 3.2 INSTALLATION

##### General

Requirement: Install signage and graphic items level and plumb, securely mounted, with concealed corrosion and theft-resistant fasteners.

##### Self-adhesive signs

Requirement: Fix free of bubbles and creases.

##### Aluminium and stainless steel signs

Pin fixing: Epoxy fix to substrate.

##### Illuminated signs

Electrical fittings: Provide a junction box for power connection, necessary lamps with proper mountings, protection and accessories including wiring transformers and insulators. Install signs and conceal cabling to *0921 Low voltage power systems*.

##### Digital signs

Display panel mountings: Provide wall mountings compatible with the Documented display panels.

Power and data: Provide the necessary power and data cabling to display panels and media players. Conceal cabling and terminate with plugs and outlets appropriate to the Documented hardware.

Wireless network: Provide a wireless network system compatible with the display system's hardware.

##### Testing

Requirement: Test illuminated and digital signage systems after installation to confirm that they operate as intended.

### 3.3 COMPLETION

#### **Cleaning**

General: Remove protective coverings, replace damaged signage and leave the Work clean, polished, free from Defects and in good condition.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturers' data as follows:

- Recommendations for service use, care and maintenance.
- List of manufacturers and suppliers of replacement parts.

#### **Warranties**

Type: Against failure of materials and installation under normal environmental and use conditions.

Period: as offered by the manufacturer and installer.

## 4 SELECTIONS

---

**Refer to Appendix A – T-Sheet.**

|                                   |
|-----------------------------------|
| <b>0612 CEMENTITIOUS TOPPINGS</b> |
|-----------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide cementitious toppings, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Appendix D – Waterproofing Specification.

**Performance**

Requirements:

- Consistent in level, finish, colour and texture.
- Free of discontinuities.
- Resistant to impacts and loads expected in use.
- Resistant to environmental degradation within the manufacturer's stated life span.
- Accommodating movement in the substrate between control joints.
- If unbonded, without edge curl.
- If bonded, compatible with the substrate and without drummy areas.
- Without obvious shrinkage cracks.

Self-levelling screeds: Provide if substrates for the various overlaid flooring types cannot meet the performance requirements.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.
- 0612 Cementitious toppings.
- 0651 Resilient finishes

**1.3 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Concrete class - normal: Concrete that is specified primarily by a standard compressive strength grade up to 50 MPa and otherwise in conformance with AS 1379 (2007) clause 1.5.3.
- Granolithic topping: A topping mix with the coarse aggregate restricted to between 2 and 3 mm.
- Laitance: Scum or whitish deposit that rises to the surface of newly placed, over-wet concrete or over-trowelled mortar.
- Self-levelling screed: Setting mixture used to make a surface smooth or horizontal to align with another surface.
- Substrate: The surface to which a material or product is applied.
- Topping: Mixture of binders, aggregate and water applied to substrates in a plastic state and dried and cured to a hard surface.
- Topping function:
  - . Levelling: Topping placed to receive applied floor finishes.
  - . Wearing: Topping placed to act as the finished floor.
- Topping method:
  - . Bonded or post-applied: Topping that is bonded to a hardened substrate from which laitance has been removed and to which a bonding agent has been applied.
  - . Monolithic or wet applied: Topping placed on a plastic substrate so that a chemical bond is created between the substrate and the topping.
  - . Separated: Topping that is separated from a concrete subfloor by a membrane.

- . Unbonded: Topping that is separated from a hardened substrate by a resilient layer.

## 1.4 TOLERANCES

### Unformed surfaces

Thickness: To the **Thickness tolerance table**.

Flatness: To the **Flatness tolerance class table**, using a straightedge placed anywhere on the surface in any direction, for the Documented class of finish.

#### Thickness tolerance table

| Thickness (mm) | Maximum deviation (mm) |
|----------------|------------------------|
| < 15           | 2                      |
| ≥ 15 and < 30  | 5                      |
| ≥ 30           | 10                     |

#### Flatness tolerance class table

| Class | Measurement         | Maximum deviation (mm) |
|-------|---------------------|------------------------|
| A     | 2 m straightedge    | 4                      |
| B     | 3 m straightedge    | 6                      |
| C     | 600 mm straightedge | 6                      |

## 1.5 SUBMISSIONS

### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

### Products and materials

Manufacturer's data: Submit manufacturer's product data for the following:

- Admixtures.
- Bonding products.
- Colouring products.
- Curing products.
- Sealant products.
- Slip resistance products.
- Surface treatment products.

### Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

### Tests

Site Tests: Submit results, as follows:

- Flatness.
- In situ crushing resistance/soundness.

### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 1.6 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrates ready for laying of toppings.

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide samples of the following:

- Colouring products.

- Control joint products.
- Surface treatment products.

### Storage and handling

General: Deliver, unload and store products and accessories in unbroken manufacturer's packaging and containers in a dry, well-ventilated and secure storage area, unaffected by weather.

## 2.2 MATERIALS

### Aggregates

Standard: To AS 2758.1 (2014).

Coarse aggregate: Nominal size not more than 0.33 of the topping thickness.

Fine aggregate: Fine, sharp, well-graded sand with a low clay content and free from efflorescing salts. Nominal size not more than 0.25 of the topping thickness for toppings less than 40 mm thick.

### Bonding products

General: Proprietary products manufactured for bonding cement-based toppings to concrete substrates.

### Separating layer

General: Provide a bond breaker to separate the topping from the substrate, as Documented.

### Cement

Standard: To AS 3972 (2010).

Type: SL.

### Colouring products

General: Provide proprietary products manufactured for colouring cement toppings.

Integral pigment proportion: 10% maximum by weight of cement.

### Concrete

Standard: To AS 1379 (2007).

Unreinforced topping: Normal-class.

### Reinforced topping table

| Above-ground exposure location                                                                                                                                                                                           | Compressive strength (MPa) | Cover to reinforcement (mm) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|-----------------------------|
| Internal residential<br>External more than 50 km from coastline and non-industrial and arid                                                                                                                              | 20                         | 20                          |
| Internal non-residential<br>External more than 50 km from coastline and non-industrial and temperate                                                                                                                     | 25                         | 30                          |
| External more than 50 km from coastline and non-industrial and tropical<br>External more than 50 km from coastline and industrial in any climatic zone<br>External between 1 to 50 km from coastline in any climate zone | 32                         | 40                          |
| External coastal less than 1 km from coastline                                                                                                                                                                           | 40                         | 45                          |

### Reinforcement

Standard: To AS/NZS 4671 (2019).

Mesh: SL 62.

Maximum joint spacing: 6 m internal, 4 m external.

**Curing products**

General: Provide proprietary products manufactured for use with cement-based toppings and with the floor finish to be laid on the toppings.

**Water**

General: Clean and free from any deleterious matter.

**Mixes**

General: Select mix proportions to the **Mix proportion table**, or alternatively provide pre-mixed concrete for toppings as follows:

- Air entrainment:  $\leq 3\%$ .
- Nominal coarse aggregate size:  $\leq 0.3 \times$  topping thickness.
- Slump: 80 mm.
- Standard strength grade: N25.

Water quantity: The minimum necessary to achieve full compaction and prevent excessive water being brought to the surface during compaction.

**Mix proportion table**

| Mix type                  | Thickness (mm)              | Upper and lower limits of proportions by weight |                |                  |
|---------------------------|-----------------------------|-------------------------------------------------|----------------|------------------|
|                           |                             | Cement                                          | Fine aggregate | Coarse aggregate |
| Bonded – cement and sand  | 35                          | 1<br>1                                          | 3<br>4.5       | 0<br>0           |
| Bonded – fine concrete    | 40                          | 1<br>1                                          | 3<br>3         | 1<br>2           |
| Unbonded – fine concrete  | 100                         | 1<br>1                                          | 3<br>3         | 1<br>2           |
| Granolithic               | Floors: 25<br>Skirtings: 13 | 1                                               | 2              | 1                |
| Separated – fine concrete | 70                          | 1<br>1                                          | 3<br>3         | 1<br>2           |

**Self-levelling screed**

Type: Proprietary self-levelling liquid cement-based screed.

**Slip resistance products**

General: Provide proprietary products manufactured to improve the wet slip resistance of toppings.

- Silicon carbide granules:
  - . Granule size: Between 300 to 600  $\mu\text{m}$ .
- Silicon carbide two-part resin:
  - . Granule size:  $\geq 300 \mu\text{m}$ .

**Surface treatment products**

General: Provide proprietary products manufactured for use with cement-based toppings to change the characteristics of the surface of the finished topping.

**2.3 CONTROL JOINTS****Control joint materials**

General: As Documented.

Control joint strip: A proprietary expansion joint consisting of a neoprene filler sandwiched between plates with lugs or ribs for mechanical keying. Set flush with the finished surface.

Proprietary slide plate divider strip: An arrangement of interlocking metal plates grouted into pockets formed in the concrete joint edges.

Sealant: One-part self-levelling non-hardening mould-resistant, silicone or polyurethane sealant applied over a backing rod. Finish flush with the surface.

- Floors: Trafficable, Shore hardness greater than 35A.

Backing rod: Compressible closed cell polyethylene foam with a bond breaking surface.

### 3 EXECUTION

#### 3.1 PREPARATION

##### Substrates

General: Prepare substrates as follows:

- Clean and remove any deposit that may impair adhesion of monolithic or bonded toppings.
- Remove excessive projections.
- Voids and hollows greater than 10 mm depth, with abrupt edges: Fill with a sand/cement mix as follows:
  - . Greater than or equal in strength to the topping.
  - . Less than or equal in strength to the substrate.
- Voids and hollows less than 10 mm depth: Scabble edges to eliminate a featheredge and apply a latex modified cementitious product.

##### Substrates for bonded toppings

Hardened concrete: Roughen by scabbling to remove 2 mm of the surface and expose the aggregate.

#### 3.2 APPLICATION

##### Installation

General: Spread the mix and compact. Strike off, consolidate and level surfaces to finished levels and to the Documented flatness tolerance class.

Bonding product: Before laying topping wash the substrate with water and provide a bonding product, or treat as follows:

- Keep wet for 2 hours or more.
- Remove surplus water and brush on neat cement or a clean slurry of cement and water.
- Place the topping while the slurry is wet.

Monolithic toppings: Lay while concrete subfloor is plastic and the surface water is no longer visible.

Unbonded toppings: Lay separating layer.

Toppings 50 to 75 mm thick:

- Lay in two layers of equal thickness.
- Place a layer of reinforcement between the topping layers. Lap reinforcement 200 mm and tie. Do not create four-way laps.

##### Temperature control

General: Make sure that the temperature of mixes, substrates and reinforcement are not less than 5°C or greater than 35°C at the time of application.

Severe temperature: If the ambient shade temperature is greater than 38°C, do not mix topping.

#### 3.3 CONTROL OF MOVEMENT

##### General

Requirement: Provide control joints as Documented and as follows:

- Location:
  - . Over structural control joints.
  - . To divide complex room plans into rectangles.
  - . Around the perimeter of the floor.
  - . At junctions between different substrates.
  - . To divide large topping-finished areas into bays.
  - . At abutments with the building structural frame and over supporting walls or beams if flexing of the substrate is anticipated.
- Depth of elastomeric sealant: Half the joint width, or 6 mm, whichever is the greater.

Control joints to divide topping into bays: Provide joints using one of the following methods:

- Form in situ using square edge steel forms and trowelling a 3 mm radius to edges.
- Form a groove, extending at least one-quarter the depth of the section, either by using a grooving tool, by sawing, or by inserting a premoulded strip.

- Install a control joint proprietary product, to manufacturer's recommendations.

### **3.4 SELF-LEVELLING SCREED**

#### **Installation**

Requirement: Pour on and spread in conformance with the manufacturer's recommendations. Finish without any bubbles and make sure screed is correctly installed into corners and re-entrant angles.

### **3.5 JOINT ACCESSORIES**

#### **Floor finish dividers**

General: Provide a corrosion-resistant metal dividing strip suitably fixed to the substrate, at junctions with differing floor finishes, with top edge flush to the finished floor. If changes of floor finish occur at doorways, make the junction directly below the centre of the closed door.

#### **Weather bars**

General: Provide corrosion-resistant weather bars or threshold plates under hinged external doors, located under the centres of closed doors or to manufacturer's recommendations.

### **3.6 TESTING**

#### **Flatness Tests**

Method: If flatness properties are required, Test to ASTM E1155/E1155M (2023).

#### **In situ crushing resistance/soundness Tests**

Method: To BS 8204-1 (2003).

#### **Slip resistance Tests**

Slip resistance of completed installation: To AS 4663 (2013).

### **3.7 COMPLETION**

#### **Curing**

General: Prevent premature or uneven drying out and protect from the sun and wind.

Keeping moist: If a proprietary curing agent is not used, as soon as toppings have set sufficiently, keep them moist by water spraying or covering with polyethylene film for 7 days.

#### **Joint sealant**

General: If required, seal joints as follows:

- Formed joints: ≤ 25 mm deep with filler and bond breaker.
- Sawn joints: Full depth of cut.

#### **Protection**

General: Protect finished Work from damage during construction.

#### **Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's published use, care and maintenance instructions.

#### **Warranties**

Type: Installer warranty against failure of materials or systems under normal environment or use conditions.

Period: As offered by installer.

## **4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                                       |
|---------------------------------------|
| <b>0621 WATERPROOFING - WET AREAS</b> |
|---------------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide wet area waterproofing systems, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Appendix D – Waterproofing Specification.

**Performance**

Requirements:

- Graded to floor wastes, to dispose of water without ponding.
- Able to prevent moisture entering the substrate or adjacent areas.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****Waterproofing wet areas**

Standard: To AS 3740 (2021).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS 3740 (2021) and the following apply:

- Membranes (waterproof): Impervious barriers to liquid water, which may be:
  - . Installed below floor finishes.
  - . Installed behind the wall sheeting or render.
  - . Installed to the face of the wall sheeting or render.
  - . Applied in liquid or gel form and air cured to form a seamless film.
  - . Applied in sheet form with joints lapped and sealed.
- Waterproofing system: Combinations of membranes, flashings, drainage and accessories that form waterproof barriers and that may be:
  - . Loose-laid.
  - . Bonded to substrates.
- Wet area: An area within a building supplied with a floor waste required by the NCC.

**1.5 SUBMISSIONS****Products and materials**

Manufacturer's data: Submit product data sheets.

Type Tests: Submit Test results for the following:

- Membranes: To **MEMBRANES, Tests**.

**Records**

General: Submit photographic records to EXECUTION, **GENERAL, Reporting**.

Flood Tests: Submit photographic records to **TESTING, Flood Tests**.

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers as recommended by the manufacturer.

Evidence of experience: QBCC Licence and CV.

**Substrate acceptance**

Requirement: Submit evidence of installer's acceptance of the flooring substrate before starting installation.

**Tests**

Site Tests: Submit results, as follows:

- Substrate moisture content Test.
- Flood Test.
- Electronic leak detection Test.
- Seam probe Test.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Substrates prepared and ready for installation of the wet area waterproofing systems.
- Secondary layers prepared and ready for subsequent layers.
- Membranes after installation and before concealment.
- Flood Tests, if applicable.
- After flood Testing, if applicable.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide 300 x 300 mm samples of each type of membrane.

**Storage and handling**

General: Store and handle to the manufacturer's recommendations and as follows:

- Protect materials from damage.

**2.2 MEMBRANES****Standards**

Standard: To AS/NZS 4858 (2004).

**Membrane system**

Requirement: Proprietary membrane system suitable for the intended internal wet area waterproofing.

Certification: provide installation certificates.

**Tests**

Standard: To AS/NZS 4858 (2004) Table 8.1.

**Total VOC limits**

Requirement: Conform to the following maximum total VOC content:

- Waterproof membrane: 250 g/L.

**2.3 ACCESSORIES****Waterstop angles**

Material: Rigid, corrosion-resistant angles compatible with the waterproof membrane system.

**Bond breakers**

Requirement: Compatible with the extensibility class of the membrane to be used.

Material: Purpose-made bond breaker tapes or fillets of sealant.

**Flashings**

Requirement: Flexible waterproof flashings compatible with the waterproof membrane system.

**Liquid membrane reinforcement**

Requirement: Flexible fabric compatible with the waterproof membrane system.

**Sealants**

Requirement: Waterproof or water resistant, flexible, mould-resistant and compatible with the waterproofing system and to the manufacturer's recommendations.

**Adhesives**

Requirement: Waterproof and compatible with the waterproofing system.

---

**3 EXECUTION**

---

**3.1 GENERAL**

General

Refer to Appendix D – Waterproofing Specification.

**Reporting**

General: Make progressive photographic records of the waterproofing installation. Label photographs with the date and location.

Timing: Record at the following stages:

- After substrate preparation.
- After primer application.
- After membrane installation.
- After protection from traffic provided.

Liquid applied membranes:

- Record wet film thickness once every 10 m<sup>2</sup> and compare to the manufacturer's requirements.
- On completion of every 100 m<sup>2</sup> of each coat, compare the amount of membrane used with the manufacturer's application rate and record the result.

Leak control flanges: Conform to BCA (2022) F2D2.

**3.2 COMPLETION****Reinstatement**

Extent: Repair or replace faulty or damaged Work. If the Work cannot be repaired satisfactorily, replace the whole area affected.

**Warranties**

Type: Interlocking supplier and installer warranty.

Period: Refer to Appendix A – T-Sheet.

---

**4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

Refer to Appendix D – Waterproofing Specification.

## 0632 STONE AND TERRAZZO TILING INCL CONCRETE PAVERS

## 1 GENERAL

## 1.1 RESPONSIBILITIES

**General**

Requirement: Provide tiling systems to floors and other substrates, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**Performance**

Requirements:

- Consistent in colour and finish.
- Firmly bonded to substrates for the expected life of the installation.
- Set out with joints as Documented, accurately aligned in both directions and wall tiling joints level and plumb.
- Direct all water flowing from supply points to drainage outlets without leakage to the substrate or adjacent areas.

## 1.2 CROSS REFERENCES

**General**

Requirement: Conform to the following:

- 0171 General requirements.

## 1.3 STANDARDS

**Tiling**

General: To AS 3958 (2023).

## 1.4 INTERPRETATION

**Definitions**

General: For the purposes of this worksection, the following definitions apply:

- Acoustic underlay: A resilient material laid between the subfloor and the flooring material to provide sound isolation.
- Adhesives - cementitious (C): Adhesive in which the binders are hydraulic, e.g. General purpose cement, with aggregates and organic additives.
- Adhesives - dispersion (D): Adhesives in which the binders are in the form of aqueous polymer dispersion with mineral fillers and organic additives.
- Adhesives - reaction resin (R): Adhesives in which the binders are synthetic resins with mineral fillers and organic additives. The curing occurs by chemical reaction.
- Artificial stone: An artificial product made by combining natural stone materials with chemical constituents, containing not more than 1% crystalline silica as a weight/weight concentration and conforming to the requirements of the applicable *Work Health and Safety* regulations.
- Bedding: Mixtures of materials that are applied to substrates in a plastic state and which dry, cure and adhere tiles to substrates:
  - . Adhesive bedding: Paving/tiling adhered by adhesives.
  - . Mortar bedding: Paving/tiling adhered in a cementitious mortar bed.
- Fixture: Fixed or permanently attached items in a building that cannot be removed without causing damage and would remain upon a change in occupancy, such as a bath, water closet suite, stove, built-in cupboards.
- Lippage: Height deviation between adjacent units.
- Stepping: The relative surface level of adjacent paving elements within the expanse of the main pavement.
- Substrate: The surface to which a material or product is applied.

- Tile: for the purpose of this specification “tile” is interchangeable with “paver” for the concrete paver specified in Appendix A – T-Sheet.
- Wet area: An area within a building supplied with a floor waste.

## 1.5 TOLERANCES

### Completed tiling

Requirement: To AS 3958 (2023) clause 5.4.8.

## 1.6 SUBMISSIONS

### Execution details

Grouting: Submit proposals for grouting methods and materials.

Margins: If a minor modification in the joint width or overall dimensions will avoid cut tiles, submit a proposal.

### Operation and maintenance manuals

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

### Products and materials

Type Tests: Submit Test results for the following:

- Tiles:
  - . Slip resistance: To **TILES, Tests**
  - . Accelerated wear: To **TILES, Tests**
- Acoustic underlay: Weighted normalised impact sound pressure level to **UNDERLAY, Acoustic underlay**.

### Samples

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

### Subcontractors

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC Licence and CV.

### Warranties

Requirement: Submit warranties to **COMPLETION, Warranties**.

## 1.7 INSPECTION

### Notice

Inspection: Give notice so that inspection may be made of the following:

- Substrate immediately before tiling.
- Trial set-outs before execution.
- Control joints before sealing and grouting.
- Grout and sealant colours before application.

---

## 2 PRODUCTS

### 2.1 GENERAL

#### Samples

Requirement: Provide labelled samples of tiles, including accessories, grout and sealants, showing the range of variation in colour and finish.

Sample panels: Prepare a sample panel of each type of finish as follows:

- Size:  $\geq 2 \text{ m}^2$ .
- Include samples of junction details and trim.
- Preserve each panel until related Work is complete.

#### Storage and handling

Tiles: Deliver, unload and store tiles in unbroken manufacturer's packaging. Inspect for damage upon delivery.

## 2.2 UNDERLAY

### **Compressed Fibre cement underlay**

Standard: To AS/NZS 2908.2 (2000), Type B, category 2 minimum.

Thickness: as Documented.

## 2.3 PAVERS

### **Concrete pavers**

General: As Documented.

#### **Tests**

Slip resistance: To AS 4586 (2013).

Accelerated wear: Tested for slip resistance after being subjected to accelerated wear conditioning, as evidence of the permanence of slip resistance.

## 2.4 ADHESIVES

### **General**

Standard: To AS ISO 13007.1 (2020).

Labelling: Marked to AS ISO 13007.1 (2020) Section 6.

### **Type**

General: Provide adhesives compatible with the materials and surfaces to be adhered, and as Documented.

Prohibited uses: Do not provide the following combinations:

- Cement-based adhesives on wood, metal, painted or glazed surfaces, gypsum-based plaster.
- Organic solvent-based adhesives on painted surfaces.
- Organic PVC-based adhesives and organic natural rubber latex adhesives in damp or wet locations.
- PVA (polyvinyl acetate) based adhesives in wet areas or externally.

## 2.5 MORTAR

### **Materials**

Cement type to AS 3972 (2010): GP.

- White cement: Iron salts content  $\leq 1\%$ .
- Off-white cement: Iron salts content  $\leq 2.5\%$ .

Lime: To AS 1672.1 (1997).

Sand: Fine aggregate with a low clay content selected for grading, sharp and free from efflorescing salts.

Measurement of volume: Measure binders and sand by volume using buckets or boxes. Do not allow sand to bulk by absorption of water.

### **Bedding mortar**

Mix proportion (cement:sand), by volume: Select proportions from the range 1:3 to 1:4 for satisfactory adhesion. Provide minimum water.

Mixing: To AS 3958 (2023) clause 5.5.

### **Water**

General: Clean and free from any deleterious matter.

## 2.6 GROUT

### **General**

Standard: To AS ISO 13007.3 (2013).

### **Type**

Epoxy grout: As Documented.

### **Pigments**

Pigments for coloured grout: Provide colourfast fillers compatible with the grout material. For cement-based grouts, provide lime-proof natural or synthetic metallic oxides compatible with cement.

## 2.7 CONTROL JOINTS

### Control joint materials

Control joint strip: A proprietary control joint consisting of a neoprene core sandwiched between metal plates with lugs or ribs for mechanical keying. Set flush with the finished surface.

Proprietary slide plate divider strip: An arrangement of interlocking metal plates grouted into pockets formed in the concrete joint edges.

Sealant: One-part self-levelling non-hardening mould resistant, silicone or polyurethane sealant applied over a backing rod. Finish flush with the finished surface.

- Floors: Trafficable, Shore hardness greater than 35A.

Backing rod: Compressible closed cell polyethylene foam with a bond breaking surface.

## 2.8 ACCESSORIES

### Accessories

Requirement: Provide tile accessories that match the composition, colour and finish of the surrounding tiles, as Documented.

Tile trim: Provide proprietary trim for wall tiles and floor tiles, as Documented.

### Tactile ground surface indicators

Standard: To AS/NZS 1428.4.1 (2009).

## 3 EXECUTION

---

### 3.1 SUBSTRATES

#### Drying and shrinkage

General: Before tiling, allow at least the following times to elapse (for initial drying out and shrinkage) for these substrates:

- Concrete slabs: 42 days.
- Concrete blockwork: 28 days.
- Toppings on slabs and rendering on brick or blockwork: A further 21 days.

### 3.2 PREPARATION

#### Ambient temperature

General: If the ambient temperature is less than 5°C or greater than 35°C, do not lay tiles.

#### Substrates without wet area membranes

General: Make sure substrates are as follows:

- Clean and free of any deposit or finish that may impair adhesion or location of tiles.
- If framed or discontinuous, support members are in full lengths without splicing.
- If solid or continuous:
  - . Remove excessive projections.
  - . Fill voids and hollows greater than 10 mm with abrupt edges with a cement:sand mix not stronger than the substrate or weaker than the bedding.
  - . Fill depressions less than 10 mm with a latex modified cementitious product and eliminate feathering by scabbling the edges.

Absorbent substrates: If suction is excessive, control it by dampening but avoid over-wetting and do not apply mortar bedding to substrates showing surface moisture.

Dense concrete: If not sufficiently rough to provide a mechanical key, roughen by scabbling to remove 3 mm of the surface and expose the aggregate; then apply a bonding treatment.

#### Substrates with wet area membranes

General: Make sure substrates are as follows:

- Clean and free of any deposit or finish that may impair adhesion or location of tiles.
- Compatible with all components of the floor system.

#### Trial set-out

General: Unless Documented: prepare a trial tile set-out of each area, as follows:

- Maximise the size of equal margins of cut tiles.
- Locate control joints.
- Locate fixtures on walls.
- Align floor and wall tile joints if possible.

### 3.3 FIXING UNDERLAY

#### Installation

Requirement: Lay and fix in staggered (brick) pattern, perpendicular to the direction of the subfloor, with joints in the underlay not coinciding with joints in the subfloor. If panels are not tongue and grooved, make sure edges are fully supported.

Fasteners: Type and spacing to the manufacturer's recommendations.

Membranes: If sheet flooring is the substrate for a wet area membrane, fix with stainless steel countersunk screws.

### 3.4 TILING GENERALLY

#### Cutting and laying

Cutting: Cut tiles neatly to fit around fixtures and at margins where necessary. Drill holes without damaging tile faces. Cut recesses for fixtures such as soap holders. Rub edges smooth without chipping.

Laying: Return tiles into sills, reveals and openings. Butt up to returns, frames, fixtures and other finishes. Strike and point up beds if exposed. Remove tile spacers before grouting.

#### Variations

General: Distribute variations in hue, colour, or pattern uniformly, by mixing tiles or tile batches before laying.

#### Protection

Floor tiles: Keep traffic off floor tiles until the bedding has set and attained its working strength.

Cleaning: Keep the Work clean as it proceeds and protect finished Work from damage.

### 3.5 SETTING OUT

#### Tile layout

Requirement: Set out tiles as Documented, allowing for control joints, or as follows if desired layout is undocumented:

#### Tile joints

Joint widths: Set out tiles to give uniform joint widths as follows:

- Floors:
  - . Stone pavers 400 x 400 mm or less : 3 mm.

#### Fixtures

General: If possible, position tiles so that holes for fixtures and other penetrations occur at the intersection of horizontal and vertical joints or on the centrelines of tiles. Continue tiling fully behind fixtures that are not built in to the tiling surface. Before tiling make sure fixtures interrupting the tile surfaces are accurately positioned in their designed or optimum locations relative to the tile layout.

### 3.6 FALLS AND LEVELS

#### Grading

Requirement: Grade floor tiling to even and correct falls to floor wastes and elsewhere as required. Make level junctions with walls. If falls are not required, lay level.

Fall: Conform to falls as Documented and the following:

- Falls to floor wastes: 1:80 minimum.
- Continuous fall of floor plane to floor waste for NCC Classes 1, 2, 3 and 4 parts of a building: 1:50 maximum.

Change of finish: Maintain finished floor level across changes of floor finish including carpet.

### 3.7 BEDDING

#### Preparation of tiles

Adhesive bedding: Fix tiles dry; do not soak.

Mortar bedding: Soak porous tiles in water for half an hour and then drain until the surface water has disappeared.

**Bedding**

General: Use bedding methods and materials that are appropriate to the tile, the substrate and the conditions of service, and which leave the tile firmly and solidly bedded in the bedding material and adhered to the substrate. Form falls integral with the substrate.

**Thin adhesive beds**

General: Provide only if the substrate deviation is less than 3 mm, Tested with a 3 m straightedge. Cover the entire tile back with adhesive when the tile is bedded.

Thickness: 1.5 to 3 mm.

**Thick adhesive beds**

General: Provide on substrates with deviations up to 6 mm, Tested with a 3 m straightedge, and with tiles having deep keys or frogs.

Nominal thickness: 6 mm.

**Adhesive bedding application**

General: Apply adhesive by notched trowel to walls and floors and direct to tiles if required, to provide evenly distributed coverage after laying as follows:

- Other walls and floors: > 90%.
- Wet areas and benchtops: 100%.

Pattern of distribution of adhesive: To AS 3958 (2023) clause 5.6.5. Verify by examining one tile in ten as WUC proceeds.

Curing: Allow the adhesive to cure for the period nominated by the manufacturer before grouting or allowing foot traffic.

**Mortar beds**

Floor tiles: Either lightly dust the screeded bed surface with dry cement and trowel level until the cement is damp, or spread a thin slurry of neat cement, or cement-based thin bed adhesive, on to the tile back. Do not use mortar after initial set has occurred.

- Nominal thickness: 20 to 40 mm.

Thick reinforced beds: Place mortar bed in two layers and incorporate the mesh reinforcement in the first layer.

**Mechanical fixing**

General: Provide a proprietary support and fixing system appropriate to the type of tile and the substrate conditions.

**3.8 CONTROL OF MOVEMENT****General**

Requirement: Provide control joints carried through the tile and the bedding to AS 3958 (2023) clause 5.4.7 and as follows:

- Floor location:
  - . Over structural control joints.
  - . To divide complex room plans into rectangles.
  - . Around the perimeter of the floor.
  - . At junctions between different substrates.
  - . To divide large tiled areas into bays.
  - . At abutments with the building structural frame and over supporting walls or beams if flexing of the substrate is anticipated.
- Wall location:
  - . Over structural control joints.
  - . At junctions with different substrate materials when the tiling is continuous.
  - . At vertical internal corners.
- Depth of joint: Right through to the substrate.
- Sealant width: 6 to 25 mm.
- Depth of elastomeric sealant: Half the joint width, or 6 mm, whichever is the greater.

### 3.9 GROUTED AND SEALANT JOINTS

#### Grouted joints

General: Commence grouting as soon as practicable after bedding has set. Clean out joints as necessary and remove any tile spacers before grouting.

Face grouting: Fill the joints solid and tool flush. Clean off surplus grout. Wash down when the grout has set. When grout is dry, polish the tiled surface with grout film remover and a clean cloth.

Edges of tiles: Grout exposed edge joints.

Epoxy grouted joints: Make sure tile edge surfaces are free of extraneous matter such as cement films or wax, before grouting.

Grouting cementitious terrazzo tiles: Thoroughly work grout into tile joints by flood grouting.

#### Sealant joints

General: Provide joints filled with sealant and finished flush with the tile surface as follows:

- Where tiling is cut around sanitary fixtures.
- At internal corners of walls.
- Around fixtures interrupting the tile surface, for example pipes, brackets, bolts and nibs.
- At junctions with elements such as window and door frames and built-in cupboards.

Material: Anti-fungal modified silicone.

Width: 5 mm.

Depth: Equal to the tile thickness.

### 3.10 JOINT ACCESSORIES

#### Floor finish dividers

General: Finish tiled floors at junctions with differing floor finishes with a corrosion-resistant metal dividing strip fixed to the substrate using mechanical fasteners, with top edge flush with the finished floor. If changes of floor finish occur at doorways, make the junction directly below the closed door.

Grout up underneath to provide continuous support.

#### Adjustments

Requirement: Check that the height of the floor finish divider is sufficient for the topping and tile thickness. Adjust as required with a matching flat bar adhesive fixed to the divider angle.

#### Weather bars

General: Provide corrosion-resistant metal weather bars or threshold plates under hinged external doors, located under the centres of closed doors or to manufacturer's recommendations.

### 3.11 COMPLETION

#### Concrete paver tiled surfaces

General: sealer to be applied as Documented.

#### Cleaning

General: Clean tiled surfaces using an appropriate tile cleaning agent, and polish.

#### Spare tiles

General: Supply spare matching tiles and accessories of each type for future replacement purposes. Store the spare materials on Site.

Quantity: At least 1% of the quantity installed.

#### Operation and maintenance manuals

Requirement: Prepare a manual describing care and maintenance of the tiling, including procedures for maintaining the slip-resistance classification stating the expected life of the slip-resistance classification.

#### Warranties

Type: against manufacturing Defects for materials, and interlocking warranty against installation Defects including adhesive materials and grouts.

Period: as offered by paver manufacturers and installers.

## 4 SELECTIONS

---

Refer to Appendix A – T-Sheet

|                                   |
|-----------------------------------|
| <b>0641 APPLIED WALL FINISHES</b> |
|-----------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide applied wall finishes to substrates, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 *General requirements.*
- 0185 *Timber products, finishes and treatment.*

**1.3 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the following definitions apply:

- High pressure decorative laminates (HPDL):
  - Panels consisting of core layers impregnated with phenolic and/or aminoplastic resins and a surface layer(s) impregnated with aminoplastic resins (mainly melamine resins).
  - Sheets consisting of a decorative face and layers of fibrous sheet material (e.g. Paper) impregnated with thermosetting resins and bonded together under heat and pressure of at least 5 MPa.

**1.4 TOLERANCES****Wall panels**

Deviation from true grid lines and planes: 1:1000 to maximum 3 mm.

Misalignment with adjoining surfaces at grid junctions: 1 mm.

Panel thickness deviation: +1 mm, -0 mm.

Length and width of panels:  $\pm 0.1\%$  of the dimension or 0.5 mm, whichever is the greater.

Flatness, twist, winding and bow: 1.5 mm from a 1500 mm straightedge placed in any position.

Deviation of edges:  $\pm 1$  mm.

**1.5 SUBMISSIONS****Fire performance**

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties.**

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Products and materials**

Requirement: For each type of finish, submit the following:

- Product data sheets.

Pest resistant materials: Submit evidence that textile components have been treated for resistance to fabric pests.

Type Tests: Submit Test results to **PRODUCTS, GENERAL, Tests** for the following:

- Weighted sound absorption coefficient.
- Weighted sound reduction index.

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples.**

**Shop drawings**

General: Submit shop drawings to a scale that best describes the detail, showing the following:

- Procedures for shop and Site assembly and fixing.
- Proposals for the break-up of large items into modules for delivery to the Site.
- Proposed module jointing method of large items.

**Substrate acceptance**

Requirement: Submit evidence of installer's acceptance of the walling substrate before starting installation.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Pre-installation meeting**

Requirement: Before installing wall panels, arrange for an inspection by the panel installer and by the substrate/support framing installer to examine substrate/frame tolerances and other conditions affecting panel installation.

**Notice**

Inspection: Give notice so that inspection may be made of the following:

- Items fabricated off Site before delivery.
- Completed substrate/framing preparation.
- Completed lighting installation for backlit wall panels.
- Wall panel set-out before fixing panels.
- Completed wall panel installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide samples of each Documented type of the following finishes:

- Proprietary board: Complete with finish and edge strips.
- Timber including veneers: Showing the maximum expected variation.

Numbers of samples: Provide the following:

- Generally: Minimum 3 samples.
- Timber: 9 samples, showing 3 variants of timber type.

**Storage and handling**

General: Deliver, unload and store panel units and accessories in unbroken wrapping or containers. Handle and install panels without distorting panels or exposing panels to surface damage from the weather or construction activities. Avoid contaminating panel faces with adhesives, solvents or cleaners during installation.

Storage: Keep storage period to a minimum by delivering items only when required for installation.

Store panels to the manufacturer's recommendations and as follows:

- Store materials flat, in dry, well-ventilated space.
- Do not stand or drop panels on ends or corners.
- Protect edges from damage.

Exposure to direct sunlight: If recommended by the manufacturer, shield panels from direct sunlight until installation.

**Tests**

Weighted sound absorption coefficient: To AS ISO 11654 (2002), as Tested to AS ISO 354 (2006).

Weighted sound reduction index: To the NCC cited AS/NZS ISO 717.1 (2004).

**2.2 FIRE PERFORMANCE****Fire hazard properties**

Group number: To AS 5637.1 (2015).

## 2.3 TIMBER

### Joinery timber

Hardwood: To AS 2796.2 (2006).

Hardwood grade: Select.

## 2.4 LAMINATE

### High pressure decorative laminate (HPDL) sheets

Standard: To AS/NZS 2924.1 (2024).

Minimum thickness: Conform to the following:

- Horizontal surfaces fixed to a continuous substrate: 1.2 mm.
- Vertical surfaces fixed to a continuous substrate: 0.8 mm.
- Post formed laminate fixed to a continuous substrate: 0.8 mm.
- Vertical surfaces fixed intermittently (e.g. to studs): 3.0 mm.

## 2.5 GLASS

### Standards

Safety glass: To AS 2208 (2023).

### Colourback glass lining

Requirement: Minimum 6 mm toughened glass with a factory applied opaque coating to the back.

## 2.6 FABRIC PANELS

### General

Facing material: Fabric from the same dye lot and of the same colour and pattern.

Insect attack: Provide textiles, either inherently resistant to insect attack or treated against insect attack by application of insect resist agents (IRA), Tested to AS 2001.6.1 (1980).

## 3 EXECUTION

---

### 3.1 PREPARATION

#### Substrate condition

Requirement: Make sure support framing, insulation or substrate lining conforms to the wall finish manufacturer's recommended requirements including moisture content, smoothness and other surface imperfections. Correct out of tolerance Work and other deficient conditions before installing panels.

Substrate surface: Smooth, clean, dry, free of dust, Defects or imperfections that will affect the installation. Fill dents or cracks as recommended by the manufacturer. Tape or seal joints in the substrate to prevent the ingress of dust and foreign particles travelling from the wall cavities to the applied wall finish.

Cleaning: Wipe down substrate surface with a damp cloth before installing applied wall finish.

Cleaning substrate surfaces: Damp clean and vacuum substrate surfaces that will be permanently concealed.

#### Acclimatisation

Requirement: Acclimatise panels by stacking them in the ambient temperature and humidity in which they will be installed, with air circulation to all surfaces after the following have been installed and are operational:

- Air conditioning.
- Lighting.
- Site drainage and stormwater works.
- Space fully enclosed and secure.
- Wet work: Completed and dry.

Storage: Remove packaging and store panels in the installation space not less than 48 hours before installing.

### 3.2 INSTALLATION GENERALLY

#### Wall panels

Accuracy: Construct and install panels and components plumb, level and square, as Documented.

Joints: Provide materials in single lengths if possible. If joints are required, install over framing supports.

Framing: Provide additional framing members at openings and penetrations, including those required by other trades.

#### Accessories and trim

General: Provide accessories and trim as Documented and as required to complete the installation.

#### Fasteners

Exposed fastenings: Provide as Documented.

Concealing exposed fastenings: If fastenings are unavoidable on the panel face, countersink the fastener heads and fill the sinking flush with a material compatible with the final finish. For timber surfaces requiring a clear or tinted finish, provide matching wood plugs showing face grain. Do not use plugs showing end grain.

#### Adhesives

Requirement: Provide adhesives to transmit the loads imposed and achieve assembly rigidity, without discolouring the finished panel face.

#### Finishing

Corners: Scribe internal and mitre external corners.

Edge strips for panel edges: As Documented. If required, finish exposed edges of panels with edge strips matching the panel face.

Matching adjoining panels: Arrange and match panels as follows:

- Generally: If panel matching is required, install panels so that the finish, texture, pattern and grain matches that of the adjoining panel, as appropriate.
- Timber panels with clear/tinted finish: Install panels to match the grain and colour of the adjoining panel.

### 3.3 LAMINATE

#### Installation

Preparation: Before fixing the laminate, make sure the adhering surface is sufficiently smooth to prevent telegraphing the transfer of Defects through to the decorative laminate surface and that it is compatible with the laminate sheet.

Fixing laminate sheet to backing/lining: Adhesive fix to the laminate manufacturer's recommendations.

### 3.4 GLASS

#### Installation

Fixing to backing panels: Fix glass to backing panels in conformance with the following:

- Adhesive: Use non-acidic silicone adhesive. Apply at a rate recommended by the sealant and glass manufacturer.
- Installation: Clean the back of the glass panel and apply walnuts of adhesive, together with double-sided adhesive tape for temporary support and fix directly to the backing panel.

Panel edge treatment: as Documented.

### 3.5 COMPLETION

#### General

Requirement: Clean finished surfaces to the panel manufacturer's recommendations. Remove adhesives, sealants and other stains. Replace damaged panels and accessories that cannot be repaired.

#### Resin panels

Light scratches and scuffs: Repair with a plastic polish only if recommended by the panel manufacturer.

Cleaning: Rinse the sheets with lukewarm water and soft cloth or sponge to remove dust and dirt. Do not expose edges of organic or fabric interlayers to water. If required, use a mild soap and/or liquid detergent in the water. Blot dry with a soft, damp cloth.

**Operation and maintenance manuals**

Requirement: Prepare a manual that includes the manufacturer's recommendations for the care and maintenance of each type of finish. Include a program for regular cleaning and instructions for removing, repairing and replacing of the finish or panel.

**Warranties**

Type: interlocking manufacturer and installer warranty.

Period: as offered by the manufacturer and installer.

---

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet**

|                                |
|--------------------------------|
| <b>0651 RESILIENT FINISHES</b> |
|--------------------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide resilient floor finishes to substrates, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 STANDARDS****General**

Installation: To AS 1884 (2021).

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS 1884 (2021) and the following apply:

- Substrate: The surface to which a material or product is applied.

**1.5 SUBMISSIONS****Fire performance**

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties.**

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals.**

**Products and materials**

Manufacturer's data: Submit the manufacturer's product data sheets for each type of finish, and the manufacturer's recommendations for its application including the following, as appropriate:

- Thickness and width of sheet, or size of tile or plank.
- Adhesive and jointing method.
- Resistance to wear, indentation, chemicals, light and fire.
- Flexibility and bending strength.

Type Tests: Submit Test results for the following:

- Slip resistance: To PRODUCTS, **GENERAL, Tests.**

**Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples.**

**Subcontractors**

General: Submit names and contact details of proposed suppliers and installers.

Evidence of experience: QBCC licence and CV.

**Substrate acceptance**

Requirement: Submit evidence of installer's acceptance of the flooring substrate before starting installation.

**Tests**

Site Tests: Submit results, as follows:

- Substrate moisture content Test.
- Surface pH Test.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Substrate immediately before fixing resilient finishes or underlay.
- Trial set-out before execution.
- Completed underlay, if any.
- Finished surface before applying sealers or polishes, if any.
- Completed installation.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide labelled samples of resilient finishes illustrating the range of colour, pattern or texture of the product.

Minimum size per sample:

- Sheets: 450 x 450 mm.
- Tiles: A whole tile or 0.09 m<sup>2</sup>, whichever is the greater.
- Planks: A whole plank.
- Linear accessories, including coving, skirting, stair nosing and protection strips: 300 mm long.
- Welded joints: 300 mm long.

Identification: Label each sample with brand, product name and manufacturer's code reference, including the code for each coat of multi-coat Work.

**Storage and handling**

Requirement: Store and handle to the manufacturer's recommendations.

**Tests**

Slip resistance: To AS 4586 (2013).

**2.2 FIRE PERFORMANCE****Fire hazard properties**

Critical radiant flux: Tested to AS ISO 9239.1 (2003).

**2.3 UNDERLAYS****General**

Preparation: Prepare underlay to flooring manufacturer's recommendations.

**Cementitious**

General: Polymer modified cementitious smoothing and self-levelling compound.

Thickness: 3 mm minimum.

**Compressed Fibre cement hard underlay**

Standard: To AS/NZS 2908.2 (2000). Type B, category 2 minimum.

Sheet thickness: as Documented.

**2.4 ADHESIVES****General**

Requirement: To the resilient finishes manufacturer's recommendations.

**2.5 SHEETS, TILES AND PLANKS****Corklinoleum**

Standard: To EN 688 (2011).

Static control flooring

General: as Documented.

## 2.6 OTHER MATERIALS

### Edge strips and threshold strips

General: Heavy duty metal moulding or extruded edge strip appropriate to the floor covering type, capable of accommodating different levels of adjacent floor finishes, if necessary.

Type: As Documented.

Location: At exposed edges of the floor covering, and at junctions with differing floor finishes or finishes of a different thickness. If edge strips occur at doorways, locate the junctions directly below the closed door.

## 3 EXECUTION

### 3.1 GENERAL

#### Subcontractors

Requirement: Use specialist installers recommended by the material manufacturers.

### 3.2 PREPARATION

#### Substrates

General: To AS 1884 (2021) Section 3.

#### Substrate tolerance table

| Property                   | Length of straightedge laid in any direction (mm) | Maximum deviation under the straightedge (mm) |
|----------------------------|---------------------------------------------------|-----------------------------------------------|
| Planeness                  | 2000                                              | 4                                             |
| Abrupt deviation tolerance | 150                                               | 0.5                                           |

#### Timber, plywood, particleboard and fibre cement sheet substrates

Requirement: Do not start installation of the resilient finishes until the timber, plywood, particleboard and fibre cement substrate conforms to AS 1884 (2021) clause 3.6.

Substrate rectification: Remove projections. If conformance to the **Substrate tolerance table** cannot be achieved, provide an underlay in a staggered pattern with joints avoiding substrate joints.

Cleaning: Remove oil, grease, traces of applied finishes and loose materials or dust.

#### Working environment

General: Do not start work before the building is enclosed, wet work is complete and dry, overhead work is complete and good lighting is available. Protect adjoining surfaces.

#### Conditioning

Conditioning of floor covering and subfloor: To AS 1884 (2021) clause 4.1 and manufacturer's recommendations.

#### Trial set-out

General: Prepare a trial sheet set-out to each area.

### 3.3 TESTING

#### Moisture content Tests

General: Test substrate for suitability for the installation of resilient floor coverings to AS 1884 (2021) Appendix A.

- Maximum relative humidity of concrete: To AS 1884 (2021) Appendix A3.2.
- Moisture content of timber, plywood and particleboard subfloors: To AS 1884 (2021) Appendix A3.3.

#### Surface pH Tests

General: Test concrete subfloor for suitability for the installation of resilient floor coverings to AS 1884 (2021) Appendix C.

- Maximum pH: 10.

#### Electrical resistance Tests

Electro-static discharge, static dissipative and electro-static conductive floors: Test installation to the recommendations of AS 1884 (2021) Appendix G. Do not carry out point-to-ground electrical

resistance Tests earlier than 14 days after installation. Make first random control measurements 24 hours after installation.

#### **Slip resistance Tests**

Slip resistance of completed installation: To AS 4663 (2013).

### **3.4 INSTALLATION**

#### **General**

Requirement: To AS 1884 (2021) Section 5 and the manufacturer's recommendations.

#### **Sheet set-out**

General: Set out sheets to give the minimum number of joints. Position joints away from areas of high stress. Run sheet joints parallel with the long sides of floor areas, vertically on non-horizontal surfaces.

#### **Change of finish**

General: As Documented. Maintain finished floor level across changes of floor finish including carpet.

#### **Cleaning**

General: Keep the surface clean as the Work proceeds.

#### **Edges**

General: Make sure edges are firm, unchipped and machine-cut accurately to size and square to the face, and that edges are square to each other before installation.

#### **Expansion joints**

General: To the manufacturer's recommendations for joint widths, and area and length limitations.

#### **Joints**

Non-welded: Butt edges together to form tight neat joints showing no visible open seams.

Doorways: Unless Documented otherwise, if changes of floor finish occur at doorways, locate the joint on the centreline of the door leaf in the closed position.

#### **Junctions**

General: Scribe neatly up to returns, edges, fixtures and fittings. Finish flush with adjoining surfaces.

#### **Rolling**

General: If rolling is required, roll the finish in multiple directions before the adhesive sets.

### **3.5 SHEETING**

#### **Welded joints**

Thermal welding: After fixing, groove the seams using a grooving tool and weld the joints with matching filler rod, using a hot air welding gun. When the weld rod has cooled, trim off flush.

Chemical welding: Apply seaming compound 100 mm wide to the substrate centrally under the seam. Roll the seam until the compound is forced up into the joint. Clean off flush with a damp cloth.

Epoxy jointing: Join seams with epoxy adhesive.

### **3.6 JOINTS AND ACCESSORIES**

#### **Accessories**

General: Provide purpose-made matching moulded accessories for nosings, coves, skirtings, edge cover strips and finishes at junctions, margins and angles, if available. Otherwise, form accessories from the sheet material. Provide solid backing for radiused coves and nosings.

#### **Control joints**

Location: Provide control joints as follows:

- Over structural control joints.
- At junctions between different substrates.

Depth of joint: Right through to the substrate.

Sealant width: 6 to 25 mm.

Depth of elastomeric sealant: Half the joint width, or 6 mm, whichever is the greater.

#### **Control joint materials – sheet flooring**

Proprietary slide plate divider strip: Provide interlocking metal plates grouted into pockets formed in the concrete joint edges to finish flush with the flooring surface.

#### **Edge strips and threshold strips**

Installation: To AS 1428.1 (2021) and the manufacturer's recommendations.

**Coved skirtings**

Site formed coving: As Documented, carry the flooring material up over a profiled coving section to form the skirting and mitre and weld all joints. Make sure the radius of the coving section conforms to the floor finish manufacturer's recommendations for sheeting material and thickness. Taper coving to meet door trims as Documented.

**3.7 COMPLETION****Cleaning**

General: Clean the finished surface. Buff and polish. Before the Date for Practical Completion, mop and leave the finished surface clean and undamaged on completion.

**Cleaning static control flooring**

General: Do not use sealers, wax or floor polish. Clean using a mild neutral detergent and lukewarm water. Dry buff clean floor using a scrubbing machine with a white nylon pad.

**Operation and maintenance manuals**

General: Prepare a manual that includes manufacturer's recommendations for care and maintenance for each type of finish.

**Protection**

Finished floor surface: Keep traffic off floors for a minimum of 24 hours after laying or until bonding has set, whichever period is longer. Avoid contact with water for minimum 7 days after laying.

**Reinstatement**

Extent: Repair or replace faulty or damaged Work. If the Work cannot be repaired satisfactorily, replace the whole area affected.

**Spare materials**

General: Supply spare matching resilient finishes and accessories of each type for future replacement purposes. Store the spare materials on Site where directed.

Quantity: At least 1% of the quantity installed.

**Warranties**

Type: Interlocking manufacturer and installer warranty.

Period: As offered by the manufacturer and supplier.

---

**4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                     |
|---------------------|
| <b>0652 CARPETS</b> |
|---------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide textile floor coverings to subfloors, as Documented. Refer to Architectural Drawings and Appendix A – T-Sheet.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- 0171 General requirements.

**1.3 INTERPRETATION****Abbreviations**

General: For the purposes of this worksection, the following abbreviations apply:

- ACCS: Australian Carpet Classification Scheme.
- CIAL: Carpet Institute of Australia Limited.
- ECS: Environmental Certification Scheme.

**Definitions**

General: For the purposes of this worksection, the definitions given in AS 2454 (2007), AS 2455.1 (2019) and AS 2455.2 (2019) apply.

**1.4 TOLERANCES****General**

Requirement: To AS/NZS 1385 (2007).

**1.5 SUBMISSIONS****Fire performance**

Fire hazard properties: Submit evidence of conformity to **FIRE PERFORMANCE, Fire hazard properties**.

**Operation and maintenance manuals**

Requirement: Submit manual to **COMPLETION, Operation and maintenance manuals**.

**Products and materials**

Manufacturer's Documentation: Submit copies of the following data:

- Product data sheets.

Type Tests: Submit Test results of the following:

- Carpet properties: To **PRODUCTS, TESTS**.
- Slip resistance: To **PRODUCTS, TESTS, Slip resistance**.

**Samples**

Requirement: Submit samples to **PRODUCTS, GENERAL, Samples**.

**Shop drawings**

General: Submit a floor covering plan to a scale that best describes the detail, including underlay, set-out point, location of joints and conformance to AS 2455.1 (2019) clauses 2.2.2 and 2.2.3.

**Subcontractors**

General: Submit name and contact details of proposed suppliers and installers.

Evidence of experience: QBCC licence and CV.

**Substrate acceptance**

Requirement: Submit evidence of installer's acceptance of the substrate before starting installation.

**Tests**

Site Tests: Submit results, as follows:

- Substrate moisture content Test.
- Substrate alkalinity Test.

**Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties.**

**1.6 INSPECTION****Notice**

Inspection: Give notice so that inspection may be made of the following:

- Each batch of material upon delivery.
- Substrate immediately before fixing underlay.
- Trial set-out before execution.
- Fastenings, edge strips and underlay installed ready to lay carpet.
- Completed carpet after cleaning and before covering for protection.

---

**2 PRODUCTS****2.1 GENERAL****Samples**

Requirement: Provide labelled production run samples demonstrating the range of colour, pattern, texture and pile yarn available in each Documented carpet type.

Sample size: Provide the following:

- Carpet: Manufacturer's standard swatch.
- Carpet tiles: 4 x tile size.
- Edge strip, trim, extrusions, and stair and landing nosings: 300 mm length of each type.

Penetrations: Provide one production carpet sample with a penetration access cut as Documented in EXECUTION, **INSTALLATION - CARPET.**

**Storage and handling**

Requirement: Store on a flat, clean, dry, well-ventilated and secure storage area, elevated above the subfloor and unaffected by weather.

**2.2 FIRE PERFORMANCE****Fire hazard properties**

Critical radiant flux: Tested to AS ISO 9239.1 (2003).

**2.3 CARPET****Batching**

Requirement: Carpet from one manufacturing batch and dye lot.

**Insect resistance**

Requirement: Carpets and underlays comprising materials either inherently resistant to insect attack or treated against insect attack by moth and carpet beetle, by application of insect resist agents (IRA) to the wool and wool blend yarns during wet processing at the manufacturing stage.

Insect resist treatment of wool: Application Level 4 to the recommendations of Woolmark Specification CP-4 (2016).

**Electrostatic propensity**

Maximum electrostatic propensity value: 2500 V at a relative humidity of 25% Tested to AATCC TM 134 (2019).

**Stain and soil resistance**

Requirement: Colourless acid-based dye stain blocker applied to dyed fibres.

**Total VOC**

Total VOC emission Tested to ISO 10580 (2010): < 0.5 mg/m<sup>2</sup>/h.

## 2.4 MODULAR CARPET TILES

### General

Type: Non-stick, non-curling carpet tiles capable of being taken up without damage and then re-laid in different positions.

Marking: On the back, showing manufacturer's recommendations or directional arrow for laying.

### Tolerances

Requirement: Conform to the following:

- Dimensional tolerance: 0.2%.
- Squareness: Maximum difference of 2 mm between lengths of diagonals.

### Sustainable carpet tile backing

Re-usable backing: Proprietary vinyl backing to carpet tiles capable of separation and recycling in new carpet tiles.

## 2.5 RECESSED MATS

### General

Requirement: Provide mats made to fit each designated mat recess, as Documented.

### Linked metal mats

Type: Proprietary hinged extruded aluminium slatted mat with capability of accepting different types of inserts.

Frames: If the mat is recessed, provide frames to match the mat material and finish.

Drainage: Provide tundish to drained mat frames to take moisture to outside the building line of enclosure.

## 2.6 OTHER MATERIALS

### Adhesives

General: Compatible with the floor covering material, and suitable for bonding it to the subfloor to AS 2455.1 (2019) clause 1.5.3.

Friction compound: Suitable for holding carpet tiles in position without permanent sticking.

### Hot-melt adhesive tapes

General: Commercial grade glass fibre and cotton thermoplastic adhesive-coated tape 60 mm wide on a 90 mm wide metal foil base and backed with silicone-coated release paper.

### Preformed carpet grippers

General: Architectural plywood carpet grippers with 3 rows of corrosion-resistant angled pins of length appropriate to the carpet type to AS 2455.1 (2019) clause 1.5.4.

Size (minimum): 33 mm wide x 7 mm thick.

Location: At edges, except where edge strips are used. Provide double grippers to edges where recommended by the manufacturer.

### Edge strips and threshold strips

General: Heavy duty metal moulding or extruded edge strip appropriate to the floor covering type and capable of accommodating varying levels of adjacent floor finishes, if necessary.

Type: As Documented.

## 2.7 TESTS

### Slip resistance

Standard: To AS 4586 (2013).

---

## 3 EXECUTION

### 3.1 PREPARATION

#### General

Pre-installation requirements: To AS 2455.1 (2019) Section 2.

- Modular carpet tiles: Pre-laying requirements including access panel floors to AS 2455.2 (2019) Section 4.

Working environment: Do not start work before the building is enclosed, wet work is complete and dry, overhead work is complete and good lighting is available.

Protection: Protect adjoining surfaces.

### Substrate

General: Conform to the following:

- To AS 2455.1 (2019) or AS 2455.2 (2019), as appropriate.
- Clean and free of any deposit or finish that may impair adhesion or location and functioning of control joints.
- Free of any imperfections, including ridges, indentations and projections that may adversely affect the installed carpet.

Concrete substrate rectification: Remove projections, grind as necessary and fill voids and hollows with a levelling compound compatible with the adhesive to achieve the required tolerance.

Timber substrate rectification: Remove projections. If conformance with the **Substrate tolerance table** cannot be achieved, fix a hard underlay in a staggered pattern. Make sure joints do not coincide with substrate joints.

Moisture content and alkalinity of concrete substrate: Do not start installation until the moisture content and alkalinity of the concrete substrate has been Tested to **TESTING, Substrate Tests** and conforms to the values in AS 2455.1 (2019) Appendix B.

Moisture content of timber, plywood or particleboard substrate: Do not start installation until the moisture content of the substrate has been Tested to **TESTING, Substrate Tests** and conforms to the values as follow:

- Air conditioned buildings: 8 to 10%.
- Intermittently heated buildings: 10 to 12.5%.
- Unheated buildings: 12 to 15%.

Fixtures: Remove door stops and other fixtures. Refix in position undamaged on completion of the installation. Make sure fasteners penetrate substrate and are stable.

### Substrate tolerance table

| Property   | Length of straightedge laid in any direction (mm) | Maximum deviation under the straightedge (mm) |
|------------|---------------------------------------------------|-----------------------------------------------|
| Planeness  | 2000                                              | 4                                             |
| Smoothness | 150                                               | 1                                             |

### Conditioning

General: Stabilise the room temperature for 7 days before and 2 days after laying carpet as follows:

- Areas with air conditioning installed: Run air conditioning at operational temperature.
- Air conditioned areas not operational: Maintain a room temperature range between 10°C and 35°C.
- Underfloor heating: Turn off heating and allow substrate to stabilise at the temperature recommended by the carpet manufacturer.
- Non-air conditioned areas: Install at room temperature between 10°C and 35°C.

Carpet: Cut to length and lay in position, 24 hours before installation.

Carpet tiles: Unpack tiles at least 48 hours before installation.

Hardboard underlay: Expose both faces of each sheet for more than 24 hours before fixing.

Soft underlay: Unroll the carpet and allow to come to the temperature of the in-service environment before laying.

### Trial set-out

General: Prepare a trial set-out of each type of carpet with underlay, minimum 10 m<sup>2</sup>, including accessories and 3000 mm of typical seam.

Location: to be agreed with the Superintendent on Site.

## 3.2 TESTING

### Substrate Tests

Moisture content and alkalinity of concrete substrate: Test substrate to AS 2455.1 (2019) Appendix B.

Moisture content of timber, plywood or particleboard substrate: Test substrate to AS/NZS 1080.1 (2012) for timber and particleboard or AS/NZS 2098.1 (2006) for plywood.

### 3.3 INSTALLATION - CARPET

#### General

Requirement: To AS 2455.1 (2019) Section 3 and the manufacturer's recommendations.

#### Batching

Requirement: In a single area and for each Documented type, quality, or colour, use carpet from one manufacturing batch and dye lot.

#### Setting out

Joints in underlay: Make sure joints in underlay do not coincide with carpet joints. Do not carry underlay over carpet grippers or edge strips.

Partition layout: Confirm that permanent partitions have been installed before starting carpet laying.

#### Seaming methods

Woven carpet: Machine or hand sew. Do not provide glued taped seams unless selvages are woven to suit and recommended by manufacturer.

Tufted carpet: Seam with hot-melt adhesive tape.

Seam sealing: Apply appropriate seam sealer to each cut edge.

#### Cutting laid carpet

Method: If penetrations through laid carpet are necessary for electrical, telephone or other outlets, cut the carpet either by cross cutting or by cutting rectangular or circular openings.

Cutting holes in concrete floors: Protect the carpet and remove concrete particles and dust on completion. Replace the cut carpet over the opening without any signs of fraying or other damage, and fix with a peel-up adhesive, or resew.

#### Edge strips and threshold strips

Installation: To AS 1428.1 (2021) and the manufacturer's recommendations.

Location: At exposed edges of the carpet, and at junctions with differing floor finishes or finishes of a different thickness. If edge strips occur at doorways, locate the junctions directly below the closed door.

### 3.4 INSTALLATION - CARPET TILES

#### General

Installation: To AS 2455.2 (2019) and the manufacturer's recommendations.

### 3.5 COMPLETION

#### Spares

Spare material: Supply spare matching materials of each type, colour and design of carpet from the same batch for future replacement purposes. Store the spare materials on Site where directed.

Offcuts: Retain carpet offcuts exceeding 0.5 m<sup>2</sup> in area and 450 mm in both length and width.

Labelling: Label spare and offcut material appropriately, including the location of the laid area corresponding to each batch. Securely and separately package each batch in a suitable wrapping.

Quantity of spare material: At least 1% of the quantity installed, in full or part length rolls.

#### Cleaning

Requirement: Progressively clean the Work. Remove waste, excess materials and adhesive.

Final cleaning: When the installation is complete, clean the carpet as necessary to remove extraneous matter, marks and soiling and to lift the pile where appropriate.

#### Protection

Requirement: Provide fabric drop sheets. Do not use plastic sheeting. If wheeled traffic is to follow carpet installation, protect with hardboard sheets butted and fixed with adhesive tape.

#### Operation and maintenance manuals

Requirement: Prepare a manual that includes the following:

- A technical specification of the carpet installation.
- The manufacturer's recommendations for use, care and maintenance of the carpet to AS/NZS 3733 (2018).

- The names and address of the supplier and manufacturer of each component.

**Warranties**

Type: interlocking manufacturer and installer warranty.

Period: as offered by the manufacturer and supplier.

---

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet**

|                      |
|----------------------|
| <b>0671 PAINTING</b> |
|----------------------|

**1 GENERAL****1.1 RESPONSIBILITIES****General**

Requirement: Provide coating systems to substrates, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Appendix E – Paint Specification.

**Performance**

Requirement:

- Consistent in colour, gloss level, texture and dry film thickness.
- Free of runs, sags, blisters or other discontinuities.
- Opaque paint finishes that are fully opaque or at the Documented level of opacity.
- Clear finishes at the level of transparency consistent with the product.
- Fully adhered.
- Resistant to environmental degradation within the manufacturer's stated life span.

**1.2 CROSS REFERENCES****General**

Requirement: Conform to the following:

- *0171 General requirements.*

**1.3 STANDARDS****Painting**

General: To the recommendations of those parts of AS/NZS 2311 (2017) referenced in this worksection.

**1.4 INTERPRETATION****Definitions**

General: For the purposes of this worksection, the definitions given in AS/NZS 2310 (2002) and the following apply:

- Gloss unit: Numerical value for the amount of specular reflection relative to that of a standard surface under the same geometric conditions.
- Levels of gloss finish: When the specular direction is 60 degrees, surfaces with the following specular gloss reading are defined as follows:
  - . Full gloss: > 85 gloss units.
  - . Gloss: > 50 and ≤ 85 gloss units.
  - . Semi-gloss (satin): > 20 and ≤ 50 gloss units.
  - . Low gloss (low sheen): > 5 and ≤ 20 gloss units.
  - . Flat finish (matt): ≤ 5 gloss units.
- Paint: A product in liquid form that, when applied to a surface, forms a dry film having protective, decorative or other specific technical properties.
- Primer, prime coat: The first coat of a coating system that helps bind subsequent coats to the substrate and which may inhibit its deterioration.
- Substrate: The surface to which a material or product is applied.
- Undercoat: An intermediate coat formulated to prepare a primed surface or other prepared surface for the finishing coat.

**1.5 SUBMISSIONS****Products and materials**

General: Submit the following at least 3 weeks before the paint is required:

- Paint brand name and product range quality statement.

- The published recommendations for maintenance.

### **Samples**

Requirement: Submit samples to PRODUCTS, **GENERAL, Samples**.

### **Subcontractors**

Specialist applicators: Submit names and contact details of proposed specialist applicators.

Evidence of experience: QBCC licence and CV.

### **Substrate acceptance**

Requirement: Submit evidence of the applicator's acceptance of the substrate before starting application.

### **Warranties**

Requirement: Submit warranties to **COMPLETION, Warranties**.

## **1.6 INSPECTION**

### **Notice**

Inspection: Give notice so that inspection may be made of the following:

- Opaque paint finishes:
  - . After surface preparation.
  - . After application of final coat.
- Clear finishes:
  - . Before surface preparation of timber.
  - . After surface preparation.
  - . After application of final coat.

## **2 PRODUCTS**

### **2.1 GENERAL**

#### **Samples**

Requirement: Provide samples, as follows:

- Clear finishes: Labelled samples of each clear finish on representative timber or timber veneer substrates, including putty, stain, seal and coat, showing surface preparation, gloss level and other physical properties.
- Opaque paint finishes: Labelled samples of each paint system on representative substrates, showing surface preparation, colour, gloss level, texture and other physical properties.

#### **Storage and handling**

General: Store materials not in use in tightly covered containers in well-ventilated areas with temperatures maintained at the manufacturer's recommendations.

Delivery: Deliver paints to the Site in the manufacturer's labelled and unopened containers.

### **2.2 PAINTING MATERIALS**

#### **Standards**

Paint types: To AS/NZS 2311 (2017) Table 4.2 and the following:

- Metal primer - general purpose for iron and steel: To AS/NZS 3750.19 (2008).
- Metal primer - latex for metallic zinc surfaces: To AS 3730.15 (2006).
- Metal primer - solvent-borne for ferrous metallic surfaces: To AS 3730.21 (2006).
- Metal primer - zinc-rich organic for iron and steel: To AS/NZS 3750.9 (2009).

#### **Combinations**

General: Do not combine products from different manufacturers in a paint system.

Clear timber finish systems: Provide only the combinations of putty, stain and sealer recommended by the manufacturer of the topcoat.

#### **Putty and fillers**

Material: To the recommendations of the paint system manufacturer, suitable for the substrate and compatible with the primer.

**Tinting**

General: Provide only products that are colour tinted by the manufacturer or supplier.

**Toxic ingredients**

General: To the *Therapeutic Goods (Poisons Standard) Instrument (2025)* Part 2 Division 9.

---

**3 EXECUTION**

---

**3.1 PREPARATION****General**

Refer to Appendix E – Paint Specification.

**Order of Work**

Other trades: Complete the work of other trades as far as practicable within the area to be painted, except for the installation of fittings, floor sanding and laying flooring materials.

Clear finishes: Complete clear timber finishes before applying opaque paint finishes in the same area.

**3.2 PAINT SYSTEMS****General**

Refer to Appendix E – Paint Specification.

**3.3 APPLICATION****General**

Refer to Appendix E – Paint Specification.

**Doors**

Drying: Maintain door leaf in the open position during drying. Do not allow door hardware or accessories to damage the door finish during the drying process.

**Wet paint warning**

Notices: Place notices in a conspicuous location and do not remove until the paint is dry.

**3.4 COMPLETION****General**

Protection and masking: Remove masking and protection coverings before paint has dried.

Cleaning: Remove splatters by washing, scraping or other methods that do not scratch or damage the surface.

Reinstatement: Repair, replace or refinish any damage, including damage made by other trades.

Touch up new damaged paintwork or misses with the paint batch used in the original application.

Fittings: Refix removed and undamaged fittings in the original locations. Make sure they are properly fitted and in proper working order.

**Disposal of paint and waste materials**

Requirement: Conform to requirements of the local authority.

**Spares**

Spare material: Supply clearly labelled sealed containers of each type, coat and colour of paint/coating from the same batch.

**Warranties**

Type: Interlocking manufacturer and supplier warranty.

Period: As offered by the manufacturer and supplier.

---

**4 SELECTIONS**

---

**Refer to Appendix A – T-Sheet.**

**Refer to Appendix E – Paint Specification.**

|                             |
|-----------------------------|
| <b>0673 POWDER COATINGS</b> |
|-----------------------------|

## 1 GENERAL

### 1.1 RESPONSIBILITIES

#### General

Requirement: Provide powder coating systems to substrates, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Appendix E – Paint Specification.

### 1.2 CROSS REFERENCES

#### General

Requirement: Conform to the following:

- 0171 General requirements.

### 1.3 STANDARDS

#### General

Application to aluminium and aluminium alloy substrates for architectural applications: To AAMA 2603 (2022), AAMA 2604 (2022) and AAMA 2605 (2022), as appropriate, and AS 3715 (2025).  
Application to metal substrates other than aluminium for architectural applications: To AS 4506 (2024).

### 1.4 INTERPRETATION

#### Definitions

General: For the purposes of this worksection, the following definitions apply:

- Powder coating: The process of preparing, applying, fusing and curing a thermoset powder coating material to a substrate.
- Thermoset powder coating: A mixture of finely ground particles of pigment and resin sprayed onto a prepared substrate. Charged powder particles adhere to electrically grounded surfaces until heated and fused into a smooth coating in a curing oven.
- Substrate: The surface to which a material or product is applied.

### 1.5 SUBMISSIONS

#### Products and materials

Coating manufacturer: Submit the following details at least 3 weeks before fabrication:

- Recommended coating system for the nominated service condition.
- Brand name.
- Storage and handling recommendations.
- Product data sheets.
- Maintenance recommendations.

#### Samples

Requirement: Submit samples to PRODUCTS, GENERAL, Samples.

#### Subcontractors

Specialist applicators: Submit name and contact details of proposed specialist applicators as registered by the coating manufacturer.

Evidence of experience: QBCC licence and CV.

#### Warranties

Requirement: Submit warranties to COMPLETION, Warranties.

## 2 PRODUCTS

### 2.1 GENERAL

#### Storage and handling

General: To AS 3715 (2025) Appendix D for powder coated aluminium and AS 4506 (2024) Appendix D for all other powder coated materials.

## 2.2 MATERIALS

### Performance requirements

Powder coating to aluminium and aluminium alloy: To AS 3715 (2025) Section 8.

Powder coating to metals, other than aluminium: To AS 4506 (2024) Section 2.

## 3 EXECUTION ---

### 3.1 PREPARATION

#### General

Refer to Appendix E – Paint Specification.

#### Substrate pre-treatment

Powder coating to aluminium: To AS 3715 (2025) Appendix G.

Powder coating to metals, other than aluminium: To AS 4506 (2024) Appendix I.

### 3.2 APPLICATION

#### General

Refer to Appendix E – Paint Specification.

Requirement: To manufacturer's recommendations.

### 3.3 COMPLETION

#### Cleaning

Aluminium architectural applications: Clean completed assembly to AS 3715 (2025) Appendix C.

Metal, other than aluminium, architectural applications: Clean completed assembly to AS 4506 (2024) Appendix E.

#### Warranties

Type: Interlocking manufacturer and supplier warranty.

Period: As offered by the manufacturer and supplier.

## 4 SELECTIONS ---

Refer to Appendix A – T-Sheet.

Refer to Appendix E – Paint Specification.

|                                 |
|---------------------------------|
| <b>0702 MECHANICAL SERVICES</b> |
|---------------------------------|

**1 MECHANICAL SYSTEMS****1.1 RESPONSIBILITIES****General**

Requirement: Provide mechanical services, as Documented. Refer to Mechanical Engineer's drawings for preliminary design pending coordination with designed and approved trusses.

**1.2 DESIGN****General**

Requirement: To DESIGN in *0171 General requirements*.

**Designer**

Requirement: To **DESIGNER** in *0171 General requirements*.

**Scope of Design**

The design is to allow for the coordination of the system as designed and Documented by the Mechanical Engineer to the extent that is required for coordination with the final truss design. The design should not diminish the performance requirements as established by the designed system, including but not limited to the acoustic performance specifications; refer to the Acoustic Engineer's performance specification.

**Seismic restraint**

Requirement: To **RESTRAINT OF NON-STRUCTURAL PARTS AND COMPONENTS, Seismic restraint** in *0171 General requirements*.

**1.3 CROSS REFERENCES****General**

Requirement: Conform to the following:

- Engineering Specification for Mechanical Services prepared by JHA, latest version.
- *0171 General requirements*.
- *0201 Demolition*.

**1.4 SUBMISSIONS****General**

Requirement: Conform to Engineering Specification for Mechanical Services prepared by JHA, and *0171 General requirements*.

Submissions for Contractor design: Before starting WUC, submit the following:

- Outdoor design conditions, selected geographic location for these conditions and source of data.
- Calculated total and sensible cooling capacities and heating capacity.
- Name of calculation method used.
- Makes and model numbers of proposed equipment.
- Total and sensible cooling capacities and heating capacity of the proposed equipment, adjusted for the Documented outdoor and indoor conditions and any effects of the proposed plant configuration.
- Any assumptions on which the calculations are based.
- Details of any departures from this specification.
- Method of heating and % of total heating load in the case of electric heating.
- Mechanical subcontractor's provision for attendance to breakdowns.
- Availability of service network of plant manufacturer.
- Licence numbers and type of licences held by persons responsible for the installation.

**Authority approvals**

Requirement: Submit evidence of approval from authorities relating to the Works.

**Baseline data**

Requirement: Submit baseline data to **BASELINE DATA** in *0171 General requirements*.

**Certification**

Requirement: To **BUILDING APPROVAL AND CERTIFICATE OF OCCUPATION** in *0171 General requirements*.

**Drawings**

Requirement: Conform to Stanwell Engineering Requirements documents. Submit drawings at minimum 1:100 scale, if not stated otherwise, showing the following:

- Areas to be air conditioned.
- Areas to be mechanically ventilated including car parks, toilets and kitchens.
- Details demonstrating compliance with mechanical ventilation requirements of NCC.
- Details demonstrating compliance with energy requirements of NCC.
- Make, model and principal performance parameters of each item heating and cooling equipment.
- Make, model and principal performance parameters of ventilation fans.
- Make, model and principal performance parameters of air filtration equipment and details of compliance with statutory requirements.
- Air conditioning zones and sensor locations and methods of heating.
- Functional descriptions of automatic control systems.
- Duct, pipe, equipment layouts and plant locations. Show proposed zoning and methods of heating.
- Details of equipment supports, including seismic restraints to *0171 General requirements*.
- Mechanical services electrical diagrams and details.
- Provisions for maintenance access, balancing and commissioning.
- If necessary to convey the requirements of the design, provide additional schematic diagrams of air, control and refrigeration systems.
- Details of fire provisions demonstrating compliance with NCC.
- Other details and information to meet statutory requirements.

**Subcontractors**

General: Submit names, contact details, licence numbers and type of licence of proposed suppliers and installers.

**1.5 SUPPORT OF PLANT AND EQUIPMENT****General**

Requirement: To **SUPPORT OF PLANT AND EQUIPMENT** in *0171 General requirements*.

|                                |
|--------------------------------|
| <b>0802 HYDRAULIC SERVICES</b> |
|--------------------------------|

## **1 HYDRAULIC SYSTEMS**

### **1.1 RESPONSIBILITIES**

#### **General**

Requirement: Provide hydraulic services, as Documented. Refer to Hydraulic Engineer's Documents. Other than verification of the Box Gutter Performance Solution and design for seismic restraint, variations from the Documented design are to be approved by the Superintendent.

### **1.2 DESIGN**

#### **General**

Requirement: To DESIGN in *0171 General requirements*.

Design criteria: Not less than the PCA (2022).

#### **Designer**

Requirement: To **DESIGNER** in *0171 General requirements*.

#### **Scope of Design**

The design is to allow for the verification of the Valley Gutter Performance Solution. The design should not diminish the performance requirements as established by the designed system; refer to the Hydraulic Engineer's performance specification. Any other design work provided by the Contractor that departs from the Documents shall be treated as a request for substitution.

#### **Seismic restraint**

Requirement: To **RESTRAINT OF NON-STRUCTURAL PARTS AND COMPONENTS, Seismic restraint** in *0171 General requirements*.

### **1.3 CROSS REFERENCES**

#### **General**

Requirement: Conform to the following:

- Hydraulic Services Notes: Drawing no. H0002: latest revision.
- Stanwell Engineering Requirements documents.
- *0171 General requirements*.
- *0201 Demolition*.
- *0223 Service trenching*.

### **1.4 STANDARDS**

#### **General**

Plumbing and drainage: To the AS/NZS 3500 series and the PCA (2022).

Microbial control: To AS/NZS 3666.1 (2011), AS/NZS 3666.2 (2011) and the recommendations of SA/SNZ HB 32 (1995).

### **1.5 SUBMISSIONS**

#### **General**

Requirement: Conform to *0171 General requirements*.

#### **Authority approvals**

Requirement: Submit evidence of approval from authorities relating to the Works. Refer also to Hydraulic Services Notes.

#### **Certification**

Requirement: To **BUILDING APPROVAL AND CERTIFICATE OF OCCUPATION** in *0171 General requirements*.

#### **Operation and maintenance manuals**

Requirement: Conform to **OPERATION AND MAINTENANCE MANUALS** in *0171 General requirements*.

Lead content: Submit evidence that copper alloy plumbing products in contact with drinking water have a weighted average lead content not more than 0.25%.

### **Drawings**

Requirement: For Contractor Design conform with Stanwell Engineering Requirements documents. Submit drawings showing the following:

- PIPework and equipment layout and sections showing the Work to be installed on the level that the services are installed.
- Long sections of below ground drainage.
- Riser layouts and sections.
- Downpipes.
- Rainwater harvesting.
- Stormwater connection.
- Stormwater drainage stacks.
- Stormwater overflow path.
- Evidence of compliance with special planning conditions.
- If not obvious from other drawings, provide single line diagrams for rainwater and stormwater infrastructure mains plans and building servicing.
- Piping and other schematic drawings including numbering of each valve to correspond to valve tags notation.
- Access openings, cover plates, valve boxes and access pits.
- Facilities for conforming to AS/NZS 3666.2 (2011).
- Details of control panels including control and power diagrams.
- Insulation of piping, fittings and tanks.
- Penetrations and associated building work. If penetrations are through external walls, detail flashing and weatherproofing at 1:10 scale.
- Fire-stopping locations and details.
- Location, capacity, type and other relevant details of water heaters, including supports and safe trays.
- Location, type, grade and finish of piping, fittings, valves, meters and pipe supports.
- On-Site detention pondage areas.
- Provision of blue metal backfill to seepage drain system.
- Provision of erosion control measures.
- Provision of road barriers and lighting.
- Provision of temporary sanitary accommodation for construction workers.
- Provision of trafficable cover plates in the public domain.
- Relevant survey levels.
- Site and floor set-out points.
- Tank stands and supporting structures.

### **Subcontractors**

General: Submit names, contact details, licence numbers and type of licence of proposed suppliers and installers.

## **1.6 INSPECTION**

### **Notice**

Inspection: Give notice so that inspection may be made of the following:

- Excavated surfaces.
- Installed services before being concealed.

## **1.7 COMMISSIONING**

### **General**

Requirement: Provide commissioning as Documented. Conform to *0171 General requirements* and SA TS 5342 (2021).

## **2 STORMWATER – BUILDINGS**

---

### **2.1 DESIGN**

#### **General**

Requirement: Design is limited to the verification for the Box Gutter Performance Solution.

### **2.2 COMPLETION**

#### **Cleaning**

General: Clean and flush the whole installation.

|                                |
|--------------------------------|
| <b>0811S SANITARY FIXTURES</b> |
|--------------------------------|

---

**1 GENERAL**

---

**1.1 RESPONSIBILITIES****General**

Requirement: Provide sanitary fixtures, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Hydraulic Engineer's Documents.

**1.2 STANDARDS****General**

Design for access and mobility: To AS 1428.1 (2021) and AS 1428.2 (1992).

---

**2 PRODUCTS**

---

**2.1 GENERAL****Authorised products**

Requirement: Listed in the WaterMark Product Database, unless otherwise required by the Network Utility Operator.

**Labelling**

Water efficiency labelling: Provide products conforming to and labelled to the Water Efficiency Labelling Scheme (WELS).

---

**3 EXECUTION**

---

**3.1 SANITARY FIXTURES****General**

Requirement: Install to manufacturer's recommendations.

---

**4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                      |
|----------------------|
| <b>0812S TAPWARE</b> |
|----------------------|

---

**1 GENERAL**

---

**1.1 RESPONSIBILITIES****General**

Requirement: Provide tapware, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Hydraulic Engineer's Documents.

**1.2 STANDARDS****General**

Design for access and mobility: To AS 1428.1 (2021) and AS 1428.2 (1992).

---

**2 PRODUCTS**

---

**2.1 GENERAL****Authorised products**

Requirement: Listed in the WaterMark Product Database, unless otherwise required by the Network Utility Operator.

**Labelling**

Water efficiency labelling: Provide products conforming to and labelled to the Water Efficiency Labelling Scheme (WELS).

---

**3 EXECUTION**

---

**3.1 TAPWARE****General**

Requirement: Install to manufacturer's recommendations.

---

**4 SELECTIONS**

---

Refer to Appendix A – T-Sheet.

|                                 |
|---------------------------------|
| <b>0902 ELECTRICAL SERVICES</b> |
|---------------------------------|

**1 ELECTRICAL SYSTEMS****1.1 RESPONSIBILITIES****General**

Requirement: Provide electrical services, as Documented. Refer to Architectural Drawings, Appendix A – T-Sheet, and Electrical Engineer's Documents. Coordinate installation of parallel contract for Audio Visual services provided by the Principal.

**1.2 DESIGN****General**

Requirement: To DESIGN in *0171 General requirements*.

**Designer**

Requirement: To **DESIGNER** in *0171 General requirements*.

**Design for durability and maintainability**

Design for durability: Develop the design so the systems achieve the Documented performance, reliability, service life, energy efficiency and safety requirements, and are easily maintainable.

Access for maintenance: Develop the design so the systems conform to **ACCESS FOR MAINTENANCE** in *0171 General requirements*.

**Seismic restraint**

Requirement: To **RESTRAINT OF NON-STRUCTURAL PARTS AND COMPONENTS, Seismic restraint** in *0171 General requirements*.

**1.3 CROSS REFERENCES****General**

Requirement: Conform to the following:

- Engineering Specification for Electrical Services prepared by JHA: latest revision.
- Stanwell Engineering Requirements documents.
- *0171 General requirements*.
- *0201 Demolition*.
- *0223 Service trenching*.

**1.4 SUBMISSIONS****General**

Requirement: Conform to *0171 General requirements*.

**Authority approvals**

Requirement: Submit evidence of approval from authorities relating to the Works.

**Baseline data**

Requirement: Submit baseline data to **BASELINE DATA** in *0171 General requirements*.

**Certification**

Requirement: To **BUILDING APPROVAL AND CERTIFICATE OF OCCUPATION** in *0171 General requirements*.

**Operation and maintenance manuals**

Requirement: Conform to **OPERATION AND MAINTENANCE MANUALS** in *0171 General requirements*.

**Subcontractors**

General: Submit names, contact details, licence numbers and type of licence of proposed suppliers and installers.

## **1.5 COMMISSIONING**

### **General**

Requirement: Pre-commission, Test and commission to AS/NZS 4509.1 (2009), AS/NZS 5033 (2021) and the manufacturer's recommendations. Obtain Test reports from manufacturers or suppliers verifying the performance of safety and control functions of each system.

## **2 LIGHTING**

---

### **2.1 GENERAL**

#### **Samples**

Requirement: Provide samples of luminaires and accessories.

### **2.2 COMMISSIONING**

#### **General**

Requirement: Before the Date for Practical Completion, carry out the following:

- Programme and tune controls and preset controls with the Principal or Principal's Representative.
- Verify the operation of all luminaires.
- Adjust aiming and controls for all luminaires under night time conditions.

Digital control system: Commission to the manufacturer's recommendations and to the Documented control requirements.